

**Patient-Specific Virtual Surgical Planning for Tongue Reconstruction:
Biomechanical Optimization, Surgical Simulation, and Guided Execution**

by

AMIR REZA ISAZADEH

A thesis submitted in partial fulfillment of the requirements for the degree of

DOCTOR OF PHILOSOPHY

in

Rehabilitation Science

Faculty of Rehabilitation Medicine

University of Alberta

© 2025 Amir Reza Isazadeh

This work is licensed under CC Attribution-ShareAlike 4.0 International

Abstract

Background In tongue reconstruction following tumor removal, the absence of a unified digital workflow for surgical planning and guided execution perpetuates inconsistent anatomical restoration and functional outcomes. Standard practice for surgical planning relies on template-based geometric heuristics and remains an intuition-driven, freehand, surgeon-dependent paradigm that lacks data-driven surgical design optimization and precision execution. **Objective** This thesis establishes a virtual surgical planning (VSP) framework for tongue reconstruction by addressing three integrated pillars: biomechanical optimization for tailored free flap design, systematic outcome simulation, and guided surgical execution. **Method** An inverse algorithm (hyperelastic inverse finite element) generates biomechanically optimized flap designs that require the least strain energy and tissue stretch while precisely closing the defect and restoring anatomy. A robust patient-specific reconstruction simulator was developed and used in a 120-virtual-case factorial analysis to isolate the effects of donor-site stiffness and overbulking strategy on post-atrophy morphology and imposed strains on the neotongue. A patient-specific surgical toolkit with integrated guides for resection, flap harvesting, and independent volume control was demonstrated through benchtop experiments on soft-tissue phantoms. **Results** The biomechanically optimized flap designs required considerably less tissue deformation than geometric flattening methods, suggesting reduced tension-related complications. The factorial simulation quantified a physics-based trade-off: overbulking enhanced anatomical restoration yet increased imposed native-tongue stretches. The automated framework demonstrated robustness with convergence across all 120 simulations. Finally, benchtop guided execution reproduced pre-operative anatomy with maximum surface deviations of 3.37–4.13 mm, comparable to reported guided osseous workflow fidelity. **Significance** Coupling free flap optimization, scalable outcome simulation, and millimeter-scale guided translation, the framework lays the groundwork for risk-aware surgical planning, prospective validation studies, and eventual digital twin augmentation. Altogether, it enables a standardized digital ecosystem for soft-tissue reconstruction VSP that could benefit surgeons and patients.

Keywords: Tongue Reconstruction, Virtual Surgical Planning, Biomechanical Modeling, Patient-Specific Optimization, Precision Medicine

Preface

This thesis is an original work by Amir Reza Isazadeh (ORCID: [0009-0004-8462-890X](https://orcid.org/0009-0004-8462-890X)). The research utilized MRI data acquired under Health Research Ethics Board of Alberta approval, Protocol No. HREBA.CC-17-0597.

Chapter 3 has been published as: Isazadeh AR, Seikaly H, Westover L, Aalto D (2024). Algorithmically designed flaps in tongue reconstruction: a feasibility analysis. *International Journal of Computer Assisted Radiology and Surgery* 19:735–746. DOI: 10.1007/s11548-024-03062-w. Reproduced with permission from Springer Nature.

Chapter 4 has been published as: Isazadeh AR, Zenke J, Westover L, Seikaly H and Aalto D (2025). Patient-specific virtual surgical planning for tongue reconstruction: evaluating hyperelastic inverse FEM with four simulated tongue cancer cases. *Physics in Medicine & Biology* 70:145017. DOI: 10.1088/1361-6560/adebd8. Licensed under Creative Commons Attribution 4.0.

Chapter 5 has been submitted as: Isazadeh AR, Westover L, Seikaly H, Aalto D. Computational analysis of tongue reconstruction surgery: the impact of free flap site and volume on the tongue anatomy and biomechanics. Submitted to *PLOS ONE*.

Chapter 6 has been submitted as: Isazadeh AR, Chen S, Seikaly H, Zheng B, Westover L, Aalto D. Guided tongue reconstruction: feasibility analysis of additively manufactured surgical toolkit on a silicone tongue model. Submitted to *Surgical Oncology*.

For all chapters, I was responsible for the conceptualization and design of the studies, development and implementation of computational algorithms and experimental methodologies, execution of all simulations and experiments, data collection and analysis, interpretation of results, and preparation of manuscripts. Siming Chen (Chapter 6) assisted with the fabrication of silicone

tongue phantoms and provided technical support during experimental procedures. The co-authors contributed as follows: Dr. Daniel Aalto served as supervisory author, was involved in concept formation and research direction, provided guidance throughout the research process, and contributed to manuscript composition and revision; Dr. Hadi Seikaly provided clinical and surgical expertise, guidance on surgical relevance and clinical applications, and critical review of manuscripts; Dr. Lindsey Westover contributed expertise in biomechanical modeling and finite element methods, provided technical consultation on computational implementation, and offered critical manuscript review; Julianna Zenke (Chapter 4) contributed clinical interpretation of results from a surgical perspective and provided manuscript review; and Dr. Bin Zheng (Chapter 6) contributed expertise in surgical evaluation methodologies and manuscript review. I also gratefully acknowledge the Institute for Reconstructive Sciences in Medicine (iRSM) for providing workspace and equipment access for benchtop experiments, and clinical consultation, and the Alberta Cancer Foundation for funding this research (Grant No. 27 601). Large language models were used to enhance text fluency and grammatical accuracy.

Dedication

*To my beautiful wife,
whose patience rivals that of saints and whose love carried me through the storms.
They say it takes a village to raise a PhD—you were my entire village.*

*To my beloved family,
whose sacrifices paved the path that led me here.
This work stands on the foundation you built.*

Acknowledgments

This journey would not have been possible without the guidance, support, and generosity of many remarkable individuals.

First and foremost, I am profoundly grateful to my supervisor, Dr. Daniel Aalto. From the very beginning, you saw potential in me that I hadn't yet recognized in myself. Your unwavering belief in my abilities gave me the confidence to venture into the uncharted territory of soft tissue biomechanical optimization—a field where engineering meets medicine in the most demanding ways. Beyond your technical brilliance, you understood the human side of this journey. You supported me not just intellectually, but mentally and financially, always ensuring I had the resources I needed to pursue bold ideas. Perhaps most importantly, you were a connector—always thoughtful about introducing me to the right people at the right time, opening doors I didn't know existed. Your genuine care for my future, far beyond the confines of this thesis, has shaped not just my research but my trajectory as a scholar. Thank you for being the kind of mentor every graduate student hopes to find.

To my supervisory committee, Dr. Hadi Seikaly and Dr. Lindsey Westover—you were perfectly assembled to guide this multidisciplinary work. Dr. Seikaly, your clinical insight grounded this research in reality, constantly reminding me what truly matters for patients and surgeons. Dr. Westover, your biomechanical rigor pushed me to think deeper and build better. Your collective wisdom on research direction transformed this work from promising to impactful. Thank you for your mentorship and for challenging me to reach higher. I am also grateful to Heather Logan, a former collaborator in iRSM, whose guidance at the onset of this PhD helped me find the right path forward.

To my colleagues and friends in the lab—Gillian de Boer, Mitchell Holmes, Praveen Prem, and Siming Chen—you made the journey not just bearable but joyful. Your support, compassion, and friendship created a space where difficult work felt lighter and challenging days felt manageable. I am grateful for every shared laugh, every encouraging word, and every moment of camaraderie. I also extend my gratitude to Dr. Jacklin Cummine, Dr. Bill Hodgetts, and of course Dr Aalto for fostering an environment where productivity and warmth coexist. You helped create a lab culture that celebrates both excellence and humanity.

Finally, to my espresso machine, my overworked computer, and the algorithm that finally converged at 3 AM on a Saturday—you deserve co-authorship.

Contents

Abstract	ii
Preface	iii
Dedication	v
Acknowledgments	vi
1 Introduction	1
1.1 Clinical Context of Tongue Reconstruction	1
1.2 Challenges in Current Surgical Practice	2
1.3 Virtual Surgical Planning (VSP): A Paradigm Shift	3
1.4 Overall Research Objective	6
2 Related Works	7
2.1 Preoperative Planning in Tongue Reconstruction	7
2.1.1 Evolution from Generic Flaps to Defect-Oriented Strategies	7
2.1.2 Template-Based Approaches	8
2.2 Advanced Imaging Technologies for Surgical Planning	8
2.3 Biomechanical Modeling: A Promising Solution	10
2.4 Tongue Biomechanical Modeling	11

2.5	Optimal Surgical Planning using High-Fidelity Simulations	13
2.6	Data-Driven Approaches and the Challenge of Data Scarcity	14
2.7	Practical Advantages of Simplified Models	15
2.8	Synthesis and Identified Gaps	16
3	Algorithmically designed flaps in tongue reconstruction: a feasibility analysis	18
	Abstract	19
3.1	Introduction	19
3.2	Materials and methods	23
3.3	Results	31
3.4	Discussion	33
3.5	Conclusion	38
	References	39
4	Patient-specific virtual surgical planning for tongue reconstruction: evaluating hyperelastic inverse FEM with four simulated tongue cancer cases	42
	Abstract	43
4.1	Introduction	44
4.2	Method	47
	4.2.1 Patient Anatomy Acquisition	47
	4.2.2 Algorithmic Flap Design with hiFEM	48
	4.2.3 Comparative Virtual Flap Design Methods	52
	4.2.4 Evaluation Metrics	53
4.3	Results	55
4.4	Discussion	60
4.5	Conclusion	65

References	66
5 Computational Analysis of Tongue Reconstruction Surgery: The Impact of Free Flap Site and Volume on the Tongue Anatomy and Biomechanics	73
Abstract	74
5.1 Introduction	75
5.2 Method	78
5.2.1 Patient-Specific Tongue Model	78
5.2.2 Biomechanically Optimized Flap Design	79
5.2.3 Simulation of Tongue Reconstruction Surgery	81
5.2.4 Simulation of postoperative flap atrophy	83
5.2.5 Boundary Conditions and Contact Mechanics	84
5.2.6 Constitutive Models	84
5.2.7 Soft Tissue Parameterization	85
5.2.8 Simulation Stability and Robustness	87
5.2.9 Outcome Metrics	89
5.2.10 The Factorial Study Design	90
5.2.11 Statistical Analysis	91
5.2.12 Data and Code Availability	92
5.3 Results	92
5.4 Discussion	95
5.5 Conclusion	102
References	105
6 Guided tongue reconstruction: feasibility analysis of additively manufactured surgical toolkit on a silicone tongue model	113

Abstract	114
6.1 Introduction	115
6.2 Method	118
6.2.1 Tongue Reconstruction Virtual Surgical Planning	118
6.2.2 The Patient-Specific Surgical Toolkit	119
6.2.3 Benchtop Study	122
6.2.4 Quantitative Evaluation of Anatomical Accuracy	125
6.3 Results	126
6.4 Discussion	126
6.5 Conclusion	132
References	133
7 Discussion & Conclusion	141
7.1 Introduction	141
7.2 Surgical Simulation: New Scientific and Clinical Paradigm	142
7.3 Shifting the Flap Planning Objective from Defect-Fitting to Biomechanical Opti- mization	144
7.4 Clinically Feasible Virtual Framework for Surgical Decision-Making	145
7.5 Fidelity in Modeling the Flap-Tongue Interface, Reconstruction, and Flap Atrophy	146
7.6 Patient-Specific Parameters	147
7.7 The Path to Clinical Impact: Implementation, Validation, and Human Factors . . .	149
7.8 Limitations of the Thesis	150
7.9 Conclusion and Future Vision	154
Bibliography	156

List of Tables

3.1	Mean, median, and standard deviation of results	37
4.1	Parameters of material models used in hiFEM and iFEM.	53
4.2	Maximum Hausdorff distance comparing flap boundary geometries generated by hiFEM versus alternative methods for four reconstruction cases.	55
5.1	Material parameters used in the finite element simulations.	88
5.2	OLS regression results for maximum surface deviation (interaction model). Depen- dent variable: maximum surface deviation (mm). Reference: <i>Case A</i> ; $is_thigh = 0$ (Forearm).	94
5.3	OLS regression results for maximum Effective Strain (main effects model). De- pendent variable: maximum Effective Strain. Reference: <i>Case A</i> ; $is_thigh = 0$ (Forearm).	95
6.1	Summary of surface deviation metrics for the four reconstructed tongue phantoms.	126

List of Figures

1.1	Key elements of tongue reconstruction and free-flap design: (A) the resected tongue, representing the remaining tissue and the resulting defect; (B) the resection, or the cancerous tissue removed; (C) the radial forearm free flap (RFFF) harvesting site; (D) the planar geometry of the harvested skin paddle, configured to conform to the defect surface; (E) additional bulk provided by the beavertail—subcutaneous adipose tissue rolled to restore tongue volume precisely; and (F) the vascular pedicle used for microvascular anastomosis.	4
3.1	Schematic representation of resection and free flap in oral hemiglossectomy reconstruction. (a) indicates the defect requiring surgical removal, while (b) shows the remaining native portion of the tongue. (c) Represents the required beavertail bulk for reconstruction, and (d) shows the outer surface that needs to be recovered with a free flap (e).	24
3.2	The overall workflow: from geometry pre-processing to optimal flap design using the IFEM algorithm	25
3.3	Flow diagram of IFEM algorithm used in this study	26
3.4	Iterative convergence process for the corrected Young’s modulus	30

3.5	Depiction of the artificially simulated hemiglossectomy defect and the corresponding free flap design in this case study. (a) provides an isometric view of the tongue with the defect displayed in a different color. (b) and (c) display the same geometry from coronal and sagittal views, respectively. (d) illustrates the designed flap using IFEM algorithm. (I) indicates the anterior mid-sagittal margin corresponding to the tip of the tongue. (II) indicates the anterior–inferior margin on the mid-sagittal plane. (III) indicates the posterior–lateral margin of the resection. All dimensions are in millimeters	32
3.6	Thickness distribution of the designed flap after deformation. a Illustrates the color-coded thickness distribution, b depicts this distribution mapped onto the 2D flap, c presents a histogram of the distribution, and d shows a statistical boxplot of the distribution. The reader is referred to the web version of this paper for the color representation of this figure	33
3.7	Equivalent strain distribution of the designed flap after deformation. a Illustrates the color-coded equivalent strain distribution, b depicts this distribution mapped onto the 2D flap, c presents a histogram of the distribution, and d shows a statistical boxplot of the distribution. The reader is referred to the web version of this paper for the color representation of this figure	34
3.8	Equivalent stress distribution of the designed flap after deformation. a Illustrates the color-coded equivalent stress distribution, b depicts this distribution mapped onto the 2D flap, c presents a histogram of the distribution, and d shows a statistical boxplot of the distribution. The reader is referred to the web version of this paper for the color representation of this figure	35
4.1	Artificially simulated glossectomy cases. The first and second rows show axial and isometric views with the resected area outlined. The third row shows the isometric view after resecting the outlined region.	48
4.2	Flowchart summarizing the virtual surgical planning process for tongue reconstruction, from MRI image segmentation to optimal flap design.	49
4.3	Comparison of 2D flap designs generated by hiFEM, iFEM, NURBS, and BFF for simulated cases A-D.	56

4.4	Area Stretch (J_{area}) distribution histograms for flap designs derived from hiFEM and alternative methods in simulated Cases A and B.	57
4.5	Area Stretch (J_{area}) distribution histograms for flap designs derived from hiFEM and alternative methods in simulated Cases C and D.	58
4.6	Convergence rate comparison between hiFEM and iFEM, showing the L2 Norm per iteration for cases A-D.	59
5.1	Patient-Specific Tongue Model, Resection Scenarios, and Flap Designs. Overview of the four simulated tongue cancer cases along with their biomechanically optimized flap designs. The texture mapping is generated with hiFEM and maps each point in the resection model to its corresponding point in the planar flap design. . .	80
5.2	Workflow for free flap model generation and surgical simulation pre-processing. Panel (A) shows the two primary inputs: the “Resected Tongue Model” (red) and “Resection model” (textured). First, the hiFEM algorithm creates an optimized 2D planar skin template for the flap (C). Using the resulting deterministic 3D-to-2D mapping, the resection volume is computationally unfolded to generate a two-layer free flap model (B), consisting of skin (orange) and adipose (yellow). To simulate the clinical practice of overbulking, flap volume is increased by scaling the thickness of the adipose layer (D). Next, the composite flap is positioned at the resection site (E), and one-dimensional tension elements (blue lines) are defined to connect corresponding suture points on the flap and the resected tongue. Anatomical boundary conditions are established by fixing nodes (yellow dots) that represent the tongue’s muscular attachments to the mandible and hyoid bone (F, G).	81
5.3	Surgical simulation pipeline: reconstruction to long-term remodeling. (i) Initial simulation setup, (ii) The suture model brings the flaps proximal to the resected tongue model and simulates suturing. (iii) After the 5-minute idling for stresses to relax with the viscoelastic model present, this is “Day 0” immediately after reconstruction. (iv) shows the tongue model after one year of post-surgery atrophy modeling. At the very end, the effective strain (v) and Hausdorff distance (vi) are calculated and used for the statistical analysis.	82

5.4	Literature review of experimental data on the material parameters for the Forearm skin (i) and the Thigh skin (ii) represented as the stress-stretch response to a uniaxial tensile test.	86
5.5	Visualization of the sensitivity of dependent variables to different skin material properties and different levels of overbulking: overbulking increases Effective Strain (top) while decreasing anatomical deviation (middle); final tongue volume varies linearly with initial overbulking (bottom).	93
5.6	Mesh convergence analysis—Maximum equivalent strain. Maximum equivalent strain of the neotongue at the final post-remodeling equilibrium versus total mesh nodes. Each curve is one of 12 representative parameter combinations and is labeled with a three-letter code in the form [Case][Donor/Stiffness][Overbulking], where Cases A–D are the four defect geometries; Donor/Stiffness A–C = Forearm (soft→stiff), D–F = Thigh (soft→stiff); Overbulking A–E = 0%, 22.2%, 44.4%, 66.7%, 88.9%.	103
5.7	Mesh convergence analysis—Maximum displacement. Maximum nodal displacement of the neotongue at the final post-remodeling equilibrium (mm) versus total mesh nodes for the same 12 representative parameter combinations used in Figure 5.6. Codes follow the [Case][Donor/Stiffness][Overbulking] convention described there. Two panels are shown for readability with a shared horizontal axis.	104
6.1	Fabrication of the soft silicone tongue phantom. (A) Two-part 3D-printed mold with alignment/fastening bosses. (B) Cast, pigmented elastomer tongue phantom after demolding. (C) Final positioning of the phantom within the 3D-printed jaw model	119
6.2	Patient-specific resection guidance toolkit and workflow. (A) Resection Marking Guide positioned on the tongue phantom for surface incision delineation. (B) Marked the resection boundary on the tongue surface using the conforming guide. (C) Solid Resection Model providing tactile reference for three-dimensional cut depth and contour. (D) Completed resection. (E) Excised tissue specimen ready for volumetric inspection. (F) Resection Volume Inspection Guide with excised tissue positioned for intraoperative volume verification against planned resection.	121

6.3	Patient-specific flap harvesting guidance toolkit and workflow. (A) Planar template guide for outlining skin perimeter geometry on the donor site phantom. (B) Volumetric guide for verifying beavertail SAT volume against planned requirements. (C and D) Multi-layered donor site phantom with skin and SAT layers. (E) 3D-printed Flap Model providing tactile reference for target volume and shape. (F) Harvested beavertail SAT positioned in the volumetric guide for intraoperative volume verification.	122
6.4	Final reconstructed tongue phantoms for the four VSP cases. Each phantom demonstrates successful translation of the virtual surgical plan to physical execution using the patient-specific toolkit.	127
6.5	Post-reconstruction surface accuracy for the four simulated cases (A-D). Color map (0-4.13 mm) shows pointwise surface deviation between the executed reconstruction and the reference pre-operative tongue model.	128

Chapter 1

Introduction

1.1 Clinical Context of Tongue Reconstruction

Glossectomy, the surgical removal of all or part of the tongue, is the primary treatment for tongue cancer (Ansarin et al., 2019; Cai et al., 2019; De Berardinis et al., 2022; Mannelli et al., 2018; Rui et al., 2022). Tumor resections vary widely, ranging from mucosal excisions to complex combined resections. Small ones can be closed primarily by suturing, whereas medium to large resections require reconstruction to restore tongue anatomy and function (Urken, 2012). Microvascular surgery has popularized immediate free-flap transfer as the most effective method for post-resection tongue reconstruction (Lam and Samman, 2013; Marra et al., 2023; Rui et al., 2022). However, re-establishing the tongue's anatomy and vital functions remains a significant challenge (Amer, 2021).

Tongue reconstruction is challenging because of the tongue's unique anatomy and physiological roles (Mannelli et al., 2018). Restoring its functions for speech, swallowing, and airway protection is particularly challenging, making the tongue one of the most complex structures to reconstruct in the oral cavity (Amer, 2021; Mannelli et al., 2018). Reduced tongue mobility impairs chewing, bolus propulsion, oral hygiene, articulation, and speech. Base-of-tongue dysfunction can result in dysphagia (swallowing disorder) and aspiration (Amer, 2021; Lam and Samman, 2013; Urken, 2012). Other associated problems include limited mouth opening, xerostomia, and thick or viscous

1.2. Challenges in Current Surgical Practice

saliva (Dzioba et al., 2017). These disorders substantially affect the quality of life (QoL) of patients undergoing surgical resection and reconstruction (Dzidek et al., 2017; Yi et al., 2020), and may lead to malnutrition and depression (Rui et al., 2022). Therefore, advanced tongue cancer treatments are shifting away from a survival-only focus (Dziegielewski et al., 2019; Urken, 2012).

Contemporary surgical approaches increasingly prioritize restoring anatomy and maximizing the mobility and function of the remaining tongue tissue to reduce complications (Seikaly et al., 2009; Amer, 2021; Idris et al., 2018; Lam and Samman, 2013; Leymarie et al., 2012; Mannelli et al., 2018; Rui et al., 2022). The extent of resection must, however, adhere to oncologic principles: tumor margins of at least 3-5 mm are generally required for malignancy clearance and to reduce recurrence rates (Chang et al., 2013; Maxwell et al., 2015; Mitchell et al., 2018). Nevertheless, reconstruction methods and adjuvant therapy strategies can still be improved. Suggested improvements include developing a standardized method and a clear decision-making algorithm for flap selection (Marra et al., 2023), and creating guidelines for flap design — the surgical planning of the donor soft tissue’s geometrical characteristics — to reconstruct the resected area (Baskin et al., 2019; Idris et al., 2018).

1.2 Challenges in Current Surgical Practice

Effective surgical planning for tongue reconstruction should restore not only the essential bulk of the tongue but also its complex structural anatomy as closely as possible (Yi et al., 2020). Achieving a near-normal neotongue structure after surgery significantly improves speech intelligibility and swallowing function (Amer, 2021; Koumoullis et al., 2020; Leymarie et al., 2012). This relationship stems from two primary factors. First, the tongue’s unique structure enables effective contact with the palate and dental arches, which is essential for both swallowing and speech articulation (Idris et al., 2018; Zhou et al., 2020; Han et al., 2022; Wang et al., 2013; Marra et al., 2023). A poor design may lead to a flattened or depressed shape, compromising these functions. Second, maintaining a near-normal neotongue reduces tethering of the tongue remnant to the floor of the mouth and

1.3. Virtual Surgical Planning (VSP): A Paradigm Shift

preserves maximal mobility (Urken, 2012; Urken and Biller, 1994). This is particularly important in complex three-dimensional (3D) defects, such as those involving the floor of the mouth, where poor design can severely restrict tongue mobility (Liu et al., 2023a).

Free-flap design—specifically flap volume and planar geometry—directly influences neotongue structure and, consequently, the restoration of oral function (Amer, 2021; Baskin et al., 2019; Dziegielewski et al., 2019; Hanubal et al., 2023; Idris et al., 2018; Seikaly et al., 2003; Tarsitano et al., 2016; Urken, 2012). Achieving optimal outcomes requires experienced surgeons who can precisely control flap bulk (beavertail tissue, Figure 1.1E) and carefully configure the harvested skin-paddle geometry (Figure 1.1D) to contour the defect’s outer surface (Urken, 2012). The goal is to restore tongue anatomy as closely as possible (Marra et al., 2023). Nevertheless, translating the three-dimensional (3D) characteristics of tongue defects into planar flap templates remains challenging (Baskin et al., 2019; Idris et al., 2018). Several studies have linked tongue-defect patterns to two-dimensional geometric shapes to guide flap design (Baskin et al., 2019; Fan et al., 2019a; Fan et al., 2019b; Han et al., 2022; Idris et al., 2018; Koumoullis et al., 2020), yet a comprehensive, standardized flap-design method remains elusive. Defect morphology varies substantially in position, size, and 3D geometry, and each patient has unique anatomical demands. As a result, these frameworks still rely heavily on surgeon expertise to adapt and refine designs for each case.

1.3 Virtual Surgical Planning (VSP): A Paradigm Shift

Over the past decade, VSP and computer-aided design and manufacturing (CAD/CAM) have revolutionized surgical planning in reconstructive surgery. In maxillary and mandibular reconstruction, VSP has significantly improved precision and reliability in preoperative planning, yielding more predictable results than free-hand surgery (Chuka et al., 2017; Logan et al., 2013; Nayar et al., 2024; Pucci et al., 2020; Serrano et al., 2019). VSP provides a high degree of spatial accuracy for precise resections, implant positioning, and reconstruction of jaw defects (Chuka et al., 2017; Idris et al., 2022; Salinero et al., 2022), while reducing the need for intraoperative adjustments (Idris et al., 2022). These technologies enhance communication within surgical teams (Nayar

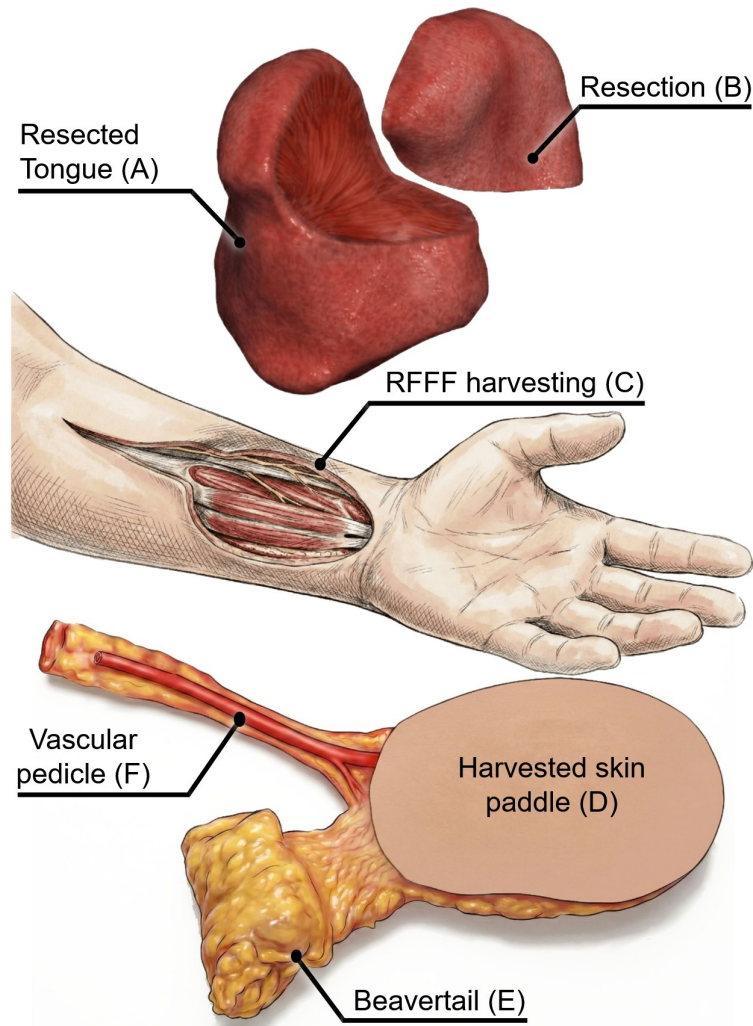


Figure 1.1 Key elements of tongue reconstruction and free-flap design: (A) the resected tongue, representing the remaining tissue and the resulting defect; (B) the resection, or the cancerous tissue removed; (C) the radial forearm free flap (RFFF) harvesting site; (D) the planar geometry of the harvested skin paddle, configured to conform to the defect surface; (E) additional bulk provided by the beavertail—subcutaneous adipose tissue rolled to restore tongue volume precisely; and (F) the vascular pedicle used for microvascular anastomosis.

et al., 2024), leading to statistically and clinically significant reductions in operating time (Idris et al., 2022; Padilla et al., 2021; Serrano et al., 2019) and in time to prosthetic connection (Chuka et al., 2017). They are also more cost-effective than traditional techniques (Kurlander et al., 2023). Their implementation has been associated with improved aesthetic outcomes (Chuka et al., 2017; Serrano et al., 2019), better functional rehabilitation results (Nayar et al., 2024; Papadopoulos-Nydam et al., 2017), and an enhanced quality of life for patients (Nayar et al., 2024). This digital transformation is now extending beyond osseous reconstruction into plastic surgery, including

1.3. Virtual Surgical Planning (VSP): A Paradigm Shift

cranial vault remodeling for craniosynostosis (Podolsky et al., 2024b; Robertson et al., 2023) and autologous breast reconstruction (Carrion et al., 2024). Patient-specific CAD-based modeling and surgical simulation platforms have demonstrated particular value in accurately representing complex soft-tissue anatomy, and in enabling preoperative optimization of reconstruction strategies through physical models derived from patient-specific imaging (Podolsky and Borschel, 2019; Podolsky et al., 2021; Podolsky et al., 2024a; Podolsky et al., 2025). This demonstrates the potential for patient-specific digital planning to improve outcomes in complex reconstructive procedures.

Soft-tissue reconstruction for complex functional organs like the tongue remains dominated by freehand surgical techniques. While emerging computational approaches such as NURBS-based surface flattening have shown promise (Xu et al., 2022), the current standard practice still relies heavily on subjective intraoperative adjustments guided by surgeon expertise. This can lead to inconsistent outcomes. A shift toward standardized, biomechanically informed VSP for soft tissue is essential to overcome this limitation. VSP enables proactive reconstruction planning by integrating computational modeling of dynamic tissue behavior, defect-specific anatomy, and patient-specific priorities. This would align soft-tissue reconstruction with the precision already achieved in mandibular and maxillary reconstruction VSP, enabling surgeons to achieve truly individualized flap geometry optimization.

Applying existing CAD/CAM technologies to soft-tissue reconstruction is biomechanically challenging because those technologies were designed for rigid skeletal structures. Unlike bone, soft tissues are hyperelastic and deform significantly under physiological loads (Hermant et al., 2017). Traditional CAD tools are therefore inadequate for simulating postoperative flap morphology or tissue interactions. Current VSP platforms lack the capacity to model these nonlinear deformations or to predict their anatomical and functional consequences (Heredero et al., 2024). Addressing this problem requires merging defect-specific 3D planning with computational biomechanics to simulate flap-tongue interactions during the procedure. By complementing CAD/CAM with finite element analysis (FEA), VSP can transition from rigid geometric planning to deformable, soft-tissue optimization. Such a development would be a fundamental step toward predictable, patient-tailored tongue reconstruction.

1.4 Overall Research Objective

The central objective of this thesis is to adapt and extend established computational and biomechanical methods for tongue reconstruction by developing a new integrated framework that shifts the procedure from a craft-based art to a data-driven science. This involves creating a VSP framework that leverages patient-specific data to optimize surgical planning, prospectively evaluate surgical strategies, and ensure their precise execution. With this in mind, a series of interrelated research questions arises:

- How can we redefine the objective of algorithmic free flap design to move beyond geometric pattern-matching and incorporate the underlying biomechanical principles that govern post-operative anatomy, mobility, function, and tissue viability? How can we develop such an individualized free flap design algorithm? (Chapters 3 & 4)
- What level of model fidelity and automation is required to create a clinically feasible surgical simulation framework capable of prospectively quantifying the long-term anatomical and biomechanical consequences of interdependent surgical decisions for an individual patient? (Chapter 5)
- What methodologies and tools are required to bridge the critical “virtual-to-physical” gap in soft tissue surgery, ensuring that a computationally optimized reconstruction plan can be translated with high fidelity and precision onto a dynamic surgical field? (Chapter 6)

The proposed research questions guide the development of a VSP framework for tongue reconstruction, with the broader goal of advancing personalized surgical care. Chapter 2 will contextualize this work within existing literature and critically evaluate prior approaches to biomechanical modeling and preoperative planning in tongue reconstruction. The four studies addressing these research questions are presented in Chapters 3 through 6, followed by a synthesis of findings, clinical implications, and future directions in Chapter 7.

Chapter 2

Related Works

2.1 Preoperative Planning in Tongue Reconstruction

2.1.1 Evolution from Generic Flaps to Defect-Oriented Strategies

Tongue reconstruction has undergone a transformation from generalized “one-flap-fits-all” approaches to defect-oriented strategies that prioritize the restoration of natural morphology and function of the tongue (Baskin et al., 2019; Cai et al., 2019; Dzioba et al., 2017; Engel et al., 2010; Idris et al., 2018; Liu et al., 2023a; Mannelli et al., 2018; Marra et al., 2023; Rui et al., 2022; Wang et al., 2016; Yi et al., 2020). Individualized and defect-specific flap design has shown to be significantly beneficial for improving tongue mobility and functional recovery (Fan et al., 2019a). A lack of personalization can lead to inconsistent outcomes, which stems from insufficient anatomical mapping and over-reliance on the surgeon’s expertise (Fan et al., 2019b). Additionally, it often leads to suboptimal flap use, the need for flap trimming, excessive pressure and tension during suturing, and poor functional outcomes (Liu et al., 2023b; Rui et al., 2022). Tailoring flap design to match precise defect sizes and shapes can enhance patient-specific precision (Ansarin et al., 2019; Koumoullis et al., 2020; Liu et al., 2023b; Yi et al., 2020). Studies consistently show improved swallowing (Fan et al., 2019a; Fan et al., 2019b; Han et al., 2022; Wang et al., 2016), speech intelligibility (Han et al., 2022; Liu et al., 2023b), and overall QoL (Liu et al., 2023b) when the patient’s unique anatomical characteristics and the defect geometry inform the flap design.

Increasing emphasis is placed on individualized treatments that align with patient-specific anatomy and defect characteristics to optimize functional outcomes (Dzioba et al., 2017; Engel et al., 2010; Fan et al., 2019a; Fan et al., 2019b; Liu et al., 2023a; Mannelli et al., 2018; Marra et al., 2023).

2.1.2 Template-Based Approaches

Contemporary approaches to tailored flap design utilize adaptable templates customized to patient anatomy based on the defect's shape and dimensions. Wang et al. (2016) described tailoring the anterolateral thigh (ALT) flap by evaluating tongue defect dimensions between key anatomical landmarks, including the tip, root, and lateral edges. These measurements guide the design of the "ABC" flap sections. Similarly, Rui et al. (2022) developed an L-shaped ALT flap that integrates preoperative measurements of critical tongue landmarks. This approach aligns the flap with the defect to restore the tongue's natural form while minimizing tissue trimming and reducing pressure and tension during suturing. Yi et al. (2020) used a paper ruler to measure defect dimensions (for example, the distance from the tongue tip to its base and the circumferential width) to inform their flap design template. Baskin et al. (2019) adopted a defect-oriented strategy, creating two-dimensional (2D) templates based on common anatomical landmarks to capture the unique geometry of each tongue region and defect type. Specific shapes, such as a pear for the total tongue and an egg for the oral tongue, were used to guide flap design. These studies report that patient- and defect-specific flap design is associated with significantly improved mobility (Rui et al., 2022; Yi et al., 2020), swallowing capacity (Rui et al., 2022; Wang et al., 2016), and speech outcomes (Rui et al., 2022; Yi et al., 2020). Despite these advances, reliance on manual measurements with basic tools such as a flexible ruler limits precision and consistency, leading to variability in outcomes.

2.2 Advanced Imaging Technologies for Surgical Planning

Advanced imaging and preoperative planning technologies can improve accuracy by enabling computational modeling of tongue defects (Liu et al., 2023b). Magnetic resonance imaging (MRI) and computed tomography (CT) have contributed to individualized flap design by providing detailed

2.2. Advanced Imaging Technologies for Surgical Planning

anatomical information (Fan et al., 2019a; Fan et al., 2019b; Han et al., 2022; Liu et al., 2023b; Liu et al., 2023a; Podolsky et al., 2025). MRI and CT are essential preoperative imaging tools for assessing the extent of tumor involvement (De Berardinis et al., 2022; Marra et al., 2023). This information helps plan accurate resections and can preserve as much healthy tissue as possible, thereby improving functional outcomes (Ansarin et al., 2019). However, intraoperative decisions about resection margins still largely rely on clinical judgment, which can lead to over-resection or inadequate margins (Miyota et al., 2014). Koumoullis et al. (2020) and Sinha et al. (2022) have suggested 3D modeling of the tumor and 3D printing of silicone molds to guide surgeons during resection. While these techniques enhance the clinical use of surgical guides for soft tissue reconstruction, accurately following these guides remains challenging because they rely solely on visual alignment of the mold with the resection area.

Imaging has also been used to measure distances between anatomical landmarks to inform flap templates. Fan and colleagues (2019a; 2019b) used imaging tools to create templates for hemi-oral tongue reconstruction. They identified key anatomical points and translated the tongue defect's 3D structure into a template by measuring distances between these landmarks. This approach reduced the need for flap trimming and improved tongue mobility by avoiding tissue tethering. Patients showed significantly better swallowing and speech outcomes compared to the control group, who received conventional freehand flap design without anatomical landmark measurements (Fan et al., 2019a; Fan et al., 2019b; Han et al., 2022). Similarly, Liu et al. (2023b) measured the dimensions of the oral tongue and mapped the topology and geometry of anatomical landmarks onto paper. By cutting, folding, and splicing this model, they created a 3D representation of the tongue, which could then be unfolded into a flat template to guide flap harvesting. Similar approaches to flap design were subsequently developed to mimic the natural anatomical structure of the anterior floor of the mouth (Liu et al., 2023a). Their image-guided approach to flap design minimized the need for trimming during surgery, prevented dead space, and resulted in enhanced speech intelligibility, swallowing function, and cosmetic outcomes.

Despite these advances, a universally individualized approach remains elusive. Current flap templates, even with personalized adjustments, rely heavily on the surgeon's expertise and intuition (Fan et al., 2019b; Ansarin et al., 2019), owing to substantial anatomical variability and the diverse

range of defect shapes, sizes, and complex 3D structures (Ansarin et al., 2019; Engel et al., 2010; Koumoullis et al., 2020; Liu et al., 2023b; Yi et al., 2020). A comprehensive approach that integrates advanced imaging techniques and computational modeling is needed to achieve precise patient- and defect-specific virtual flap design (Koumoullis et al., 2020). Such a framework would accommodate a wide range of defect types and geometries, reduce reliance on the surgeon's expertise, and enable consistent outcomes across cases.

2.3 Biomechanical Modeling: A Promising Solution

Soft-tissue biomechanical modeling using the Finite Element Method (FEM) offers a promising solution to challenges in individualized free-flap design. Biomechanical simulations create a controlled environment for analyzing the mechanical behavior of soft tissue under simulated surgical conditions. These models are valuable for evaluating new surgical techniques. They enable assessment of soft-tissue dynamics during interventions without exposing patients to the risks of clinical trials (Anderson et al., 2017; Chabanas et al., 2003; Stavness, 2010). Computer models can replicate the biomechanical and biological interactions among soft tissues, bones, and joints with a level of insight often unattainable in experimental or clinical settings (Magenat Thalmann et al., 2014; Sorouri et al., 2018). For example, while measuring stress during surgery is impractical, biomechanical models provide a reliable alternative for informing surgical planning (Chaudhry et al., 1998; Lott-Crumpler and Chaudhry, 2001; Tepole et al., 2014a). By estimating procedural outcomes, these simulations enable clinicians to anticipate post-surgical functional impairments, identify potential complications, and optimize surgical approaches and rehabilitation exercises (Al-Zanoon et al., 2024; Buchaillard et al., 2007; Kappert et al., 2019; Maguenat Thalmann et al., 2014; Tepole et al., 2014a; Tepole et al., 2014b; van Alphen et al., 2013). The utility of biomechanical models in surgical planning is particularly promising for reducing reliance on trial-and-error approaches in clinical practice (Chabanas et al., 2003). Integrating individualized computational models into routine preoperative planning could advance patient-specific decision-making (Kappert et al., 2019; Kappert et al., 2021a; Tepole et al., 2014a; Tepole et al., 2014b).

2.4 Tongue Biomechanical Modeling

Modeling the biomechanics of the tongue using FEM presents multiple challenges. The high stretch values observed in tagged MRI scans during speech highlight the complexity of its behavior (Harandi et al., 2017a; Harandi et al., 2017b). The muscular and connective tissues that constitute the tongue are complex and exhibit nonlinear, time-dependent, inhomogeneous, and anisotropic properties (Anderson et al., 2017). Capturing the tongue’s mechanical properties with the constitutive models proposed in the literature is very challenging, and the mechanical behavior of soft tissue can differ between in vivo and ex vivo conditions (Hermant et al., 2017). Empirical studies also reveal substantial variability in mechanical properties across different oral-cavity tissue types (Sorouri et al., 2018). High nonlinearity requires a high-quality volumetric mesh—preferably hexahedral or higher-order elements—to capture large nonlinear deformations without locking; generating such a discretization is notably difficult (Anderson et al., 2017). Additionally, modeling interactions between hard and soft tissues in the oral cavity, particularly during speech and swallowing, adds another layer of complexity. These processes require multi-scale and multi-component integration to achieve reasonable accuracy while maintaining computational efficiency (Stavness et al., 2011; Stavness et al., 2014b; Stavness et al., 2014b).

Despite the challenges, biomechanical simulations have explored the effects of tongue surgery on postoperative mobility. Buchaillard et al. (2007) used a generic FEM tongue model to simulate hemiglossectomy and floor-of-mouth reconstruction by modifying stiffness in affected areas. They deactivated muscles and adjusted mechanical properties to represent surgical removal and free-flap reconstruction. The simulation predicted that low-stiffness flaps preserved mobility but lacked the support needed for effective tongue elevation, leading to a theoretical increase in torsion and less controlled movements. Conversely, simulated high-stiffness flaps provided better stability but restricted mobility. While these predictions qualitatively aligned with clinical observations of speech deficits, the model was not quantitatively validated against postoperative patient data. This model demonstrated how flap stiffness theoretically affects outcomes; however, its reliance on a generalized tongue structure limits its applicability to patient-specific cases. It also reduced free-flap

2.4. Tongue Biomechanical Modeling

reconstruction to merely adjusting the stiffness of healthy tongue tissue in the affected area. van Alphen et al. (2013) implemented a similar model to simulate postoperative scarring with stiffened elements, but it shares the same limitations.

Building on the generic tongue model (Buchaillard et al., 2009; Buchaillard et al., 2007), recent efforts have integrated rigid- and soft-tissue dynamics to enhance understanding of the jaw-tongue-hyoid complex (Stavness et al., 2011; Stavness et al., 2014a). Multibody dynamics frameworks that incorporate rigid–soft-tissue interactions; transversely isotropic or orthotropic hyperelastic and visco-hyperelastic material models; point-to-point or continuum-based (FEM) muscle representations; and dynamic models that predict muscle activation all serve to improve the fidelity of biomechanical models in predicting the complex biomechanics of the oropharynx (Anderson et al., 2017; Hermant et al., 2017). Furthermore, studies have applied skinning techniques to simulate fluid-particle dynamics and acoustic phenomena, expanding the potential applications for modeling swallowing and sound production (Harandi et al., 2017a; Stavness et al., 2014b).

To streamline patient-specific simulation of tongue biomechanics, Harandi and colleagues (2017a; 2017b) developed a method for patient-specific oropharyngeal biomechanical modeling based on the patient’s MRI data. They used elastic registration to adapt generic tongue, mandible, and hyoid models to individual anatomy. This process aligned muscle fiber orientations from the generic model to the patient-specific geometry, but it often degraded mesh quality and required remeshing. They also applied an incompressible deformation algorithm to integrate 2D tagged MRI motion with Cine MRI, producing a 3D deformation field that was used to estimate muscle activation patterns. To evaluate the model, their simulations of individualized tongue dynamics were coupled with an acoustic simulator, producing spectral outputs (formants) with an error of less than 10% compared to the subject’s audio recordings.

Kappert et al. (2021a) advanced this approach by developing an atlas model that incorporated detailed muscle fiber orientations from diffusion-weighted MRI (DW-MRI). This enabled the identification and segmentation of distinct muscle bundles through fiber tracking, enhancing anatomical fidelity. They employed a mapping technique to transfer the fiber-orientation field to each participant’s tongue model. The results demonstrated improved alignment with observed tongue movements in healthy volunteers compared with generic models. The importance of such

patient-specific modeling is further underscored by recent findings showing that mandibular and hyoid morphology alter tongue muscle architecture in healthy adults (Lloyd et al., 2025). While these approaches have been validated and proven effective for participants without tongue defects, their predictive accuracy for post-surgical outcomes and surgical planning requires further development and validation. Notably, the impact of free-flap reconstruction on post-surgical outcomes using these patient-specific modeling approaches has yet to be addressed.

2.5 Optimal Surgical Planning using High-Fidelity Simulations

Soft tissue biomechanical simulations have been used for surgical planning. Chabanas et al. (2003) developed a FEM model tailored to individual facial anatomy, providing a predictive framework for soft tissue deformation in maxillofacial surgical planning. Kappert et al. (2019) extended this to tongue glossectomy, simulating small resections with primary closure using sutures. Their approach replicated surgical suturing by pulling resection edges together and creating a new FE model from the deformed surface mesh, resetting all surgery-induced stretches and stresses. Next, they simulated fibrotic tissue effects, showing that higher stiffness levels further restricted movement, consistent with post-operative video observations. While simplified, their method established a foundation for modeling the surgical procedure of glossectomy and surgical closures.

Biomechanical simulations have demonstrated the potential for personalized surgical planning, but current models face limitations in accurately replicating critical surgical aspects. Addressing these limitations is essential; further improvements in robustness and efficiency are needed for clinical adoption. Key challenges include the complexity of preparing individualized models (Harandi et al., 2017a; Harandi et al., 2017b), managing contact formulations that are often complex and unstable between the tongue, flap, and surrounding oral structures (Stavness et al., 2011), and handling large soft-tissue deformations that can lead to element inversion and numerical instability (Anderson et al., 2017).

Moreover, identifying the optimal free-flap design for a specific defect often requires an optimization framework that involves numerous simulations. Each potential flap must be evaluated for its ability to restore anatomical integrity, minimize internal stresses, and maximize functional

outcomes. This process is computationally intensive and may involve hundreds of simulations to explore the vast parameter space of flap shapes, sizes, and positions (Tepole et al., 2014a). High computational cost, numerical instability, and a lack of automation for patient-specific model generation currently limit clinical use (Anderson et al., 2017; Tepole et al., 2014b). Enhancing mesh quality through novel automatic hexahedral meshing (Brückler and Campen, 2023) or higher-order tetrahedral meshing (Hu et al., 2020; Schneider et al., 2022; Maas et al., 2016), developing more robust numerical solvers (Maas et al., 2012; Martin et al., 2024; Lloyd et al., 2012), and implementing robust, efficient contact algorithms such as Incremental Potential Contact (Li et al., 2020) can improve stability. By addressing these challenges in modeling, computational efficiency, and simulation stability, biomechanical simulations can evolve into practical tools for individualized VSP.

2.6 Data-Driven Approaches and the Challenge of Data Scarcity

Recently, data-driven approaches, specifically machine learning (ML) and deep learning (DL), have emerged as powerful alternatives to address the limitations of high-fidelity 3D FEM simulations. Unlike traditional biomechanical models that rely on explicitly solving constitutive equations, deep learning architectures can learn complex, non-linear tissue behaviors directly from large datasets. These surrogate models offer a distinct advantage: once trained, they can predict tissue deformations in near real-time, effectively bypassing the computational bottleneck of non-linear solvers (Tonutti et al., 2017). For instance, generative deep learning has demonstrated potential in predicting post-surgical anatomical outcomes without explicit biomechanical modeling. In domains such as breast reconstruction, Generative Adversarial Networks (GANs) have been successfully trained on pre-operative and post-operative imaging cohorts to predict anatomical outcomes, essentially generating the post-surgical result based on learned morphological patterns (Chartier et al., 2022).

While the application of these data-driven methods to tongue reconstruction holds promise, it is currently stalled by a fundamental pragmatic barrier: data scarcity. Deep learning models are notoriously data-hungry, typically requiring thousands of standardized, labeled data pairs to generalize effectively. In the specific domain of glossectomy, large-scale, standardized datasets of

pre- and post-operative MRI scans with consistent resection geometries do not exist. Consequently, the field faces a cold start problem: the clinical data does not exist in sufficient volume. To unlock the potential of ML for this domain, the primary requirement is a mechanism to generate large-scale, physically realistic synthetic training data.

2.7 Practical Advantages of Simplified Models

Recent advances in individualized VSP for soft-tissue reconstruction leverage efficient and robust flattening methods. Image-guided surgical planning for breast reconstruction was the first to gain attention (Huang et al., 2007). After imaging and modeling a desired geometry for the breast, the problem was similar to challenges in tongue reconstruction flap design: converting complex 3D geometries into planar shapes with minimal deformation. They proposed a surface flattening method based on a mass-spring model to transform 3D breast shapes into 2D templates that guide surgical planning. Xu et al. (2022) adopted a similar approach. Using CT imaging, they segmented and generated a 3D model of the tongue and specified the resection area. They extracted the defect's outer surface and transformed it into a planar flap-cutting guide using Non-Uniform Rational B-Splines (NURBS) transformations. To assess the effectiveness of this computerized planning method, they conducted a case series. Fifteen patients who received the 3D-printed guide experienced significantly reduced flap trimming, shorter operative times, shorter postoperative hospital stays, and improved accuracy in defect matching compared with patients treated using traditional flap-design methods without virtual planning. This study underscores the potential of VSP and 3D-printed guides to enhance precision and efficacy in reconstructing soft-tissue defects.

To enhance the precision of 3D-to-2D transformations for wound reconstruction surgical planning, Ji et al. (2024) adopted a flattening technique grounded in FEM principles. This approach translates complex 3D wound geometries into individualized planar flap shapes by minimizing deformation energy in the flattened configuration. Unlike geometric flattening methods, this FEM-based approach integrates specific material properties of the skin, such as anisotropy and elasticity, to reduce distortions in area and angle. The method demonstrated close alignment between the flap designs and the wound's unique contours. Flattening methods simplify flap design by reducing 3D

problems to 2D, avoiding contact handling, and eliminating trial-and-error optimization (Huang et al., 2007; Ji et al., 2024; Xu et al., 2022). These methods offer efficient, robust, and practical solutions, making them better suited for clinical use than high-fidelity 3D simulations (Ji et al., 2024).

Flattening techniques, however, are fundamentally misaligned with the surgical sequence. In tongue reconstruction, surgeons begin with a planar free flap and carefully deform it to recreate the defect’s 3D structure. The primary objective is to achieve the intended 3D shape while minimizing stretches, tensions, and internal stresses. This approach reduces the risks of complications such as compromised blood supply, tissue necrosis, impaired healing, scar formation, tongue tethering, and restricted mobility (Benjamin et al., 2020; Lee et al., 2019; Rui et al., 2022; Tepole et al., 2014b; Tepole et al., 2014a). Effective flap design must therefore prioritize minimizing deformation energy in the final, deformed 3D configuration where the flap is sutured and integrated into the neotongue. In contrast, flattening techniques follow the opposite workflow: they begin with the defect’s 3D surface and computationally “flatten” it into a 2D template, aiming to minimize deformation energy in the flattened state (Ji et al., 2024). This divergence underscores the need for a new approach that aligns with the surgical intervention sequence and focuses on minimizing deformation energy in the flap’s deformed configuration. Moreover, existing flattening techniques have yet to incorporate hyperelastic material models (Ji et al., 2024), despite the critical importance of such models for accurately representing the nonlinear behavior of soft tissue (Anderson et al., 2017; Chagnon et al., 2015; Hermant et al., 2017; Mostafavi Yazdi and Baqersad, 2022).

2.8 Synthesis and Identified Gaps

Current advancements in image-guided, template-based flap design enhance defect measurement precision, yet they fall short in addressing the complexity of soft-tissue biomechanics and patient-specific variations; these methods still require the surgeon’s experience for effective free-flap design and execution. While biomechanical simulations offer granular insights by modeling tissue biomechanics, their clinical adoption is hindered by high computational demands, instability in

2.8. Synthesis and Identified Gaps

large-deformation regimes, and the labor-intensive preparation of patient-specific models. Simplified models, such as flattening methods, though efficient, oversimplify biomechanical intricacies and neglect the true surgical sequence of intervention, which limits their accuracy and effectiveness.

Bridging this gap requires a hybrid strategy that merges computational efficiency with biomechanical fidelity. By integrating patient-specific imaging with optimized simulation techniques, such an approach could enable precise flap design while remaining feasible within real-world resource constraints. This synthesis promises a scalable framework for personalized tongue reconstruction that prioritizes functional restoration and practicality, ultimately advancing outcomes for patients undergoing glossectomy.

Chapter 3

Algorithmically designed flaps in tongue reconstruction: a feasibility analysis

Amir Reza Isazadeh^{1,*}, Hadi Seikaly^{2,3}, Lindsey Westover⁴ and Daniel Aalto^{1,2}

* A version of this chapter has been published as: Isazadeh AR, Seikaly H, Westover L, Aalto D (2024) *Algorithmically designed flaps in tongue reconstruction: a feasibility analysis*. *International Journal of Computer Assisted Radiology and Surgery*. DOI: [10.1007/s11548-024-03062-w](https://doi.org/10.1007/s11548-024-03062-w). Reproduced with permission from Springer Nature.

¹ Department of Communication Sciences and Disorders, Faculty of Rehabilitation Medicine, University of Alberta, 6-129 Clinical Sciences Building, 11304 - 83 Ave NW, Edmonton, AB T6G 2G3, Canada.

² Institute for Reconstructive Sciences in Medicine (iRSM), Misericordia Community Hospital, Edmonton, Canada.

³ Division of Otolaryngology, Department of Surgery, Faculty of Medicine and Dentistry, University of Alberta, Edmonton, Canada.

⁴ Department of Mechanical Engineering, Faculty of Engineering, University of Alberta, Edmonton, Canada.

Abstract

Purpose Despite the significance and complexity of tongue reconstruction surgery, a digital tool for flap design is currently lacking. This study investigates the effectiveness of employing inverse finite element method (IFEM) for meticulously designing the geometric characteristics of harvested tissue (free flap) for tongue reconstruction. **Methods** In the case of an artificially simulated hemiglossectomy, IFEM algorithm was applied for algorithmic flap design. The method's effectiveness was evaluated by assessing flap deformation in a simplified virtual reconstruction, focusing on parameters such as stress, strain, and thickness. **Results** The IFEM algorithm successfully generated an optimal flap design for the intended surgical removal. Analysis of the flap's overall surface area, deformation characteristics, and safety margins demonstrated the feasibility of the deformation. Notably, the stress and thickness assessments suggested that the flap's tension post-surgery would not adversely affect the mobility of the reconstructed tongue, suggesting a positive outcome for functional recovery. **Conclusion** The IFEM demonstrates significant potential as a tool for precise free flap design in tongue reconstruction surgeries. Its application could lead to improved surgical accuracy and better quality of life for patients undergoing such procedures.

Keywords Head and neck cancer · Oral cancer · Tongue reconstruction · Precision health · Personalized medicine

3.1 Introduction

Oral and pharyngeal cancer constitutes the sixth most common cancer worldwide, posing a substantial and escalating concern (Warnakulasuriya, 2009). Treating advanced oral cancer can be a challenge, particularly in the case of tongue reconstruction surgery. Surgical resection and reconstruction of the tongue is an accepted procedure to restore the function of the tongue after cancer removal, yet due to the numerous factors involved, the surgical design of flaps in tongue reconstruction is complicated.

3.1. Introduction

In tongue reconstruction, sufficient volume and tongue surface recovery are reported to be necessary for acceptable outcomes (Hanubal et al., 2023). For instance, treatment protocol design is especially important for the reconstruction of the base of the tongue since it requires a precise bulk recovery (Seikaly et al., 2009). Chepeha et al. (2008) highlight that for a successful premaxillary and palatal contact recovery, it is crucial to design the anterior tongue with enough bulk and surface area. This is essential for effective movement and control of food and liquid in the mouth, as well as for clear speech production. Conversely, inefficient bulk and surface recovery can cause disorders and negatively impact functional outcomes. For instance, excessive bulk can cause choking, sleep apnea, and mastication disorders, while insufficient bulk can interrupt the swallowing mechanism and speech production (Hanubal et al., 2023). Moreover, a poorly designed flap could impede the essential mobility of the neotongue by tethering the remaining portion of the tongue (Fan et al., 2019a). Therefore, precise design of flaps is required to ensure acceptable functional outcomes.

The precise design of free flaps results in sufficient bulk and surface area after reconstruction. Currently, there are three commonly used donor sites for tissue transplantation: Anterolateral Thigh (ALT), Rectus Abdominis (RA), and Radial Forearm Free Flap (RFFF) (Hanubal et al., 2023). The ALT and RA donor tissues often provide excessive volume. Although this volume can be trimmed down, the trimming also reduces the skin surface area. This coupling makes precise flap design extremely difficult. Additionally, the muscle of the neotongue might atrophy during the recovery with the ALT and RA donor tissues (Hashida et al., 2017), which makes RFFF an ideal option to achieve the desired functional outcomes (Baskin et al., 2019). RFFF results in less bulk of tissue, and the remaining bulk required can be tailored by augmenting the fibroadipose tissue of the forearm with the beavertail radial forearm free flap method (BT-RFFF, (Hanubal et al., 2023)). This means that one can design the geometry of the flap without worrying about its bulk and then adjust the beavertail fat tissue to tailor the volume of the entire donor tissue.

The BT-RFFF technique allows for the independent design of harvested volume and skin surface area. However, determining the precise geometry of free flaps remains a challenge. Current methods focus on the topology of the defect and attempt to establish a relationship between specific defects and flap shape. Baskin et al. (2019) conducted a cadaver study to further investigate this relationship by examining the most common types of tongue reconstruction. They incised the cadaver tongue

according to the reconstruction type, resected the region of interest, and debulked it to create an unfolded topology suitable for free flap design. For example, they proposed that a pear-shaped free flap is appropriate for total oral tongue reconstruction, while a bullet-shaped free flap is suitable for oral hemiglossectomy reconstruction. After planning for two surgeries using the predetermined flap shapes, they concluded that these shapes could improve functional outcomes. This study provided valuable insights for further geometrical investigation into the precise design of free flaps.

The literature on patient-specific geometrical flap design in tongue reconstruction is limited, and to our knowledge, only Fan et al. (2019a; 2019b) have proposed a method for such a design. In their study, they introduced a novel anatomy-based flap design to improve functional and cosmetic outcomes in subtotal tongue reconstruction. To devise this design, they conducted a virtual geometrical study and collected magnetic resonance imaging (MRI) data from a volunteer's tongue. They identified five landmarks for hemiglossectomy reconstruction and virtually segmented and incised the tongue based on these landmarks. Through geometrical manipulations, they concluded that the optimal shape for a hemiglossectomy free flap is a pentagon. The dimensions of this shape can be determined based on the anatomical landmarks on the patient's tongue. To evaluate their method, they conducted a cohort study involving thirty-five patients divided into two groups. They designed free flaps for the second group based on the actual anatomical landmarks and compared aesthetic outcomes, swallowing function, and speech intelligibility. They presented their method as a more precise approach to designing flaps in hemiglossectomy reconstruction, demonstrating that even simple dimension measurements can enhance outcomes. This suggests that computer-assisted free flap design in tongue reconstruction could lead to more precise surgery. Therefore, further studies are needed to investigate the effectiveness of digital tools for precise flap design.

Surface flattening refers to the process of transforming a three-dimensional (3D) curved surface into a two-dimensional (2D) domain, aiming to preserve the original surface's geometric and topological characteristics as faithfully as possible. This involves geometric considerations, where the inherent properties of the 3D surface, such as shape, size, and connectivity, are retained during the transformation. In computer graphics, surface flattening is utilized for texture mapping, shape analysis, and mesh parameterization (Sawhney and Crane, 2018). The fabrication industries use it in car body manufacturing (Bostan Shirin and Assempour, 2014), furniture making, and

clothing (garment) design (Liu et al., 2010). It also forms the backbone of flattening-based medical visualizations (Kreiser et al., 2018). Wound reconstruction is the most relevant field where computer-assisted surgical planning has been introduced for precise flap transplantation. To this end, Ji et al. (2024) employed a surface flattening technique based on the finite element method (FEM) and utilized the minimization of potential energy to reduce element-wise deformations. Given the successful implementation of surface flattening in similar contexts, it is a promising technique for the objectives of this study.

Although computer-assisted jaw reconstruction has revolutionized surgical practice by reducing operation time, decreasing treatment costs, and improving reconstruction precision (Serrano et al., 2019; Logan et al., 2013), there is currently no digital tool available for the algorithmic design of flaps in tongue reconstruction. This makes the surgery heavily dependent on the experience of the surgeons. This study evaluates the feasibility of using a surface flattening algorithm to determine the 2D geometry of the flap required for reconstructing a patient-specific tongue defect. An optimal flap design will enable more accurate surgical planning and increase confidence during the procedure. By standardizing the flap design process, this tool could reduce reliance on individual surgeons and ultimately improve both functional and aesthetic outcomes for patients with head and neck cancer.

Exploring algorithmic strategies for optimal flap design

Central to the surface flattening process is the challenge of minimizing distortion: ensuring that angles, areas, and distances are represented as accurately as possible on the 2D plane. Achieving an optimal balance in this regard is a complex task, often necessitating a strategic compromise between conformal (angle-preserving) and authalic (area-preserving) distortions. It is impossible to fully preserve both area and angle during flattening, so the selection of a technique depends on the specific application and the required qualities. For instance, conformal mapping is preferred in texture mapping to preserve angles and shapes, while optimization-based methods are chosen in fabrication to minimize area distortion and better replicate the physical behavior of materials under deformation.

To achieve an optimal design for free flaps, it is crucial to select a surface flattening technique that minimizes area and angle distortions. These distortions indicate the severity of deformation, and a suitable technique should aim to reduce them to more accurately imitate the real-world behavior of flap deformation. While recent developments in computer graphics, such as boundary first flattening, BFF (Sawhney and Crane, 2018), have been effective in preserving area and angles, they do not account for the underlying physics of deformation, surface thickness, and material properties. In physical deformation problems, such as sheet metal forming for car body fabrication, the quality of the final product heavily depends on the initial flat sheet's shape, thickness, and material properties. Therefore, engineers often prefer energy-based algorithms, such as the inverse finite element method (IFEM), which incorporate all these factors. IFEM employs an iterative algorithm to estimate an efficient flat surface, and recent developments have demonstrated its effectiveness in sheet metal forming applications (Bostan Shirin and Assempour, 2014). Nevertheless, it is necessary to evaluate the algorithm's performance specifically in the context of free flap design for tongue reconstruction.

3.2 Materials and methods

In the present pre-clinical study, we introduce the IFEM as a promising algorithm for free flap design and investigate its applicability to tongue reconstruction. To demonstrate this, we selected a common oral hemiglossectomy reconstruction as a case study (depicted in Figure 3.1), allowing us to showcase the algorithm's potential in a representative scenario. For this study, a standard and well-established tongue model, based on previous modeling works (Buchaillard et al., 2007; Buchaillard et al., 2009; Stavness et al., 2011), was deemed sufficient. Consequently, we adopted this standard model to artificially simulate hemiglossectomy surgical removal and then design an optimal flap for tongue reconstruction. It is important to note that this phase of the research involved no real patients and solely utilized the simulated model to assess the algorithm's effectiveness. The model exists in hexahedral discretization form; consequently, our first step was to extract its boundary representation as an STL file (Figure 3.2a). Subsequently, we utilized the smoothing and

remeshing algorithms available in Autodesk MeshMixer software (www.meshmixer.com) to process this boundary representation and retrieve its organic form. Furthermore, a uniform triangular mesh made the geometry suitable for the flattening algorithms (Figure 3.2b).

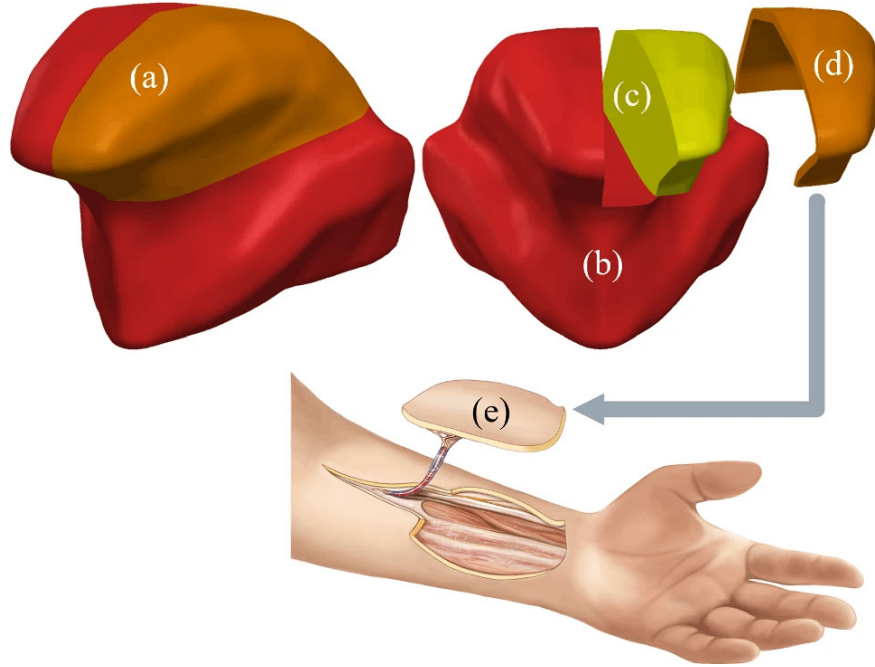


Figure 3.1 Schematic representation of resection and free flap in oral hemiglossectomy reconstruction. (a) indicates the defect requiring surgical removal, while (b) shows the remaining native portion of the tongue. (c) Represents the required beavertail bulk for reconstruction, and (d) shows the outer surface that needs to be recovered with a free flap (e).

The next step involves planning the surgical removal and separating this geometry from the rest of the tongue (Figure 3.1a). The BT-RFFF can be tailored to fit the specific defect, allowing for a more precise and independent design of free flaps (Seikaly et al., 2009). This flexibility allows for dividing the defect into two subsections, as illustrated in Figure 3.1c,d. In this figure, (c) denotes the internal bulk, which is comprised of the fat underneath the skin, the beavertail fat tissue, or a combination of both, while (d) represents the outer surface, which is reconstructed using the skin of the free flap (e). This surface is subsequently re-meshed with uniform membrane triangle elements using Autodesk MeshMixer software (Figure 3.2d). The surface is then introduced to the BFF algorithm to generate a reasonable initial guess (Figure 3.2e), serving as an input for the IFEM algorithm.

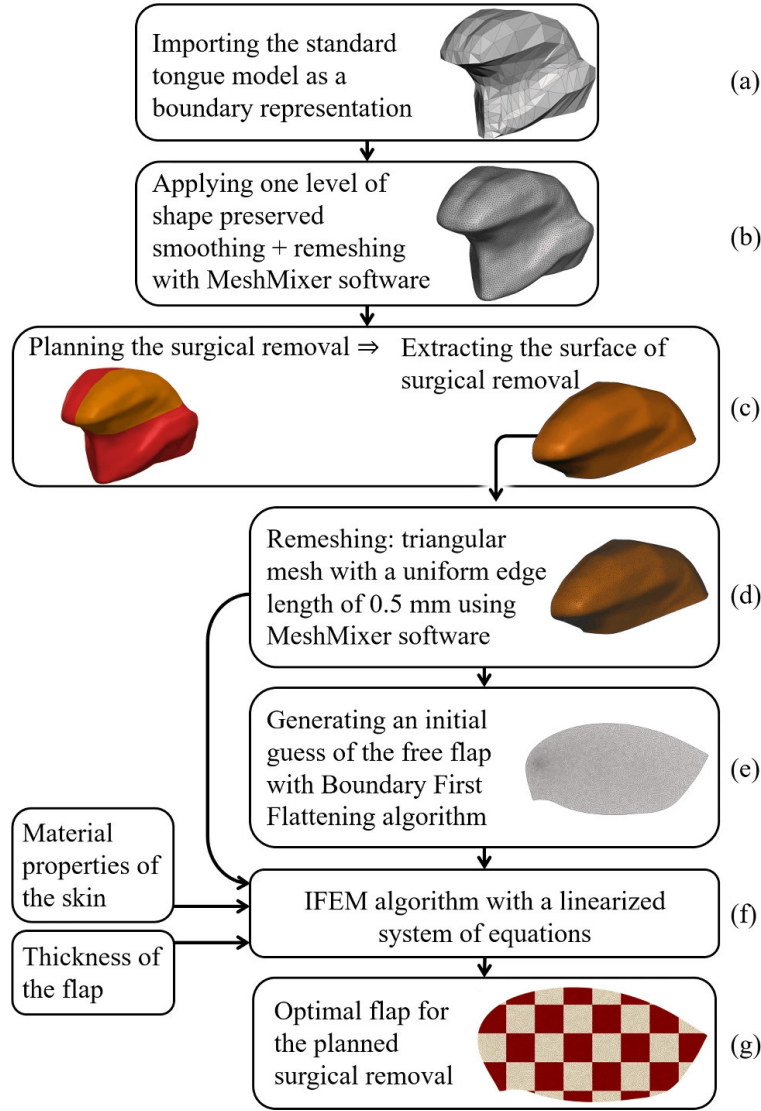


Figure 3.2 The overall workflow: from geometry pre-processing to optimal flap design using the IFEM algorithm

Figure 3.3 illustrates the fundamental concept of the IFEM algorithm. In this inverse approach, we start with the known final deformed state of the flap as a 3D surface (Figure 3.1d). Additionally, the thickness of the undeformed free flap is assumed to be known; for this study, we adopted a thickness of two millimeters for the flap as suggested by Baskin et al. (2019). The objective is to estimate an optimal flap that can be deformed into the final 3D shape while minimizing deformation energy. The specific IFEM algorithm utilized in our study is adapted from Bostan Shirin and Assempour (2014) with a linearized system of equations. Such linearization enhances the

algorithm's robustness and ensures convergence, without compromising on accuracy (Isazadeh et al., 2021). This approach implements linearization in each iteration, adeptly capturing the material's nonlinear behavior through the iterative inverse process.

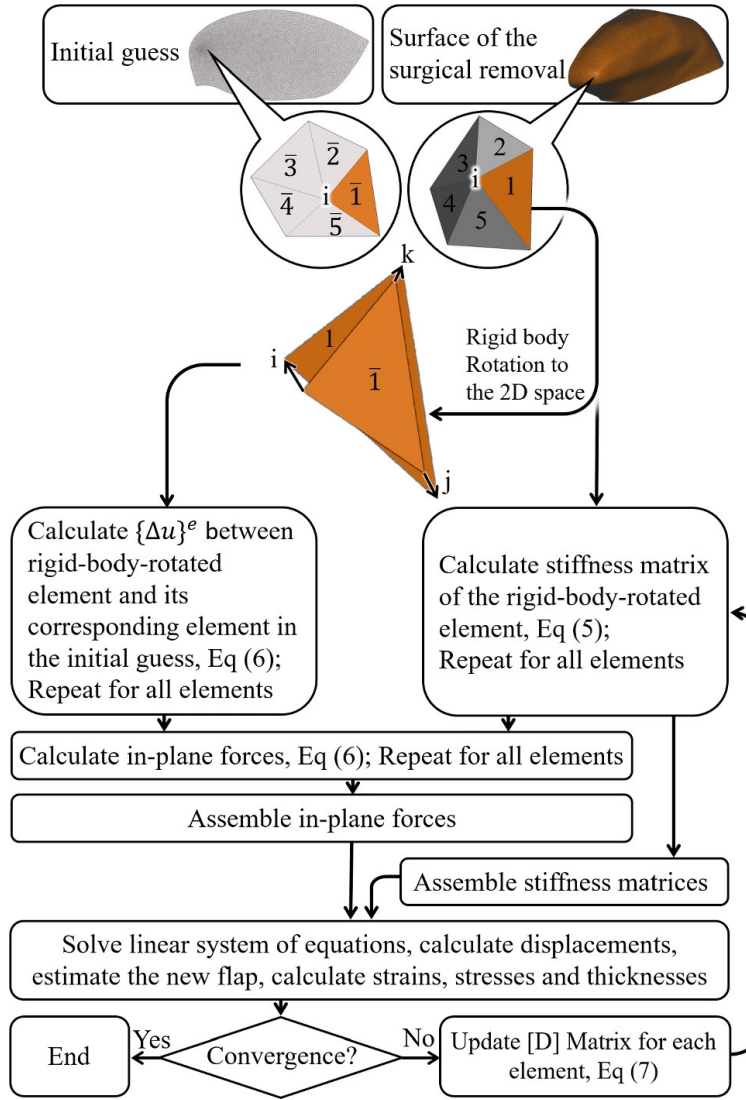


Figure 3.3 Flow diagram of IFEM algorithm used in this study

The IFEM algorithm employed in our study is formulated based on the principle of minimum potential energy and operates under several key assumptions. Firstly, it conceptualizes the deformation as a plane stress problem, focusing solely on the initial and final states of the deformation without considering intermediate stages. This approach is iterative: At the outset, the undeformed body is taken as the initial guess, and through successive iterations, this model is refined until it converges to the optimal flap design. One of the algorithm's significant strengths is its capacity to

calculate strains, stresses, and thickness distribution within the deformed flap. This capability is instrumental in evaluating the severity of deformation and ensuring that the designed flap closely mimics the physical behavior of natural tissues. However, it is important to note a critical limitation: The IFEM algorithm in its current form is exclusively focused on the deformation of the free flap. It does not account for the deformation of other components of the tongue, such as the muscles and the beavertail fat tissue.

What follows is a brief explanation of the kinematics and finite element formulation of the IFEM algorithm. In the final configuration, the increment of potential energy is formulated as follows (Bostan Shirin and Assempour, 2014):

$$\delta\psi = \sum_{e=1}^n \int_{V^e} \{\delta\boldsymbol{\varepsilon}^e\}^T \{\boldsymbol{\sigma}^e\} dV - \sum_{e=1}^n \int_{S^e} \{\delta\mathbf{u}^e\}^T \{\mathbf{T}^e\} dS \quad (3.1)$$

where $\{\delta\mathbf{u}\}$, $\{\boldsymbol{\sigma}\}$, $\{\boldsymbol{\varepsilon}\}$, and $\{\mathbf{T}\}$ represent the incremental displacement of the nodes, the stress vector, the strain vector, and the boundary tractions, respectively.

The relationship between the stress vector and the displacement vector is expressed as follows:

$$\{\boldsymbol{\sigma}^e\} = [D^e]\{\boldsymbol{\varepsilon}^e\} = [D^e][B^e]\{\mathbf{u}^e\} \quad (3.2)$$

where $[D]$ represents the material property matrix, and $[B]$ is the strain operator, as defined below for a membrane triangular element in the two-dimensional space, with y_{jk} representing the difference $y_k - y_j$. The indices i, j, k denote the local nodal numbering on the triangular element.

$$[B^e] = \begin{bmatrix} -y_{jk} & 0 & -y_{ki} & 0 & -y_{ij} & 0 \\ 0 & x_{jk} & 0 & x_{ki} & 0 & x_{ij} \\ x_{jk} & -y_{jk} & x_{ki} & -y_{ki} & x_{ij} & -y_{ij} \end{bmatrix} \quad (3.3)$$

As stated, $[B^e]$ is considered linear in each iteration. However, ultimately, the IFEM captures the nonlinearity of the material behavior through an iterative inverse approach. Based on the principle of virtual work and considering triangular membrane elements, Eq. (3.1) yields the following (Bostan Shirin and Assempour, 2014):

$$\sum_{e=1}^n [B^e]^T [D^e] [B^e] \{\mathbf{u}^e\} A^e t^e - \sum_{e=1}^n \{F^e\} = 0 \quad (3.4)$$

in which $\{F\}$ are the external forces, calculated in each iteration based on the updated material property matrix, area A^e , and thickness t^e of each element. In every iteration, nodal displacements are determined by solving this linear system of equations.

Figure 3.3 briefly describes the iterative approach in the IFEM algorithm. It first takes the 3D surface and an initial guess for the solution as two inputs. Since equilibrium should be satisfied in the deformed body, all elements of the 3D surface are transformed to the 2D space through a rigid body rotation and are subsequently considered for stiffness matrix calculations as below:

$$\mathbf{k}^e = [B^e]^T [D^e] [B^e] A^e t^e \quad (3.5)$$

External forces F are then assumed to be planar and are generated during the flattening of the deformed body. The calculation of the external forces requires determining the in-plane forces for each element:

$$\begin{aligned} \{f_{\text{in-plane}}^e\} &= [k^e] \{\Delta u^e\}, \\ \{\Delta u^e\} &= \{x_{\text{initial guess}}^e\} - \{x_{\text{transferred}}^e\} \end{aligned} \quad (3.6)$$

in which $\{x_{\text{transferred}}^e\}$ are the nodal coordinates of the transferred element, and $\{x_{\text{initial guess}}^e\}$ denotes the nodal coordinates of the corresponding element in the initial guess. Having determined the in-plane forces for all elements, the external force at each node is calculated by summing the contributions from all elements sharing that node. For instance, in Figure 3.3, the external force at node i is calculated as the resultant of adding the contributing in-plane forces from elements 1 to 5 at node i . Therefore, in Eq. (3.4) the only unknowns are the material properties matrix $[D]$ and the thickness (t^e) for each element. Initially, the thickness of all elements is set equal to the thickness of the free flap. This thickness is updated at the end of each iteration. Moreover, the material properties matrix is calculated for a plane stress problem as follows:

$$[D]^e = \frac{\hat{E}^e}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} \quad (3.7)$$

where $\hat{E}^e = f(\bar{\sigma}, \bar{\varepsilon}) = \frac{\bar{\sigma}}{\bar{\varepsilon}}$ is the corrected Young's modulus used to capture each element's nonlinear behavior through iterations. Here, $\bar{\varepsilon}$, $\bar{\sigma}$, and ν represent the equivalent strain, equivalent stress, and Poisson ratio of the skin, respectively. For the incompressible hyperelastic skin, $\nu = 0.5$ is assumed, and the equivalent strains and stresses are updated at the end of each iteration. For the initial step, $\hat{E}^e = 1$ MPa is selected for all elements. This arbitrary initial value does not affect the solution, and the iterative approach converges to its solution regardless of this value (Bostan Shirin and Assempour, 2014).

Assembling the in-plane forces and stiffness matrices allows us to solve the linear system of Eq. (3.4) and calculate the nodal displacements. The initial guess of the flap is then updated, resulting in a new estimation. Given the initial and final configurations of the deformation, we measure its severity through element-wise calculation of stretches using the continuum mechanics of large deformations, as explained in (Bonet and Wood, 2008). This process yields the equivalent strain $\bar{\varepsilon}$ for each element, which is then used to calculate the equivalent stress $\bar{\sigma}$ based on the actual material behavior of the skin. We adopted this relationship from an experimental cadaver study by Ní Annaidh et al. (2012), considering the average of all measurements and approximating the stress-strain relationship with a power function ($\sigma = K\varepsilon^n$, where σ , ε , K , and n are true stress, true strain, first, and second constant, respectively). For this feasibility study, we assumed that the anisotropic behavior of the forearm skin is not predominant and only considered the isotropic behavior. The constants were approximated to values of 140.9 MPa and 2.6, with the coefficient of determination (R^2) for this approximation being 99.14%. They also reported a mean failure true strain (0.43) and mean ultimate tensile strength (21.6 MPa), which we considered as thresholds for safety factor calculation.

The calculated ε and σ are then used to update each element's material properties matrix (Eq. (3.7)), the thickness distribution on the final body, and the stiffness matrix of all elements (Eq. (3.5)). The iterative process recalculates the in-plane forces and solves the new linear system

of equations to estimate a new flap design (see Figure 3.3). Convergence is achieved when for all elements there is a negligible change in the corrected Young's modulus between two subsequent iterations:

$$\frac{\hat{E}_{\text{iteration}(n)} - \hat{E}_{\text{iteration}(n-1)}}{\hat{E}_{\text{iteration}(n-1)}} < 0.001 \quad (3.8)$$

Figure 3.4 demonstrates this iterative process up to the convergence point for one element. Initially, the value of $\hat{E}^e$ results in a calculated value for $\bar{\epsilon}$, which is then used to find the corresponding $\bar{\sigma}$ based on the actual material behavior. These two values are used to calculate the slope for the next iteration ($\hat{E}^e = \bar{\sigma}/\bar{\epsilon}$).

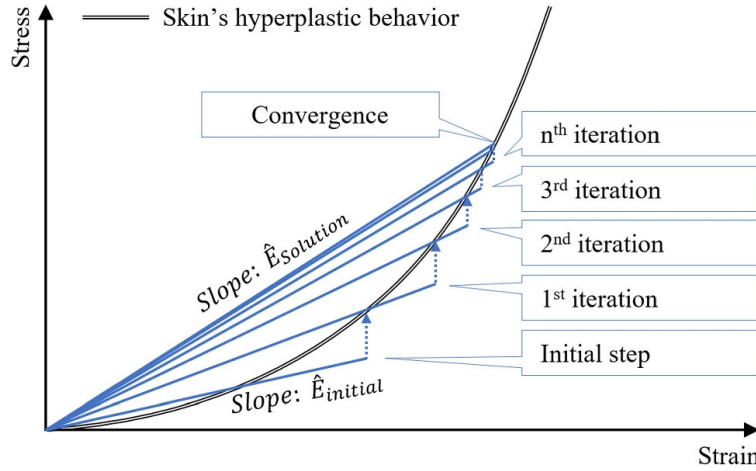


Figure 3.4 Iterative convergence process for the corrected Young's modulus

IFEM is an iterative algorithm that employs the principle of minimizing potential energy to estimate the optimal undeformed configuration. Consequently, the converged step of the IFEM procedure yields an optimal flap design. In practical terms, this means that the deformation of the designed flap (Figure 3.1e) to the curved surface (Figure 3.1d) results in minimal deformation energy, closely mimicking the skin's physical behavior.

To assess the effectiveness of the designed flap, we investigated the strain, stress, and thickness distribution on the deformed body. These calculated outcome variables are indicative of the deformation's severity and provide insights into the potential tethering of the tongue post-surgery. The inclusion of thickness distribution as a criterion is crucial because thinning of the flap signals bi-axial stretches in the deformed body. These stretches can lead to tensions, potentially resulting in

the tethering of the tongue, thereby reducing mobility and causing functional disorders. For example, stretching a flap in two perpendicular directions can lead to it becoming thinner, generating stresses, and ultimately resisting further stretching. Such resistance might cause tethering in the new tongue. Additionally, local thickening of the flap could lead to wrinkles, which are aesthetically undesirable and may cause accidental injuries due to occlusion of the jaw. Therefore, post-reconstruction, it is desirable to achieve minimal stress and an even thickness distribution across the flap to mitigate any unwanted tensions. The visual and statistical representations of these field variables are detailed in the Results section.

3.3 Results

In this study, we focused on a hemiglossectomy reconstruction as a representative case study. Initially, we artificially simulated a hemiglossectomy surgical removal, followed by designing an optimal flap for tongue reconstruction. To accurately define the defect, two incisions were made in the tongue model. Figure 3.5b displays the coronal view of the tongue, illustrating a planar cut intersecting with the sagittal plane. The second incision, which follows a curved path, is depicted as a curved line in the sagittal view (Figure 3.5c), effectively separating the defect from the rest of the tongue. We discretized the surface of this defect using triangular membrane elements, resulting in a model with 10,532 nodes and 20,698 elements.

Employing the IFEM algorithm, we generated an optimal free flap for the planned surgical removal. The computational process was executed on a standard desktop computer, taking approximately 90 s to reach a converged state after 8 iterations. The designed flap's surface area closely matched that of the resection area (0.7% difference). The outcome of this algorithmic design process is presented in Figure 3.5d, including isometric, coronal, and sagittal views of the neotongue in Figure 3.5a-c. For illustrative purposes, we applied a one-to-one mapping of a chess pattern to demonstrate the transformation from the 2D flap to the surgical removal. Additionally, the inclusion of landmarks (I), (II), and (III) across different views aids in orienting the reader and correlating various regions of the flap with the surgical removal.

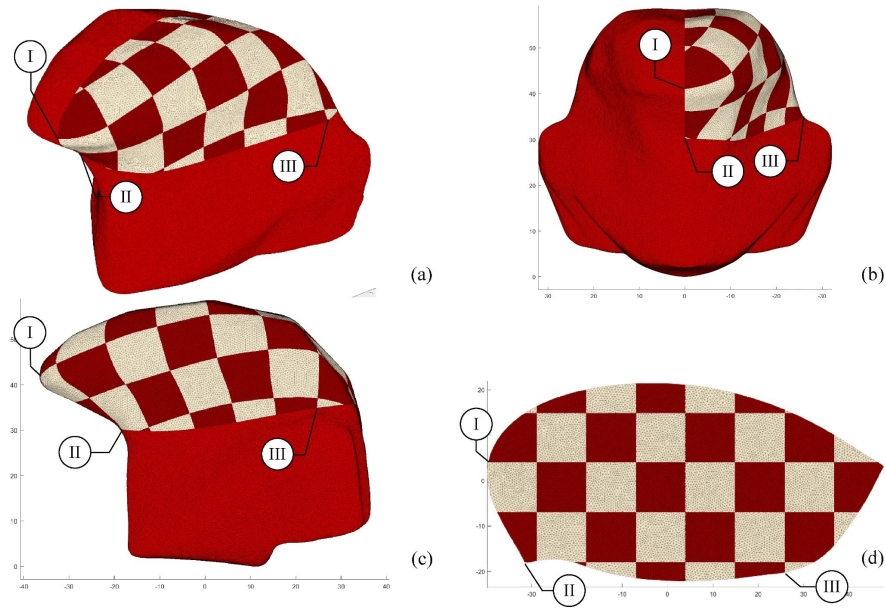


Figure 3.5 Depiction of the artificially simulated hemiglossectomy defect and the corresponding free flap design in this case study. (a) provides an isometric view of the tongue with the defect displayed in a different color. (b) and (c) display the same geometry from coronal and sagittal views, respectively. (d) illustrates the designed flap using IFEM algorithm. (I) indicates the anterior mid-sagittal margin corresponding to the tip of the tongue. (II) indicates the anterior–inferior margin on the mid-sagittal plane. (III) indicates the posterior–lateral margin of the resection. All dimensions are in millimeters

Figures 3.6, 3.7, and 3.8 illustrate the results of the deformation from the 2D flap to the spatially curved surface. Figure 3.6a shows the color-coded thickness distribution of the deformed flap on the neotongue. Accompanying this, Figure 3.6c,d displays the histogram and statistical box plot of the thickness distribution, respectively. In these plots, the dashed line indicates the average, while the solid line represents the median of the distribution. Figures 3.7 and 3.8 follow a similar presentation format, showcasing the equivalent strain and stress distributions, respectively. Since the heatmaps on the 3D geometry present some blind spots, we have adapted these heatmaps for 2D representation, effectively mapping them onto the 2D flap layout. This modification not only illuminates the previously obscured areas but also provides a clearer and more comprehensive view of which regions on the flap are likely to undergo the most significant deformation post-reconstruction. All these plots were generated using MATLAB (MathWorks, Inc., Natick, MA, USA). Additionally, Table 3.1 provides a summary of the mean, median, and standard deviation of these distributions.

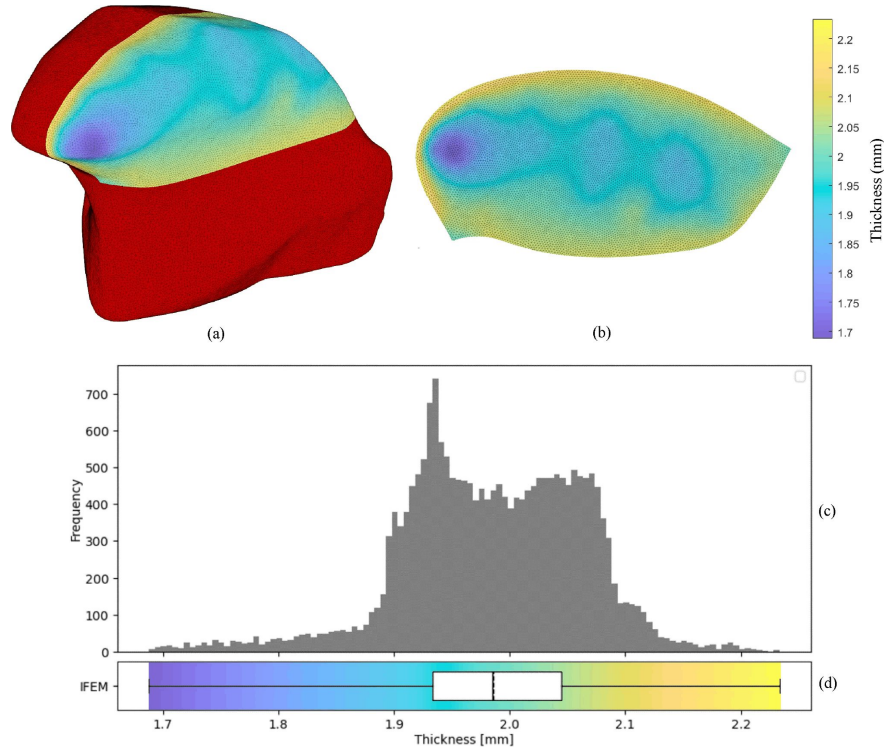


Figure 3.6 Thickness distribution of the designed flap after deformation. a Illustrates the color-coded thickness distribution, b depicts this distribution mapped onto the 2D flap, c presents a histogram of the distribution, and d shows a statistical boxplot of the distribution. The reader is referred to the web version of this paper for the color representation of this figure

3.4 Discussion

The importance of accurately designing the shape and dimensions of the flap is paramount, as imprecise designs can lead to undesirable aesthetic outcomes and diminished functionality, particularly due to the tethering of the tongue. In this work, the success of the flap design was evaluated by measuring the thickness, stress, and strain distribution post-reconstruction. Ideally, after reconstruction, all stress and strain values would be zero, and the flap would maintain its initial thickness. However, achieving such a scenario is practically unattainable when reconstructing a 3D curved surface using a flat flap. A certain degree of deformation is inevitable in order to mold the flat flap to the complex topography of the 3D surface, which naturally leads to some compromise in stress and strain distribution. The novel method of this work is designed to minimize these compromises. We compared the observed results with both the theoretical ideal and the known thresholds of skin tissue behavior. Equally crucial is the flap's overall surface area, which should closely mirror that

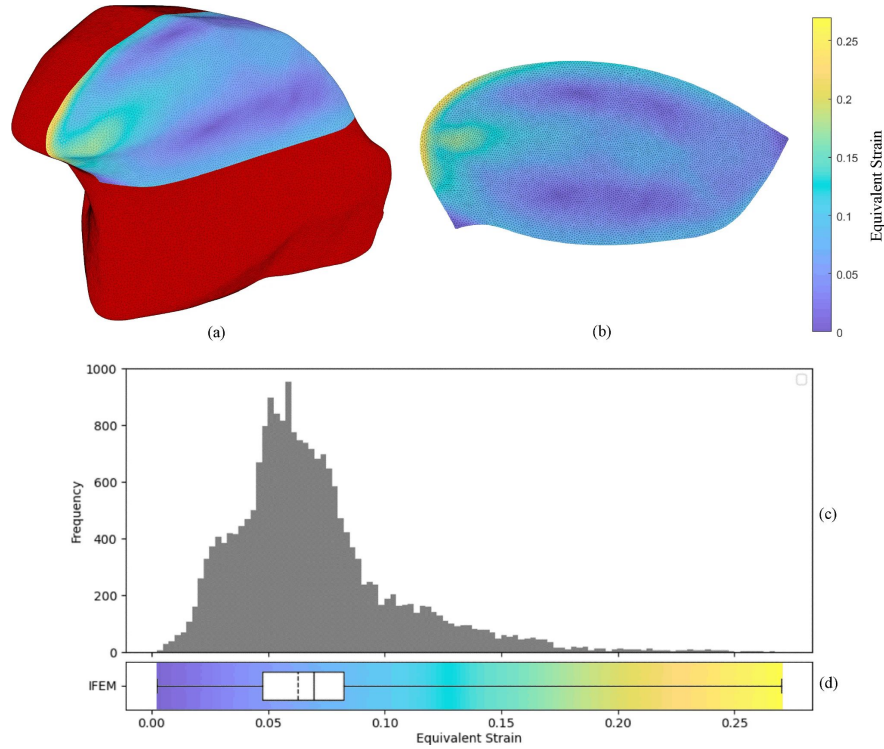


Figure 3.7 Equivalent strain distribution of the designed flap after deformation. a Illustrates the color-coded equivalent strain distribution, b depicts this distribution mapped onto the 2D flap, c presents a histogram of the distribution, and d shows a statistical boxplot of the distribution. The reader is referred to the web version of this paper for the color representation of this figure

of the surgical removal. This agreement is vital for facilitating the recovery of essential interactions with other oral structures. In assessing the design's effectiveness, we took into account all these outcomes, striking a balance between the ideal and the practical.

One of the key achievements of this study is the demonstrated capability of the IFEM in designing a free flap for the planned surgical removal. The chess pattern on the flap (Figure 3.5) aids in visualizing the flap's deformation after reconstruction. This visual aid suggests that the flap's deformation aligns well with our intuitive understanding of it. Furthermore, the measurement of the flap's overall surface area supports the rationale that the deformation is reasonable. The surface area of the designed flap (2964 mm²) closely matches that of the surgical removal (2943 mm²). Such a close match is a strong indicator that the deformation of the designed flap into its final shape is consistent with the incompressibility properties of skin tissue. This finding supports the feasibility of the flap conforming to the curved surface of the resection area without significant compression or

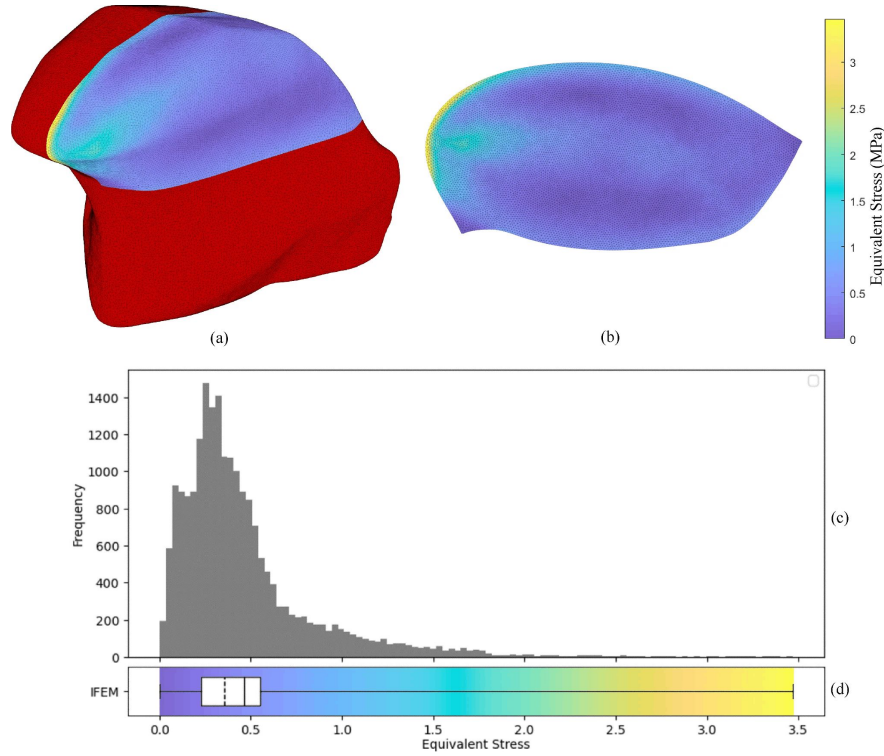


Figure 3.8 Equivalent stress distribution of the designed flap after deformation. a Illustrates the color-coded equivalent stress distribution, b depicts this distribution mapped onto the 2D flap, c presents a histogram of the distribution, and d shows a statistical boxplot of the distribution. The reader is referred to the web version of this paper for the color representation of this figure

expansion, which is a critical consideration in the success of such reconstructive procedures. It is worth noting that the maximum deviation of the flap designed using IFEM from the initial guess is only 3 mm, suggesting that even the boundary first flattening (BFF) algorithm surpasses mere guesswork in flap design. However, the most precise flap design is achievable with IFEM. While these results appear acceptable, we conducted further investigations to quantitatively assess the quality of this deformation.

The success of the flap design can be gauged by measuring the thickness distribution after reconstruction, which is the most tangible metric. The ideal scenario is where the flap's thickness remains consistently close to 2 mm—the initial thickness—showcasing a narrow distribution and minimal deviation. Such uniformity is indicative of minimal bi-axial stretching, which is crucial as it suggests a reduced likelihood of post-surgical tethering of the tongue. The histogram in Figure 3.6b shows a concentration of thickness values around two millimeters with a small standard

deviation, as detailed in Table 3.1. A balanced distribution without significant skewness is key. Should the histogram show a skewness to the left or right, it would indicate excessive thinning or thickening of the flap post-deformation. This is a critical point as it implies an inadequate design, potentially leading to tethering and wrinkling in the reconstructed tongue. In essence, an ideal thickness distribution is one that closely mirrors the original thickness of the flap, ensuring both functional efficiency and aesthetic integrity in the postoperative outcomes.

The thinnest area appears in the region of high curvature, specifically at the anterior–lateral tip of the reconstructed tongue, where it thins down to 1.7 mm. Several factors could contribute to this thinning. First, our simulation only accounted for the deformation of the flap. A significant portion of the thinning could be counterbalanced by including the deformation of the native tongue and the fat tissue in a more comprehensive simulation. Secondly, the material properties used in this research were adopted from a study using cadavers that were, on average, 4 days old and from a very elderly population. This suggests that the skin material properties in our study might be stiffer than those of fresh skin. Incorporating more accurate skin properties could potentially address this thinning issue. It is worth mentioning that the thickness of the majority of elements is very close to the initial thickness of the flap, with 85% of elements falling within a range of 1.9 to 2.1 mm. This indicates that the overall deformation is within acceptable limits.

Equivalent strains and stresses serve as vital indicators of the severity of deformation. Lower values are preferred to prevent tissue damage and undesired tensions in the neotongue, as they may result in reduced mobility. According to the histogram of the equivalent strain distribution in Figure 3.7c, the maximum strain (0.27) is significantly less than the threshold (0.43), indicating that the overall deformation is within safe limits. Equivalent stress is a more reliable predictor of tissue rupture. As shown in Figure 3.8a, high stresses occur at the tip of the tongue on the mid-sagittal plane. These stresses are compressive and occur due to the high curvature of the deformed flap in that area. In other words, the flap's border is shaped into a high curvature, resulting in compression. The thickening of the flap in that area also confirms this observation. It should be noted that in a more comprehensive simulation, which accounts for factors such as the deformation of the beavertail fat tissue and the native portion of the tongue, most of these compressive stresses would likely be alleviated. However, this hypothesis warrants further investigation in future studies.

3.4. Discussion

Maximum tensile stresses do not exceed 2 MPa, which is ten times lower than the threshold for damaging the tissue. This safety factor of ten assures that the deformation is safe. Additionally, these relatively low stresses imply that only minor deformations are anticipated when the neotongue is in a relaxed state. This means that, upon completion of the reconstruction, the relatively low stresses would result in minimal deviations from the intended final shape of the neotongue, thereby enhancing its aesthetics and mobility. It is vital to reiterate that including the native tongue's deformation could significantly reduce extreme conditions and further alleviate strains and stresses. Moreover, using material properties that accurately reflect fresh skin would simulate a less stiff skin model, leading to reduced stress levels and improved mobility in the reconstructed tongue. Considering these factors, we can deduce that the simplified simulation used in this study provides the most conservative, yet effective, evaluation of flap deformation in a virtual reconstruction simulation.

Table 3.1 Mean, median, and standard deviation of results

Descriptive statistic	Value
Thickness (mm)	
Mean	1.985
Median	1.986
Standard deviation	0.006
Equivalent strain	
Mean	0.069
Median	0.063
Standard deviation	0.0012
Equivalent stress (MPa)	
Mean	0.464
Median	0.357
Standard deviation	0.156

We acknowledge several limitations to this study, which could be seen as opportunities for future research. Firstly, we relied solely on a standard, well-established tongue model from the literature, the anatomical accuracy of which leaves room for improvement. Ideally, we would collect

multiple images from patients undergoing tongue reconstruction surgery. Secondly, the mid-sagittal glossectomy resection line typically extends inferiorly to the floor of the mouth. Though this surgical perspective is important for a real hemiglossectomy reconstruction, for the purposes of this feasibility study, our example is sufficient. Additionally, we examined flap design only for hemiglossectomy; future studies should consider other types of tongue reconstruction. In our virtual reconstruction, the only deformable continuum was the flap. A more comprehensive simulation could model the deformation of the remainder of the neotongue and investigate the resting state of the neotongue. While our virtual reconstruction examined three measurements and interpreted the results in terms of their impact on the quality of the neotongue, additional measurements—such as the relative values of the principal strains and their potential to predict rupture and wrinkles during reconstructions—may also be beneficial. Further objectives for future research could include considering the non-homogeneity and directional properties of the flap, measuring the stiffness of the reconstructed neotongue, conducting mobility analysis of the neotongue, examining more tongue postures, and carrying out experimental validations.

3.5 Conclusion

The choice of IFEM for achieving an acceptable flap design has shown effective, as the algorithm precisely designed a free flap while preserving the required surface area for reconstruction. The results of the virtual reconstruction confirmed this, as the thickness, strain, and stress analyses demonstrated acceptable flap deformations. The safety factor of ten implies that the flap designed is safe for use in reconstruction, and the conservative overestimation of strains and stresses in the simplified simulation adds a layer of assurance. IFEM, taking into consideration the thickness and material properties of the flap along with the underlying physics of deformation, proves to be an apt digital tool for accomplishing optimal flap design in tongue reconstruction. The development of digital tools for precise flap design has the potential to enhance the precision and efficiency of tongue reconstruction, possibly leading to improved functional outcomes.

Funding This research has been funded by Alberta Cancer Foundation, Grant Number 27601.

Conflict of interest The authors declare that they have no conflict of interest.

References

- Baskin, R. M., Seikaly, H., Sawhney, R., Danan, D., Burt, M., Idris, S., Shama, M., Boyce, B., and Dziegielewska, P. T. (2019). “Tongue Reconstruction: Rebuilding Mobile Three-Dimensional Structures from Immobile Two-Dimensional Substrates, a Fresh Cadaver Study.” In: *Head Neck-J. Sci. Spec. Head Neck* 41.10, pp. 3693–3699. ISSN: 1043-3074, 1097-0347. DOI: [10.1002/hed.25889](https://doi.org/10.1002/hed.25889).
- Bonet, J. and Wood, R. D. (2008). *Nonlinear Continuum Mechanics for Finite Element Analysis*. 2nd ed. Cambridge University Press. ISBN: 978-0-521-83870-2 978-0-511-75544-6. DOI: [10.1017/CB09780511755446](https://doi.org/10.1017/CB09780511755446).
- Bostan Shirin, M. and Assempour, A. (2014). “Some Improvements on the Unfolding Inverse Finite Element Method for Simulation of Deep Drawing Process.” In: *Int J Adv Manuf Technol* 72.1, pp. 447–456. ISSN: 1433-3015. DOI: [10.1007/s00170-014-5663-4](https://doi.org/10.1007/s00170-014-5663-4).
- Buchaillard, S., Brix, M., Perrier, P., and Payan, Y. (2007). “Simulations of the Consequences of Tongue Surgery on Tongue Mobility: Implications for Speech Production in Post-Surgery Conditions.” In: *Int. J. Med. Robot. Comput. Assist. Surg.* 3.3, pp. 252–261. ISSN: 1478-5951, 1478-596X. DOI: [10.1002/rcs.142](https://doi.org/10.1002/rcs.142).
- Buchaillard, S., Perrier, P., and Payan, Y. (2009). “A Biomechanical Model of Cardinal Vowel Production: Muscle Activations and the Impact of Gravity on Tongue Positioning.” In: *J. Acoust. Soc. Am.* 126.4, pp. 2033–2051. ISSN: 0001-4966, 1520-8524. DOI: [10.1121/1.3204306](https://doi.org/10.1121/1.3204306).
- Chepeha, D. B., Teknos, T. N., Shargorodsky, J., Sacco, A. G., Lyden, T., Prince, M. E., Bradford, C. R., and Wolf, G. T. (2008). “Rectangle Tongue Template for Reconstruction of the Hemiglossectomy Defect.” In: *Arch. Otolaryngol. Head Neck Surg.* 134.9, pp. 993–998. ISSN: 0886-4470. DOI: [10.1001/archotol.134.9.993](https://doi.org/10.1001/archotol.134.9.993).
- Fan, S., Li, Q.-X., Zhang, H.-Q., Liang, M.-J., Tian, T., Wang, Y.-Y., Lin, Z.-Y., Chen, W.-X., Pan, G.-K., Ahn, M. H.-Y., He, L., Sun, S., Wu, B.-H., Yu, P., and Li, J.-S. (2019a). ““Five-point Eight-Line” Anatomic Flap Design for Precise Hemitongue Reconstruction.” In: *Head Neck-J. Sci. Spec. Head Neck* 41.5, pp. 1359–1366. ISSN: 1043-3074, 1097-0347. DOI: [10.1002/hed.25571](https://doi.org/10.1002/hed.25571).

- Fan, S., Zhang, H., Li, Q., Liang, F., Bai, Z., Chen, W., Sun, S., Yu, P., and Li, J. (2019b). “A Novel Anatomy-Based Five-Points Eight-Line-Segments Technique for Precision Subtotal Tongue Reconstruction: A Pilot Study.” In: *Oral Oncol.* 89, pp. 1–7. ISSN: 13688375. DOI: [10.1016/j.oraloncology.2018.12.004](https://doi.org/10.1016/j.oraloncology.2018.12.004).
- Hanubal, K. S., Reschly, W. J., Conrad, D., Festa, B. M., Weiss, J. P., Shama, M., Danan, D., Hughley, B., and Dziegielewski, P. T. (2023). “The Beavertail Modified Radial Forearm Free Flap: Retrospective Review of a Versatile Technique to Increase Flap Bulk in the Head and Neck.” In: *Microsurgery* 43.8, pp. 767–774. ISSN: 0738-1085, 1098-2752. DOI: [10.1002/micr.31028](https://doi.org/10.1002/micr.31028).
- Hashida, N., Shamoto, H., Maeda, K., Wakabayashi, H., Suzuki, M., and Fujii, T. (2017). “Rehabilitation and Nutritional Support for Sarcopenic Dysphagia and Tongue Atrophy after Glossectomy: A Case Report.” In: *Nutrition* 35, pp. 128–131. ISSN: 08999007. DOI: [10.1016/j.nut.2016.11.003](https://doi.org/10.1016/j.nut.2016.11.003).
- Isazadeh, A. R., Shamloofard, M., and Assempour, A. (2021). “Some Improvements on the One-Step Inverse Isogeometric Analysis by Proposing a Multi-Step Inverse Isogeometric Methodology in Sheet Metal Stamping Processes.” In: *Applied Mathematical Modelling* 91, pp. 476–492. ISSN: 0307904X. DOI: [10.1016/j.apm.2020.09.032](https://doi.org/10.1016/j.apm.2020.09.032).
- Ji, X., Wen, G., Gong, H., Sun, R., and Li, H. (2024). “Three-Dimensional Wound Flattening Method for Mapping Skin Mechanical Properties Based on Finite Element Method.” In: *Comput. Methods Biomech. Biomed. Eng.* 27.2, pp. 237–250. ISSN: 1025-5842, 1476-8259. DOI: [10.1080/10255842.2023.2183347](https://doi.org/10.1080/10255842.2023.2183347).
- Kreiser, J., Meuschke, M., Mistelbauer, G., Preim, B., and Ropinski, T. (2018). “A Survey of Flattening-Based Medical Visualization Techniques.” In: *Computer Graphics Forum* 37.3, pp. 597–624. ISSN: 0167-7055, 1467-8659. DOI: [10.1111/cgf.13445](https://doi.org/10.1111/cgf.13445).
- Liu, Y.-J., Zhang, D.-L., and Yuen, M. M.-F. (2010). “A Survey on CAD Methods in 3D Garment Design.” In: *Computers in Industry* 61.6, pp. 576–593. ISSN: 01663615. DOI: [10.1016/j.compind.2010.03.007](https://doi.org/10.1016/j.compind.2010.03.007).
- Logan, H., Wolfaardt, J., Boulanger, P., Hodgetts, B., and Seikaly, H. (2013). “Exploratory Benchtop Study Evaluating the Use of Surgical Design and Simulation in Fibula Free Flap Mandibular Reconstruction.” In: *J. Otolaryngol. Head Neck Surg.* 42.1, p. 42. ISSN: 1916-0216. DOI: [10.1186/1916-0216-42-42](https://doi.org/10.1186/1916-0216-42-42).

- Ní Annaidh, A., Bruyère, K., Destrade, M., Gilchrist, M. D., and Otténio, M. (2012). “Characterization of the Anisotropic Mechanical Properties of Excised Human Skin.” In: *J. Mech. Behav. Biomed. Mater.* 5.1, pp. 139–148. ISSN: 17516161. DOI: [10.1016/j.jmbbm.2011.08.016](https://doi.org/10.1016/j.jmbbm.2011.08.016).
- Sawhney, R. and Crane, K. (2018). “Boundary First Flattening.” In: *ACM Trans. Graph.* 37.1, pp. 1–14. ISSN: 0730-0301, 1557-7368. DOI: [10.1145/3132705](https://doi.org/10.1145/3132705).
- Seikaly, H., Rieger, J., O’Connell, D., Ansari, K., AlQahtani, K., and Harris, J. (2009). “Beavertail Modification of the Radial Forearm Free Flap in Base of Tongue Reconstruction: Technique and Functional Outcomes.” In: *Head Neck* 31.2, pp. 213–219. ISSN: 1043-3074, 1097-0347. DOI: [10.1002/hed.20953](https://doi.org/10.1002/hed.20953).
- Serrano, C., Van Den Brink, H., Pineau, J., Prognon, P., and Martelli, N. (2019). “Benefits of 3D Printing Applications in Jaw Reconstruction: A Systematic Review and Meta-Analysis.” In: *J. Cranio-MaxilloFac. Surg.* 47.9, pp. 1387–1397. ISSN: 10105182. DOI: [10.1016/j.jcms.2019.06.008](https://doi.org/10.1016/j.jcms.2019.06.008).
- Stavness, I., Lloyd, J. E., Payan, Y., and Fels, S. (2011). “Coupled Hard–Soft Tissue Simulation with Contact and Constraints Applied to Jaw–Tongue–Hyoid Dynamics.” In: *Numer Methods Biomed Eng* 27.3, pp. 367–390. ISSN: 2040-7939, 2040-7947. DOI: [10.1002/cnm.1423](https://doi.org/10.1002/cnm.1423).
- Warnakulasuriya, S. (2009). “Global Epidemiology of Oral and Oropharyngeal Cancer.” In: *Oral Oncol.* 45.4-5, pp. 309–316. ISSN: 13688375. DOI: [10.1016/j.oraloncology.2008.06.002](https://doi.org/10.1016/j.oraloncology.2008.06.002).

Chapter 4

Patient-specific virtual surgical planning for tongue reconstruction: evaluating hyperelastic inverse FEM with four simulated tongue cancer cases

Amir R Isazadeh^{1,*}, Julianna Zenke², Lindsey Westover³, Hadi Seikaly^{2,4} and Daniel Aalto^{1,4}

* This chapter is a reprint of an open access article originally published as: Isazadeh AR, Zenke J, Westover L, Seikaly H and Aalto D (2025) *Patient-specific virtual surgical planning for tongue reconstruction: evaluating hyperelastic inverse FEM with four simulated tongue cancer cases*, *Physics in Medicine & Biology* 70:145017. DOI: [10.1088/1361-6560/adebd8](https://doi.org/10.1088/1361-6560/adebd8). This work is licensed under the Creative Commons Attribution 4.0 licence (CC BY 4.0, <https://creativecommons.org/licenses/by/4.0/>). Any further distribution must maintain attribution to the author(s), the title of the work, journal citation and DOI.

¹ Department of Communication Sciences and Disorders, Faculty of Rehabilitation Medicine, University of Alberta, Edmonton, AB, Canada.

² Division of Otolaryngology, Department of Surgery, Faculty of Medicine and Dentistry, University of Alberta, Edmonton, AB, Canada.

³ Department of Mechanical Engineering & Department of Biomedical Engineering, Faculty of Engineering, University of Alberta, Edmonton, AB, Canada.

⁴ Institute for Reconstructive Sciences in Medicine (iRSM), Misericordia Community Hospital, Edmonton, AB, Canada.

Abstract

Objective. Anatomically and functionally optimal tongue reconstruction after tumor removal presents significant challenges. Current virtual surgical planning (VSP) utilizes patient-specific data with geometric algorithms for free flap design. However, these geometric approaches often inadequately account for complex soft tissue biomechanics. This study introduces a biomechanics-informed VSP algorithm and computationally compares its flap designs against those derived from purely geometric methods. **Approach.** Hyperelastic inverse finite element method (hiFEM) was developed by integrating an Ogden hyperelastic constitutive model into a predecessor algorithm. The planar flap shape is determined by minimizing potential energy when tissue deforms to match patient-specific MRI-derived 3D defect geometry. Four clinically plausible tongue cancer cases were simulated, and resection regions were delineated. For each case, flap designs were generated using hiFEM, its predecessor iFEM, and two geometric flattening techniques: NURBS surface flattening and boundary first flattening (BFF). Intrinsic tissue deformation for these designs was compared across methods and quantified using area stretch metric. **Main results.** Across all simulated cases, hiFEM-generated flap designs required less intrinsic tissue deformation. Maximum area stretch ranged from 1.10–1.12 for hiFEM designs, versus 1.19–1.38 for NURBS flattening and 1.54–1.74 for BFF designs. Furthermore, hiFEM’s area stretch distribution was tighter, centered around one (ideal, no stretch). Geometric comparison showed hiFEM yields flap designs similar to the clinically validated geometric algorithm, NURBS flattening, with an average Hausdorff distance of only 1.3 mm. hiFEM’s distinct advantage is its core objective of minimizing tissue stretch, which has clinical relevance and suggests potential for improved patient outcomes. Computationally, hiFEM demonstrated robustness and efficiency. It converged rapidly (8 to 10 iterations; less than 0.3 s/case), even for complex geometries where iFEM failed. **Significance.** hiFEM offers a biomechanically informed and computationally robust tool for tongue VSP, showing potential for broader application in breast, nasal, and other soft tissue reconstructions.

Keywords: plastic and reconstructive surgery, tongue reconstruction, precision medicine, patient-specific virtual surgical planning, soft tissue modeling, image-guided surgery

4.1 Introduction

Glossectomy, the surgical removal of soft tissue, is the primary treatment modality for tongue cancer (Mannelli et al., 2018; Ansarin et al., 2019; De Berardinis et al., 2022). While small resections may permit primary closure, medium-to-large defects necessitate reconstruction to restore critical functions (Ansarin et al., 2019; Mallet et al., 2009). Microvascular free flap transplantation represents the preferred reconstructive method whenever feasible (Mallet et al., 2009). This procedure involves harvesting a two-dimensional (2D) section of skin and subcutaneous tissue, often from the forearm or thigh, and meticulously shaping it to replicate the complex three-dimensional (3D) anatomy of the resected tongue defect (Baskin et al., 2019; Idris et al., 2018).

Free flap planning, particularly its volume and planar geometry, directly influences the final structure of the post-surgery tongue (neotongue) and the subsequent restoration of oral functions (Seikaly et al., 2003; Seikaly et al., 2009; Urken, 2012; Idris et al., 2018; Baskin et al., 2019). Achieving optimal functional outcomes demands precise control over not only the flap's bulk but also the geometric configuration of the harvested skin. This configuration must accurately contour the defect's outer surface to restore the tongue's anatomical integrity as closely as possible (Idris et al., 2018; Baskin et al., 2019; Fan et al., 2019b; Fan et al., 2019a). Individualized and defect-specific flap planning demonstrates significant benefits for improving tongue mobility and functional recovery. Studies consistently report improved swallowing (Wang et al., 2016; Fan et al., 2019b; Fan et al., 2019a), speech intelligibility (Han et al., 2022; Liu et al., 2023b; Yi et al., 2020), and overall quality of life (QoL) (Liu et al., 2023b) when the flap planning incorporates the patient's unique anatomical characteristics and defect dimensions.

Effective flap planning, however, faces considerable hurdles. Accurately matching the intricate 3D characteristics of diverse tongue defects to a planar flap template remains a significant challenge. Several studies have explored associations between common tongue defects and specific planar shapes to guide flap planning (Idris et al., 2018; Fan et al., 2019b; Fan et al., 2019a; Baskin et al., 2019; Koumoullis et al., 2020; Han et al., 2022). These traditional approaches often involve manual measurements between anatomical landmarks, which are then used to adapt predefined generic

templates (Fan et al., 2019b; Fan et al., 2019a; Wang et al., 2016; Liu et al., 2023a). However, substantial variability exists in defect position, size, and spatial geometry, compounded by individual anatomical differences (Ansarin et al., 2019; Engel et al., 2010; Koumoullis et al., 2020; Liu et al., 2023b). Consequently, these methods can lead to suboptimal flap utilization, intraoperative flap trimming, excessive tensions, and inconsistent functional outcomes, relying heavily on the surgeon's experience and intuition (Fan et al., 2019b; Liu et al., 2023b). This underscores the need for innovative, patient-specific VSP techniques that can provide precise planning and predictable outcomes.

Digital tools and 3D modeling are driving transformative changes in plastic and reconstructive surgery (Logan et al., 2013; Chuka et al., 2017; Papadopoulos-Nydam et al., 2017; Serrano et al., 2019; Idris et al., 2022). These technologies can accurately capture a patient's unique anatomy from medical images, such as MRI or CT scans (Fan et al., 2019b; Fan et al., 2019a; Han et al., 2022; Liu et al., 2023b; Xu et al., 2022). This anatomical data can then inform flap planning, either by adapting predefined templates (Fan et al., 2019b; Fan et al., 2019a; Han et al., 2022; Liu et al., 2023b) or enabling the independent algorithmic design of customized flaps through surface flattening algorithms (Xu et al., 2022).

An alternative perspective on optimizing flap planning considers the intrinsic deformation when shaping a 2D flap into a 3D geometry. Specifically, designing a flap shape that minimizes the required tissue stretch during this transformation is biomechanically desirable for several reasons. Lower intrinsic stretch implies less internal tension within the shaped flap. This reduction potentially reduces the risks of complications such as compromised blood supply, tissue necrosis, impaired healing, and scar formation (Tepole et al., 2014a; Tepole et al., 2014b; Ogawa et al., 2012). It also minimizes tethering to surrounding structures, thereby preserving mobility (Urken, 2012; Fan et al., 2019b; Benjamin et al., 2020; Rui et al., 2022). Furthermore, minimizing the stored elastic energy should reduce the flap's tendency to recoil after suturing, which allows for better conformity of the flap to the defect. Therefore, designing a flap shape that intrinsically requires less stretch is a key consideration for optimizing reconstructive outcomes.

To achieve this, current VSP tools often employ flattening algorithms. These algorithms map the 3D defect surface onto a 2D plane while aiming to minimize some form of geometric distortion (e.g., changes in area, length, or angle) (Huang et al., 2007; Xu et al., 2022). For instance, the Computer-Aided Reconstruction of Soft Tissue (CARST) method, utilizing Non-Uniform Rational B-Splines (NURBS) surface flattening, demonstrated significant clinical advantages (Xu et al., 2022). Their NURBS-based method flattens a digital representation of the resection surface while attempting to preserve geometric properties during the transformation to create individualized flap-cutting guides. A study involving 30 patients showed significant clinical improvements compared to traditional methods, including reduced flap trimming, shorter operative times, shorter hospital stays, and better defect matching (Xu et al., 2022). This work validates the substantial value of virtual surgical planning.

While powerful for geometric translation and computer graphics applications, such methods optimize mapping properties instead of the physical work involved in deformation. Addressing this limitation requires algorithms that accurately account for the biomechanics of soft tissue deformations. Unlike rigid skeletal structures where geometric planning is often sufficient (Serrano et al., 2019; Logan et al., 2013), soft tissues undergo complex, nonlinear, large deformations during surgical manipulation. Predictive surgical planning must therefore incorporate realistic models of the tissue behavior. To bridge this gap, we introduce hiFEM. This method builds upon its predecessor –iFEM (Isazadeh et al., 2024)– but crucially integrates a hyperelastic constitutive model, which is essential for accurately capturing the large, nonlinear mechanical response of the skin under surgical manipulations (Mostafavi Yazdi and Baqersad, 2022). The core principle involves finding the optimal planar flap shape by minimizing the deformation energy within the tissue as it deforms to the 3D defect geometry. We hypothesize that optimizing based on the physical deformation work allows hiFEM to yield a flap design that inherently requires less intrinsic stretch to achieve the target shape, compared to designs derived from geometric flattening models.

This study: (1) implements the hiFEM algorithm, incorporating a validated Ogden hyperelastic model, to generate patient-specific flap designs for four simulated tongue reconstruction cases, (2) quantitatively compares the intrinsic flap deformation required to map the planar flap to the target 3D shape for hiFEM designs against those generated by the prior iFEM method (Isazadeh et al., 2024),

NURBS surface flattening (Xu et al., 2022), and BFF (Sawhney and Crane, 2018); (3) assesses the resultant effective stress within hiFEM-designed flaps against relevant physiological thresholds, and (4) evaluates the computational robustness and convergence rate of hiFEM in comparison to the predecessor iFEM algorithm.

4.2 Method

4.2.1 Patient Anatomy Acquisition

Following approval by the Health Research Ethics Board of Alberta (HREBA.CC-17-0597), a male participant in his early 40s provided written informed consent for magnetic resonance imaging (MRI) of the head and neck region. A Siemens Magnetom Prisma 3T MRI scanner (Siemens Medical Solutions, Erlangen, Germany) acquired a T1-weighted structural scan of the head and neck. This included the cervical spine, mandible, submental tissues, and palate. The imaging parameters include a 1 mm axial slice thickness, a 0.5 mm in-plane resolution, a 12° flip angle, a repetition time (TR) of 4.72 ms, an echo time (TE) of 2.46 ms, a receiver bandwidth of 650 Hz/pixel, a 256 mm field of view (FOV), and a 512×512 matrix size. The resulting images were segmented to produce a 3D model of the tongue using Mimics Medical 17.0 software (Materialise, Leuven, Belgium. 2017).

The 3D model served as the basis for defining four artificially simulated tumor resection cases. These scenarios aligned with Urken’s classification (Urken, 2012) for class I and II tongue cancers. Figure 4.1 illustrates the four cases: (A) a hemiglossectomy that leaves the tip of the oral tongue intact, (B) a glossectomy involving half of the oral tongue, (C) a hemiglossectomy extending to the base of tongue posterior side but sparing the root, and (D) an asymmetric partial glossectomy that includes the anterior two-thirds of the oral tongue but extends across the midline. The use of the asymmetric case aims to demonstrate the algorithm’s capability in handling complex resections. Two perspectives—top view and isometric view—are shown in Figure 4.1 to aid visualization. The geometric complexity of the required reconstructions increases from case A to D, providing

a systematic test of the algorithm’s robustness. Preprocessing involves separating the simulated defect from the tongue. The resection surface was then discretized into triangular elements of approximately 0.8 mm size using MeshMixer software (Autodesk Inc. 2018).

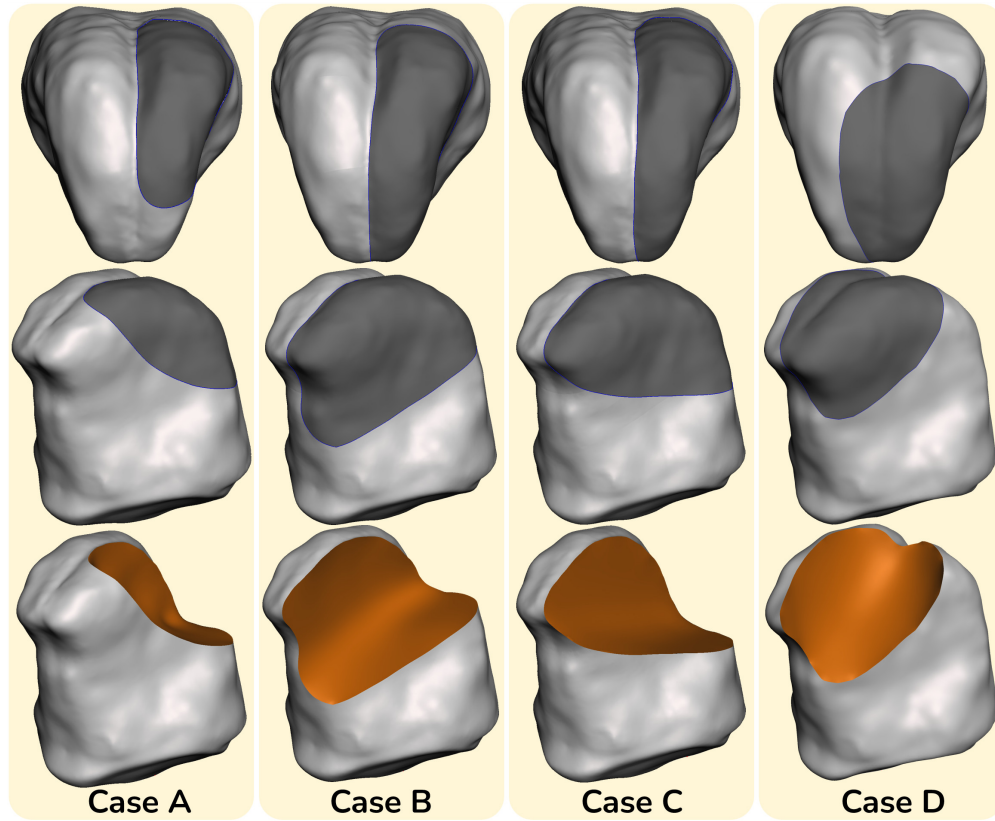


Figure 4.1 Artificially simulated glossectomy cases. The first and second rows show axial and isometric views with the resected area outlined. The third row shows the isometric view after resecting the outlined region.

4.2.2 Algorithmic Flap Design with hiFEM

Virtual free flap planning for tongue reconstruction can be systematically addressed using an inverse approach, introduced as iFEM in our previous work (Isazadeh et al., 2024). Unlike traditional forward Finite Element Methods that simulate deformation from known initial conditions, our inverse strategy (workflow summarized in Figure 4.2) begins with the desired anatomical outcome: the 3D geometry of the resected tongue defect serves as the target configuration. The algorithm then works backward. It iteratively refines an initial guess to ultimately determine the optimal planar

flap shape, defined as the one that will transform into the target 3D configuration with minimal deformation energy. The initial guess is often derived from a geometric flattening technique such as BFF (Sawhney and Crane, 2018).

A key challenge in this iterative optimization arises from the soft tissues' inherent material nonlinearity: their mechanical response, and therefore stiffness, depends on the applied deformation. To computationally manage this complexity within the inverse framework, the algorithm employs an iterative scheme centered on the effective secant modulus ($\hat{E}$). Within each iteration, the deformation state for every element must first be determined. This is typically achieved by quantifying the principal stretches required to map that element from its current 2D configuration to its corresponding state on the fixed 3D target surface. This element-specific deformation is then used for calculating the $\hat{E}$ for that element in the current iteration, based on the chosen nonlinear constitutive model.

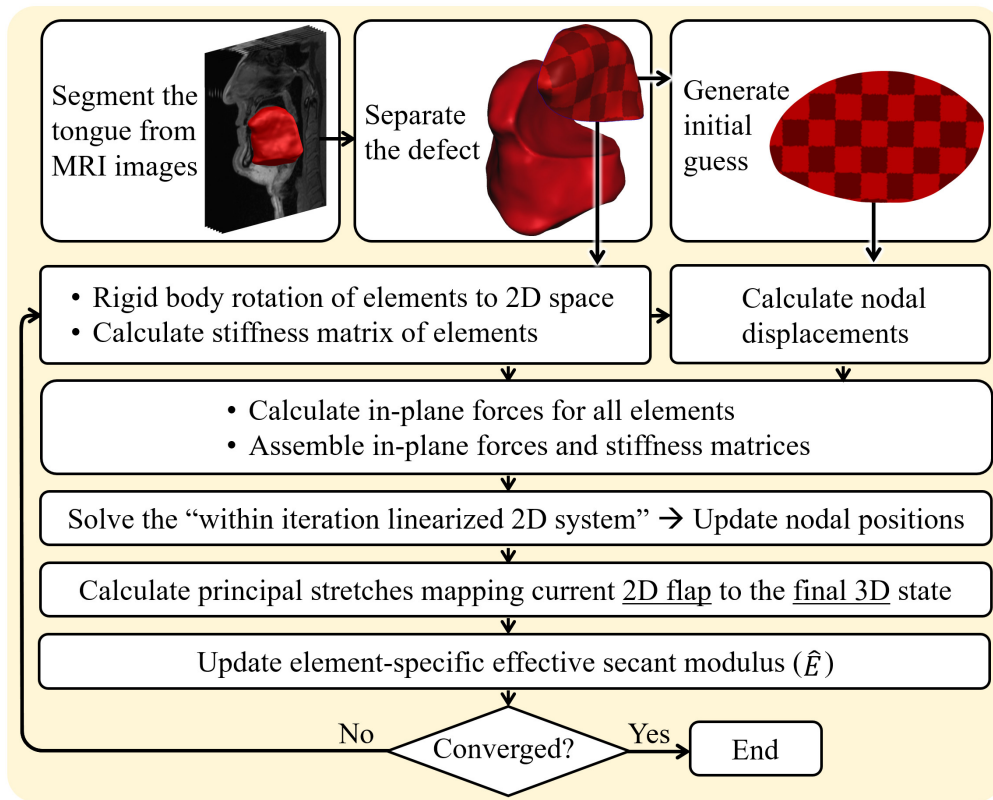


Figure 4.2 Flowchart summarizing the virtual surgical planning process for tongue reconstruction, from MRI image segmentation to optimal flap design.

The assignment of an element-specific $\hat{E}$ permits a temporary linearization of the problem within each iteration. By utilizing the calculated $\hat{E}$, each element can be treated, for the purpose of that computational step, as if it possessed a constant stiffness equal to its current $\hat{E}$. A global system matrix, incorporating the effective secant stiffness values ($\hat{E}$) of all elements, is assembled under plane stress assumptions in the 2D domain. Solution of this linear system yields nodal displacements, which are used to update the 2D planar shape of the flap geometry. This iterative cycle—calculating element deformations, determining element-specific $\hat{E}$ from the constitutive model, linearizing the step and solving the linearized 2D system, and updating the planar geometry—is repeated. Convergence occurs when the changes in the calculated $\hat{E}$ across all elements become negligible between successive iterations.

While the inverse approach offers a structured way to optimize flap shapes, the accuracy and robustness, particularly for complex defect geometries, depend on the fidelity of the material model used to determine the effective secant modulus. The earlier implementation (iFEM) utilized simplified models like the power law (Isazadeh et al., 2024), which may not fully capture the complex, nonlinear stress-strain response of soft tissue under large deformations. This limitation can lead to a failure to converge in challenging cases. To address this, we developed the hyperelastic iFEM (hiFEM). The core advancement in hiFEM is the direct integration of a validated hyperelastic constitutive model into the inverse iterative framework.

Specifically, hiFEM employs the Ogden (N=2) hyperelastic model, chosen because it accurately describes the in-vivo mechanical behavior of human skin across a wide range of deformations (Flynn et al., 2011). For the material properties, we adopted validated isotropic parameters derived from in-vivo tests on 21 participants' forearm skin (Flynn et al., 2011) (Table 1), a common donor site for tongue reconstruction. Furthermore, we assumed a uniform flap thickness of 2 mm, consistent with clinical observations (Baskin et al., 2019; Laurent et al., 2007; Iivarinen et al., 2011; Groves et al., 2013).

Within each iteration of the hiFEM, the process begins by quantifying the physical deformation required to map each element from its current 2D configuration to its designated state on the fixed target 3D surface. This kinematic analysis yields the principal stretches ($\lambda_1, \lambda_2, \lambda_3$) for the element. The third principal stretch, λ_3 , perpendicular to the flap surface, is determined by the

incompressibility constraint ($\lambda_1 \lambda_2 \lambda_3 = 1$). Once the principal stretches characterizing the element's deformation are known, hiFEM calculates the actual strain energy density stored within that element using the integrated hyperelastic constitutive law. For this work, we employ the Ogden (N=2) model, described by the strain energy function (Holzapfel, 2002):

$$W_{Ogden} = \sum_{i=1}^2 \frac{2\mu_i}{\alpha_i^2} (\lambda_1^{\alpha_i} + \lambda_2^{\alpha_i} + \lambda_3^{\alpha_i} - 3) - p(J - 1) \quad (4.1)$$

Here, μ_i and α_i are the Ogden material parameters. The hydrostatic pressure term, $p(J - 1)$, computationally enforces the simplifying assumption of perfect incompressibility ($J = \lambda_1 \lambda_2 \lambda_3 = 1$). Under this common simplification for soft tissues, the strain energy is expressed directly in terms of the total principal stretches (λ_i) (Holzapfel, 2002). This calculation directly incorporates the nonlinear, large-deformation mechanical behavior of the skin tissue into the assessment of the deformation energy.

To make the inverse problem computationally tractable using efficient linear solvers, hiFEM derives the element-specific effective secant modulus ($\hat{E}$) required for the iterative linearization step. The value of $\hat{E}$ is determined by relating the calculated hyperelastic strain energy density (W_{Ogden}) to the equivalent energy expression from linear elasticity under incompressible conditions. In principal directions, the strain energy density for a linear elastic, isotropic, and incompressible material ($\epsilon_1 + \epsilon_2 + \epsilon_3 = 0$) subjected to principal strains ($\epsilon_1, \epsilon_2, \epsilon_3$) and Hooke's law ($\sigma_{ij} = \frac{E\nu}{(1+\nu)(1-2\nu)} \epsilon_{kk} \delta_{ij} + \frac{E}{1+\nu} \epsilon_{ij}$) is given by:

$$W_{linear-incomp} = \frac{1}{2} \sigma_{ij} \epsilon_{ij} = \frac{E}{2(1+\nu)} (\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2) \quad (4.2)$$

where E , ν , σ_{ij} and ϵ_{ij} are Young's modulus, Poisson's ratio (taken as 0.5 for incompressibility), the components of the Cauchy stress tensor, and the components of linearized strain tensor, respectively. By substituting the hyperelastic energy density (W_{Ogden}) calculated from Equation 4.1 in place of $W_{linear-incomp}$ in Equation 4.2, and solving for E , we obtain the effective secant modulus $\hat{E}$ for that element in the current iteration:

$$\hat{E} = \frac{2W_{Ogden}(1 + \nu)}{\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2} \quad (4.3)$$

Note that the principal strains ϵ_i in Equation 4.3 correspond to the large deformation state defined by the principal stretches ($\epsilon_i = \ln(\lambda_i)$).

This calculated $\hat{E}$ provides the necessary element-specific secant stiffness for the iterative linearization step. A global system matrix is assembled based on these effective secant stiffness values ($\hat{E}$) for all elements, applying plane stress assumptions within the 2D domain. Solving this linear system yields nodal displacements, which are then used to update the 2D planar shape of the flap geometry. The algorithm iterates through this cycle—calculating stretches, determining W_{Ogden} , deriving $\hat{E}$, solving the linearized system, and updating the 2D geometry—until the changes in the effective secant moduli across all elements become negligible between successive iterations, indicating convergence.

Crucially, because $\hat{E}$ derives directly from a validated hyperelastic model (Equation 4.1), the overall optimization process genuinely minimizes the physical potential energy associated with the tissue’s hyperelastic deformation. This physics-based energy minimization distinguishes hiFEM from methods that rely on minimizing purely geometric distortion metrics or employ less accurate material models. The result is a 2D flap design that is biomechanically optimized to conform to the target 3D defect with minimal intrinsic tissue stretches and stored strain energy.

4.2.3 Comparative Virtual Flap Design Methods

To evaluate how effectively the proposed hiFEM algorithm minimizes intrinsic tissue deformation, we compared its generated flap designs against those from three alternative methods. The first comparison method was the immediate predecessor to hiFEM based on our previous work (Isazadeh et al., 2024). While iFEM utilizes similar inverse iterative framework to minimize deformation energy, it differs fundamentally by employing a simplified power-law constitutive model to approximate the tissue’s stress-stretch response ($\sigma_{eq} = k_1 \epsilon_{eq}^{n_1} + k_2 \epsilon_{eq}^{n_2}$, where σ_{eq} and ϵ_{eq} are the equivalent von Mises stress and strain respectively). For a direct comparison, the power-law parameters (k_1, n_1, k_2, n_2) were determined by fitting this model to the uniaxial stress-stretch data derived from

the Ogden hyperelastic model used in hiFEM (Table 1). Once the iFEM algorithm converged or reached its iteration limit, the resulting 2D flap geometry was used for subsequent deformation analysis.

The other two comparison methods were representative geometric flattening techniques, reflecting approaches used clinically or established in computer graphics. The first, NURBS surface flattening, represents the methodology of the clinically validated CARST method (Xu et al., 2022). This technique uses NURBS to represent the 3D defect surface, which is then computationally flattened into a 2D pattern. While the specific flattening function within the commercial CAD software (Rhino, Robert McNeel Associates) used in the CARST method was not detailed, it likely employed the “Squish” function, designed for non-developable NURBS surfaces, aiming to minimize geometric distortions like changes in length and area. The second geometric method was BFF (Sawhney and Crane, 2018), chosen as a state-of-the-art algorithm from computer graphics known for its robustness in producing low-distortion 2D mappings by balancing angle and area preservation. Crucially, both NURBS flattening and BFF optimize the geometric mapping based solely on geometric principles, without incorporating material properties or the physics of tissue deformation.

Table 4.1 Parameters of material models used in hiFEM and iFEM.

Material modeling	Constant parameters
Hyperelastic Ogden model (N=2)	$\mu_1 = 29.159 \text{ kPa}$, $\mu_2 = 2.147 \text{ Pa}$, $\alpha_1 = 1.708$, $\alpha_2 = 40.087$
Power law model	$k_1 = 1000 \text{ MPa}$, $n_1 = 8.802$, $k_2 = 0.0982 \text{ MPa}$, $n_2 = 1.043$

4.2.4 Evaluation Metrics

The primary metric for comparing hiFEM against alternative flap design algorithms is the intrinsic flap deformation. We quantify the deformation required to shape the planar flap, generated by each of the four methods (hiFEM, iFEM, NURBS surface flattening, BFF), into the target 3D surface. This comparison assesses how effectively each method minimizes the inherent tissue stretch necessary for the reconstruction. For every triangular element within the 2D flap design produced by each method, we first determine the principal stretches, λ_1 and λ_2 , that occur when mapping this planar

element to its corresponding state on the target 3D surface. We then calculate the Area Stretch, $J_{area} = \lambda_1 \lambda_2$, for each element. This metric measures the local change in surface area during the 2D-to-3D transformation; a value of $J_{area} = 1$ signifies perfect local area preservation, indicating no in-plane stretching or compression, while deviations from unity quantify the magnitude of local deformation. To compare the methods across the four simulated reconstruction cases (A-D), we analyze the distributions of J_{area} using histograms and statistical box plots.

A unique advantage of the physics-based hiFEM approach is its ability to estimate the resulting mechanical stresses within the flap due to the required deformation. Using the converged principal stretches (λ_1, λ_2 , and $\lambda_3 = 1/J_{area}$) obtained from the final hiFEM solution, we calculate the Cauchy principal stresses ($\sigma_1, \sigma_2, \sigma_3$) within each element using the integrated Ogden hyperelastic constitutive law (Eq. 1). From the principal stresses, we compute the effective (von Mises) stress, σ_{vM} , for each element and report the maximum value across the four cases.

The tertiary metric compares the computational efficiency and robustness of the two inverse methods, hiFEM (using the hyperelastic model) and iFEM (using the power-law model). We assess robustness by evaluating the ability of each algorithm to successfully reach the predefined convergence criterion within a maximum number of 30 iterations. Success or failure to converge is recorded. The convergence rate, indicating the speed at which each algorithm stabilizes, is visualized. This involves calculating the L2 Norm of the vector that contains the relative changes in the effective secant modulus for all elements at each iteration. Specifically, this norm is computed as the square root of the sum of the squared relative changes across all elements ($\sqrt{\sum_{el=1}^n (\Delta \hat{E}_{el} / \hat{E}_{el})^2}$). Faster decay of this norm indicates faster convergence. Convergence is defined as the point where this L2 Norm falls below a specified threshold (10^{-10}). Reaching this criterion signifies that the iterative updates to the element's effective secant modulus, and consequently the flap geometry, have become negligible, indicating that a stable solution has been found.

The hiFEM and iFEM algorithms were developed in Python, leveraging the NumPy and SciPy libraries for numerical computations. Geometric processing, mesh handling, and visualization tasks were managed using the PyVista library (Sullivan and Kaszynski, 2019). To ensure transparency and facilitate reproducibility, the complete source code and all input data files necessary to replicate

the findings are publicly available as an open-source package in the hiFEM repository (Isazadeh et al., 2025a). All computational simulations reported in this study were executed on a standard desktop computer (Core™ i5-12400 2.50 GHz with 16 GB installed RAM).

4.3 Results

The hiFEM successfully generated optimized planar flap designs for all four simulated tongue reconstruction cases (A-D). Figure 4.3 presents the resulting 2D flap geometries obtained from hiFEM alongside those generated by the comparative methods. The figure also includes visualizations mapping the planar flap texture onto the corresponding 3D defect surface for easier orientation.

To quantify the geometric differences between the flap shapes, the maximum Hausdorff distance was calculated between the boundary of the hiFEM-designed flap and the boundaries of the flaps from the other methods (Table 2). The geometric difference between hiFEM and iFEM flaps was negligible (0.09–0.14 mm) for the cases A, B, and C; iFEM failed to converge for the complex asymmetric Case D. Larger geometric deviations were observed when comparing hiFEM to the geometric flattening techniques.

Table 4.2 Maximum Hausdorff distance comparing flap boundary geometries generated by hiFEM versus alternative methods for four reconstruction cases.

Case	hiFEM vs iFEM (mm)	hiFEM vs NURBS flattening (mm)	hiFEM vs BFF (mm)
A	0.09	0.90	1.50
B	0.12	1.20	2.40
C	0.14	2.30	2.40
D	–	0.90	2.00

Figure 4.4 and Figure 4.5 focus on quantifying the intrinsic tissue deformation, measured by the Area Stretch ($J_{area} = \lambda_1 \lambda_2$), required to map the 2D planar flap designs onto the target 3D defect surfaces for each of the four reconstruction cases (A-D). They present comparative histograms and statistical box plots illustrating the distributions of J_{area} for flaps designed using hiFEM, iFEM, NURBS surface flattening, and BFF.

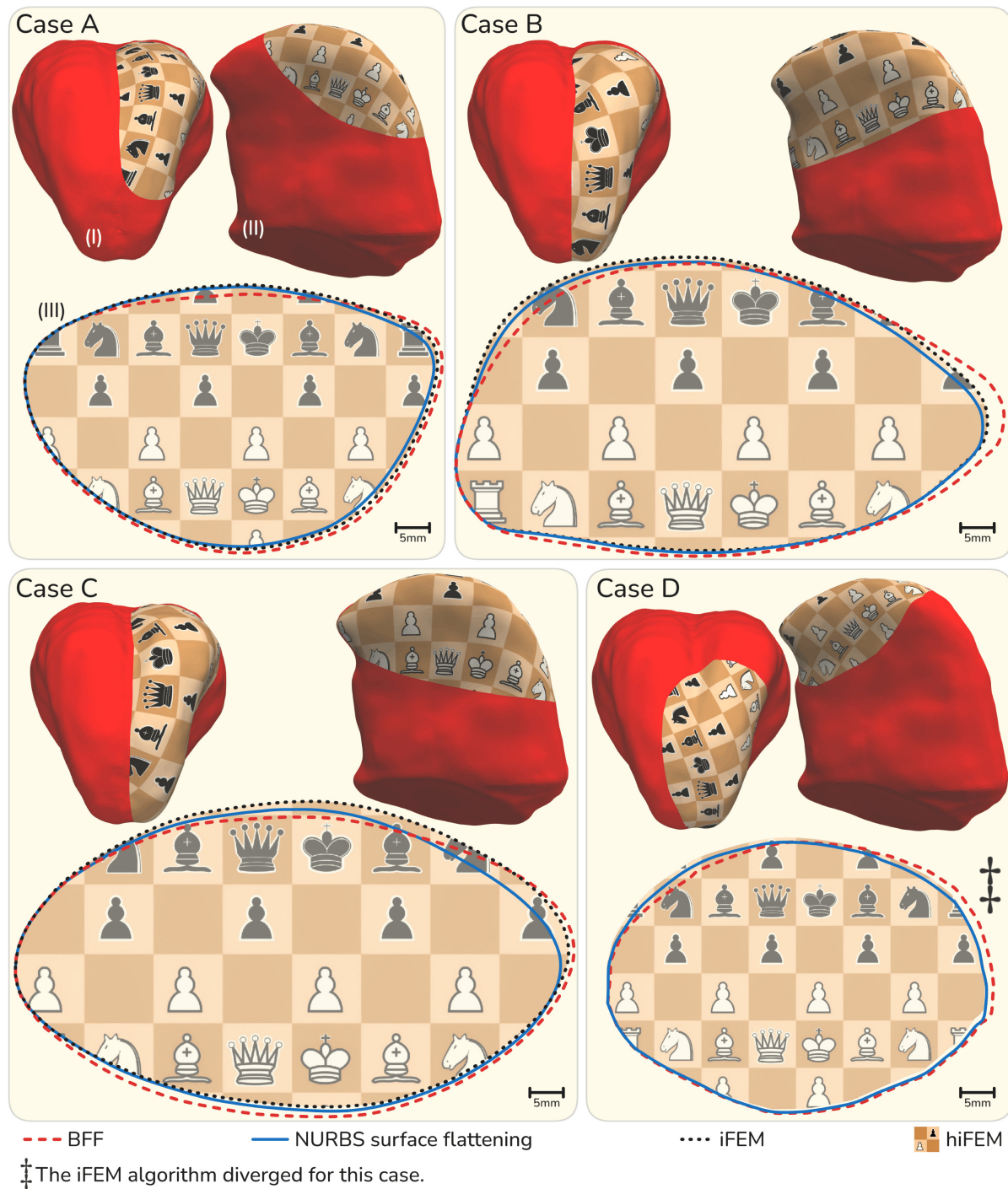
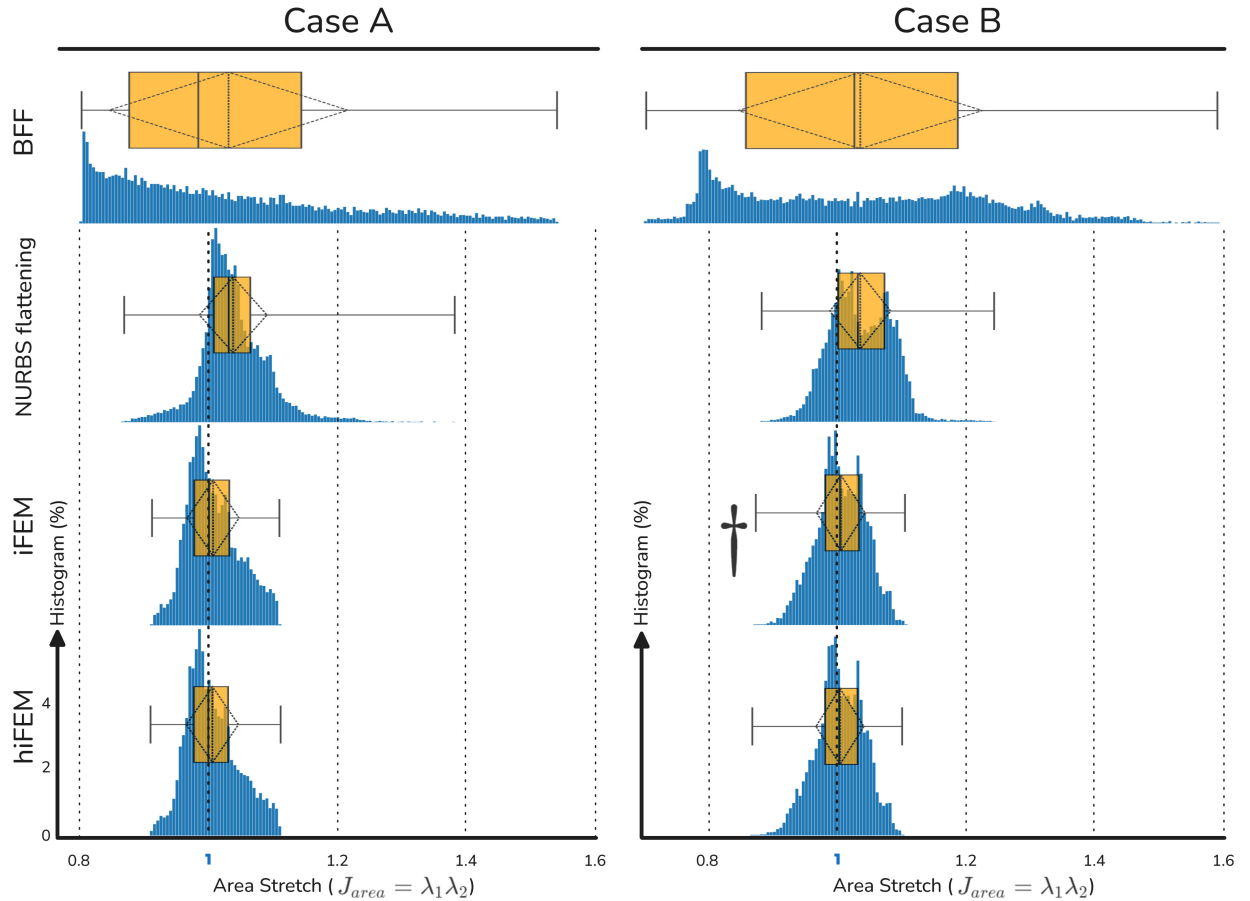


Figure 4.3 Comparison of 2D flap designs generated by hiFEM, iFEM, NURBS, and BFF for simulated cases A-D.

A consistent pattern emerged across all four simulated cases: flap designs generated by hiFEM required less intrinsic deformation compared to those produced by the geometric flattening techniques. Specifically, the maximum observed area stretches for hiFEM designs ranged from 1.10–1.12. In

4.3. Results

contrast, maximum J_{area} values were higher for NURBS flattening, ranging from 1.19–1.38, and substantially higher for BFF, ranging from 1.54–1.74. The average J_{area} values showed a similar trend, indicating that hiFEM designs necessitate lower overall stretch to achieve the target 3D geometry. Furthermore, the distribution of J_{area} values for hiFEM designs was generally tighter and more concentrated around the ideal value of 1.0 (no area change), suggesting a more uniform and less severe deformation pattern compared to the geometric methods (Figure 4.4 and Figure 4.5). While hiFEM and iFEM yielded similar J_{area} distributions for Cases A–C, hiFEM proved more robust. It successfully converged for all four cases, including the complex Case D, whereas iFEM failed to converge for Cases B and C and diverged for Case D. This demonstrates the hyperelastic model’s enhanced capability to handle challenging deformations.



† Final iteration result: The algorithm terminated before reaching convergence.

Figure 4.4 Area Stretch (J_{area}) distribution histograms for flap designs derived from hiFEM and alternative methods in simulated Cases A and B.

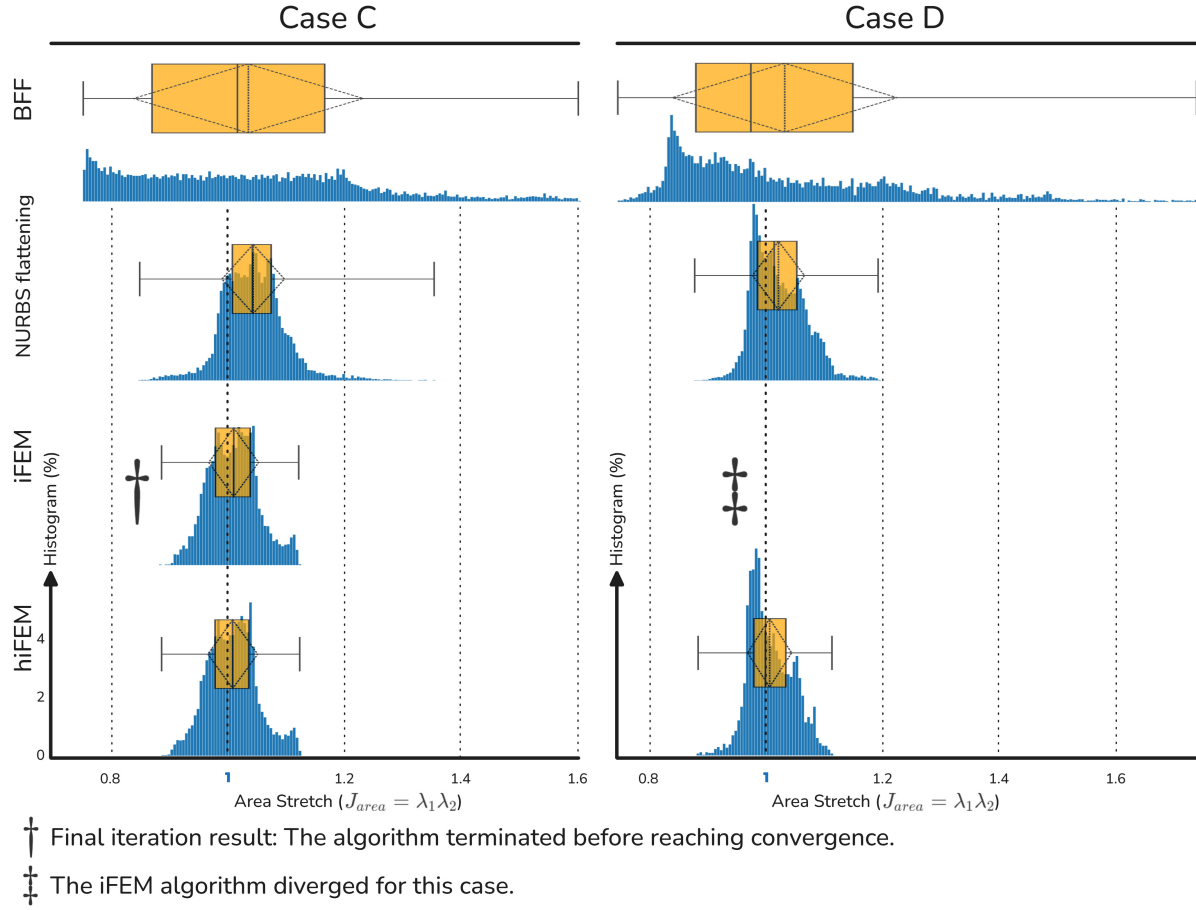


Figure 4.5 Area Stretch (J_{area}) distribution histograms for flap designs derived from hiFEM and alternative methods in simulated Cases C and D.

hiFEM's physics-based nature uniquely enables estimation of mechanical stress (σ_{vM}) within the flap. The analysis revealed low resultant stress levels for all flap designs generated by hiFEM. The maximum effective stresses (σ_{vM}) calculated for Cases A, B, C, and D were 18.0 kPa, 21.5 kPa, 21.3 kPa, and 24.5 kPa, respectively.

Another critical check enabled by the physics-based hiFEM is the verification of material volume preservation, consistent with the near-incompressible nature of skin tissue (Mostafavi Yazdi and Baqersad, 2022). The underlying hyperelastic model within hiFEM enforces this constraint. For all four cases, the volume change between the 2D planar flap and the deformed flap was less than 10^{-5} percent, confirming the algorithm's adherence to the incompressibility constraint.

4.3. Results

Lastly, computational performance of hiFEM was compared to the prior iFEM algorithm. Figure 4.6 illustrates the convergence behavior by plotting the L2 Norm of relative changes in effective secant moduli against iteration count. hiFEM consistently reached the convergence criterion (L2 Norm $< 10^{-10}$) within 8–10 iterations for all four simulated cases A-D. In contrast, iFEM failed to converge within 30 iterations for Cases B and C and diverged for Case D. On the standard desktop hardware, hiFEM completed the optimal flap design calculation in under 0.3 seconds per case.

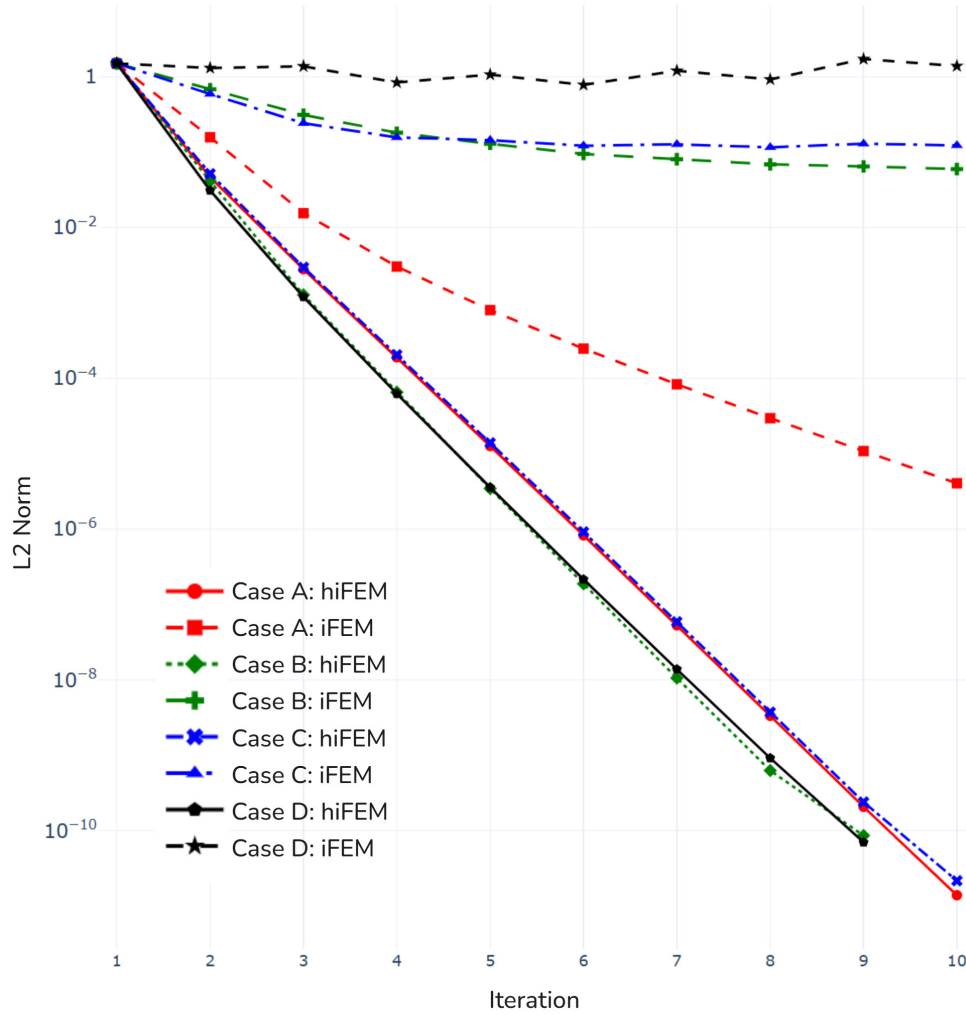


Figure 4.6 Convergence rate comparison between hiFEM and iFEM, showing the L2 Norm per iteration for cases A-D.

4.4 Discussion

This study introduces hiFEM, a digital tool for algorithmic free flap planning in soft tissue reconstructive surgeries. It advances an earlier algorithm by integrating Ogden hyperelastic model consistent with human skin's mechanical properties (Flynn et al., 2011; Mostafavi Yazdi and Baqersad, 2022). Consequently, hiFEM generates biomechanically optimized flap designs by accurately simulating large tissue deformations.

Unlike traditional patient-specific approaches that rely on manual measurement with instruments like flexible rulers to inform and adapt flap templates (Wang et al., 2016; Idris et al., 2018; Baskin et al., 2019; Liu et al., 2023a), hiFEM leverages MRI data for virtual surgical planning. MRI-guided models provide detailed anatomical information and improve the precision of the flap planning process. Past studies have shown that MRI-based, patient-specific models enhance surgical precision and functional outcomes (Fan et al., 2019b; Fan et al., 2019a; Xu et al., 2022; Han et al., 2022; Liu et al., 2023b).

However, a digital tongue model alone is insufficient for VSP; determining the corresponding planar free flap geometry is crucial. Current methods like ICTM (Liu et al., 2023b) and FILEPS (Fan et al., 2019b; Fan et al., 2019a; Han et al., 2022) achieve this by manually identifying anatomical landmarks on MRI scans, measuring distances between them, and using these measurements to guide predefined flap templates based on resection type. While systematic, these approaches require significant surgical expertise, particularly for selecting the appropriate template and accurately translating complex 3D anatomy into a 2D design. In contrast, hiFEM automates this critical translation using an inverse, physics-informed algorithm. By incorporating hyperelastic skin behavior, hiFEM produces flaps that are anatomically tailored and mechanically optimized for the specific resection.

While methods like CARST (Xu et al., 2022) represent a significant, clinically validated improvement over traditional techniques by leveraging patients' imaging data, digital modeling of their tongue, and computer-aided individualized flap planning, hiFEM provides incremental refinement to this approach through its different optimization objective. Rather than minimizing

purely geometric distortions, hiFEM optimizes the flap design based on the underlying mechanics of soft tissue deformation; it focuses on designing flap shapes that require less internal tissue stretching when being shaped into the desired 3D geometry.

This optimization objective is clinically relevant because excessive tensile stretch ($\lambda > 1$) leads to high tissue tension and stored energy, which can have immediate adverse consequences. High tension can impede blood flow within the delicate flap vasculature, increasing the risk of ischemia, partial necrosis, and impaired healing (Tepole et al., 2014a; Tepole et al., 2014b; Rui et al., 2022). Studies also suggest a correlation between mechanical tension, particularly near suture lines, and the development of keloids and hypertrophic scarring (Ogawa et al., 2012; Van De Water et al., 2013). Furthermore, excessive tension can pull on surrounding native tongue tissue, restricting the mobility crucial for speech and swallowing (Urken, 2012; Rui et al., 2022; Fan et al., 2019b; Liu et al., 2023b; Benjamin et al., 2020). Maintaining a near-normal neotongue structure, facilitated by appropriate flap design, is known to reduce tethering and allow maximum mobility of the tongue remnant (Urken, 2012; Urken and Biller, 1994).

Beyond these immediate effects, the stored elastic energy within a stretched flap influences post-suturing outcomes, particularly regarding the conformity of the flap to the defect. After suturing, the flap tissue naturally seeks to release this stored energy and recoil toward its unstretched state. The magnitude of this recoil is directly related to the initial stretch and the tissue's stiffness. A common challenge in tongue reconstruction is the stiffness mismatch, where the harvested skin flap is often considerably stiffer than the underlying native tongue muscle or adipose tissue graft bed (Buchaillard et al., 2007). Consequently, the recoil forces exerted by a suboptimally designed (highly stretched) flap can disproportionately deform the softer surrounding tissues. Instead of the flap conforming precisely to the defect contours, its recoil pulls the neotongue structure away from the intended anatomical shape, leading to deviations from the surgical plan and potentially compromising function. Therefore, designing a flap shape that intrinsically requires less stretch is crucial not only for immediate tissue viability, improved healing, and minimizing tethering but also essential for achieving accurate anatomical reconstruction. Lower, more uniform stretches promote

better, more natural conformity of the flap to the defect shape with less adverse deformation of the neotongue structure post-surgery. By minimizing the hyperelastic potential energy, hiFEM directly addresses these key biomechanical factors.

Comparing the intrinsic deformation (J_{area}) required by hiFEM flap designs to alternatives highlights the benefit of its physics-based optimization. Across all four simulated cases, hiFEM consistently designed flaps requiring less intrinsic area stretch (lower maximum J_{area} values and distributions tighter around 1.0) compared to established geometric flattening methods (Figures 4, 5). This superiority stems from hiFEM's optimization objective.

Regarding the comparison with NURBS-based flattening, it is crucial to first acknowledge the significant clinical success and validation of the CARST approach presented by Xu et al. (2022). Their work clearly demonstrated the substantial value of computer-aided surgical planning over traditional, non-image-guided methods, showing measurable improvements in operative time, flap utilization, defect matching, and functional outcomes. While NURBS flattening represents a proven advance in leveraging patient-specific geometry, hiFEM offers a potential next step in refinement by optimizing based on tissue biomechanics rather than purely geometric distortion metrics.

While hiFEM produces flap shapes geometrically distinct from those generated by NURBS flattening (0.9–2.3 mm maximum Hausdorff distance, Table 2), achieving such fine precision consistently during surgical resection and flap harvesting remains practically challenging. Therefore, the full realization of the clinical benefit derived purely from this subtle shape difference might be influenced by the accuracy of surgical execution. However, the rationale for optimizing the design itself remains compelling. Surgical planning and execution involve multiple stages where inaccuracies can accumulate. By starting with a flap design that is biomechanically optimized, we minimize a known source of potential suboptimality inherent in the plan itself. Minimizing error at the design stage is crucial to avoid compounding inaccuracies that could negatively impact the reconstruction. Thus, while NURBS surface flattening signifies a major, clinically validated improvement, hiFEM's physics-based approach offers a theoretically sound refinement aimed at further enhancing biomechanical outcomes. Additionally, it is worth noting that while the specific

NURBS flattening tool used in the CARST study is part of commercial software (Rhino 3D), our hiFEM implementation, along with documentation and application examples, is available as an open-source package (Isazadeh et al., [2025a](#)).

Computationally, hiFEM outperforms its predecessor (Isazadeh et al., [2024](#)), demonstrating enhanced robustness by converging reliably across all cases, even complex geometries where iFEM failed. The key advantage stems from hiFEM's use of the Ogden hyperelastic model. This model not only represents skin's large deformations more accurately than the previous power-law approach but also features a well-behaved strain energy function. This property likely improves the numerical stability of the iterative secant modulus updates within the inverse algorithm. With optimal designs generated in under 0.3 seconds, hiFEM achieves practical clinical feasibility. It offers an efficient, direct optimization, which avoids the prohibitive computational cost and instability associated with trial-and-error forward simulation approaches (Tepole et al., [2014b](#); Anderson et al., [2017](#)). Consequently, hiFEM's speed and reliability position it as a promising tool for VSP integration.

Lastly, the physics-based nature of hiFEM uniquely allows for the estimation of the resultant stresses within the flap after conforming to the defect shape. This provides an additional layer of design evaluation unavailable from purely geometric approaches. The hiFEM stress estimations for all designs are considerably below thresholds associated with compromised tissue viability—around 1 MPa (Tepole et al., [2014a](#)). They are also lower than physiological stresses during functional activities like swallowing—around 100 kPa (Hermant et al., [2017](#)). These estimations, upon proper future validation studies, could provide independent, physics-based evidence corroborating the primary finding of minimized intrinsic deformation. This further supports the biomechanical soundness of hiFEM designs and their potential to reduce tension-related complications.

Despite its promise, this study has limitations that guide future research directions. The current work was based on MRI data from a single healthy participant; validation across a diverse patient cohort, encompassing a range of anatomies and demographics, as well as a broader spectrum of defect types—including complex geometries involving the floor of the mouth or those requiring total/subtotal reconstructions—is a crucial next step. Furthermore, the current implementation employs validated isotropic hyperelastic parameters (Flynn et al., [2011](#)), a simplification that neglects known biomechanical complexities such as anisotropy, pre-stress, and the significant

variations in skin properties due to anatomical donor site, age, and sex. Therefore, a critical next step is to perform a sensitivity analysis to quantify how material property variations impact the optimized flap design. Ultimately, the long-term vision for VSP in tongue reconstruction involves incorporating patient-specific material properties—an enhancement uniquely achievable with physics-based methods like hiFEM.

Beyond refining the material model and chosen parameters, future computational studies must account for the mechanical interplay between the flap and the recipient site. Such simulations would be essential to predict the final post-suturing anatomical conformity. These predictive models could eventually be extended to incorporate long-term biological processes such as flap atrophy (Gewirtz et al., 2024) and radiation-induced tissue fibrosis (Al-Zanoon et al., 2024) as well. Crucially, the computational studies must be followed by benchtop experiments and prospective clinical studies. These are essential to rigorously evaluate the real-world efficacy of hiFEM-guided reconstructions against standard-of-care techniques. This future work must also address the critical challenge of translating the virtual plan to the operating room, particularly given the tongue’s mobility, by developing and validating patient-specific surgical guides that ensure accurate registration of the planned resection onto the intraoperative anatomy.

A critical consideration for clinical translation is the placement of the vascular pedicle. The hiFEM algorithm accommodates this surgical requirement within an integrated VSP workflow. First, the surgeon identifies the optimal pedicle location on the 3D defect model. The algorithm’s deterministic 3D-to-2D mapping then precisely transfers this point to the biomechanically optimized 2D flap template. This marked template guides the surgeon during harvest to align the flap with the donor vessel. In this manner, hiFEM integrates, rather than automates, the surgeon’s crucial anatomical judgment; it provides a biomechanically sound design that respects clinical constraints. Future work could also incorporate pedicle length and orientation as explicit optimization constraints.

The fundamental principle of hiFEM renders it a versatile tool with broad applicability for VSP across various soft tissue reconstruction domains. It offers potential biomechanical advantages in applications such as breast reconstruction (Huang et al., 2007), nasal reconstruction (Wei et al., 2022), and the planning of complex wound closures (Ji et al., 2024), where minimizing tissue

tension and ensuring conformational accuracy are similarly crucial. The capacity to integrate patient-specific anatomy and biomechanics provides a pathway to enhance precision and predictability in these challenging reconstructive scenarios, moving beyond purely geometric planning towards physics-informed optimization.

4.5 Conclusion

This study introduced hiFEM, a VSP tool that incorporates tissue biomechanics into algorithmic free flap design. Our comparative analysis across four simulated reconstruction cases demonstrated that hiFEM-generated flaps require less intrinsic tissue deformation compared to designs derived from a clinically validated flattening algorithm. In theory, this algorithmic improvement carries clinical relevance. Reduced internal stretches may improve tissue viability and lower tissue tethering to surrounding structures, thereby enhancing mobility. Furthermore, the minimization of stored elastic energy can mitigate flap recoil, which promotes better anatomical conformity of the flap to the defect. Optimizing for minimal tissue stretches, hiFEM offers a physics-informed VSP method yielding biomechanically sound flap designs.

Beyond the biomechanical advantages, hiFEM demonstrated superior robustness compared to its predecessor (iFEM). Its rapid convergence and enhanced robustness, even for complex cases where the previous iFEM failed, establish its practical feasibility for integration into clinical VSP workflows. This holds promise for improving the predictability and functional effectiveness not only of tongue reconstruction but potentially extending to other challenging soft tissue procedures, such as breast reconstruction, nasal reconstruction, and complex wound closure. Ultimately, this method aims to enhance patient-specific surgical precision.

Data availability statement The data that support the findings of this study are openly available at: <https://github.com/is-amir/hifem>.

Funding This research has been funded by Alberta Cancer Foundation, Grant Number 27 601.

Conflict of interest The authors declare that they have no conflict of interest.

References

- Al-Zanoon, N., Cummine, J., Jeffery, C. C., Westover, L., and Aalto, D. (2024). “The Effect of Simulated Radiation Induced Fibrosis on Tongue Protrusion.” In: *Biomech. Model. Mechanobiol.* 23, pp. 1649–1660. ISSN: 1617-7940. DOI: [10.1007/s10237-024-01860-4](https://doi.org/10.1007/s10237-024-01860-4).
- Anderson, P., Fels, S., Harandi, N. M., Ho, A., Moisik, S., Sánchez, C. A., Stavness, I., and Tang, K. (2017). “FRANK: A Hybrid 3D Biomechanical Model of the Head and Neck.” In: *Biomechanics of Living Organs*. Ed. by Y. Payan and J. Ohayon. Vol. 1. Translational Epigenetics. Oxford: Academic Press, pp. 413–447. DOI: [10.1016/B978-0-12-804009-6.00020-1](https://doi.org/10.1016/B978-0-12-804009-6.00020-1).
- Ansarin, M., Bruschini, R., Navach, V., Giugliano, G., Calabrese, L., Chiesa, F., Medina, J. E., Kowalski, L. P., and Shah, J. P. (2019). “Classification of GLOSSECTOMIES: Proposal for Tongue Cancer Resections.” In: *Head Neck-J. Sci. Spec. Head Neck* 41.3, pp. 821–827. ISSN: 1043-3074, 1097-0347. DOI: [10.1002/hed.25466](https://doi.org/10.1002/hed.25466).
- Baskin, R. M., Seikaly, H., Sawhney, R., Danan, D., Burt, M., Idris, S., Shama, M., Boyce, B., and Dziegielewski, P. T. (2019). “Tongue Reconstruction: Rebuilding Mobile Three-Dimensional Structures from Immobile Two-Dimensional Substrates, a Fresh Cadaver Study.” In: *Head Neck-J. Sci. Spec. Head Neck* 41.10, pp. 3693–3699. ISSN: 1043-3074, 1097-0347. DOI: [10.1002/hed.25889](https://doi.org/10.1002/hed.25889).
- Benjamin, M., Aliano, K., Davenport, T., and Frank, D. (2020). “Functional Outcomes Regarding Facial Artery Musculomucosal Flap for Reconstruction of Partial Glossectomy Defects.” In: *Ann. Plast. Surg.* 85, S76–S79. ISSN: 0148-7043, 1536-3708. DOI: [10.1097/SAP.0000000000002346](https://doi.org/10.1097/SAP.0000000000002346).
- Buchaillard, S., Brix, M., Perrier, P., and Payan, Y. (2007). “Simulations of the Consequences of Tongue Surgery on Tongue Mobility: Implications for Speech Production in Post-Surgery Conditions.” In: *Int. J. Med. Robot. Comput. Assist. Surg.* 3.3, pp. 252–261. ISSN: 1478-5951, 1478-596X. DOI: [10.1002/rcs.142](https://doi.org/10.1002/rcs.142).
- Chuka, R., Abdullah, W., Rieger, J., Nayar, S., Seikaly, H., Osswald, M., and Wolfaardt, J. (2017). “Implant Utilization and Time to Prosthetic Rehabilitation in Conventional and Advanced Fibular Free Flap Reconstruction of the Maxilla and Mandible.” In: *Int. J. Prosthodont.* 30.3, pp. 289–294. ISSN: 0893-2174, 1942-4426. DOI: [10.11607/ijp.5161](https://doi.org/10.11607/ijp.5161).

- De Berardinis, R., Tagliabue, M., Belloni, P., Gandini, S., Scaglione, D., Maffini, F., Margherini, S., Riccio, S., Giugliano, G., Bruschini, R., Chu, F., and Ansarin, M. (2022). “Tongue Cancer Treatment and Oncological Outcomes: The Role of Glossectomy Classification.” In: *Surg. Oncol.-Oxf.* 42, p. 101751. ISSN: 0960-7404, 1879-3320. DOI: [10.1016/j.suronc.2022.101751](https://doi.org/10.1016/j.suronc.2022.101751).
- Engel, H., Huang, J. J., Lin, C.-Y., Lam, W., Kao, H.-K., Gazyakan, E., and Cheng, M.-H. (2010). “A Strategic Approach for Tongue Reconstruction to Achieve Predictable and Improved Functional and Aesthetic Outcomes.” In: *Plast. Reconst. Surg.* 126.6, p. 1967. ISSN: 0032-1052. DOI: [10.1097/PRS.0b013e3181f44742](https://doi.org/10.1097/PRS.0b013e3181f44742).
- Fan, S., Li, Q.-X., Zhang, H.-Q., Liang, M.-J., Tian, T., Wang, Y.-Y., Lin, Z.-Y., Chen, W.-X., Pan, G.-K., Ahn, M. H.-Y., He, L., Sun, S., Wu, B.-H., Yu, P., and Li, J.-S. (2019a). ““Five-point Eight-Line” Anatomic Flap Design for Precise Hemitongue Reconstruction.” In: *Head Neck-J. Sci. Spec. Head Neck* 41.5, pp. 1359–1366. ISSN: 1043-3074, 1097-0347. DOI: [10.1002/hed.25571](https://doi.org/10.1002/hed.25571).
- Fan, S., Zhang, H., Li, Q., Liang, F., Bai, Z., Chen, W., Sun, S., Yu, P., and Li, J. (2019b). “A Novel Anatomy-Based Five-Points Eight-Line-Segments Technique for Precision Subtotal Tongue Reconstruction: A Pilot Study.” In: *Oral Oncol.* 89, pp. 1–7. ISSN: 13688375. DOI: [10.1016/j.oraloncology.2018.12.004](https://doi.org/10.1016/j.oraloncology.2018.12.004).
- Flynn, C., Taberner, A., and Nielsen, P. (2011). “Modeling the Mechanical Response of In Vivo Human Skin Under a Rich Set of Deformations.” In: *Ann. Biomed. Eng.* 39.7, pp. 1935–1946. ISSN: 0090-6964, 1573-9686. DOI: [10.1007/s10439-011-0292-7](https://doi.org/10.1007/s10439-011-0292-7).
- Gewirtz, J. I., Zhao, S., Brock, G., Luttrull, M. D., Sethuraman, S., Kang, S. Y., Vankoeverying, K. K., and Seim, N. B. (2024). “A Pilot Study: Free Flap Atrophy in Tongue Reconstruction Using 3D Volumetric Analysis.” In: *Ann. Otol. Rhinol. Laryngol.* 133.3, pp. 253–260. ISSN: 0003-4894, 1943-572X. DOI: [10.1177/00034894231204720](https://doi.org/10.1177/00034894231204720).
- Groves, R., Coulman, S., Birchall, J., and Evans, S. (2013). “An Anisotropic, Hyperelastic Model for Skin: Experimental Measurements, Finite Element Modelling and Identification of Parameters for Human and Murine Skin.” In: *J. Mech. Behav. Biomed. Mater.* 18, pp. 167–180. ISSN: 1878-0180. DOI: [10.1016/j.jmbbm.2012.10.021](https://doi.org/10.1016/j.jmbbm.2012.10.021).

- Han, S.-W., Hong, X.-W., Li, H., and Han, W. (2022). “Utility of a Palatal Speech Appliance Combined with the Five-Point Eight-Line Segment Technique in Total Tongue Reconstruction.” In: *Int. J. Oral Maxillofac. Surg.* 51.8, pp. 1016–1021. ISSN: 0901-5027, 1399-0020. DOI: [10.1016/j.ijom.2022.01.001](https://doi.org/10.1016/j.ijom.2022.01.001).
- Hermant, N., Perrier, P., and Payan, Y. (2017). “Human Tongue Biomechanical Modeling.” In: *Biomechanics of Living Organs*. Ed. by Y. Payan and J. Ohayon. Vol. 1. Translational Epigenetics. Oxford: Academic Press, pp. 395–411. DOI: [10.1016/B978-0-12-804009-6.00019-5](https://doi.org/10.1016/B978-0-12-804009-6.00019-5).
- Holzappel, G. A. (2002). *Nonlinear Solid Mechanics: A Continuum Approach for Engineering Science*. Kluwer Academic Publishers Dordrecht. ISBN: 0025-6455.
- Huang, P., Gu, L., Xu, H., Dong, J., Liu, J., Zhang, J., Song, J., Yu, X., Chen, W., Zhou, H., Li, B., and Pei, W. (2007). “Image-Guided Surgery Planning for Breast Reconstruction Flap Design.” In: *IEEE Nucl. Sci. Symp. Conf. Rec.* New York: IEEE. ISBN: 978-1-4244-0922-8. DOI: [10.1109/NSSMIC.2007.4436997](https://doi.org/10.1109/NSSMIC.2007.4436997).
- Idris, S., Logan, H., Tabet, P., Osswald, M., Nayar, S., and Seikaly, H. (2022). “The Accuracy of 3D Surgical Design and Simulation in Prefabricated Fibula Free Flaps for Jaw Reconstruction.” In: *J. Pers. Med.* 12.11, p. 1766. ISSN: 2075-4426. DOI: [10.3390/jpm12111766](https://doi.org/10.3390/jpm12111766).
- Idris, S., Mlynarek, A. M., Ansari, K., Harris, J. R., Rizk, N., Cote, D., O’Connell, D. A., Allen, H., Dziegielewski, P., and Seikaly, H. (2018). “Multi-Dimensional Analysis of Oral Cavity and Oropharyngeal Defects Following Cancer Extirpation Surgery, a Cadaveric Study.” In: *J. Otolaryngol. Head Neck Surg.* 47, p. 27. ISSN: 1916-0216. DOI: [10.1186/s40463-018-0276-9](https://doi.org/10.1186/s40463-018-0276-9).
- Iivarinen, J. T., Korhonen, R. K., Julkunen, P., and Jurvelin, J. S. (2011). “Experimental and Computational Analysis of Soft Tissue Stiffness in Forearm Using a Manual Indentation Device.” In: *Med. Eng. Phys.* 33.10, pp. 1245–1253. ISSN: 13504533. DOI: [10.1016/j.medengphys.2011.05.015](https://doi.org/10.1016/j.medengphys.2011.05.015).
- Isazadeh, A. R., Seikaly, H., Westover, L., and Aalto, D. (2024). “Algorithmically Designed Flaps in Tongue Reconstruction: A Feasibility Analysis.” In: *Int. J. Comput. Assist. Radiol. Surg.* 19, pp. 735–746. ISSN: 1861-6410, 1861-6429. DOI: [10.1007/s11548-024-03062-w](https://doi.org/10.1007/s11548-024-03062-w).
- Isazadeh, A. R., Zenke, J., Westover, L., Seikaly, H., and Aalto, D. (2025a). *hiFEM: Hyperelastic Inverse Finite Element Method*. DOI: [10.5281/zenodo.17427896](https://doi.org/10.5281/zenodo.17427896).

- Ji, X., Wen, G., Gong, H., Sun, R., and Li, H. (2024). “Three-Dimensional Wound Flattening Method for Mapping Skin Mechanical Properties Based on Finite Element Method.” In: *Comput. Methods Biomech. Biomed. Eng.* 27.2, pp. 237–250. ISSN: 1025-5842, 1476-8259. DOI: [10.1080/10255842.2023.2183347](https://doi.org/10.1080/10255842.2023.2183347).
- Koumoullis, H., Burley, O., and Kyzas, P. (2020). “Patient-Specific Soft Tissue Reconstruction: An IDEAL Stage I Report of Hemiglossectomy Reconstruction and Introduction of the PANSOFOS Flap.” In: *Br. J. Oral Maxillofac. Surg.* 58.6, pp. 681–686. ISSN: 0266-4356. DOI: [10.1016/j.bjoms.2020.04.017](https://doi.org/10.1016/j.bjoms.2020.04.017).
- Laurent, A., Mistretta, F., Bottiglioli, D., Dahel, K., Goujon, C., Nicolas, J. F., Hennino, A., and Laurent, P. E. (2007). “Echographic Measurement of Skin Thickness in Adults by High Frequency Ultrasound to Assess the Appropriate Microneedle Length for Intradermal Delivery of Vaccines.” In: *Vaccine* 25.34, pp. 6423–6430. ISSN: 0264-410X. DOI: [10.1016/j.vaccine.2007.05.046](https://doi.org/10.1016/j.vaccine.2007.05.046).
- Liu, M.-D., Liu, D.-Z., Al-Aroomi, M. A., Xiong, J.-J., Liu, S., Sun, C.-F., and Liu, F.-Y. (2023a). “Reconstruction of Large Defects of Anterior Floor of Mouth with Free Flaps Using a Novel Individualized Flap Design Method.” In: *J. Cranio-MaxilloFac. Surg.* 51.9, pp. 574–579. ISSN: 1010-5182, 1878-4119. DOI: [10.1016/j.jcms.2023.08.006](https://doi.org/10.1016/j.jcms.2023.08.006).
- Liu, M.-D., Xue, X.-M., Al-Aroomi, M. A., Xiong, J.-J., Liu, S., Sun, C.-F., and Liu, F.-Y. (2023b). “A Novel Flap Design Technique for Subtotal Tongue Reconstruction with an “Individualized and Convenient Tongue Model”.” In: *Oral Oncol.* 145, p. 106531. ISSN: 1368-8375. DOI: [10.1016/j.oraloncology.2023.106531](https://doi.org/10.1016/j.oraloncology.2023.106531).
- Logan, H., Wolfaardt, J., Boulanger, P., Hodgetts, B., and Seikaly, H. (2013). “Exploratory Benchtop Study Evaluating the Use of Surgical Design and Simulation in Fibula Free Flap Mandibular Reconstruction.” In: *J. Otolaryngol. Head Neck Surg.* 42.1, p. 42. ISSN: 1916-0216. DOI: [10.1186/1916-0216-42-42](https://doi.org/10.1186/1916-0216-42-42).
- Mallet, Y., El Bedoui, S., Penel, N., Ton Van, J., Fournier, C., and Lefebvre, J. L. (2009). “The Free Vascularized Flap and the Pectoralis Major Pedicled Flap Options: Comparative Results of Reconstruction of the Tongue.” In: *Oral Oncol.* 45.12, pp. 1028–1031. ISSN: 1368-8375. DOI: [10.1016/j.oraloncology.2009.05.639](https://doi.org/10.1016/j.oraloncology.2009.05.639).

- Mannelli, G., Arcuri, F., Agostini, T., Innocenti, M., Raffaini, M., and Spinelli, G. (2018). “Classification of Tongue Cancer Resection and Treatment Algorithm.” In: *J. Surg. Oncol.* 117.5, pp. 1092–1099. ISSN: 0022-4790, 1096-9098. DOI: [10.1002/jso.24991](https://doi.org/10.1002/jso.24991).
- Mostafavi Yazdi, S. and Baqersad, J. (2022). “Mechanical Modeling and Characterization of Human Skin: A Review.” In: *J. Biomech.* 130. ISSN: 0021-9290. DOI: [10.1016/j.jbiomech.2021.110864](https://doi.org/10.1016/j.jbiomech.2021.110864).
- Ogawa, R., Okai, K., Tokumura, F., Mori, K., Ohmori, Y., Huang, C., Hyakusoku, H., and Akaishi, S. (2012). “The Relationship between Skin Stretching/Contraction and Pathologic Scarring: The Important Role of Mechanical Forces in Keloid Generation.” In: *Wound Repair Regen.* 20.2, pp. 149–157. ISSN: 1067-1927, 1524-475X. DOI: [10.1111/j.1524-475X.2012.00766.x](https://doi.org/10.1111/j.1524-475X.2012.00766.x).
- Papadopoulos-Nydam, G., Wolfaardt, J., Seikaly, H., O’Connell, D., Harris, J., Osswald, M., Nayar, S., and Rieger, J. (2017). “Comparison of Speech and Resonance Outcomes across Three Methods of Treatment for Maxillary Defects.” In: *Int. J. Maxillofac. Prosthet.* 1.1, pp. 2–8. ISSN: 2432-3993. DOI: [10.26629/ijmp.2017.02](https://doi.org/10.26629/ijmp.2017.02).
- Rui, X., Huang, Z., Zuo, J., Wang, Y., Liang, Q., Jin, T., Wang, J., Chang, S., and Huang, Z. (2022). “Application of an L-shaped Anterolateral Thigh Flap in Reconstruction after Hemiglossectomy.” In: *BMC Surg.* 22.1, p. 32. ISSN: 1471-2482. DOI: [10.1186/s12893-022-01473-7](https://doi.org/10.1186/s12893-022-01473-7).
- Sawhney, R. and Crane, K. (2018). “Boundary First Flattening.” In: *ACM Trans. Graph.* 37.1, pp. 1–14. ISSN: 0730-0301, 1557-7368. DOI: [10.1145/3132705](https://doi.org/10.1145/3132705).
- Seikaly, H., Rieger, J., O’Connell, D., Ansari, K., AlQahtani, K., and Harris, J. (2009). “Beavertail Modification of the Radial Forearm Free Flap in Base of Tongue Reconstruction: Technique and Functional Outcomes.” In: *Head Neck* 31.2, pp. 213–219. ISSN: 1043-3074, 1097-0347. DOI: [10.1002/hed.20953](https://doi.org/10.1002/hed.20953).
- Seikaly, H., Rieger, J., Wolfaardt, J., Moysa, G., Harris, J., and Jha, N. (2003). “Functional Outcomes After Primary Oropharyngeal Cancer Resection and Reconstruction With the Radial Forearm Free Flap.” In: *Laryngoscope* 113.5, pp. 897–904. ISSN: 0023-852X, 1531-4995. DOI: [10.1097/00005537-200305000-00023](https://doi.org/10.1097/00005537-200305000-00023).

- Serrano, C., Van Den Brink, H., Pineau, J., Prognon, P., and Martelli, N. (2019). “Benefits of 3D Printing Applications in Jaw Reconstruction: A Systematic Review and Meta-Analysis.” In: *J. Cranio-MaxilloFac. Surg.* 47.9, pp. 1387–1397. ISSN: 10105182. DOI: [10.1016/j.jcms.2019.06.008](https://doi.org/10.1016/j.jcms.2019.06.008).
- Sullivan, C. and Kaszynski, A. (2019). “PyVista: 3D Plotting and Mesh Analysis through a Streamlined Interface for the Visualization Toolkit (VTK).” In: *J. Open Source Softw.* 4.37, p. 1450. ISSN: 2475-9066. DOI: [10.21105/joss.01450](https://doi.org/10.21105/joss.01450).
- Tepole, A. B., Gosain, A. K., and Kuhl, E. (2014a). “Computational Modeling of Skin: Using Stress Profiles as Predictor for Tissue Necrosis in Reconstructive Surgery.” In: *Comput. Struct.* 143, pp. 32–39. ISSN: 0045-7949, 1879-2243. DOI: [10.1016/j.compstruc.2014.07.004](https://doi.org/10.1016/j.compstruc.2014.07.004).
- Tepole, A. B., Steinberg, J. P., Kuhl, E., and Gosain, A. K. (2014b). “Application of Finite Element Modeling to Optimize Flap Design with Tissue Expansion.” In: *Plast. Reconst. Surg.* 134.4, pp. 785–792. ISSN: 0032-1052. DOI: [10.1097/PRS.0000000000000553](https://doi.org/10.1097/PRS.0000000000000553).
- Urken, M. L. and Biller, H. F. (1994). “A New Bilobed Design for the Sensate Radial Forearm Flap to Preserve Tongue Mobility Following Significant Glossectomy.” In: *Arch. Otolaryngol. Head Neck Surg.* 120.1, pp. 26–31. ISSN: 0886-4470. DOI: [10.1001/archotol.1994.01880250022002](https://doi.org/10.1001/archotol.1994.01880250022002).
- Urken, M. (2012). *Multidisciplinary Head and Neck Reconstruction: A Defect-Oriented Approach*. Lippincott Williams & Wilkins. ISBN: 978-1-4511-5337-8.
- Van De Water, L., Varney, S., and Tomasek, J. J. (2013). “Mechanoregulation of the Myofibroblast in Wound Contraction, Scarring, and Fibrosis: Opportunities for New Therapeutic Intervention.” In: *Adv. Wound Care.* 2.4, pp. 122–141. ISSN: 2162-1918, 2162-1934. DOI: [10.1089/wound.2012.0393](https://doi.org/10.1089/wound.2012.0393).
- Wang, L., Liu, K., Shao, Z., and Shang, Z. (2016). “Individual Design of the Anterolateral Thigh Flap for Functional Reconstruction after Hemiglossectomy: Experience with 238 Patients.” In: *Int. J. Oral. Maxillofac. Surg.* 45.6, pp. 726–730. ISSN: 0901-5027. DOI: [10.1016/j.ijom.2015.11.020](https://doi.org/10.1016/j.ijom.2015.11.020).

- Wei, H., Zhang, A., Tao, C., Li, X., Li, Q., Shen, C., Wang, P., and Jin, P. (2022). “Three-Dimensional Rapid Printing-Assisted Individualized Total Nasal Reconstruction Based on a Database of Normal External Noses.” In: *Plast. Reconstr. Surg.* 150.4, pp. 888–898. ISSN: 0032-1052, 1529-4242. DOI: [10.1097/PRS.00000000000009566](https://doi.org/10.1097/PRS.00000000000009566).
- Xu, J., Lai, F., Liu, Y., Tan, Z., Zheng, C., Wang, J., Guo, H., Jiang, L., Ge, X., Lan, X., Chen, C., and Ge, M. (2022). “Novel Computer-Aided Reconstruction of Soft Tissue Defects Following Resection of Oral and Oropharyngeal Squamous Cell Carcinoma.” In: *World J. Surg. Oncol.* 20.1, p. 196. ISSN: 1477-7819. DOI: [10.1186/s12957-022-02654-7](https://doi.org/10.1186/s12957-022-02654-7).
- Yi, C. R., Jeong, W. S., Oh, T. S., Koh, K. S., and Choi, J.-W. (2020). “Analysis of Speech and Functional Outcomes in Tongue Reconstruction after Hemiglossectomy.” In: *J. Reconstr. Microsurg.* 36.07, pp. 507–513. ISSN: 0743-684X, 1098-8947. DOI: [10.1055/s-0040-1709493](https://doi.org/10.1055/s-0040-1709493).

Chapter 5

Computational Analysis of Tongue

Reconstruction Surgery: The Impact of Free Flap Site and Volume on the Tongue

Anatomy and Biomechanics

Amir Reza Isazadeh^{1,*}, Lindsey Westover², Hadi Seikaly^{3,4} and Daniel Aalto^{1,4}

* Corresponding author; submitted to PLOS ONE.

¹ Department of Communication Sciences and Disorders, Faculty of Rehabilitation Medicine, University of Alberta, Edmonton, AB, Canada.

² Department of Mechanical Engineering & Department of Biomedical Engineering, Faculty of Engineering, University of Alberta, Edmonton, AB, Canada.

³ Division of Otolaryngology, Department of Surgery, Faculty of Medicine and Dentistry, University of Alberta, Edmonton, AB, Canada.

⁴ Institute for Reconstructive Sciences in Medicine (iRSM), Misericordia Community Hospital, Edmonton, AB, Canada.

Abstract

Objective Tongue reconstruction requires a series of decisions—including donor site and flap volume—tailored to patient needs to restore anatomy and preserve speech and swallowing. The impact of these interdependent choices is difficult to evaluate, as clinical outcomes depend on case-specific factors. This study presents an automated framework that simulates free flap tongue reconstruction and quantifies the anatomical and biomechanical effects of key surgical decisions.

Method The framework leverages biomechanically optimized flap design to generate a multi-component virtual flap, which is then computationally sutured to the resection site, and its long-term tissue atrophy is simulated to predict the final neotongue state. Using this platform, we conducted a 120-simulation factorial study to systematically quantify how key surgical decisions, namely donor site selection (Radial Forearm vs. Anterolateral Thigh) and flap overbulking, influence the final anatomical and biomechanical outcomes. Four clinically plausible tongue reconstruction scenarios, six levels of flap stiffness, and five levels of flap overbulking were simulated. **Results**

The results reveal a fundamental, physics-based trade-off: while increasing flap overbulking was the dominant factor in restoring preoperative anatomy ($p < 0.001$), it came at the cost of a predictable increase in biomechanical strain imposed on the native tongue ($p < 0.001$). Furthermore, stiffer flaps induced significantly higher baseline strain ($p < 0.001$), providing a quantitative measure for tissue tethering. The anatomical benefit of overbulking was significantly modulated by tissue properties, proving more effective with more pliable forearm flaps ($p < 0.001$). These findings provide a biomechanical rationale for clinically observed functional trade-offs and support practices like the “beavertail” modification of the forearm flap, which adds volume while preserving pliability.

Conclusion This work presents an open-source robust computational testbed for systematically evaluating interdependent surgical variables. It provides quantitative, physics-based insights into the clinical trade-offs of surgical planning. Ultimately, the framework’s automation and scalability offer a pathway toward personalized, simulation-informed surgical planning for tongue reconstruction.

Keywords: Patient-specific surgical simulation, Free flap tongue reconstruction, Finite Element Analysis in surgery, Virtual Surgical Planning, Biomechanical modeling

5.1 Introduction

Surgical removal of the cancerous defect from the tongue, known as glossectomy, is a primary treatment for tongue cancer. The standard of care to address this is microvascular free flap reconstruction, where tissue is transplanted from a donor site to rebuild the tongue. However, this procedure can severely compromise vital functions such as speech and swallowing (Lam and Samman, 2013). A successful tongue reconstruction is a complex process that relies on a series of interdependent surgical decisions tailored to the individual. The surgeon's strategy is critically influenced by the defect's specific characteristics, including its size, volume, location, and geometry of resection. Key surgical considerations involve donor site selection and free flap design, both its planar geometry and the volume, to restore the tongue's intricate three-dimensional anatomy (Urken, 2012; Baskin et al., 2019). While recent research has begun to address the challenge of patient-specific free flap design in terms of its planar geometry via template-based (Baskin et al., 2019; Liu et al., 2023a; Liu et al., 2023b) or algorithmic approaches (Xu et al., 2022; Isazadeh et al., 2025b), other critical decisions that significantly impact surgical outcomes remain largely guided by surgeon experience.

The preservation of residual tongue mobility is the foremost determinant of functional recovery after reconstruction—mobility for speech and swallowing precedes bulk. Consistent with this priority, the Radial Forearm Free Flap (RFFF) is often preferred for defects where pliability and fine shape conformity are paramount: it is thin, pliable, and readily contours to the mobile anterior tongue, supporting articulation (Marra et al., 2023; Seikaly et al., 2009; Seikaly et al., 2003). At the same time, the Anterolateral Thigh (ALT) flap remains a strong option when substantial bulk is needed, particularly for base-of-tongue defects where palate contact is essential for swallowing, or when forearm donor-site morbidity to wrist function should be avoided, as many patients would prioritize hand function and cosmesis (Wang et al., 2013; Marra et al., 2023; Sepúlveda et al., 2025; Cai et al., 2019). This choice entails a familiar trade-off: the greater stiffness of a bulky ALT flap can tether the residual mobile tongue and impair speech articulation, whereas the more compliant RFFF mitigates tethering but may provide insufficient long-term volume for optimal swallowing (Seikaly et al., 2009; Sepúlveda et al., 2025). Potential donor-site morbidity further complicates decision-making. This decision is therefore not merely about defect size but is a complex, experience-based judgment on the long-term anatomical and functional consequences.

5.1. Introduction

Several studies have investigated the impact of donor site choice on the function and quality of life outcomes (Cai et al., 2019; Marra et al., 2023; Sepúlveda et al., 2025). Nevertheless, assessing the impact of donor site selection on the biomechanics and anatomy of the tongue brings granular insights.

A universal challenge in reconstructive surgery is the significant volume loss, or atrophy, that free flaps undergo in the months following transplantation, primarily due to denervation, muscle disuse, and fat resorption. To mitigate this, surgeons intentionally harvest a flap larger than the resection defect, a practice known as “overbulking.” The extent of overbulking is mostly based on the surgeon’s expertise because evidence-based recommendations are limited to a handful of studies and depend on individual circumstances (Tarsitano et al., 2016; Vincent et al., 2019; Kim et al., 2020; Razavi et al., 2022; Gewirtz et al., 2024; Zirk et al., 2024). The clinical consequences of miscalculation are severe. Insufficient overbulking can lead to a neotongue with inadequate volume, which severely impairs speech intelligibility and swallowing. Conversely, excessive overbulking can produce a bulky and immobile tongue, leading to difficulties with mastication, articulation, and even potential airway obstruction (Bashorun et al., 2023).

The interdependent choices of donor-site selection and flap overbulking are highly patient-specific, often necessitating intraoperative, experience-based judgment. The choice of donor site impacts not only the provided volume but also the extent of mechanical property mismatch between the donor tissue and the native tongue. Likewise, the decision on the extent of flap overbulking must account for patient-specific variables such as patient anatomy and the extent of the resection, BMI, age, postoperative radiation therapy (RT), and individual healing responses and expected tissue atrophy (Gewirtz et al., 2024; Razavi et al., 2022). These decisions are not made in isolation, which creates a complex, case-specific decision matrix.

Designing a prospective randomized clinical trial to systematically isolate these variables presents ethical and methodological challenges (Reynolds, 1999; Bielecki, 2017). Ethically, random assignment of patients to surgical arms that may be considered suboptimal for their specific defect is difficult to justify. Methodologically, the integrity of such a trial would be challenged by numerous confounding variables. Patient-specific factors like tumor extent and location, age, overall health, and healing rates, combined with treatment-related factors like flap design, donor site selection, and

radiation dosage, introduce significant variability that can obscure the true effect of the surgical decisions. While controlling for this array of confounders is theoretically possible, it would require a large, expensive, and logistically complex multi-center study, making a traditional clinical investigation a difficult first step.

Biomechanical modeling offers a viable and ethical alternative to investigate these variables and gain insights into their interactions. Buchaillard and colleagues (2007; 2009), for instance, utilized a generic Finite Element tongue model to simulate hemiglossectomy and floor-of-mouth reconstruction by modifying stiffness in affected areas. The approach to modeling surgical removal and free flap reconstruction, though simplified, provided insight into the functional outcome aligned with the clinical understanding. Building on their generic tongue model, several studies introduced multibody dynamics, rigid-soft tissue interactions, advanced hyperelastic and visco-hyperelastic models, and various muscle modeling techniques, all serving to improve the fidelity of the simulation of the complex biomechanics of the oropharynx as accurately as possible (Stavness et al., 2011; Stavness et al., 2014a; Anderson et al., 2017; Hermant et al., 2017). Further, to streamline the patient-specific model generation, Harandi et al. (2017a; 2017b) developed a method for patient-specific oropharyngeal modeling based on patient MRI data. They used elastic registration to adapt generic tongue and other intraoral models to individual anatomy. Kappert et al. (2021a) advanced this approach by developing an atlas model that incorporated detailed muscle fiber orientations from Diffusion-weighted MRI (DW-MRI). While effective for assessing patient-related nuances and how they impact function, refinements are needed to implement these techniques in patient-specific surgical intervention analysis.

To our knowledge, a limited number of studies have explored the simulation of surgical interventions on the tongue. Foundational work simulated the functional consequences of glossectomy by altering the material properties of the affected regions to represent the reconstructed tissue (Buchaillard et al., 2007). This approach of modifying regional soft tissue properties has since been adapted to investigate other clinical phenomena, such as the biomechanical impact of radiation-induced fibrosis (Al-Zanoon et al., 2024). To advance the simulation of surgical procedures beyond solely property modification, Kappert et al. (2019) developed a framework for simulating small

resections that are treated with primary closure. Their method replicated the surgical act of suturing by computationally pulling the resection edges together, generating a new finite element model from the deformed geometry, and increasing the stiffness in the scar region.

This study builds on these simulation works by developing an end-to-end patient-specific simulation framework. Such a high-fidelity simulation models the free flap as an independent, multi-component body that is virtually sutured to the resected tongue model. The simulation computationally replicates the surgical workflow, including the attachment and tensioning of sutures to reconstruct the tongue. Furthermore, it incorporates clinically-reported biological phenomena, such as tissue atrophy, to estimate the final, post-remodeling anatomical and biomechanical state.

The proposed robust, automated workflow serves as a computational testbed for surgical planning and systematic studies. Accordingly, we conducted a computational factorial experiment to decouple and quantify the effects of two critical surgical decisions: donor-site selection and flap overbulking. By systematically varying the flap’s biomechanical properties and its initial subcutaneous volume, this research can isolate the independent and interactive effects of these variables on the final anatomical and biomechanical outcomes. Such computational modeling allows for the simulation of numerous scenarios and provides quantitative physics-based insight that is otherwise difficult to obtain from clinical studies alone.

5.2 Method

5.2.1 Patient-Specific Tongue Model

This study is built upon a high-resolution, patient-specific 3D model of the tongue, previously derived from an MRI scan of a healthy male participant in his 40s (Isazadeh et al., 2025b) (Ethics approval: Health Research Ethics Board of Alberta, protocol HREBA.CC-17-0597; written informed consent obtained). The tongue model establishes the baseline preoperative anatomy for all simulations herein. To ensure consistency across representative cases, we adopted the same four clinically plausible tongue reconstruction scenarios and used “resection models” identical to those in our prior work. For each case, the “resected tongue model” was generated via a Boolean

subtraction of the corresponding resection model from the baseline tongue anatomy. The resulting volumetric mesh serves as the recipient site for the virtual free flap reconstruction, and the resection model provides the geometric basis for subsequent flap design (Figure 5.1).

To establish a stable anatomical reference frame and define physiological boundary conditions, a rigid jaw model was also created. This was accomplished by first extracting the mandibular surface from the patient’s MRI data. Subsequently, a high-resolution generic jaw model was registered to this surface using an elastic registration algorithm guided by manually placed anatomical landmarks. The resulting registered jaw model serves as a rigid, stationary foundation for the simulation, and defines the anatomical locations of the tongue’s muscular attachments.

5.2.2 Biomechanically Optimized Flap Design

As an input for the surgical simulation, a patient-specific three-dimensional free flap model is generated, which is composed of skin and adipose layers (Figure 5.2B–D). The process begins by employing the hiFEM algorithm, as detailed in previous work (Isazadeh et al., 2025b), to produce a biomechanically optimized planar skin template for each resection case (Figure 5.2C). A key challenge, however, is the unknown geometric model of the subcutaneous adipose tissue.

Mechanically, adipose tissue exhibits a very low shear modulus, allowing it to undergo large, quasi-fluidic deformations with relatively low elastic energy storage (Sun et al., 2021a). Its primary contribution to the flap’s integrity is volumetric bulk rather than a predefined geometric shape. This principle is clinically leveraged in techniques like “beavertail” augmentation, where supplementary adipose tissue is malleably shaped to increase flap volume (Seikaly et al., 2009). Consequently, the critical modeling requirement is to define an initial adipose geometry that has the necessary volume and maintains geometric congruency with the planar skin template.

To achieve this, a reproducible initial 3D shape for the adipose layer is generated by computationally “unfolding” the original 3D resection volume (Figure 5.2B). This transformation is guided by the deterministic 3D-to-2D nodal mapping generated by the hiFEM solution. This step ensures the resulting adipose layer is perfectly congruent with the planar design from the hiFEM algorithm, and creates a geometrically consistent foundation for assembling the composite flap model. The

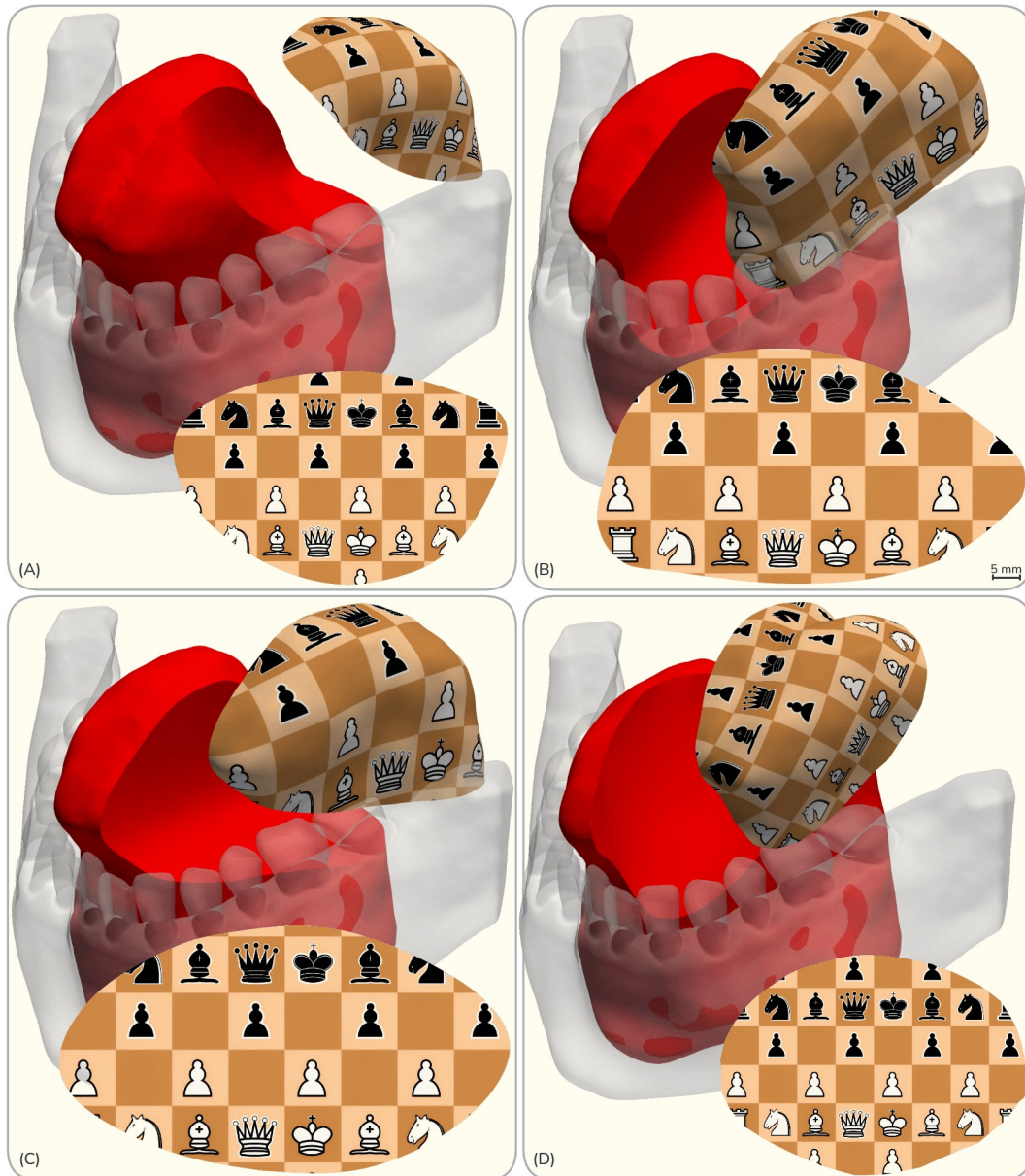


Figure 5.1 Patient-Specific Tongue Model, Resection Scenarios, and Flap Designs. Overview of the four simulated tongue cancer cases along with their biomechanically optimized flap designs. The texture mapping is generated with hiFEM and maps each point in the resection model to its corresponding point in the planar flap design.

multilayer flap model is discretized with a high-quality volumetric tetrahedral mesh using fTetWild (Hu et al., 2020), which guarantees congruent meshing and a seamless interface between the skin and fat domains.

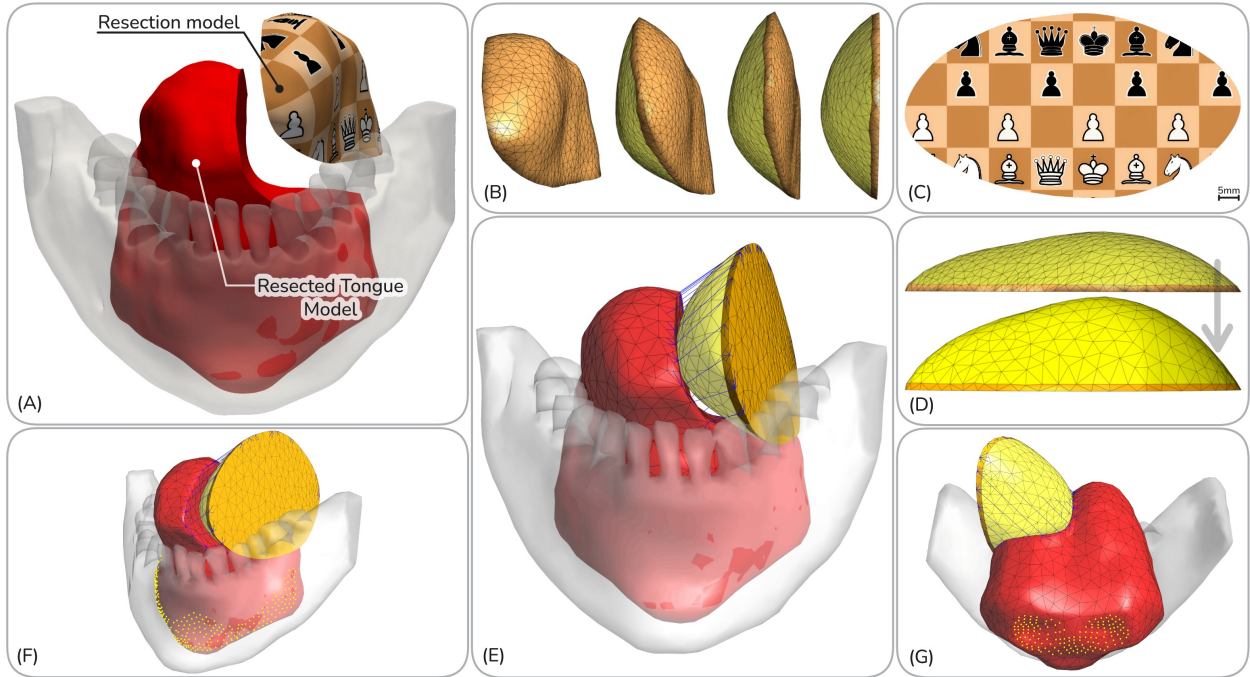


Figure 5.2 Workflow for free flap model generation and surgical simulation pre-processing. Panel (A) shows the two primary inputs: the “Resected Tongue Model” (red) and “Resection model” (textured). First, the hiFEM algorithm creates an optimized 2D planar skin template for the flap (C). Using the resulting deterministic 3D-to-2D mapping, the resection volume is computationally unfolded to generate a two-layer free flap model (B), consisting of skin (orange) and adipose (yellow). To simulate the clinical practice of overbulking, flap volume is increased by scaling the thickness of the adipose layer (D). Next, the composite flap is positioned at the resection site (E), and one-dimensional tension elements (blue lines) are defined to connect corresponding suture points on the flap and the resected tongue. Anatomical boundary conditions are established by fixing nodes (yellow dots) that represent the tongue’s muscular attachments to the mandible and hyoid bone (F, G).

5.2.3 Simulation of Tongue Reconstruction Surgery

The surgical procedure of tongue reconstruction was computationally simulated to establish the postoperative neotongue configuration (Figure 5.3i–iii). This involves inseting the multi-component flap onto the resected tongue and applying sutures to close the defect. First, a series of suture points is established along the flap’s perimeter, spaced at clinically-relevant 5 mm intervals (Jeong and Oh, 2016). To establish a corresponding set of attachment points on the resected tongue model, the simulation leverages the deterministic 3D-to-2D nodal mapping generated by the hiFEM solution. This critical step ensures that the boundary deformation is guided by the optimized geometry derived from the flap design stage.

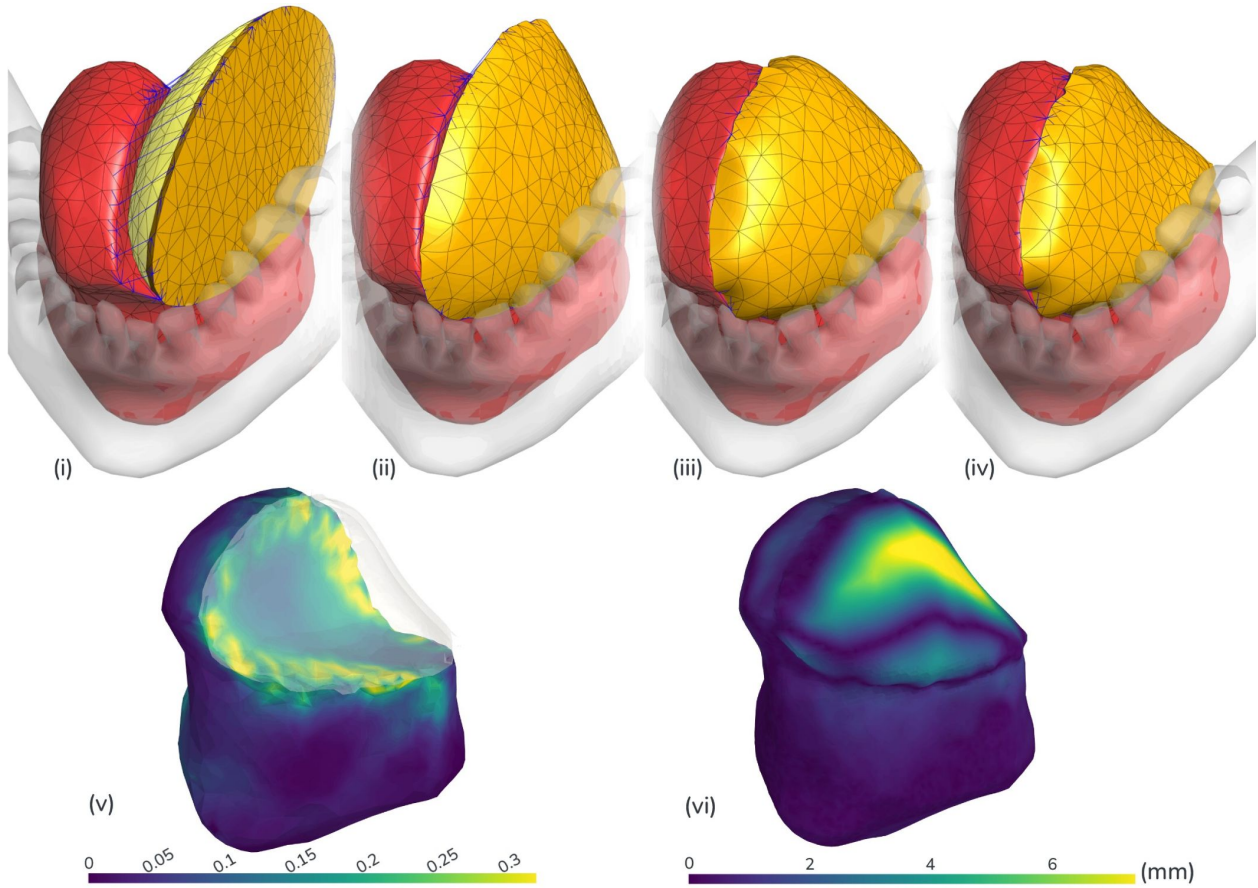


Figure 5.3 Surgical simulation pipeline: reconstruction to long-term remodeling. (i) Initial simulation setup, (ii) The suture model brings the flaps proximal to the resected tongue model and simulates suturing. (iii) After the 5-minute idling for stresses to relax with the viscoelastic model present, this is “Day 0” immediately after reconstruction. (iv) shows the tongue model after one year of post-surgery atrophy modeling. At the very end, the effective strain (v) and Hausdorff distance (vi) are calculated and used for the statistical analysis.

A multi-step numerical approach was developed to simulate the application of sutures and ensure a realistic distribution of forces. To mimic how a physical suture engages a small area of tissue and avoid artificial stress concentrations, the force at each suture point was applied to a small cluster of nodes within a 2 mm radius rather than to a single node. The force is distributed to the cluster nodes via linear-spring elements that connect all nodes in a cluster to a central point. The corresponding central points on the flap and the resected tongue are then drawn together using time-activated, one-dimensional tension elements that shorten to a near-zero length over the simulation step. This sequence computationally replicates the surgical action of tightening and securing sutures.

Following the virtual suturing, a subsequent step allows the system to equilibrate for five minutes. This simulates the immediate viscoelastic stress relaxation that occurs in soft tissues after being stretched and sutured. By modeling this phenomenon, internal stresses are allowed to redistribute throughout the neotongue. The final, equilibrated geometry after this relaxation represents the immediate, postoperative “Day 0” anatomical state (Figure 5.3iii), which provides a stable configuration for the subsequent simulation of biological remodeling.

5.2.4 Simulation of postoperative flap atrophy

A critical challenge in free flap tongue reconstruction is the significant, time-dependent volume loss that occurs postoperatively. Clinical studies have reported a wide range of atrophy, with long-term volume reduction ranging from approximately 20% to nearly 60% (Gewirtz et al., 2024; Razavi et al., 2022; Tarsitano et al., 2016; Sakakibara et al., 2019). The final flap volume is heavily influenced by factors such as tissue composition, patient age, BMI, and most critically, the administration of adjuvant radiation therapy. For instance, at one year post-surgery, ALT flaps in patients receiving radiotherapy exhibited an average volume loss of 44.2%, compared to just 19.8% in non-irradiated patients (Tarsitano et al., 2016). Similarly, radial forearm free flaps (RFFF) have shown a mean volume loss of 42.7% at ten months post-radiotherapy (Razavi et al., 2022). Given that radiotherapy is a common component of treatment for advanced cancers, and considering this body of evidence, a total volume loss of 40% was selected for this study as a clinically representative value to model the extent of flap atrophy over a one-year period. While this value serves as a clinically representative average for this systematic study, its application to an individual patient would require personalization based on known risk factors for atrophy, such as the planned radiotherapy dose, patient age, and BMI.

To simulate this phenomenon, the model is evolved from its “Day 0” state (Figure 5.3iii) to an estimated one-year postoperative state (Figure 5.3iv). This remodeling was implemented by applying a time-dependent, isotropic negative kinematic growth to the finite elements of the flap’s subcutaneous tissue. Consistent with clinical observations, atrophy was confined to the flap; the native tongue was assumed to maintain its volume. The simulation progresses over the defined

one-year timeframe, with appropriate time-scaling to ensure computational tractability. The final, equilibrated geometry represents the estimated one-year anatomical and biomechanical state of the neotongue, and this 3D model serves as the primary output for the subsequent quantitative analysis.

5.2.5 Boundary Conditions and Contact Mechanics

A stable and physiologically realistic reference frame was established by defining the anatomical boundary conditions. The jaw model was treated as a rigid body and was fully constrained in all degrees of freedom. The tongue's physiological connection to this frame was modeled by fixing the nodes of the tongue mesh at its anatomical attachment points to the mandible and hyoid bone, which include the genioglossus, mylohyoid, and hyoglossus muscle origins as illustrated in Figure 5.2F and G. This configuration ensures that all simulated deformations originate from a stable anatomical base. As the objective was to estimate the final, stable anatomy at rest, muscle activation was not included in the model; however, gravity was applied as a global body force.

The physical interface between the flap and native tongue was modeled using a robust contact formulation. A penalty-based contact algorithm was enforced between the surfaces of the flap and the resected tongue to prevent interpenetration. To computationally model the desired surgical outcome of complete tissue apposition, a “sticky” contact formulation was also specified at the interface of the flap and the resected tongue surface. The simulation results confirmed that this approach prevented the formation of gaps or voids and yielded a seamless integration at the tongue-flap interface. Additionally, a penalty contact was defined between the tongue and the jaw model to prevent interpenetration.

5.2.6 Constitutive Models

All soft tissues (tongue, skin, and subcutaneous adipose tissue) were modeled as nearly incompressible, non-linear hyperelastic materials. Native tongue adopted Yeoh hyperelastic model, while donor flap skin and adipose tissue were modeled with Ogden hyperelastic model, both following the FEBio definition as below, respectively (Maas et al., 2012).

$$\Psi_{\text{Yeoh}} = \sum_{i=1}^2 c_i (\tilde{I}_1 - 3)^i + \frac{k}{2} (\ln J)^2 \quad (5.1)$$

$$\Psi_{\text{Ogden}} = \sum_{i=1}^2 \frac{c_i}{m_i^2} \left(\tilde{\lambda}_1^{m_i} + \tilde{\lambda}_2^{m_i} + \tilde{\lambda}_3^{m_i} - 3 \right) + \frac{k}{2} (\ln J)^2 \quad (5.2)$$

Where Ψ represents the strain energy density function. In both models, the volumetric response is governed by the material's bulk modulus, k , and the volume ratio, J , which is the determinant of the deformation gradient tensor. The Yeoh model's isochoric behavior is a function of the first deviatoric strain invariant, $\tilde{I}_1$, with c_i being the material constants. In the Ogden formulation, the isochoric response is described in terms of the deviatoric principal stretches, $\tilde{\lambda}_i$, where c_i and m_i are material parameters.

To simulate the immediate postoperative behavior, the hyperelastic models were augmented with uncoupled viscoelastic properties to capture stress relaxation. The deviatoric part of the second Piola-Kirchhoff stress, $\mathbf{S}$, is calculated using the following integral:

$$\mathbf{S}(t) = \int_{-\infty}^t G(t-s) \frac{d\mathbf{S}^e}{ds} ds, \quad G(t) = 1 + \sum_{i=1}^N \gamma_i \exp\left(-\frac{t}{\tau_i}\right) \quad (5.3)$$

Where the current stress is determined by the entire history of the elastic deviatoric stress, $\mathbf{S}^e$, derived from the hyperelastic potential. The material's time-dependent behavior is characterized by the relaxation function, $G(t)$, which is represented by a Prony series. In this series, γ_i and τ_i are the relaxation constants and relaxation times for each of the N terms.

5.2.7 Soft Tissue Parameterization

To investigate the impact of donor site selection, material parameters for the Forearm skin and the Thigh skin were derived from a thorough literature review of experimental data reported in the literature (Figure 5.4). These experimental curves reveal significant variability in tissue stiffness, both within and between donor sites. To account for this clinically observed range, a

5.2. Method

bounding approach was implemented. To account for the observed biomechanical variability, three representative uniaxial stress-stretch curves were defined for each donor site, corresponding to three stiffness levels (Figure 5.4).

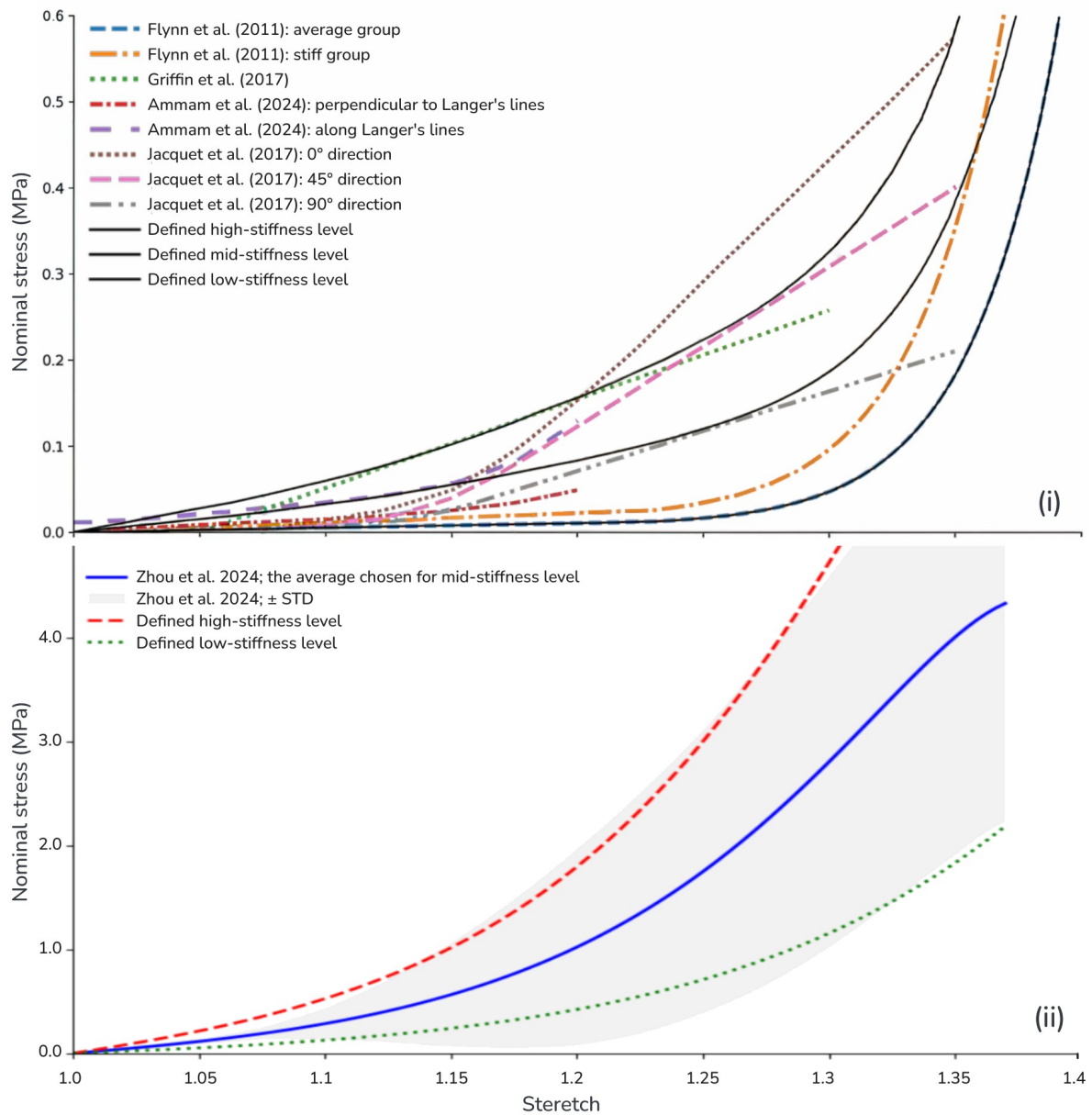


Figure 5.4 Literature review of experimental data on the material parameters for the Forearm skin (i) and the Thigh skin (ii) represented as the stress-stretch response to a uniaxial tensile test.

To incorporate the observed range of tissue properties into the simulation, a two-term Ogden material model was calibrated to each of the three stiffness levels for both the Forearm and Thigh skin. This yielded six distinct sets of material parameters that encapsulate the clinically relevant range of soft tissue properties (three for each donor site). This methodology enables the factorial study to simulate reconstructions with three levels of tissue stiffness and analyze how tissue biomechanics influence surgical outcomes.

Table 5.1 summarizes the calibrated hyperelastic parameters for the Forearm and Thigh skin across the three stiffness levels, as well as the constitutive parameters for the other soft tissues used in the simulation. All tissues were assigned viscoelastic properties using the Prony series (Eq 5.3) to model stress relaxation.

5.2.8 Simulation Stability and Robustness

To ensure the robustness and automation required for the 120-simulation factorial study, a strategy was implemented using the open-source finite element solver FEBio, which is specifically designed for biomechanical applications (Maas et al., 2012). The significant non-linearities inherent to the simulation, primarily arising from non-linear materials and contacts, were managed using an implicit dynamic solver with a robust automatic time-stepping algorithm. The highly non-linear system of equations was solved using a quasi-Newton (BFGS) method with a non-symmetric stiffness matrix.

The stability of the simulations was further enhanced through careful model setup. Contact mechanics, a common source of numerical instability, were configured by tuning the penalty parameters to ensure minimal surface penetration without inducing ill-conditioning of the stiffness matrix. To improve the numerical performance for nearly incompressible materials, uncoupled formulations were used for the hyperelastic and viscoelastic constitutive models. A critical component of the strategy was the meshing approach. Given the complex, organic geometries of the tongue and flap, tetrahedral elements were chosen over hexahedral elements due to the availability of automated and robust meshing algorithms, such as the state-of-the-art mesher fTetWild (Hu et al., 2020). To mitigate the risk of volumetric and bending locking with linear elements, higher-order quadratic tetrahedral elements (tet10) were used. These elements have been shown to reliably capture complex

Table 5.1 Material parameters used in the finite element simulations.

Soft Tissue	Model	Parameters (Values)	Source
Native Tongue	Yeoh (Hyperelastic)	$(c_1, c_2, k) = (1.037kPa, 0.486kPa, 100kPa)$	Buchaillard et al., 2009
Native Tongue	Prony (Viscoelastic)	$(\tau_1, \tau_2, \tau_3, \tau_4) = (0.05s, 1.0s, 20s, 400s);$ $(\gamma_1, \gamma_2, \gamma_3, \gamma_4) = (39.0, 4.526, 4.895, 3.211)$	Vaidya and Wheatley, 2020 [†]
Subcutaneous Adipose	Ogden (Hyperelastic)	$(c_1, m_1, k) = (9.0kPa, 6.0kPa, 100kPa)$	Sun et al., 2021c
Subcutaneous Adipose	Prony (Viscoelastic)	$(\tau_1, \tau_2, \tau_3) = (0.1s, 1.0s, 10.0s);$ $(\gamma_1, \gamma_2, \gamma_3) = (4.428, 8.692, 0.738)$	Sun et al., 2021b [†]
Forearm Skin (Stiffness Levels)	Ogden (Hyperelastic)	Level 1: $(c_1, c_2, m_1, m_2, k) = (41.5kPa, 1.39Pa, 1.223, 41.672, 200kPa);$ Level 2: $(c_1, c_2, m_1, m_2, k) = (61.3kPa, 4.38Pa, 7.059, 39.062, 200kPa);$ Level 3: $(c_1, c_2, m_1, m_2, k) = (102.1kPa, 5.20Pa, 7.660, 38.691, 200kPa)$	‡
Thigh Skin (Stiffness Levels)	Ogden (Hyperelastic)	Level 1: $(c_1, c_2, m_1, m_2, k) = (90.0kPa, -10.0Pa, 13.431, 31.000, 200kPa);$ Level 2: $(c_1, c_2, m_1, m_2, k) = (180.0kPa, -60.0Pa, 14.531, 36.333, 200kPa);$ Level 3: $(c_1, c_2, m_1, m_2, k) = (360.0kPa, -120.0Pa, 13.839, 35.274, 200kPa)$	‡
Skin	Prony (Viscoelastic)	$(\tau_1, \tau_2, \tau_3, \tau_4) = (0.3791s, 4.1987s, 6.9961s, 180.02s);$ $(\gamma_1, \gamma_2, \gamma_3, \gamma_4) = (3.649, 0.452, 0.553, 0.627)$	Then et al., 2017 [†]

Notes: [†] The referenced studies define the relaxation function differently from Eq 5.3. The reported viscoelastic parameters have been mathematically converted to be consistent with the formulation used in this study. Relaxation coefficients (γ_i) are dimensionless. [‡] Material profiles for the three stiffness levels of Forearm skin were calibrated using experimental data from several published studies (Flynn et al., 2011; Griffin et al., 2017; Ammam et al., 2024; Jacquet et al., 2017). Similarly, the profiles for Thigh skin were calibrated based on data reported by (Zhou et al., 2024).

deformations and contact mechanics without the locking artifacts, while providing solution accuracy that is comparable or superior to hexahedral elements with the same number of degrees of freedom (Maas et al., 2016).

Finally, a mesh convergence analysis was conducted to ensure that the simulation outcomes were independent of the mesh discretization. Key simulations were repeated with progressively refined mesh densities, and the results were evaluated using key outcome metrics: maximum displacement and maximum von Mises equivalent strain. The analysis demonstrated that the solution converged as the mesh was refined. The mesh density selected for the full factorial study was chosen from the converged region to maintain computational tractability. The detailed results of this convergence analysis are presented in the [Supporting information](#).

5.2.9 Outcome Metrics

To quantitatively evaluate the results of the simulations, two primary metrics were established to assess the final state of the neotongue. These metrics focus on the anatomical fidelity and the biomechanical state of the reconstructed tongue, as visualized in Figure 5.3v–vi.

The first metric, anatomical state, quantifies how closely the final, post-remodeling neotongue morphology restores the patient’s original, preoperative anatomy. The preoperative native tongue 3D model serves as the ground truth for this comparison. Since both the simulated and original tongue models are anatomically fixed to the same rigid jaw model, they share a common coordinate system, allowing for direct spatial comparison without the need for further registration. The discrepancy between the two surfaces is quantified by the maximum surface deviation, calculated as the maximum Hausdorff distance. To ensure this metric is robust against numerical artifacts or isolated mesh irregularities, the 99th percentile of the Hausdorff distances is reported, providing a stable measure of the largest regional anatomical deviation.

The second metric, imposed tongue deformation, evaluates the biomechanical strain imparted to the native tongue tissue as a result of the reconstruction and subsequent remodeling. To provide a robust and physically meaningful measure of deformation, the von Mises equivalent strain (ϵ_{eq}) was selected, hereafter referred to as Effective Strain. This scalar invariant distills the complex, three-dimensional strain state into a single value, permitting a direct and objective comparison across all simulated scenarios. The equivalent strain was calculated from the Lagrange strain tensor, $\mathbf{E}$, as follows:

$$\epsilon_{eq} = \sqrt{\frac{2}{3} \mathbf{dev} \mathbf{E} : \mathbf{dev} \mathbf{E}} \quad (5.4)$$

Where $\mathbf{dev} \mathbf{E} = \mathbf{E} - \frac{1}{3} \text{tr}(\mathbf{E}) \mathbf{I}$ is the deviatoric part of the strain tensor, $:$ denotes the double contraction, $\text{tr}(\mathbf{E})$ is the trace of $\mathbf{E}$, and $\mathbf{I}$ is the identity tensor. This formulation is particularly advantageous as it is based on the deviatoric part of the strain tensor, which mathematically isolates the distortional component of deformation (i.e., shear and tensile strain) from the hydrostatic (volumetric) component. This allows the metric to specifically quantify the mechanically significant distortion that can lead to tissue tethering and functional impairment, while remaining insensitive to the uniform volumetric shrinkage associated with the simulated flap atrophy.

5.2.10 The Factorial Study Design

A computational factorial experiment was designed to systematically investigate the anatomical and biomechanical impact of key surgical decisions. This approach allows for a controlled analysis of how different factors independently and interactively influence the outcome metrics. The study examined three primary independent variables:

- **Patient-Specific Defect Geometry:** Four distinct, clinically-plausible glossectomy scenarios were used (Figure 5.1, Cases A–D). These cases represent several distinct anatomical defects.
- **Donor Site Biomechanics:** The mechanical properties of the flap’s skin layer were varied to simulate tissue harvested from two common donor sites: the radial forearm and the thigh. To account for natural patient-to-patient variability in tissue stiffness, three distinct stiffness profiles, as detailed in section 2.7, were defined for each site.
- **Flap Overbulking:** Five percentage levels of adipose bulk increase were simulated: 0%, 22.2%, 44.4%, 66.7%, and 88.9%. Overbulking was achieved by scaling the flap’s subcutaneous adipose component in the thickness direction (Figure 5.2D). This approach reflects the clinical practice of elevating a thicker subcutaneous fat layer (for thigh flaps) or harvesting extra beavertail fat tissue (for forearm flaps). This method isolates the effect of flap’s bulk from

the flap's skin paddle size. The selected linear levels span a clinically relevant range, which includes the calculated volume needed to perfectly match the resection size after 40% atrophy (the 66.7% level).

A full factorial design was implemented by combining every level of each factor. This comprehensive approach ensures that all possible interactions between the variables can be studied. The total number of unique simulations conducted was the product of the levels for each factor: (4 defect geometries) $\times$ (6 skin stiffness levels) $\times$ (5 overbulking levels), resulting in 120 distinct simulation setups.

5.2.11 Statistical Analysis

To quantify the influence of the experimental factors on the simulated outcomes, Ordinary Least Squares (OLS) linear regression was employed for each of the two primary metrics. The independent variables from the factorial design were encoded for the regression models. *Case* was treated as a four-level categorical variable representing the defect geometries, with Case A serving as the reference. *Overbulking* was included as a continuous numerical variable. To decouple the effect of donor site biomechanics, two predictors were created: *is_thigh*, a binary variable (0 = Forearm, 1 = Thigh), and *stiffness_level*, an ordinal variable coded as -1, 0, +1 for the three stiffness levels for each of the donor sites. This encoding treats stiffness as a numerical factor and assumes a linear effect for changes in stiffness within a given donor site.

A hierarchical approach was used to build and select the most appropriate model. An initial model included only the main effects of each predictor ($Outcome \approx Case + Overbulking + is_thigh + stiffness_level$). A second, more complex model was then fitted, containing all main effects plus all two-way scientifically relevant interaction terms. The final model for interpretation was selected by comparing these nested models using an analysis of variance (ANOVA) F-test. This test determined if the interaction terms provided a statistically significant improvement in explaining outcome variance, with the adjusted R-squared also being evaluated to ensure that added complexity was justified by increased explanatory power.

Following selection, the final model for each outcome was subjected to diagnostic checks to validate the underlying assumptions of OLS regression. The normality of the model's residuals was assessed through visual inspection of quantile-quantile (Q-Q) plots. Multicollinearity was assessed using the Variance Inflation Factor (VIF), ensuring that all VIF scores were below the common threshold of 10 to confirm the stability of the model's coefficients. All statistical analyses were conducted using the *statsmodels* open-source library (Seabold and Perktold, 2010).

5.2.12 Data and Code Availability

The simulation framework is made openly available (Isazadeh, 2025) under MIT license, without restriction, enabling full reproducibility of all 120 simulations and direct application to new patient cases. Patient-specific baseline tongue MRI data cannot be shared due to privacy restrictions and ethics board requirements; however, all inputs required to reproduce the simulation outcomes and statistical results are included in this repository (hosted on GitHub at <https://github.com/is-Amir/VSP-BioSim> and archived with a citable DOI: [10.5281/zenodo.17438091](https://doi.org/10.5281/zenodo.17438091)).

This open-source release provides the research community with immediate access to a robust, scalable platform for patient-specific tongue reconstruction simulation and empowers researchers to adapt this methodology to other soft-tissue reconstruction and virtual surgical planning applications.

5.3 Results

The computational framework demonstrated high stability and robustness, successfully completing all 120 unique simulations in the full factorial study without numerical failures. The simulations provided quantitative outcomes for the final anatomy and biomechanics of the neotongue across the full spectrum of surgical variables tested.

The overall trends are visualized in Figure 5.5. A clear trade-off emerges between the two primary outcomes: increasing the level of subcutaneous tissue overbulking consistently increased the maximum Effective Strain imposed on the native tongue (top row), while simultaneously decreasing the maximum surface deviation from the original anatomy (middle row). The final, post-remodeling

5.3. Results

tongue volume was a direct and linear function of the initial overbulking percentage (bottom row). The 66.7% overbulking level consistently resulted in a final flap volume that closely compensated for the simulated 40% atrophy. At this level, the final tongue volume closely matched the preoperative native tongue volume, resulting in a percentage change close to zero.

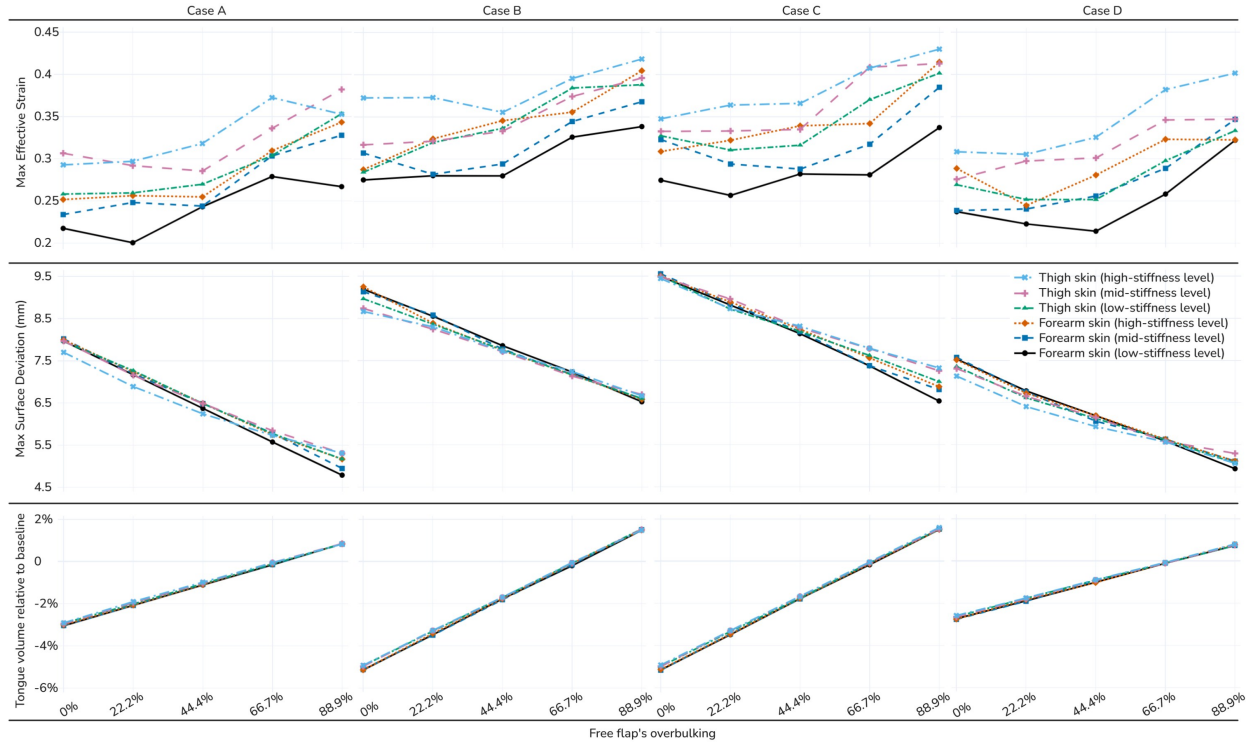


Figure 5.5 Visualization of the sensitivity of dependent variables to different skin material properties and different levels of overbulking: overbulking increases Effective Strain (top) while decreasing anatomical deviation (middle); final tongue volume varies linearly with initial overbulking (bottom).

To quantify the factors influencing the first dependent variable (maximum surface deviation) a hierarchical linear regression analysis was performed. A formal model comparison using an ANOVA F-test revealed that the model including two-way interactions provided a significantly better fit to the data than a main effects model ($F = 26.12$, $p < 0.001$). The interaction model explained a substantial portion of the variance ($\text{Adj. } R^2 = 0.993$) and was therefore selected for interpretation.

The regression results, detailed in Table 5.2, identify *overbulking* as the most dominant factor for restoring the original tongue anatomy. The significant negative coefficient (-3.420 , $p < 0.001$) indicates that increasing the flap volume consistently reduces the maximum surface deviation.

5.3. Results

However, the beneficial effect of overbulking is significantly modulated by donor site selection (coeff = 0.494, $p < 0.001$), stiffness level (coeff = 0.160, $p < 0.001$), and defect geometry (all $p < 0.001$).

Table 5.2 OLS regression results for maximum surface deviation (interaction model). Dependent variable: maximum surface deviation (mm). Reference: *Case A*; *is_thigh* = 0 (Forearm).

Predictor	Coefficient	Std. Error	t-statistic	p-value
Intercept	8.000	0.036	220.77	< 0.001
<i>Case B</i> vs. <i>A</i>	1.101	0.046	24.01	< 0.001
<i>Case C</i> vs. <i>A</i>	1.594	0.046	34.77	< 0.001
<i>Case D</i> vs. <i>A</i>	-0.578	0.046	-12.61	< 0.001
<i>overbulking</i>	-3.420	0.067	-51.38	< 0.001
<i>is_thigh</i>	-0.232	0.032	-7.15	< 0.001
<i>stiffness_level</i>	-0.035	0.023	-1.54	0.125
<i>overbulking</i> × <i>Case B</i>	0.482	0.084	5.72	< 0.001
<i>overbulking</i> × <i>Case C</i>	0.335	0.084	3.98	< 0.001
<i>overbulking</i> × <i>Case D</i>	0.625	0.084	7.42	< 0.001
<i>overbulking</i> × <i>is_thigh</i>	0.494	0.060	8.30	< 0.001
<i>overbulking</i> × <i>stiffness_level</i>	0.160	0.036	4.39	< 0.001
<i>is_thigh</i> × <i>stiffness_level</i>	-0.065	0.023	-2.82	0.006

A separate regression analysis was conducted for the second dependent variable: Maximum Effective Strain. The model comparison showed that adding interaction terms did not provide any significant improvement over the main effects model ($F = 0.069$, $p = 0.999$). The main effects model was therefore selected as the superior representation of the dependent variable, explaining 86.3% of the variance in maximum effective strain.

The results for the main effect model, presented in Table 5.3, describe a simple, additive system. All main effects—*overbulking*, *is_thigh*, and *stiffness_level*—were highly significant positive predictors of strain. Increasing flap volume, using tissue from the Thigh, and increasing skin stiffness each independently and consistently increased the maximum Effective Strain experienced

by the native tongue tissue. The defect geometry (*Case*) also had a significant effect, with Cases B and C (larger resections) resulting in higher strain than the reference Case A, while Case D was not significantly different. The lack of significant interactions implies that the strain-inducing effect of overbulking was consistent and not modulated by other independent variables.

Table 5.3 OLS regression results for maximum Effective Strain (main effects model). Dependent variable: maximum Effective Strain. Reference: *Case A*; *is_thigh* = 0 (Forearm).

Predictor	Coefficient	Std. Error	t-statistic	p-value
Intercept	0.226	0.005	49.27	< 0.001
<i>Case B</i> vs. A	0.051	0.005	10.37	< 0.001
<i>Case C</i> vs. A	0.052	0.005	10.67	< 0.001
<i>Case D</i> vs. A	0.004	0.005	0.81	0.422
<i>overbulking</i>	0.093	0.006	16.75	< 0.001
<i>is_thigh</i>	0.043	0.003	12.32	< 0.001
<i>stiffness_level</i>	0.023	0.002	10.74	< 0.001

5.4 Discussion

The fundamental goal of tongue reconstruction is twofold: to restore the anatomical volume required for effective tongue-palate contact during swallowing and speech articulation (Wang et al., 2013; Seikaly et al., 2009), and to preserve the dynamic mobility of the residual tongue essential for chewing, food propulsion, and clear speech (Marra et al., 2023; Urken, 2012; Seikaly et al., 2009). This study provides a mechanistic and quantitative explanation for the clinical challenges inherent in balancing these objectives. Our simulations reveal a fundamental, physics-based trade-off: surgical decisions aimed at restoring the neotongue’s bulk through flap overbulking invariably impose significant and predictable strain on the remaining native tongue tissue. This finding suggests that the “ideal” reconstruction is one that optimally balances volume restoration against the biomechanical cost.

These insights were enabled by the primary methodological achievement of this work: the development of a novel, end-to-end computational pipeline for surgical simulation. This framework uniquely integrates patient-specific model generation, biomechanically-optimized flap design, the simulation of tongue reconstruction, and the modeling of biological atrophy. Its power as a robust and automated virtual testbed was exemplified through the successful execution of a 120-simulation factorial study, which enabled such a systematic investigation.

The selection of a donor site is a foundational surgical decision that imposes a distinct and quantifiable biomechanical cost on the reconstructed tongue. Our regression analysis (Table 5.3) reveals a clear, quantitative relationship: the stiffer the flap tissue, the greater the biomechanical strain imposed on the residual native tongue. Specifically, the model demonstrated that Thigh flaps induced a significantly greater mechanical strain on the surrounding native tongue than their Forearm counterparts. This biomechanical penalty was further amplified by increasing stiffness levels within each donor site. This finding provides the first physics-based, quantitative measure of the clinical concept of “tethering”; the mechanical anchoring of the mobile tongue remnant by a stiff, non-compliant flap.

This quantified tethering effect provides a direct mechanical rationale for the functional trade-offs reported extensively in the clinical literature. Studies consistently associate the thin, pliable Radial Forearm Free Flap (RFFF) with superior speech articulation, a function dependent on preserving tongue mobility (Marra et al., 2023; Sepúlveda et al., 2025; Seikaly et al., 2009). Our results support this clinical observation, demonstrating that the lower material stiffness of the Forearm flap results in the lowest imposition of strain, thereby creating a more favorable biomechanical environment for the complex movements of speech. In contrast, for extensive resections creating large volumetric deficits, such as total and subtotal glossectomies and those involving the tongue base, the ALT flap is the preferred reconstructive choice to restore the required bulk (Marra et al., 2023; Cai et al., 2019). Our simulations support the rationale for this choice but also reveal its inherent biomechanical cost: it simultaneously imposes a greater mechanical strain, which can compromise the mobility required for vital functions.

The clinical practice of overbulking is a direct response to the challenge of postoperative atrophy, yet it forces surgeons into a delicate balancing act to avoid creating a neotongue that is too bulky and immobile (Bashorun et al., 2023). The computational framework provides a clear explanation for this surgical dilemma by quantifying its consequences. The results reveal a fundamental conflict: increasing flap volume is, by a significant margin, the most powerful surgical parameter for reducing maximum surface deviation, which is a direct proxy for restoring the preoperative anatomy (Table 5.2, $\text{coeff} = -3.420$, $p < 0.001$). However, this anatomical gain comes at a direct and unavoidable biomechanical cost, similar to the findings regarding stiffness levels. As shown in Table 5.3, every increase in overbulking imposes a linear and predictable increase in the strain imparted to the residual native tongue ($\text{coeff} = 0.093$, $p < 0.001$). This finding quantitatively defines the trade-off between form and function. The “ideal” reconstruction is therefore not simply a matter of perfectly matching the original volume, but rather a patient-specific compromise that optimizes anatomical fidelity against the predictable biomechanical strain that may tether the tongue and impair function; what we know as “too bulky and immobile”.

While the biomechanical cost of overbulking is straightforward and additive, its anatomical benefit is highly conditional, modulated by both the properties of the flap and the geometry of the defect. The regression analysis for anatomical deviation (Table 5.2) is characterized by significant interaction terms, showing that the effectiveness of adding volume is not constant. The most critical interaction is with the donor site (*overbulking* x *is_thigh*, $\text{coeff} = 0.494$, $p < 0.001$). The positive coefficient reveals that the anatomical benefit of adding volume is significantly less pronounced for the stiffer Thigh flaps compared to the more pliable Forearm flaps. This finding introduces a key nuance: while a Thigh flap shows a minor anatomical advantage at baseline (*is_thigh*, $\text{coeff} = -0.232$, $p < 0.001$), the Forearm flap’s final anatomy is more responsive to the surgeon’s corrective overbulking. We interpret this as a difference in “tunability”; the compliant Forearm flap is more malleable, allowing the added subcutaneous volume to be effectively shaped by the surrounding native tongue, leading to better anatomical conformity. This effect is reinforced by the *overbulking* x *stiffness_level* interaction ($\text{coeff} = 0.160$, $p < 0.001$), which reveals that for any given donor site, the anatomical return on investment from overbulking is greatest when using the most pliable tissue available. This provides a direct, physics-based rationale for the clinical preference for the beavertail modification of the RFFF, which increases the subcutaneous bulk while maintaining

the flap's overall pliability (Seikaly et al., 2009; Dziegielewski et al., 2019; Hanubal et al., 2023). Furthermore, the significant *overbulking* x *Case* interactions highlight that the benefit of adding volume is not universal; it is dictated by the unique geometrical constraints of the resection site. Taken together, the model demonstrates that while overbulking is universally beneficial for restoring anatomy, its clinical effectiveness is highly variable and depends on the flap's inherent biomechanics and the specific demands of the defect.

Together, the surgical decisions of donor site selection and flap overbulking are not independent variables but a coupled system with compounding effects. The choice of donor site establishes a baseline biomechanical cost in the form of tissue strains and tensions. Moreover, it dictates the therapeutic effectiveness of overbulking; a pliable Forearm flap imposes less initial strain and allows its final anatomy to be more effectively 'tuned' with added volume. Conversely, a stiffer Thigh flap introduces a higher baseline strain and shows diminished anatomical returns from the same corrective overbulking. Therefore, an optimal reconstructive strategy must consider these decisions in unison, recognizing that the inherent properties of the chosen flap define the potential and the limitations of subsequent efforts to restore anatomical form. This interdependence underscores the complexity of surgical planning and highlights the need for patient-specific models such as the one presented herein to navigate this complex decision-making process.

The presented framework introduces a novel approach for simulating the complex procedure of free flap reconstruction. This is achieved by modeling the flap as an independent, multi-component body that is virtually sutured to the resection site. This approach fundamentally advances two established paradigms in surgical simulation. First, it moves beyond the "property modification" approach, where the outcome of a reconstruction is simulated by altering the material properties of elements within a single, continuous mesh. Foundational work by Buchaillard et al. (2007), later adopted by van Alphen et al. (2013) and more recently Al-Zanoon et al. (2024), effectively used this simplification to explore the functional consequences of flap stiffness mismatch or fibrotic tissue. Our framework, in contrast, models the flap as a distinct biomechanical entity. This allows explicit capture of the physics of the flap-tongue interface, the induced deformations by suture tensioning, and the final postoperative equilibrium state. In essence, it simulates the surgical process, not just its endpoint material state. Second, our work extends the "primary closure" simulation paradigm,

pioneered by Kappert et al. (2019). Their method represented a major advance by computationally replicating the surgical act of suturing, successfully modeling the deformation caused by pulling native tissue edges together to close a small defect. Our framework adapts this concept to the more complex and common scenario of microvascular free flap reconstruction.

Another key innovation of this pipeline is the simulation of postoperative tissue atrophy, which enables the model to evolve from its immediate “Day 0” state to a predicted one-year post-remodeling anatomy; the state most relevant to long-term patient function. By modeling this crucial remodeling phase, the framework provides a biomechanical basis to investigate the clinical practice of flap overbulking. The focus on long-term biological change complements prior simulation efforts that have also investigated post-treatment tissue alteration. Al-Zanoon et al. (2024), for instance, modeled the development of radiation-induced fibrosis by increasing the stiffness of affected tissue elements. Our approach is complementary, modeling atrophy via a prescribed volumetric change, another critical remodeling phenomenon that dictates the final tongue morphology. Furthermore, while Wang et al. (2019) specifically studied the functional consequences of base-of-tongue volume loss on swallowing, their work focused on the hydrodynamics of bolus flow driven by a prescribed boundary motion. Our framework, in contrast, simulates the solid mechanics of how the reconstructed neotongue itself deforms and settles into a new passive equilibrium as the flap tissue atrophies. This provides direct insight into the final resting anatomy and the resulting strains imposed on the native tongue, a different set of biomechanical outcomes.

The successful execution of a 120-simulation factorial study demonstrates the framework’s core design principles: automation and robustness. This scalability marks a methodological shift. We present a high-throughput virtual testbed that contrasts with a significant body of research focused on building baseline models of the highest fidelity. Such landmark efforts include the comprehensive FRANK model, which integrates numerous head and neck components (Stavness et al., 2011; Stavness et al., 2014a; Anderson et al., 2017), and patient-specific models derived from DW-MRI that incorporate detailed muscle fiber architecture (Kappert et al., 2021a). Our framework however, makes a deliberate, pragmatic trade-off. Since our primary goal is to predict the final, passive anatomy at rest after biological remodeling, we do not model the active muscle architecture, which is critical for functional simulation but not for this specific question. Instead,

the framework employs well-established isotropic hyperelastic models (Hermant et al., 2017) rather than more complex anisotropic formulations, which have proven exceptionally difficult to parameterize from in-vivo data (Kappert et al., 2021a). This trade-off prioritizes simulation scalability over baseline complexity. The focus is therefore not on perfecting the pre-operative model but on robustly simulating the post-operative consequences that arise from interdependent surgical decisions. Ultimately, this capability distinguishes our approach from prior tongue surgery simulations, which have necessarily focused on single-case analyses or a small number of scenarios (Buchaillard et al., 2007; Kappert et al., 2019). It is this scalability that opens the door to systematic, multi-parameter investigations of surgical variables.

This degree of automation and robustness is what transforms the framework from a research platform into a viable tool for patient-specific pre-operative planning. The laborious, manual, and expert-driven process of creating high-fidelity, patient-specific biomechanical models has traditionally been a major bottleneck, limiting their scalability for large-scale or clinical applications (Anderson et al., 2017; Harandi et al., 2017b; Harandi et al., 2017a; Kappert et al., 2021a). In contrast, our framework is designed for clinical tractability. The pipeline requires only two triangulated meshes—the baseline tongue and the planned resection—which directly align with the modern VSP workflows where a surgeon delineates the resection on medical images. From this point, the entire workflow—from biomechanically optimized flap design to volumetric FEM generation and final post-atrophy analysis—is fully automated. The clinical necessity for such a tool is underscored by the results of the factorial study itself. The significant *overbulking* $\times$ *Case* interaction demonstrates that the optimal surgical strategy is dictated by the unique geometric constraints of the patient’s specific defect. The streamlined and robust nature of the pipeline makes it computationally feasible to conduct a tailored series of simulations for an individual patient to optimize a critical variable like the overbulking percentage. This offers a direct, physics-based method to navigate the complex, interdependent surgical decision and move toward data-driven, individualized surgical planning.

The current framework, while robust, relies on a simplified representation of the tongue's complex biomechanics and postoperative healing. The neotongue undergoes continuous dynamic loading during the remodeling period. Neurological compensatory mechanisms, such as the stretch-reflex response to altered tissue states and motor equivalence strategies, drive differential muscle recruitment patterns (Al-Zanoon et al., 2025). Consequently, native muscle bundles may undergo compensatory hypertrophy or atrophy. The extent and mechanism of these activity-dependent anatomical changes in the native tongue have yet to be quantified clinically and are not captured by the passive models used in this study. Future work should incorporate active muscle contraction models and muscle-fiber orientation data, potentially from DT-MRI, to simulate both the tongue's anisotropic properties and these dynamic functional and anatomical adaptations. Ultimately, such an enhanced framework would capture how these compensatory adaptations influence speech and swallowing outcomes.

Moreover, the current biological remodeling simulation can be enhanced by implementing patient-specific atrophy models that factor in variables like radiation dose, flap composition, and patient-specific factors like age, sex, and BMI. Such studies could also benefit from modeling the formation of stiffer, contracted scar tissue at the incision interfaces. Finally, refining the constitutive parameters through in vivo techniques, such as MR Elastography for the tongue (Cheng et al., 2011) and preoperative mechanical testing of donor skin (Flynn et al., 2011), will increase the model's predictive accuracy.

To translate this framework into a clinically viable tool, several validation and generalization steps are necessary. A prospective clinical study is needed to validate the simulated anatomical outcomes against long-term postoperative imaging from a cohort of patients. The model's generalizability must also be established by applying it to a diverse range of patient anatomies and tumor-specific resection geometries. The robust pipeline is also well-suited for large-scale in silico studies to systematically investigate surgical variables, such as comparing the biomechanically optimized flaps used herein against conventional flap designs sourced from surgeons. Ultimately, this framework has the potential to serve as a research platform for systematically investigating surgical variables and as a tool for conducting patient-specific preoperative simulations to compare optimal reconstructive strategies.

5.5 Conclusion

This study's primary contribution is the development and application of a robust and automated simulation pipeline capable of modeling key aspects of tongue reconstruction surgery and its long-term anatomical remodeling. This framework provides a novel platform to systematically investigate interdependent surgical variables that are otherwise challenging to assess. Our findings provide a quantitative explanation for the delicate balance surgeons must strike between restoring volume and preserving mobility. The simulations confirm that surgical decisions to increase bulk invariably impart a biomechanical cost in the form of tissue strain, which can lead to the tethering of the mobile tongue. We demonstrated that the choice of donor site and the degree of overbulking are a coupled system. The selection of a stiffer donor site, like the thigh, not only imposes a higher baseline mechanical strain but also limits the anatomical effectiveness of subsequent overbulking. This supports the clinical rationale for using pliable flaps, such as the beavertail-modified forearm flap, to achieve anatomical conformity with a lower biomechanical penalty. Crucially, the significant interaction between surgical strategy and the patient's unique defect geometry confirms that a universal approach is suboptimal. The results highlight the potential for this framework to be used for patient-specific preoperative planning, allowing surgeons to computationally explore different scenarios and gain insight into the specific case. Methodological refinements should include anisotropic material properties and active muscle modeling, and the framework's predictive accuracy must ultimately be validated through prospective clinical studies. This work lays the foundation for a powerful research platform and, following clinical validation, a potential decision-support tool to help surgeons optimize and individualize reconstructive strategies.

Funding This work was supported by the Alberta Cancer Foundation (grant number 27 601). The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Supporting information

Mesh convergence analysis. We assessed mesh-independence for 12 randomly-selected representative parameter combinations (Figure 5.6). Maximum equivalent strain (Eq 5.3) versus total mesh nodes plateaus between $\sim 24\text{k}$ and $\sim 31\text{k}$ nodes; the change across the last two refinements is $< 5\%$ for all series. Maximum displacement showed the same plateau (Figure 5.7). Consistent with Simulation Stability and Robustness, we therefore used $\sim 24\text{k}$ nodes for the full study as mesh-independent within tolerance.

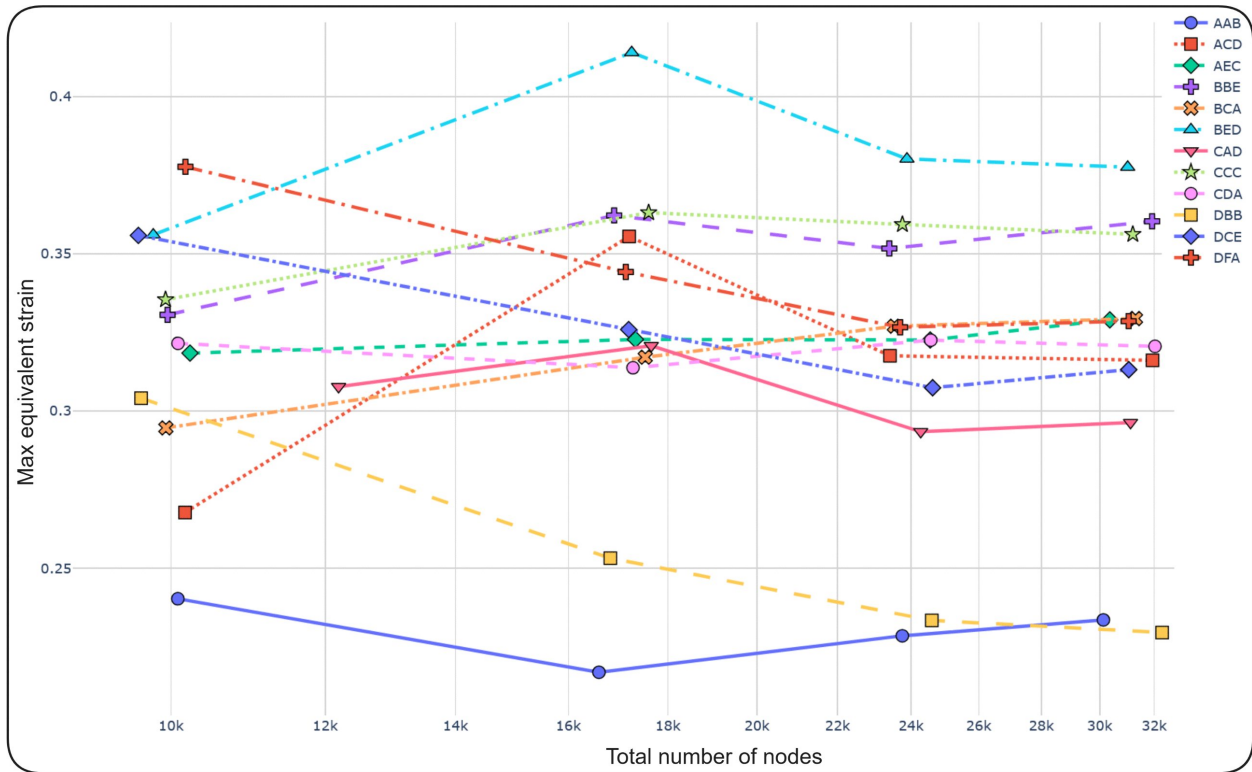


Figure 5.6 Mesh convergence analysis—Maximum equivalent strain. Maximum equivalent strain of the neotongue at the final post-remodeling equilibrium versus total mesh nodes. Each curve is one of 12 representative parameter combinations and is labeled with a three-letter code in the form [Case][Donor/Stiffness][Overbulking], where Cases A–D are the four defect geometries; Donor/Stiffness A–C = Forearm (soft→stiff), D–F = Thigh (soft→stiff); Overbulking A–E = 0%, 22.2%, 44.4%, 66.7%, 88.9%.

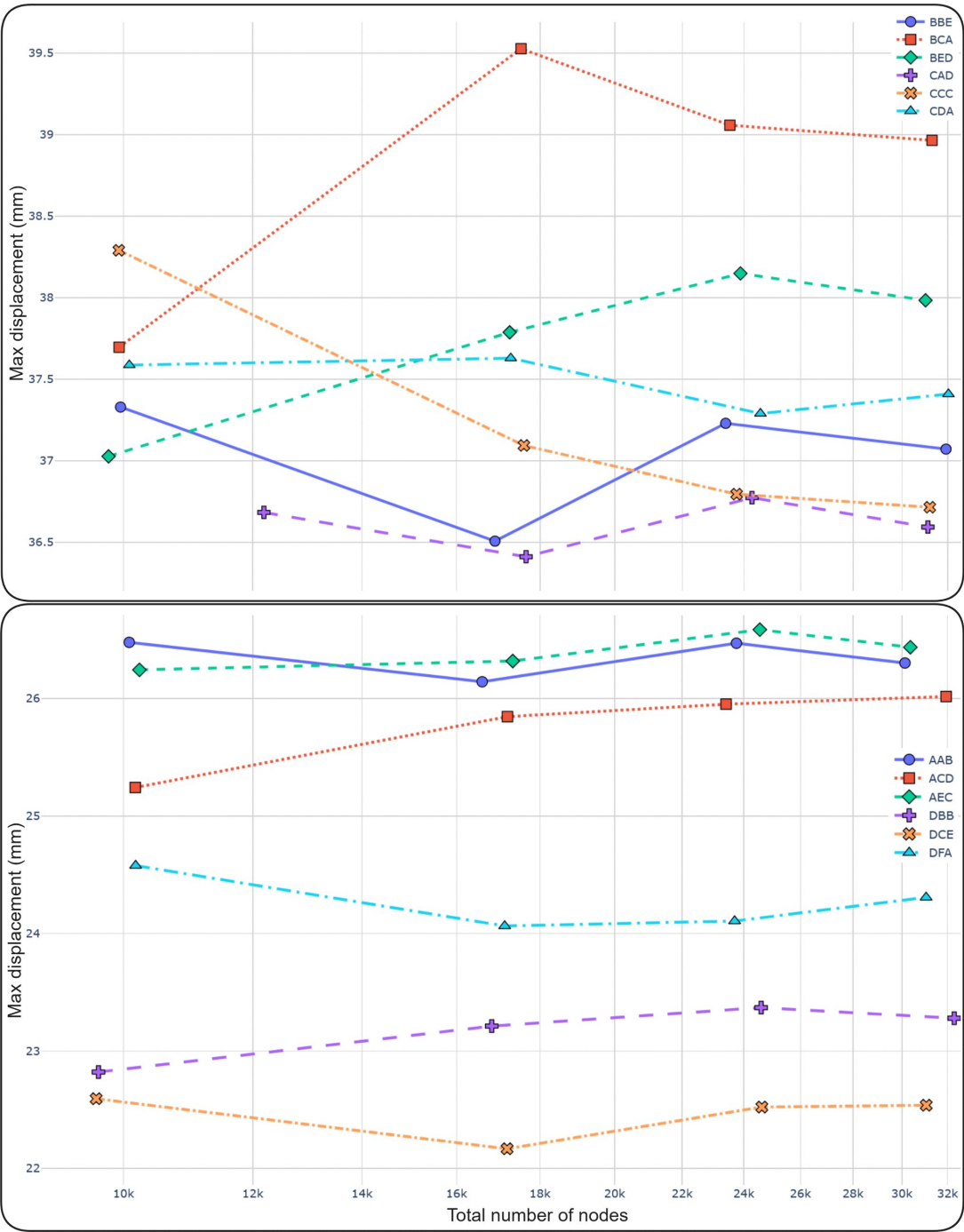


Figure 5.7 Mesh convergence analysis—Maximum displacement. Maximum nodal displacement of the neotongue at the final post-remodeling equilibrium (mm) versus total mesh nodes for the same 12 representative parameter combinations used in Figure 5.6. Codes follow the [Case][Donor/Stiffness][Overbulking] convention described there. Two panels are shown for readability with a shared horizontal axis.

References

- Al-Zanoon, N., Cummine, J., Jeffery, C. C., and Aalto, D. (2025). “Compensatory Behaviours after Oral Cancer: Stretch Reflex Improves Simulated Tongue Protrusion.” In: *Computer Methods in Biomechanics and Biomedical Engineering*, pp. 1–11. ISSN: 1025-5842, 1476-8259. DOI: [10.1080/10255842.2025.2482128](https://doi.org/10.1080/10255842.2025.2482128).
- Al-Zanoon, N., Cummine, J., Jeffery, C. C., Westover, L., and Aalto, D. (2024). “The Effect of Simulated Radiation Induced Fibrosis on Tongue Protrusion.” In: *Biomech. Model. Mechanobiol.* 23, pp. 1649–1660. ISSN: 1617-7940. DOI: [10.1007/s10237-024-01860-4](https://doi.org/10.1007/s10237-024-01860-4).
- Ammam, I., Guillermin, A., Ouillon, L., Vargiolu, R., Perrot, J.-L., and Zahouani, H. (2024). “Coupling Tensile Test with LC-OCT and Ultrasound Imaging: Investigation of the Skin Sub-layers Mechanical Behaviour.” In: *R. Soc. Open Sci.* 11.6, p. 231712. ISSN: 2054-5703. DOI: [10.1098/rsos.231712](https://doi.org/10.1098/rsos.231712).
- Anderson, P., Fels, S., Harandi, N. M., Ho, A., Moisik, S., Sánchez, C. A., Stavness, I., and Tang, K. (2017). “FRANK: A Hybrid 3D Biomechanical Model of the Head and Neck.” In: *Biomechanics of Living Organs*. Ed. by Y. Payan and J. Ohayon. Vol. 1. Translational Epigenetics. Oxford: Academic Press, pp. 413–447. DOI: [10.1016/B978-0-12-804009-6.00020-1](https://doi.org/10.1016/B978-0-12-804009-6.00020-1).
- Bashorun, O. H., Johnson, R. M., and Johnson, E. A. (2023). “Flap Debulking and Secondary Revisions in Head and Neck Reconstruction: A Systematic Review with Clinical Applications.” In: *Semin Plast Surg* 37.01, pp. 073–082. ISSN: 1535-2188, 1536-0067. DOI: [10.1055/s-0042-1760444](https://doi.org/10.1055/s-0042-1760444).
- Baskin, R. M., Seikaly, H., Sawhney, R., Danan, D., Burt, M., Idris, S., Shama, M., Boyce, B., and Dziegielewska, P. T. (2019). “Tongue Reconstruction: Rebuilding Mobile Three-Dimensional Structures from Immobile Two-Dimensional Substrates, a Fresh Cadaver Study.” In: *Head Neck-J. Sci. Spec. Head Neck* 41.10, pp. 3693–3699. ISSN: 1043-3074, 1097-0347. DOI: [10.1002/hed.25889](https://doi.org/10.1002/hed.25889).
- Bielecki, K. (2017). “Are Randomized Clinical Trials Necessary in Cancer Surgery?” In: *Nowotw. J. Oncol.* 66.6, pp. 494–498. ISSN: 2300-2115, 0029-540X. DOI: [10.5603/NJO.2016.0088](https://doi.org/10.5603/NJO.2016.0088).

- Buchaillard, S., Brix, M., Perrier, P., and Payan, Y. (2007). "Simulations of the Consequences of Tongue Surgery on Tongue Mobility: Implications for Speech Production in Post-Surgery Conditions." In: *Int. J. Med. Robot. Comput. Assist. Surg.* 3.3, pp. 252–261. ISSN: 1478-5951, 1478-596X. DOI: [10.1002/rcs.142](https://doi.org/10.1002/rcs.142).
- Buchaillard, S., Perrier, P., and Payan, Y. (2009). "A Biomechanical Model of Cardinal Vowel Production: Muscle Activations and the Impact of Gravity on Tongue Positioning." In: *J. Acoust. Soc. Am.* 126.4, pp. 2033–2051. ISSN: 0001-4966, 1520-8524. DOI: [10.1121/1.3204306](https://doi.org/10.1121/1.3204306).
- Cai, Y.-C., Li, C., Zeng, D.-f., Zhou, Y. Q., Sun, R. H., Shui, C. Y., Pei, J., Liu, W., Wang, X., Jiang, Z. H., Tang, Z. Q., Jiang, J., and Wang, W. (2019). "Comparative Analysis of Radial Forearm Free Flap and Anterolateral Thigh Flap in Tongue Reconstruction after Radical Resection of Tongue Cancer." In: *ORL-J. Oto-Rhino-Laryngol. Head Neck Surg.* 81.5-6, pp. 252–264. ISSN: 0301-1569, 1423-0275. DOI: [10.1159/000502151](https://doi.org/10.1159/000502151).
- Cheng, S., Gandevia, S., Green, M., Sinkus, R., and Bilston, L. (2011). "Viscoelastic Properties of the Tongue and Soft Palate Using MR Elastography." In: *Journal of Biomechanics* 44.3, pp. 450–454. ISSN: 00219290. DOI: [10.1016/j.jbiomech.2010.09.027](https://doi.org/10.1016/j.jbiomech.2010.09.027).
- Dziegielewski, P. T., Rieger, J., Shama, M. A., O'Connell, D. A., Harris, J. R., and Seikaly, H. (2019). "Beavertail Modification of the Radial Forearm Free Flap in Total Oral Glossectomy Reconstruction: Technique and Functional Outcomes." In: *Oral Oncol.* 96, pp. 71–76. ISSN: 1368-8375, 1879-0593. DOI: [10.1016/j.oraloncology.2019.07.004](https://doi.org/10.1016/j.oraloncology.2019.07.004).
- Flynn, C., Taberner, A., and Nielsen, P. (2011). "Modeling the Mechanical Response of In Vivo Human Skin Under a Rich Set of Deformations." In: *Ann. Biomed. Eng.* 39.7, pp. 1935–1946. ISSN: 0090-6964, 1573-9686. DOI: [10.1007/s10439-011-0292-7](https://doi.org/10.1007/s10439-011-0292-7).
- Gewirtz, J. I., Zhao, S., Brock, G., Luttrull, M. D., Sethuraman, S., Kang, S. Y., Vankoeverying, K. K., and Seim, N. B. (2024). "A Pilot Study: Free Flap Atrophy in Tongue Reconstruction Using 3D Volumetric Analysis." In: *Ann. Otol. Rhinol. Laryngol.* 133.3, pp. 253–260. ISSN: 0003-4894, 1943-572X. DOI: [10.1177/00034894231204720](https://doi.org/10.1177/00034894231204720).
- Griffin, M. F., Leung, B. C., Premakumar, Y., Szarko, M., and Butler, P. E. (2017). "Comparison of the Mechanical Properties of Different Skin Sites for Auricular and Nasal Reconstruction." In: *Journal of Otolaryngology - Head & Neck Surgery* 46.1, p. 33. ISSN: 1916-0216, 1916-0216. DOI: [10.1186/s40463-017-0210-6](https://doi.org/10.1186/s40463-017-0210-6).

- Hanubal, K. S., Reschly, W. J., Conrad, D., Festa, B. M., Weiss, J. P., Shama, M., Danan, D., Hughley, B., and Dziegielewski, P. T. (2023). “The Beavertail Modified Radial Forearm Free Flap: Retrospective Review of a Versatile Technique to Increase Flap Bulk in the Head and Neck.” In: *Microsurgery* 43.8, pp. 767–774. ISSN: 0738-1085, 1098-2752. DOI: [10.1002/micr.31028](https://doi.org/10.1002/micr.31028).
- Harandi, N., Stavness, I., Woo, J., Stone, M., Abugharbieh, R., and Fels, S. (2017a). “Subject-Specific Biomechanical Modelling of the Oropharynx: Towards Speech Production.” In: *Comput. Methods Biomech. Biomed. Eng. Imaging Vis.* 5.6, pp. 416–426. ISSN: 2168-1163. DOI: [10.1080/21681163.2015.1033756](https://doi.org/10.1080/21681163.2015.1033756).
- Harandi, N., Woo, J., Stone, M., Abugharbieh, R., and Fels, S. (2017b). “Variability in Muscle Activation of Simple Speech Motions: A Biomechanical Modeling Approach.” In: *The Journal of the Acoustical Society of America* 141.4, pp. 2579–2590. ISSN: 0001-4966. DOI: [10.1121/1.4978420](https://doi.org/10.1121/1.4978420).
- Hermant, N., Perrier, P., and Payan, Y. (2017). “Human Tongue Biomechanical Modeling.” In: *Biomechanics of Living Organs*. Ed. by Y. Payan and J. Ohayon. Vol. 1. Translational Epigenetics. Oxford: Academic Press, pp. 395–411. DOI: [10.1016/B978-0-12-804009-6.00019-5](https://doi.org/10.1016/B978-0-12-804009-6.00019-5).
- Hu, Y., Schneider, T., Wang, B., Zorin, D., and Panozzo, D. (2020). “Fast Tetrahedral Meshing in the Wild.” In: *ACM Trans. Graph.* 39.4. ISSN: 0730-0301, 1557-7368. DOI: [10.1145/3386569.3392385](https://doi.org/10.1145/3386569.3392385).
- Isazadeh, A. R. (2025). *VSP-BioSim: Patient-Specific Virtual Surgical Planning via Biomechanical Simulations*. DOI: [10.5281/zenodo.17438091](https://doi.org/10.5281/zenodo.17438091).
- Isazadeh, A. R., Zenke, J., Westover, L., Seikaly, H., and Aalto, D. (2025b). “Patient-Specific Virtual Surgical Planning for Tongue Reconstruction: Evaluating Hyperelastic Inverse FEM with Four Simulated Tongue Cancer Cases.” In: *Phys. Med. Biol.* 70, p. 145017. ISSN: 0031-9155, 1361-6560. DOI: [10.1088/1361-6560/adebd8](https://doi.org/10.1088/1361-6560/adebd8).
- Jacquet, E., Chambert, J., Pauchot, J., and Sandoz, P. (2017). “Intra- and Inter-individual Variability in the Mechanical Properties of the Human Skin from in Vivo Measurements on 20 Volunteers.” In: *Skin Research and Technology* 23.4, pp. 491–499. ISSN: 0909-752X, 1600-0846. DOI: [10.1111/srt.12361](https://doi.org/10.1111/srt.12361).

- Jeong, W. S. and Oh, T. S. (2016). “Oral and Oropharyngeal Reconstruction with a Free Flap.” In: *Arch Craniofac Surg* 17.2, p. 45. ISSN: 2287-1152, 2287-5603. DOI: [10.7181/acfs.2016.17.2.45](https://doi.org/10.7181/acfs.2016.17.2.45).
- Kappert, K. D. R., van Alphen, M. J. A., van Dijk, S., Smeele, L. E., Balm, A. J. M., and van der Heijden, F. (2019). “An Interactive Surgical Simulation Tool to Assess the Consequences of a Partial Glossectomy on a Biomechanical Model of the Tongue.” In: *Comput. Methods Biomech. Biomed. Eng.* 22.8, pp. 827–839. ISSN: 1025-5842, 1476-8259. DOI: [10.1080/10255842.2019.1599362](https://doi.org/10.1080/10255842.2019.1599362).
- Kappert, K. D. R., Voskuilen, L., Smeele, L. E., Balm, A. J. M., Jasperse, B., Nederveen, A. J., and van der Heijden, F. (2021a). “Personalized Biomechanical Tongue Models Based on Diffusion-Weighted MRI and Validated Using Optical Tracking of Range of Motion.” In: *Biomech. Model. Mechanobiol.* 20.3, pp. 1101–1113. ISSN: 1617-7959, 1617-7940. DOI: [10.1007/s10237-021-01435-7](https://doi.org/10.1007/s10237-021-01435-7).
- Kim, M.-S., Oh, K. H., Cho, J.-G., Baek, S.-K., Woo, J.-S., Jung, K.-Y., and Kwon, S. Y. (2020). “Assessment of Chronological Volume Changes in Radial Forearm Free Flaps for Tongue Cancer.” In: *ORL* 82.1, pp. 40–46. ISSN: 0301-1569, 1423-0275. DOI: [10.1159/000504196](https://doi.org/10.1159/000504196).
- Lam, L. and Samman, N. (2013). “Speech and Swallowing Following Tongue Cancer Surgery and Free Flap Reconstruction - A Systematic Review.” In: *Oral Oncol.* 49.6, pp. 507–524. ISSN: 1368-8375, 1879-0593. DOI: [10.1016/j.oraloncology.2013.03.001](https://doi.org/10.1016/j.oraloncology.2013.03.001).
- Liu, M.-D., Liu, D.-Z., Al-Aroomi, M. A., Xiong, J.-J., Liu, S., Sun, C.-F., and Liu, F.-Y. (2023a). “Reconstruction of Large Defects of Anterior Floor of Mouth with Free Flaps Using a Novel Individualized Flap Design Method.” In: *J. Cranio-MaxilloFac. Surg.* 51.9, pp. 574–579. ISSN: 1010-5182, 1878-4119. DOI: [10.1016/j.jcms.2023.08.006](https://doi.org/10.1016/j.jcms.2023.08.006).
- Liu, M.-D., Xue, X.-M., Al-Aroomi, M. A., Xiong, J.-J., Liu, S., Sun, C.-F., and Liu, F.-Y. (2023b). “A Novel Flap Design Technique for Subtotal Tongue Reconstruction with an “Individualized and Convenient Tongue Model”.” In: *Oral Oncol.* 145, p. 106531. ISSN: 1368-8375. DOI: [10.1016/j.oraloncology.2023.106531](https://doi.org/10.1016/j.oraloncology.2023.106531).
- Maas, S. A., Ellis, B. J., Ateshian, G. A., and Weiss, J. A. (2012). “FEBio: Finite Elements for Biomechanics.” In: *J. Biomech. Eng.* 134.1, p. 011005. ISSN: 0148-0731, 1528-8951. DOI: [10.1115/1.4005694](https://doi.org/10.1115/1.4005694).

- Maas, S. A., Ellis, B. J., Rawlins, D. S., and Weiss, J. A. (2016). "Finite Element Simulation of Articular Contact Mechanics with Quadratic Tetrahedral Elements." In: *J. Biomech.* 49.5, pp. 659–667. ISSN: 00219290. DOI: [10.1016/j.jbiomech.2016.01.024](https://doi.org/10.1016/j.jbiomech.2016.01.024).
- Marra, C., Pinto, V., Benanti, E., De Maria, F., Pinelli, M., Spaggiari, A., and De Santis, G. (2023). "Radial Forearm Flap versus Antero-Lateral Thigh Flap in Oral Soft Tissue Reconstruction: Update and Statistical Analysis on Our 77 Consecutive Cases for an Objective Selection Criteria." In: *Acta Biomed* 94.5, e2023252. ISSN: 2531-6745. DOI: [10.23750/abm.v94i5.15174](https://doi.org/10.23750/abm.v94i5.15174).
- Razavi, C. R., Hostetter, J., Shukla, A., Cheng, Z., Aygun, N., Boahene, K., Byrne, P. J., Richmon, J., Quon, H., and Desai, S. C. (2022). "Predictors of Free Flap Volume Loss in Nonosseous Reconstruction of Head and Neck Oncologic Defects." In: *ENT-Ear Nose Throat J.* 101.1, pp. 48–53. ISSN: 0145-5613, 1942-7522. DOI: [10.1177/0145561320938903](https://doi.org/10.1177/0145561320938903).
- Reynolds, T. (1999). "Why Randomized Surgical Oncology Trials Are So Scarce." In: *JNCI Journal of the National Cancer Institute* 91.14, pp. 1182–1183. ISSN: 0027-8874, 1460-2105. DOI: [10.1093/jnci/91.14.1182](https://doi.org/10.1093/jnci/91.14.1182).
- Sakakibara, A., Kusumoto, J., Sakakibara, S., Hasegawa, T., Akashi, M., Minamikawa, T., Furudoi, S., Hashikawa, K., and Komori, T. (2019). "Long-Term Effects on Volume Change in Musculocutaneous Flaps after Head and Neck Reconstruction." In: *J. Reconstr. Microsurg.* 35.4, pp. 235–243. ISSN: 0743-684X, 1098-8947. DOI: [10.1055/s-0038-1672134](https://doi.org/10.1055/s-0038-1672134).
- Seabold, S. and Perktold, J. (2010). "Statsmodels: Econometric and Statistical Modeling with Python." In: *9th Python Sci. Conf.*
- Seikaly, H., Rieger, J., O'Connell, D., Ansari, K., AlQahtani, K., and Harris, J. (2009). "Beavertail Modification of the Radial Forearm Free Flap in Base of Tongue Reconstruction: Technique and Functional Outcomes." In: *Head Neck* 31.2, pp. 213–219. ISSN: 1043-3074, 1097-0347. DOI: [10.1002/hed.20953](https://doi.org/10.1002/hed.20953).
- Seikaly, H., Rieger, J., Wolfaardt, J., Moysa, G., Harris, J., and Jha, N. (2003). "Functional Outcomes After Primary Oropharyngeal Cancer Resection and Reconstruction With the Radial Forearm Free Flap." In: *Laryngoscope* 113.5, pp. 897–904. ISSN: 0023-852X, 1531-4995. DOI: [10.1097/00005537-200305000-00023](https://doi.org/10.1097/00005537-200305000-00023).

- Sepúlveda, L., Peralta J., G., Ampuero E., V., and Fernández A., F. (2025). “Comparison of Functional Outcomes of Radial Forearm and Anterolateral Thigh Free Flaps after Hemiglossectomy in Patients with Tongue Cancer: A Systematic Review and Meta-Analysis.” In: *Rev. Otorrinolaringol. Cir. Cabeza Cuello* 85.1, pp. 95–104. ISSN: 0718-4816. DOI: [10.4067/s0718-48162025000100549](https://doi.org/10.4067/s0718-48162025000100549).
- Stavness, I., Lloyd, J. E., Payan, Y., and Fels, S. (2011). “Coupled Hard–Soft Tissue Simulation with Contact and Constraints Applied to Jaw–Tongue–Hyoid Dynamics.” In: *Numer Methods Biomed Eng* 27.3, pp. 367–390. ISSN: 2040-7939, 2040-7947. DOI: [10.1002/cnm.1423](https://doi.org/10.1002/cnm.1423).
- Stavness, I., Nazari, M. A., Flynn, C., Perrier, P., Payan, Y., Lloyd, J. E., and Fels, S. (2014a). “Coupled Biomechanical Modeling of the Face, Jaw, Skull, Tongue, and Hyoid Bone.” In: *3D Multiscale Physiological Human*. Ed. by N. Magnenat-Thalmann, O. Ratib, and H. F. Choi. London: Springer, pp. 253–274. ISBN: 978-1-4471-6275-9. DOI: [10.1007/978-1-4471-6275-9_11](https://doi.org/10.1007/978-1-4471-6275-9_11).
- Sun, Z., Gepner, B. D., Cottler, P. S., Lee, S.-H., and Kerrigan, J. R. (2021a). “In Vitro Mechanical Characterization and Modeling of Subcutaneous Adipose Tissue: A Comprehensive Review.” In: *J. Biomech. Eng.* 143.7, p. 070803. ISSN: 0148-0731, 1528-8951. DOI: [10.1115/1.4050286](https://doi.org/10.1115/1.4050286).
- Sun, Z., Gepner, B. D., Lee, S.-H., Rigby, J., Cottler, P. S., Hallman, J. J., and Kerrigan, J. R. (2021b). “Multidirectional Mechanical Properties and Constitutive Modeling of Human Adipose Tissue under Dynamic Loading.” In: *Acta Biomaterialia* 129, pp. 188–198. ISSN: 17427061. DOI: [10.1016/j.actbio.2021.05.021](https://doi.org/10.1016/j.actbio.2021.05.021).
- Sun, Z., Lee, S.-H., Gepner, B. D., Rigby, J., Hallman, J. J., and Kerrigan, J. R. (2021c). “Comparison of Porcine and Human Adipose Tissue Loading Responses under Dynamic Compression and Shear: A Pilot Study.” In: *J. Mech. Behav. Biomed. Mater.* 113, p. 104112. ISSN: 17516161. DOI: [10.1016/j.jmbbm.2020.104112](https://doi.org/10.1016/j.jmbbm.2020.104112).
- Tarsitano, A., Battaglia, S., Cipriani, R., and Marchetti, C. (2016). “Microvascular Reconstruction of the Tongue Using a Free Anterolateral Thigh Flap: Three-dimensional Evaluation of Volume Loss after Radiotherapy.” In: *J. Cranio-MaxilloFac. Surg.* 44.9, pp. 1287–1291. ISSN: 1010-5182. DOI: [10.1016/j.jcms.2016.04.031](https://doi.org/10.1016/j.jcms.2016.04.031).

- Then, C., Stassen, B., Depta, K., and Silber, G. (2017). “New Methodology for Mechanical Characterization of Human Superficial Facial Tissue Anisotropic Behaviour in Vivo.” In: *Journal of the Mechanical Behavior of Biomedical Materials* 71, pp. 68–79. ISSN: 17516161. DOI: [10.1016/j.jmbbm.2017.02.022](https://doi.org/10.1016/j.jmbbm.2017.02.022).
- Urken, M. (2012). *Multidisciplinary Head and Neck Reconstruction: A Defect-Oriented Approach*. Lippincott Williams & Wilkins. ISBN: 978-1-4511-5337-8.
- Vaidya, A. J. and Wheatley, B. B. (2020). “An Experimental and Computational Investigation of the Effects of Volumetric Boundary Conditions on the Compressive Mechanics of Passive Skeletal Muscle.” In: *Journal of the Mechanical Behavior of Biomedical Materials* 102, p. 103526. ISSN: 17516161. DOI: [10.1016/j.jmbbm.2019.103526](https://doi.org/10.1016/j.jmbbm.2019.103526).
- van Alphen, M. J. A., Kreeft, A. M., van der Heijden, F., Smeele, L. E., and Balm, A. J. M. (2013). “Towards Virtual Surgery in Oral Cancer to Predict Postoperative Oral Functions Preoperatively.” In: *Br. J. Oral Maxillofac. Surg.* 51.8, pp. 747–751. ISSN: 0266-4356, 1532-1940. DOI: [10.1016/j.bjoms.2013.06.012](https://doi.org/10.1016/j.bjoms.2013.06.012).
- Vincent, A., Kohlert, S., Lee, T., Inman, J., and Ducic, Y. (2019). “Free-Flap Reconstruction of the Tongue.” In: *Seminars in Plastic Surgery* 33.01, pp. 038–045. ISSN: 1535-2188, 1536-0067. DOI: [10.1055/s-0039-1677789](https://doi.org/10.1055/s-0039-1677789).
- Wang, J., Ho, A. K., Papadopoulos-Nydam, G., Rieger, J., Inamoto, Y., Fels, S., Saitoh, E., Guo, C., and Aalto, D. (2019). “Simulated Volume Loss in the Base of Tongue in a Virtual Swallowing Model.” In: *Comp. Meth. Biomech. Biomed. Eng.* 7.4, pp. 389–395. ISSN: 2168-1163, 2168-1171. DOI: [10.1080/21681163.2017.1382392](https://doi.org/10.1080/21681163.2017.1382392).
- Wang, X., Yan, G., Zhang, G., Li, J., Liu, J., and Zhang, Y. (2013). “Functional Tongue Reconstruction with the Anterolateral Thigh Flap.” In: *World J Surg Onc* 11.1, p. 303. ISSN: 1477-7819. DOI: [10.1186/1477-7819-11-303](https://doi.org/10.1186/1477-7819-11-303).
- Xu, J., Lai, F., Liu, Y., Tan, Z., Zheng, C., Wang, J., Guo, H., Jiang, L., Ge, X., Lan, X., Chen, C., and Ge, M. (2022). “Novel Computer-Aided Reconstruction of Soft Tissue Defects Following Resection of Oral and Oropharyngeal Squamous Cell Carcinoma.” In: *World J. Surg. Oncol.* 20.1, p. 196. ISSN: 1477-7819. DOI: [10.1186/s12957-022-02654-7](https://doi.org/10.1186/s12957-022-02654-7).

- Zhou, M., González, P. J., Van Haasterecht, L., Soylu, A., Mihailovski, M., Van Zuijlen, P., and Groot, M. L. (2024). “Uniaxial Mechanical Stretch Properties Correlated with Three-Dimensional Microstructure of Human Dermal Skin.” In: *Biomech Model Mechanobiol* 23.3, pp. 911–925. ISSN: 1617-7959, 1617-7940. DOI: [10.1007/s10237-023-01813-3](https://doi.org/10.1007/s10237-023-01813-3).
- Zirk, M., Niknazemi, M., Riekert, M., Kreppel, M., Linz, C., and Lentzen, M.-P. (2024). “Comparative Analysis of Volumetric Changes between Resection Volume of Oral Tongue Cancer and Post Operative Volume of Radial Forearm Flaps.” In: *Clin Oral Invest* 28.9, p. 498. ISSN: 1436-3771. DOI: [10.1007/s00784-024-05885-y](https://doi.org/10.1007/s00784-024-05885-y).

Chapter 6

Guided tongue reconstruction: feasibility analysis of additively manufactured surgical toolkit on a silicone tongue model

Amir Reza Isazadeh^{1,*}, Siming Chen¹, Hadi Seikaly^{2,4}, Bin Zheng², Lindsey Westover³, and Daniel Aalto^{1,4}

* Corresponding author; submitted to Surgical Oncology.

¹ Department of Communication Sciences and Disorders, Faculty of Rehabilitation Medicine, University of Alberta, Edmonton, AB, Canada.

² Department of Surgery, Faculty of Medicine and Dentistry, University of Alberta, Edmonton, AB, Canada.

³ Department of Mechanical Engineering & Department of Biomedical Engineering, Faculty of Engineering, University of Alberta, Edmonton, AB, Canada.

⁴ Institute for Reconstructive Sciences in Medicine (iRSM), Misericordia Community Hospital, Edmonton, AB, Canada.

Abstract

Objective Virtual Surgical Planning (VSP) for tongue reconstruction promises patient-specific optimizations, but its utility is limited by the difficulty of translating a digital plan to a deformable, landmark-poor organ. This study evaluated the feasibility and anatomical accuracy of a patient-specific, additively manufactured surgical toolkit through benchtop experiments on soft-tissue phantoms. **Method** Four distinct hemiglossectomy reconstruction cases were planned using VSP and executed on patient-specific silicone tongue phantoms. The guided workflow, performed by an independent surgeon, incorporated: (1) resection marking guide, (2) intraoperative volume inspection guide enabling closed-loop feedback, (3) donor flap harvesting guide with decoupled planar and volumetric control, and (4) orientation-based inset. High-resolution 3D surface scans were used to quantify anatomical fidelity by measuring the surface deviation between pre- and post-procedure scans. **Results** All four guided reconstructions were successfully completed. Mean surface deviation was 0.76–1.26 mm, and the 99th percentile (robust maximum) ranged 3.37–4.13 mm. This accuracy is comparable to that reported for established patient-specific osseous reconstructions. The operating surgeon found the toolkit intuitive, the workflow sequencing clear, and the closed-loop volumetric inspection step practical. **Conclusion** This benchtop study demonstrates that a patient-specific guided surgical toolkit can feasibly and reliably translate tongue VSP into an accurate reconstruction on soft-tissue phantoms. By introducing a closed-loop volumetric feedback mechanism and decoupling the flap's planar shape from volume, the system provides adaptive, geometry- and volume-aware guidance. These findings justify progression to cadaveric and early clinical validation to evaluate overbulking strategies, intraoperative adaptability, and correlation with functional outcomes.

Keywords Virtual Surgical Planning, Patient-Specific 3D-Printed Surgical Guides, Free Flap Tongue Reconstruction, Surgical Simulation on Soft-Tissue Phantoms, Plastic and Reconstructive Surgery

6.1 Introduction

Virtual Surgical Planning (VSP) has become an established treatment method in head and neck reconstructive surgery (Rodby et al., 2014). It utilizes high-fidelity medical imaging and computational modeling to generate patient-specific 3D-printed cutting guides, fixtures, and plates, demonstrating reductions in operative time, lower costs, and improved functional outcomes (Chuka et al., 2017; Papadopoulos-Nydam et al., 2017; Serrano et al., 2019; Idris et al., 2022; Kurlander et al., 2023; Nayar et al., 2024). Building on this success, recent research has expanded to the complex and challenging domain of soft tissue reconstruction. Pioneering VSP approaches for tongue reconstruction now allow surgeons to algorithmically design patient-specific soft tissue free flaps from a 3D model of the tongue and the defect, yielding measurable clinical benefits such as reduced flap trimming, shorter operative times, and better defect matching (Xu et al., 2022). More advanced biomechanical models further refined this process by designing a flap geometry to minimize tissue stretch and tension (Isazadeh et al., 2024; Isazadeh et al., 2025b).

While tongue reconstruction VSP represents a significant leap forward, an optimal digital plan is merely a sophisticated visualization exercise if it cannot be translated accurately to the surgical field (Efanov et al., 2018). This creates a critical “Execution Challenge”; the gap between the virtual design and the physical surgical action. The innovation lies in the implementation of a guided surgical toolkit, which acts as a physical “bridge” to transfer the digital blueprint to the operating room with high fidelity. This challenge is particularly pronounced in the deformable, landmark-poor environment of soft tissue, where the functional benefits of a biomechanically superior plan are entirely dependent on high-fidelity execution; geometric inaccuracy can negate the intended biomechanical advantages (Isazadeh et al., 2025b).

In mandibular and maxillofacial reconstruction, for example, the clinical benefits frequently attributed to VSP are results of employing the guided surgical toolkit to execute that plan with a level of precision unattainable through freehand methods (Logan et al., 2013; Idris et al., 2022). The toolkit is defined as a set of instruments that translates the digital plan into precise, repeatable surgical actions. The core components of this toolkit include patient-specific, 3D-printed surgical guides for resection and osteotomy, patient-specific implants (PSIs) for fixation, and life-sized

anatomical models for preoperative rehearsal and communication. The toolkit is the critical, and often under-appreciated, component that makes VSP clinically viable and transformative (Idris et al., 2022; Nayar et al., 2024).

Developing an effective surgical guiding toolkit for soft tissue reconstruction presents unique challenges. Unlike bone, which provides a stable, landmark-rich substrate for guide registration, soft tissues are mobile, deformable, and lack consistent reference points, which makes the accurate transfer of a virtual plan a far more complex undertaking (Krauel et al., 2021). Despite these hurdles, pioneering efforts are demonstrating the value of guided techniques across various soft tissue applications. In autologous breast reconstruction, for example, 3D-printed tools function as both anatomical guides for landmark identification and physical molds for flap harvesting and shaping (Carrion et al., 2024). Patient-specific templates derived from computed tomography angiography (CTA) data have been shown to pinpoint perforator vessels, significantly reducing intraoperative search times (Chae et al., 2021). This principle of guided accuracy is shown to be effective in complex facial surgeries as well. Sterilized contour guides for rhinoplasty can translate the virtual plan with sub-millimeter accuracy (Gordon et al., 2021), multi-component systems for skin and cartilage significantly improve symmetry in heminasal reconstruction (Yen et al., 2019), and for facial fat grafting, guides with integrated volumetric containers allow surgeons to precisely quantify and place the required tissue (Arias et al., 2019). Surgical training and technology innovation also draw on validated soft-tissue simulators as controlled, high-fidelity test beds for rehearsal and evaluation (Podolsky, 2017; Podolsky et al., 2017; Podolsky et al., 2024a).

The complex, mobile, and deformable nature of the tongue, which lacks the stable bony landmarks used in osseous reconstruction, makes high-fidelity translation particularly difficult. Current literature reveals several pioneering, albeit fragmented, approaches to bridging this virtual-physical gap, primarily focusing on two distinct challenges: (1) guiding the accurate resection of the complex, three-dimensional defect on the tongue, and (2) guiding the precise harvesting of the donor flap in terms of its planar geometry and volume.

Early VSP translation methods for tongue reconstruction involve manually applying landmark-based measurements from patient-specific medical images to guide both the surgical resection and flap harvest (Fan et al., 2019b; Liu et al., 2023b). More advanced approaches leverage 3D

printing to create physical guides for the resection. One such approach uses patient-specific soft silicone resection guides that fit directly onto the tongue’s surface to physically delineate the planned resection line (Koumoullis et al., 2020). The primary challenge for such contact-based guides is to achieve accurate registration on a mobile and deformable organ. Recent VSP workflows concentrate on the second challenge: Once the flap’s planar geometry is designed virtually, it is used to create a physical template—either printed on paper or 3D-printed—which is then sterilized for the harvesting stage (Xu et al., 2022; Isazadeh et al., 2025b). VSP in tongue reconstruction can go further and enable surgeons to precisely map perforator anatomy and customize the flap’s design for optimal vascularity (Heredero et al., 2024).

Before surgical guiding toolkits for tongue reconstruction, VSP can be clinically adopted, their feasibility and potential benefit must be evaluated in a controlled, non-clinical setting. Exploratory benchtop studies provide a platform for such initial validation. The value of this approach was established in osseous reconstruction by Logan et al. (2013), who conducted a benchtop study evaluating surgical guides with 3D-printed fibula free flap and mandible models. While rigid anatomical models are sufficient for assessing osseous reconstruction, the evaluation of a toolkit for soft tissue demands a higher degree of material fidelity for the phantom; a realistic soft tissue analogue is necessary to accurately simulate the physical actions of cutting, handling, and suturing tissue. Luckily, a number of studies dedicated focused effort to study suitable silicone phantom models for a realistic skin (Chattrairat et al., 2023) and human tongue (Darmont Araya and Radhakrishnan, 2021; Eu et al., 2024).

To address the critical gap between virtual planning and execution in tongue reconstruction, this study proposes a patient-specific, additively manufactured surgical toolkit. This system is designed to act as a physical map that translates a biomechanically-optimized virtual plan to the surgical field to minimize intraoperative guesswork and promote procedural standardization. In this paper, we present the design details of this novel surgical toolkit for guiding tongue reconstruction and demonstrate its feasibility and anatomical accuracy in a complete, end-to-end workflow using soft-tissue phantoms. Accuracy is quantified from high-resolution 3D scans as pointwise surface deviation between the reconstructed neotongue and the reference pre-operative model.

6.2 Method

To address the challenges of creating a novel surgical system for such a complex soft tissue reconstruction procedure, an iterative design framework was adopted. It is inspired by its use in patient-specific design of facial prostheses (McHutchion and Aalto, 2021), while grounded in the established principles of guided surgical toolkits for mandibular and maxillary reconstruction. This strategy served as a de-risking step and enabled rapid refinement of the entire surgical workflow through a continuous feedback loop with a senior surgeon co-author acting as the end-user evaluator. This granular, feedback-driven process was applied to key aspects of the system’s development, from the selection of fabrication methods to the swift refinement of the surgical toolkit design, procedural workflow logic, clinical practicality, and intuitive integration into the procedure. This approach enabled all engineering decisions to be directly informed by clinical expertise and real-world procedural demands.

6.2.1 Tongue Reconstruction Virtual Surgical Planning

The feasibility of the proposed surgical toolkit was assessed across four clinically plausible tongue cancer cases. A previously published MRI-derived 3D tongue model from a healthy male participant in his early 40s (informed consent obtained from Health Research Ethics Board of Alberta: HREBA.CC-17-0597) was reused without modification (Isazadeh et al., 2025b). To span a representative spectrum of Urken class I-II tongue defects (Urken, 2012), four resection cases were computationally simulated as detailed in the prior work (Isazadeh et al., 2025b): tip-sparing hemiglossectomy, lateral hemiglossectomy, posteriorly extended hemiglossectomy (sparing the tongue base), and an asymmetric midline-crossing partial glossectomy.

The foundation for the surgical toolkit for each case is a patient-specific Virtual Surgical Plan (VSP), which is treated as a “black box” input for this study. The technical details of the VSP algorithm used to generate these plans are described in separate publications (Isazadeh et al., 2024; Isazadeh et al., 2025b). For the purposes of designing the guided toolkit, the VSP provides three essential 3D model outputs for each case: the pre-operative tongue model (Figure 6.1B), the

planned resection volume (Figure 6.2C), and the corresponding free flap design (Figure 6.3A and Figure 6.3E). These digital models serve as the blueprints from which the entire patient-specific surgical toolkit is engineered.

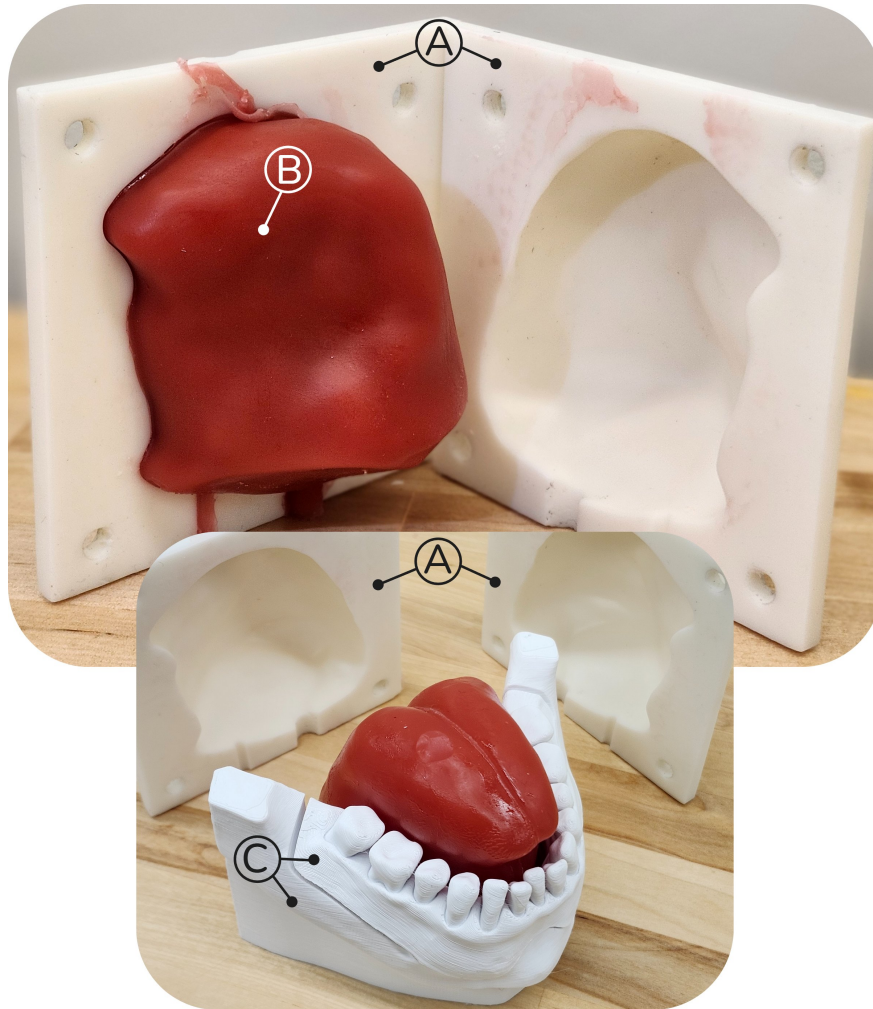


Figure 6.1 Fabrication of the soft silicone tongue phantom. (A) Two-part 3D-printed mold with alignment/fastening bosses. (B) Cast, pigmented elastomer tongue phantom after demolding. (C) Final positioning of the phantom within the 3D-printed jaw model

6.2.2 The Patient-Specific Surgical Toolkit

The design of the surgical toolkit was guided by the primary challenge of translating a digital plan onto a mobile, deformable, and landmark-poor anatomy. To be effective, the guides had to prioritize intuitive usability, ensuring stable and unambiguous registration by conforming precisely to the

patient's unique anatomy. The toolkit was engineered not only to provide clear physical boundaries for surface marking but also to offer tactile, three-dimensional references for volumetric control throughout the procedure.

The first phase of the guided procedure focuses on accurate resection of the defect. The surgeon begins by positioning the Resection Marking Guide (Figure 6.2A), a 2mm thick, shell-like instrument that conforms to the tongue's surface to delineate the precise incision boundary (Figure 6.2B). Precise placement is achieved by aligning the guide with pre-planned anatomical landmarks. To guide the cut's depth and complex three-dimensional shape, a solid Resection Model (Figure 6.2C) is also included. This model, a physical replica of the tissue being removed, acts as a tactile reference for the surgeon during the resection. After resection (Figure 6.2D), the resected tissue (Figure 6.2E) is placed into the Resection Volume Inspection Guide, a two-part negative mold that incorporates the marking guide as one of its halves (Figure 6.2F). This tool allows for an intraoperative comparison and provides immediate feedback on whether the resected tissue's volume is larger or smaller than what the surgical plan requires. The excess or the deficient volume is recorded for free flap harvesting adjustment.

The second part of the toolkit is designed for the donor site. It provides a two-part Flap Harvesting Guide that translates the exact planar geometry and volumetric requirements to the donor site phantom. It consists of a planar template (Figure 6.3A) for outlining the skin perimeter and a volumetric guide (Figure 6.3B) to verify that the volume of the beavertail tissue (Figure 6.3F) corresponds to the plan. This surgical step is complemented by a 3D-printed Flap Model (Figure 6.3E), which serves as a tactile reference for the surgeon to develop a haptic sense for the target volume and shape. Once harvested, the flap is ready for transfer and inset into the neotongue. Correct orientation of the flap is achieved using corresponding marks located on both the resection and flap harvesting guides. These marks are placed on strategic anatomical landmarks for intuitive recognition by the surgeon. The virtual plan determines the correspondence and facilitates the correct alignment for inset and suturing.

To physically realize these complex, patient-specific designs, the fabrication method had to meet stringent technical and safety requirements. The method required high-dimensional accuracy for precise registration, a smooth surface finish for atraumatic tissue contact, and sufficient rigidity to

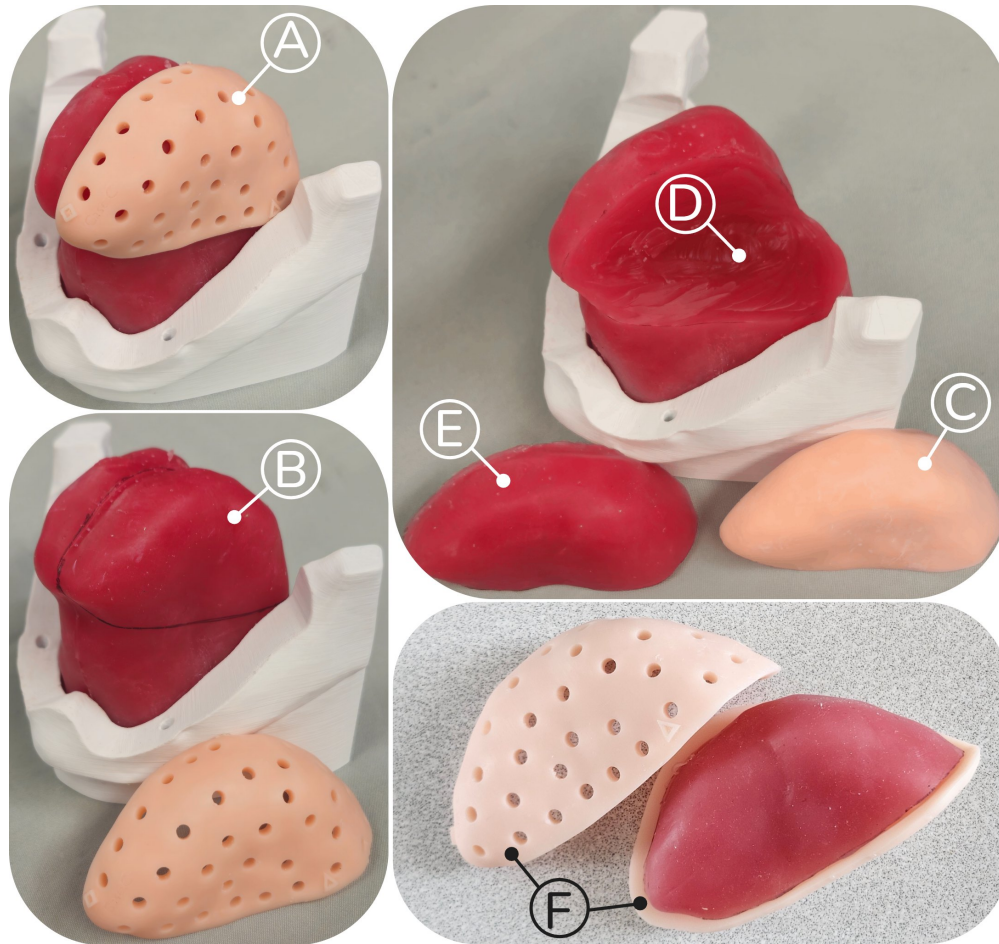


Figure 6.2 Patient-specific resection guidance toolkit and workflow. (A) Resection Marking Guide positioned on the tongue phantom for surface incision delineation. (B) Marked the resection boundary on the tongue surface using the conforming guide. (C) Solid Resection Model providing tactile reference for three-dimensional cut depth and contour. (D) Completed resection. (E) Excised tissue specimen ready for volumetric inspection. (F) Resection Volume Inspection Guide with excised tissue positioned for intraoperative volume verification against planned resection.

prevent deformation during handling. For eventual patient use, the material must also be certified as biocompatible and capable of withstanding standard sterilization protocols. Based on these criteria, Stereolithography (SLA) 3D printing was selected due to its sub-millimeter accuracy, smooth surface finish, and the availability of certified biocompatible photopolymer resins.

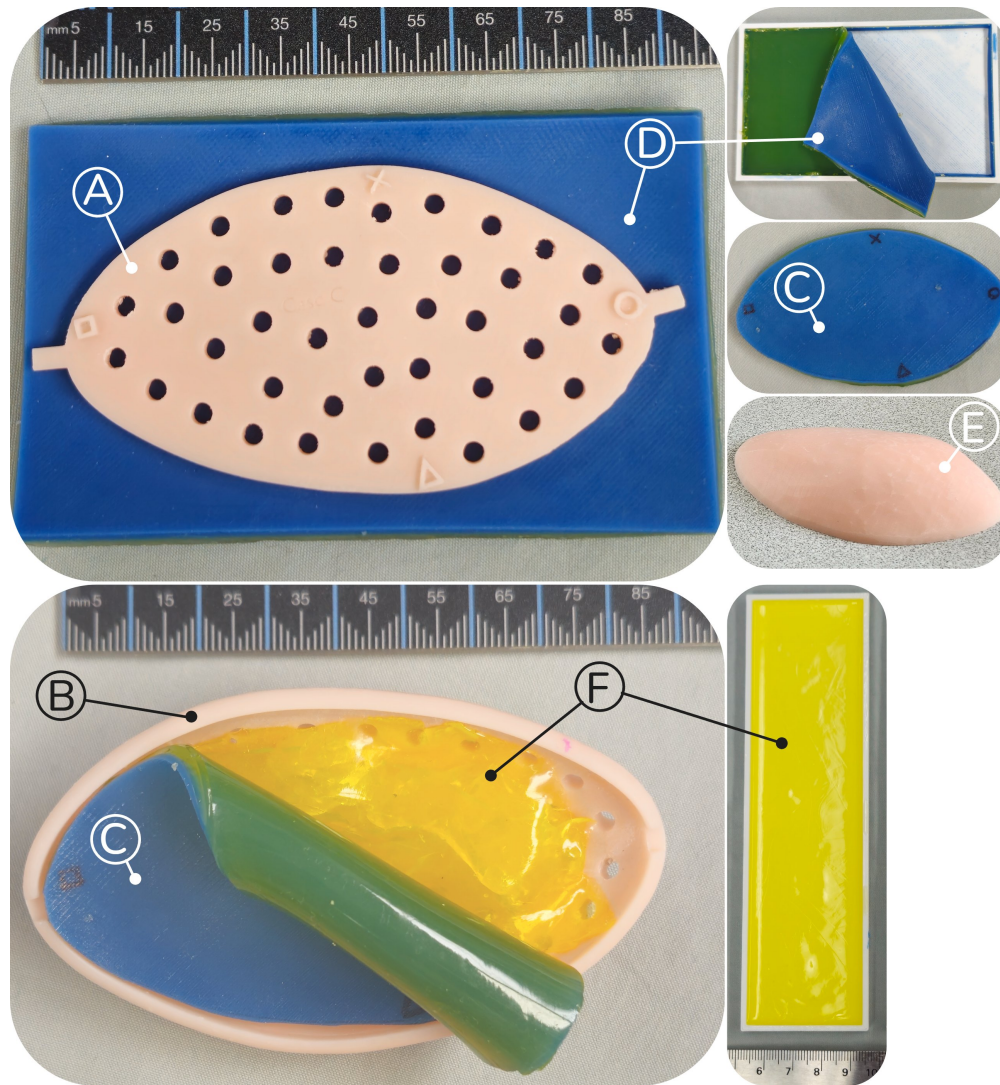


Figure 6.3 Patient-specific flap harvesting guidance toolkit and workflow. (A) Planar template guide for outlining skin perimeter geometry on the donor site phantom. (B) Volumetric guide for verifying beavertail SAT volume against planned requirements. (C and D) Multi-layered donor site phantom with skin and SAT layers. (E) 3D-printed Flap Model providing tactile reference for target volume and shape. (F) Harvested beavertail SAT positioned in the volumetric guide for intraoperative volume verification.

6.2.3 Benchtop Study

Patient-specific tongue phantoms were fabricated for the benchtop surgical simulation. A 3D-printed mold (Figure 6.1A), generated from the patient's pre-operative MRI data, was used to cast four identical, 1:1 scale tongue phantom replicas from Ecoflex™ 00-10 platinum-cure silicone (Figure 6.1B). Ecoflex™ 00-10 was selected for its ability to provide a high-fidelity haptic analogue

for the surgical tasks of dissection, retraction, and handling. The apparent elastic modulus of the fabricated phantom at 20% strain is approximately 45 kPa (Sparks et al., 2015). This value was compared against the wide range of stiffness reported for the human tongue. Published studies report the Young's modulus for a passive or fully relaxed tongue to be in the range of 1.6 kPa to 12.2 kPa (Buchaillard et al., 2007; Ishihara et al., 2013; Kappert et al., 2021b). While the phantom's modulus is higher than that of a completely relaxed tongue, this was a deliberate selection based on a critical trade-off between biomechanical fidelity and geometric stability. Softer, "gel-like" silicones that more closely match the biomechanics of passive tissue often exhibit poor form retention; they are prone to slumping under gravity and can degenerate or lose their anatomical shape during handling. Similar to other researchers (Darmont Araya and Radhakrishnan, 2021; Eu et al., 2024), a "form-retaining" material was prioritized over a more compliant but "form-forgetting" analogue. Ecoflex™ 00-10 provides a robust phantom that maintains its complex 3D shape while still offering sufficient compliance for realistic surgical simulation.

To provide a stable and anatomically relevant jig for the surgical simulation, a patient-specific jaw model was created (Figure 6.1C). The initial mandibular anatomy was derived from the patient model's MRI data; however, due to its low resolution, a high-resolution generic jaw model was elastically registered to the MRI-derived model using anatomical landmarks. The resulting jaw model was engineered to serve as a solid base structure to which the tongue phantom could be attached. Another key design feature is the jaw model's two-part construction, specifically implemented to facilitate unobstructed 3D surface scanning for post-operative quantitative evaluation. The mandible was meticulously partitioned so that the stationary lower section provides all anatomical attachment points, while the removable upper section is anatomically independent of the tongue; the removable upper section was affixed to the lower section. During the 3D scanning phase, removing the upper section grants an unobstructed line of sight to the tongue's entire superior surface, crucial for surface scanning.

A multi-layered phantom was fabricated to simulate the anatomy of the radial forearm free flap donor site (Figure 6.3D). The phantom consisted of a 1.5 mm thick superficial skin layer laminated onto a 2.0 mm thick subcutaneous adipose tissue (SAT) layer. The skin simulant was formulated according to the methods described by Chattrairat et al. (2023), comprising an 88:12 weight ratio

of Mold Star™ 30 and Ecoflex™ 00-10 platinum-cure silicones. This composite was selected to replicate the mechanical and haptic properties of human skin during surgical manipulation. The SAT layer was cast from Ecoflex™ 00-05 silicone. This material was chosen for its low shear modulus, mimicking the compliance and handling characteristics of human adipose tissue.

To simulate the “beavertail modification” technique used for tongue bulk adjustment (Seikaly et al., 2009), separate SAT phantoms were created. Consistent with the clinical procedure where subdermal fat and fascia are harvested, these phantoms were modeled as a single layer of adipose tissue simulant. To achieve the case-specific volume required by the virtual plan, each SAT phantom was cast from Ecoflex™ 00-05 silicone at a uniform thickness of 2.0 mm and predetermined adjusted bulk according to the virtual plan (Figure 6.3F). While this study utilized a uniform thickness, a clinical workflow could involve pre-operative imaging (MRI/CT) of the patient’s forearm to precisely quantify and map the available SAT for harvest. For the purposes of this feasibility study, the planned flap volume directly matched the planned resection volume (i.e., 0% overbulking), as the phantom models do not simulate postoperative tissue atrophy.

The benchtop surgical simulation was conducted in a controlled laboratory setting. Each of the four tongue phantoms underwent the complete guided reconstruction procedure using the patient-specific surgical toolkit. Figure 6.2 and Figure 6.3 showcase the process for case C. To avoid any bias from the senior surgeon who advised on the toolkit’s design, the benchtop procedure was performed by a different co-author surgeon with no prior experience in this domain. This approach isolates the performance of the surgical toolkit, which attributes the reconstruction’s accuracy to the toolkit’s guidance rather than pre-existing surgical skill. The flap was secured to the tongue phantom using 4-0 silk braided sutures (Isaac Technologies, IC-SN4002, Canada). Suturing was performed with a simple interrupted technique, placing each stitch approximately 5 mm apart and 3 mm from the wound edge. All suturing procedures were performed by the same operator to minimize variability.

6.2.4 Quantitative Evaluation of Anatomical Accuracy

To quantitatively assess the geometric fidelity of the reconstruction, high-resolution 3D surface scans of each tongue phantom were acquired both before the procedure and after the final reconstruction. The resulting pre-operative and post-operative 3D models were then digitally aligned for each case. This crucial alignment step was performed using an iterative closest point (ICP) algorithm with 100 iterations (Sullivan and Kaszynski, 2019). To ensure a repeatable and accurate registration, the ICP algorithm was applied exclusively to the surface data of the static jaw model, which remained unchanged across scans. The resulting spatial transformation was then applied to the entire post-operative scan.

Maximum Deviation, calculated using the Hausdorff distance between the aligned pre-operative and post-operative models, was selected as the primary measure of anatomical accuracy, similar to relevant studies (Duchscherer et al., 2021; Idris et al., 2022; Robertson et al., 2023). The Hausdorff distance quantifies the greatest discrepancy between two surfaces, thus providing a direct measure of the worst-case geometric error in the reconstruction. However, the standard maximum distance is highly sensitive to outliers. Such outliers are common artifacts in 3D scanning, arising from factors like specular reflections off the moist silicone surface, edge noise, or data voids from complex geometries. To enable robust and clinically meaningful assessment of the toolkit's performance, the 99th percentile of the surface deviation was used instead of the absolute maximum, which mitigates the impact of a small number of extreme, non-representative measurement errors. This approach yields a more representative measure of the overall surface conformity and the toolkit's true ability to translate the virtual plan with high fidelity. The PyVista library was utilized for these calculations and to generate surface deviation color maps for visualization.

6.3 Results

The proposed additively manufactured toolkit was used to perform a complete guided reconstruction on all four patient-specific tongue phantoms. During the surgical simulation, all toolkit components were utilized to guide their corresponding procedural steps, and the operator reported that the workflow was straightforward and the guides were easy to handle. The final sutured reconstructions for all four cases are shown in Figure 6.4.

Post-operative 3D surface scans were compared against the pre-operative scans. The mean surface deviation across all four reconstructions ranged from 0.76 mm to 1.26 mm. The 99th percentile of surface deviation, a robust measure of maximum error, ranged from 3.37 mm to 4.13 mm. Detailed metrics for each case are presented in Table 6.1. The surface deviation color maps in Figure 6.5 provide a visual representation of these measurements. The maps show large areas of low deviation (purple and blue), indicating close correspondence to the virtual plan across the majority of the reconstructed surface. Areas of higher deviation (green and yellow) were consistently localized along the suture lines at the flap-tongue interface and at resection regions with high surface curvature.

Table 6.1 Summary of surface deviation metrics for the four reconstructed tongue phantoms.

Case	Mean Deviation (mm)	99th Percentile Deviation (mm)
A	0.76	3.37
B	1.26	4.13
C	0.99	3.91
D	1.18	3.54

6.4 Discussion

The findings establish the feasibility of an additively manufactured surgical toolkit for guided tongue reconstruction and demonstrate that such a system can bridge the gap between VSP and its physical execution on a deformable soft-tissue phantom. A comprehensive, patient-specific toolkit was designed and successfully exercised in a complete end-to-end benchtop simulation.

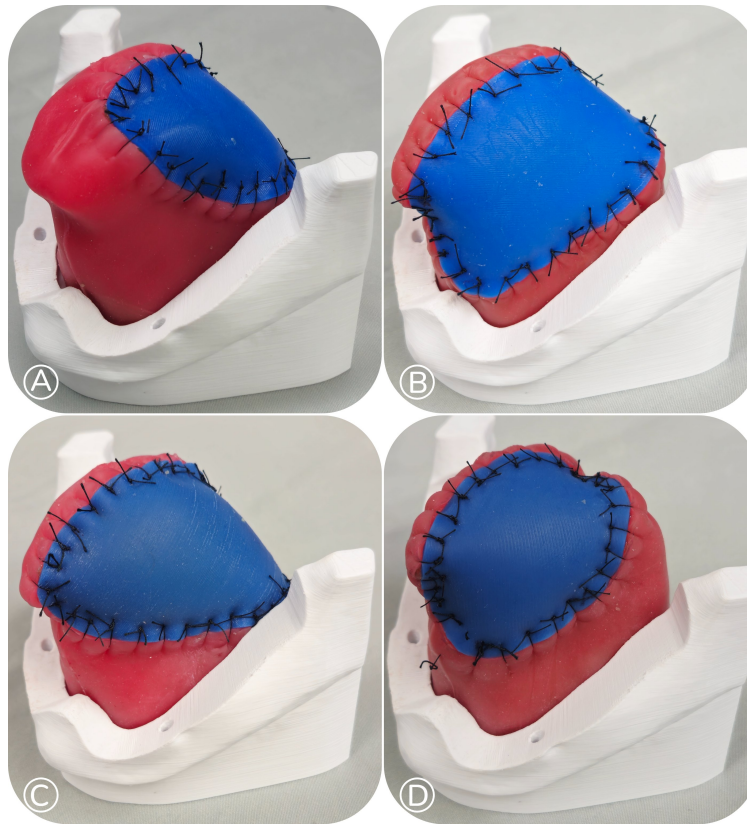


Figure 6.4 Final reconstructed tongue phantoms for the four VSP cases. Each phantom demonstrates successful translation of the virtual surgical plan to physical execution using the patient-specific toolkit.

Qualitative feedback was particularly informative: although the operator had no prior involvement in the VSP, the workflow was reported as easy to comprehend, intuitive, and predictable. Each component fulfilled a distinct role: the resection guide conformed snugly for stable surface marking; the physical models provided haptic appreciation of target geometries; and the harvesting guides afforded definitive control over flap shape, volume, and final orientation during inseting and suturing. Complementing this subjective assessment, the quantitative evaluation confirmed high anatomical accuracy: achieved surface deviations were comparable to the high-fidelity results reported for guided osseous VSP, where mandibular reconstruction deviations frequently fall within 0.1 mm to 3.8 mm (Idris et al., 2022; Suhaym et al., 2024). Attaining this precision on a mobile, landmark-poor soft tissue model underscores the toolkit’s potential to translate a sophisticated virtual plan into a physically actionable reconstruction with high fidelity. Nonetheless, validation in clinical trials beyond the benchtop setting remains a necessary next step.

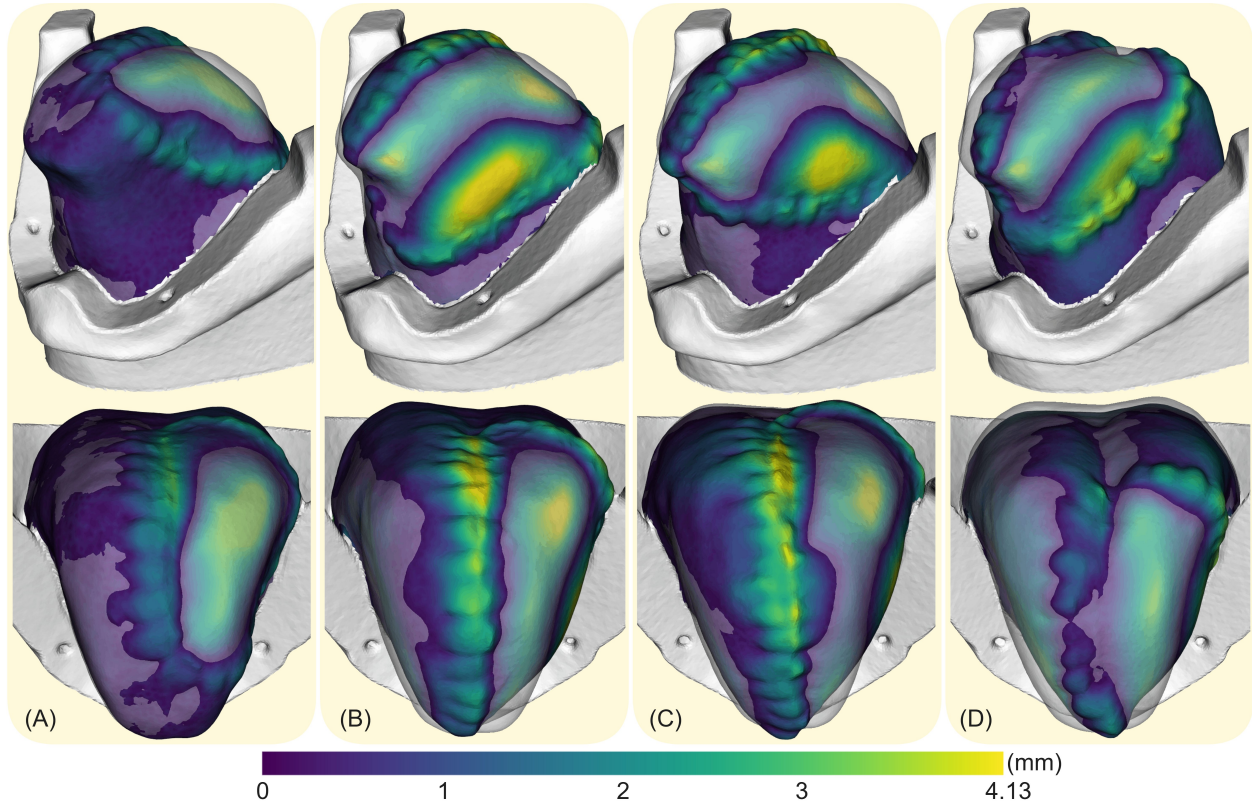


Figure 6.5 Post-reconstruction surface accuracy for the four simulated cases (A-D). Color map (0-4.13 mm) shows pointwise surface deviation between the executed reconstruction and the reference pre-operative tongue model.

The proposed surgical toolkit is grounded in the proven VSP paradigm for mandibular and maxillofacial reconstruction; it adopts shared principles of high-fidelity plan translation, comprehensive guide systems, and intraoperative inspections (Idris et al., 2022; Nayar et al., 2024). However, its design diverges to address the distinct clinical objective of tongue reconstruction. Whereas osseous reconstruction is a predominantly geometric task concerned with restoring a static structural framework and judged largely by spatial conformity (Idris et al., 2022; Logan et al., 2013), tongue reconstruction is a biomechanical challenge aimed at restoring dynamic function (speech and swallowing) by providing sufficient bulk and mobility (Urken and Biller, 1994; Buchaillard et al., 2007; Fan et al., 2019a; Choi et al., 2022). This shift in endpoint necessitates reimaging guidance: bone guides exploit “lock-and-key” registration on rigid and landmark-rich surfaces (Idris et al., 2022; Bibb et al., 2024), but the tongue’s deformable anatomy precludes such rigid fixation (Krauel et al., 2021). This distinction extends to the nature of guidance itself. While osseous toolkits often provide active guidance by physically constraining an instrument’s trajectory, such a

forceful approach is impractical and potentially traumatic for delicate soft tissue. Consequently, our toolkit shifts the paradigm to a system of referential and inspection guidance, designed to inform the surgeon's judgment, not rigidly control the instrument.

Moving beyond freehand execution, where the resection is guided solely by the surgeon's visual understanding of the VSP (Fan et al., 2019b; Liu et al., 2023b), our system provides three key instruments to enhance accuracy and confidence. First, the Resection Marking Guide provides a definitive physical boundary for the surface incision. Building on pioneering work with soft silicone guides (Koumoullis et al., 2020), our design likewise achieves a stable, conforming fit by maximizing surface contact. However, to address the critical challenge of guide registration and positional accuracy, it further incorporates major tongue landmarks denoted on the guide itself to enable precise and repeatable placement. The guide can be secured with sutures to prevent slippage during marking. Second, the Resection Model provides a tangible, three-dimensional target. This solid, 1:1 scale replica of the planned resection volume serves as an intraoperative haptic reference. Handling the model gives the surgeon a direct tactile sense of the cut's required depth and complex 3D contour during dissection. Third, the Resection Volume Inspection Guide serves as a unique feedback control, a feature absent in other VSP workflows. This two-part negative mold enables immediate, intraoperative inspection of the excised tissue against the virtual plan. It creates a direct feedback loop, which allows for assessment of actual resected bulk (i.e., excess or deficiency compared to the resection model). Most importantly, this inspection step directly informs any necessary adjustments to the subsequent flap harvesting stage; it enables informed adjustments to the flap bulk and creates an adaptive link between the resection and reconstruction phases of the procedure.

Guiding the donor free flap dissection is arguably the most critical phase of the reconstructive workflow, as its design and precise execution directly dictate functional outcomes. The challenge is twofold: first, the flap must possess optimal planar geometry to reduce tongue tethering and improve mobility (Buchillard et al., 2007; Lam and Samman, 2013; Idris et al., 2018; Vincent et al., 2019; Baskin et al., 2019; Al-Zanoon et al., 2024); second, it must provide adequate volume for proper palatal and oropharyngeal contact, which is essential for bolus propulsion, swallowing, and the prevention of aspiration (Lam and Samman, 2013; Fan et al., 2019b; Choi et al., 2022).

Recognizing these dual imperatives, recent VSP approaches have evolved beyond landmark-based measurements (Fan et al., 2019b; Liu et al., 2023b) to algorithmically design patient-specific flaps that are biomechanically optimized to match the defect (Isazadeh et al., 2025b). However, the execution of this virtual plan remains a significant challenge. One VSP technique employs 3D-printed flap-cutting guides to delineate the planar shape of the free flap, which has demonstrated clinical benefits (Xu et al., 2022). While this method offers excellent control over the planar geometry, it provides no guidance for its volume; the surgeon must still estimate the required SAT freehand. An alternative workflow addresses volumetric control by using a negative mold of the resected tissue as a guide; after harvesting, the flap is placed into this mold to assess and adjust its bulk until it matches the resected tissue (Koumoullis et al., 2020). By enforcing a 1:1 volume match with the resected tissue, this method, however, cannot accommodate clinical strategies such as planned overbulking.

The proposed surgical toolkit synthesizes the strengths of these approaches while introducing an adaptive feedback mechanism to overcome their respective limitations. The key innovation is a decoupled harvesting system that provides independent control over both flap parameters. It features a separate Planar Guide (Figure 6.3A) for geometric accuracy, analogous to the CARST method (Xu et al., 2022), and a dedicated Volumetric Guide (Figure 6.3B) for precise bulk management. This decoupling allows each guide to be specifically engineered for its distinct purpose: the Planar Guide provides an unambiguous boundary for skin marking, while the Volumetric Guide allows the clinical strategy of overbulking to be precisely executed based on a volume that is pre-calculated in the VSP. This overbulking is necessary to counteract postoperative atrophy, which can range from 20% to over 50% and is often exacerbated by radiotherapy (Tarsitano et al., 2016; Vincent et al., 2019; Gewirtz et al., 2024).

Crucially, the system introduces an intraoperative feedback loop that connects the resection and harvesting stages. A common criticism of current osseous VSP workflows is their rigidity; an intraoperative change, such as unexpected tumor spread, can render the entire patient-specific toolkit obsolete (Efanov et al., 2018). The proposed toolkit offers guides like the Resection Volume Inspection (Figure 6.2F) that provide quantitative feedback on the actual resected volume. This information allows the surgeon to make informed adjustments to the flap's volume by proportionally

adjusting the bulk of beavertail SAT. This adaptive capability creates a closed-loop process that accounts for intraoperative deviations from the preoperative plan. Finally, the inclusion of a solid Flap Model (Figure 6.3E) as a haptic reference allows the surgeon to develop a tactile appreciation for the target free flap geometry and volume, which reduces the cognitive load of translating a digital plan into a physical entity.

Despite promising accuracy, the toolkit faces key limitations in translating a virtual plan onto dynamic soft tissue. A primary challenge is the limited active guidance for resection depth and angle; the marking guide delineates the 2D incision but relies on the surgeon's tactile reference to a separate model to gauge the 3D cut. Furthermore, the system's fidelity is contingent upon the stable registration of a rigid guide onto the mobile, deformable tongue, where minor misalignments risk propagating significant errors (Krauel et al., 2021; Suhaym et al., 2024). The workflow is only partially robust to unexpected intraoperative findings; for example, the need for wider oncologic margins could render the pre-fabricated guides obsolete (Efanov et al., 2018). Future work must address these hurdles by developing more robust registration techniques, such as using dental splints as stable anchors, and enhancing adaptability through a set of guides with pre-planned alternative margins.

Methodological limitations of this benchtop validation are apparent. The static phantom cannot replicate the dynamic challenges of live surgery, such as bleeding, saliva, tissue mobility, and restricted access. Crucially, the native tongue is significantly softer and more pliable than the form-retaining phantom used in this study; consequently, achieving stable guide registration on a slippery, highly deformable organ will present greater difficulties than those observed in this controlled setting. Furthermore, the lack of a control group prevents a direct comparison with the current standard of care to quantify the toolkit's added value (Logan et al., 2013; Xu et al., 2022). The use of a single operator on only four defect geometries from one patient's anatomy also limits the assessment of inter-surgeon variability, the learning curve, and overall generalizability. Future work must therefore involve controlled studies with multiple surgeons across a wider range of patient anatomies. These studies should incorporate qualitative face-validity assessments, where experienced surgeons perform the simulated surgery and rate the tactile realism of tissue incision and suturing, before progressing toward clinical trials.

The fidelity of the silicone phantoms, while adequate for this feasibility study, is a key limitation. The material selection relied primarily on Young’s modulus; however, stiffness alone is insufficient to establish quantitative equivalence to biological tissue. The tongue is a complex muscular hydrostat with layered, anisotropic, and viscoelastic properties that our homogeneous model necessarily simplifies (Sanders and Mu, 2013; Kappert et al., 2021a; Wrench, 2024). This reflects an inherent limitation of synthetic analogues; regardless of the specific silicone formulation, they lack the complex stress relaxation, biological heterogeneity, and high non-linearity of real tissue. This impacts the realism of tissue handling, dissection, and suturing, and demands the development of higher-fidelity, multi-material phantoms (Ishikawa et al., 2025). Future validation should move beyond qualitative material selection to include a quantitative sensitivity analysis comparing various silicone grades to confirm the necessity of specific material choices. Furthermore, as a non-biological surrogate, the phantom lacks features like blood vessels and cannot model long-term postoperative atrophy. This prevented testing a critical clinical capability of the volumetric guide: the ability to execute a pre-planned “overbulking” strategy to compensate for expected tissue loss.

Finally, while this study successfully demonstrates high geometric accuracy, this endpoint is a surrogate for the true clinical goals of tongue reconstruction: improved speech, swallowing, and quality of life. The correlation between this geometric fidelity and functional outcomes is yet to be established. A prospective clinical trial comparing guided versus conventional reconstruction is the ultimate goal, but several practical hurdles must first be addressed. These include validating that sterilization protocols do not compromise the dimensional accuracy of the guides and evaluating the end-to-end turnaround time required for VSP and manufacturing.

6.5 Conclusion

A patient-specific, additively manufactured surgical toolkit translated a sophisticated virtual tongue resection-reconstruction plan into a high-fidelity physical outcome on soft-tissue phantoms. It decouples planar geometry from volumetric control at the donor site and embeds an intraoperative resection-volume inspection step. Together, these features create an adaptive, closed-loop workflow that links what is resected to how bulk is restored. This architecture converts flap design from

a sequential estimation task into a controllable process where surface shape and tissue volume become independently specified, inspected, and iteratively reconciled. The workflow advances translation fidelity for tongue reconstruction VSP and provides a reproducible structure for a procedure historically reliant on intraoperative estimation.

Current validation is bounded by a homogeneous, non-biological phantom, a single operator, and the inability to model dynamic surgical conditions or postoperative atrophy. Even so, the results adequately reflect plan execution accuracy on a tongue phantom. Next, a staged progression through multi-surgeon early clinical case studies will evaluate registration robustness, functional correlates of geometric accuracy, and the efficacy of planned overbulking, sterilization, and turnaround workflows.

Funding This work was supported by the Alberta Cancer Foundation (grant number 27 601). The funder had no involvement in the research, writing, or submission of this manuscript.

References

- Al-Zanoon, N., Cummine, J., Jeffery, C. C., Westover, L., and Aalto, D. (2024). “The Effect of Simulated Radiation Induced Fibrosis on Tongue Protrusion.” In: *Biomech. Model. Mechanobiol.* 23, pp. 1649–1660. ISSN: 1617-7940. DOI: [10.1007/s10237-024-01860-4](https://doi.org/10.1007/s10237-024-01860-4).
- Arias, E., Huang, Y.-H., Zhao, L., Seelaus, R., Patel, P., and Cohen, M. (2019). “Virtual Surgical Planning and Three-Dimensional Printed Guide for Soft Tissue Correction in Facial Asymmetry.” In: *J. Craniofac. Surg.* 30.3, pp. 846–850. ISSN: 1049-2275, 1536-3732. DOI: [10.1097/SCS.0000000000005204](https://doi.org/10.1097/SCS.0000000000005204).
- Baskin, R. M., Seikaly, H., Sawhney, R., Danan, D., Burt, M., Idris, S., Shama, M., Boyce, B., and Dziegielewska, P. T. (2019). “Tongue Reconstruction: Rebuilding Mobile Three-Dimensional Structures from Immobile Two-Dimensional Substrates, a Fresh Cadaver Study.” In: *Head Neck-J. Sci. Spec. Head Neck* 41.10, pp. 3693–3699. ISSN: 1043-3074, 1097-0347. DOI: [10.1002/hed.25889](https://doi.org/10.1002/hed.25889).

- Bibb, R., Eggbeer, D., Paterson, A., and Mohammed, M. I. (2024). *Medical Modeling: The Application of Advanced Design and Additive Manufacturing Techniques in Medicine*. Elsevier. ISBN: 978-0-323-95734-2.
- Buchaillard, S., Brix, M., Perrier, P., and Payan, Y. (2007). “Simulations of the Consequences of Tongue Surgery on Tongue Mobility: Implications for Speech Production in Post-Surgery Conditions.” In: *Int. J. Med. Robot. Comput. Assist. Surg.* 3.3, pp. 252–261. ISSN: 1478-5951, 1478-596X. DOI: [10.1002/rcs.142](https://doi.org/10.1002/rcs.142).
- Carrion, K., Akhter, M. F., Bernal, C., Cox, E., Krakauer, K., Mohan, A. T., Nazerali, R. S., Lee, G. K., and Nguyen, D. (2024). “A Systematic Review of 3D Printing in Autologous Breast Reconstruction.” In: *Plast Aesthet Res* 11, N–A. ISSN: 2349-6150. DOI: [10.20517/2347-9264.2024.56](https://doi.org/10.20517/2347-9264.2024.56).
- Chae, M. P., Hunter-Smith, D. J., Chung, R. D., Smith, J. A., and Rozen, W. M. (2021). “3D-printed, Patient-Specific DIEP Flap Templates for Preoperative Planning in Breast Reconstruction: A Prospective Case Series.” In: *Gland Surg* 10.7, pp. 2192–2199. ISSN: 2227684X, 22278575. DOI: [10.21037/gs-21-263](https://doi.org/10.21037/gs-21-263).
- Chattrairat, A., Aimmanee, S., Kandare, E., Tran, P., and Das, R. (2023). “Modelling and Characterisation of the Dynamic Behaviours of Silicone-Based Composite Skin Simulant with Short Polyethylene Fibres and Bioactive Glass Particles.” In: *Mechanics of Materials* 184, p. 104740. ISSN: 01676636. DOI: [10.1016/j.mechmat.2023.104740](https://doi.org/10.1016/j.mechmat.2023.104740).
- Choi, J.-W., Alshomer, F., and Kim, Y.-C. (2022). “Current Status and Evolution of Microsurgical Tongue Reconstructions, Part I.” In: *Arch Craniofac Surg* 23.4, pp. 139–151. ISSN: 2287-1152, 2287-5603. DOI: [10.7181/acfs.2022.00654](https://doi.org/10.7181/acfs.2022.00654).
- Chuka, R., Abdullah, W., Rieger, J., Nayar, S., Seikaly, H., Osswald, M., and Wolfaardt, J. (2017). “Implant Utilization and Time to Prosthetic Rehabilitation in Conventional and Advanced Fibular Free Flap Reconstruction of the Maxilla and Mandible.” In: *Int. J. Prosthodont.* 30.3, pp. 289–294. ISSN: 0893-2174, 1942-4426. DOI: [10.11607/ijp.5161](https://doi.org/10.11607/ijp.5161).
- Darmont Araya, F. and Radhakrishnan, P. (2021). “Investigating the Design and Manufacture of PneuNet Actuators as a Prosthetic Tongue for Mimicking Human Deglutition.” In: *J. Eng. Sci. Med. Diagn. Ther.* 4.4, p. 041003. ISSN: 2572-7958, 2572-7966. DOI: [10.1115/1.4051665](https://doi.org/10.1115/1.4051665).

- Duchscherer, J., Aalto, D., and Westover, L. (2021). "Evaluation of Facial Symmetry after Jaw Reconstruction Surgery." In: *Comput. Methods Biomech. Biomed. Engin.* 24.11, pp. 1212–1220. ISSN: 1025-5842. DOI: [10.1080/10255842.2020.1870965](https://doi.org/10.1080/10255842.2020.1870965).
- Efanov, J. I., Roy, A.-A., Huang, K. N., and Borsuk, D. E. (2018). "Virtual Surgical Planning: The Pearls and Pitfalls." In: *Plast. Reconstr. Surg. Glob. Open* 6.1, e1443. ISSN: 2169-7574. DOI: [10.1097/GOX.0000000000001443](https://doi.org/10.1097/GOX.0000000000001443).
- Eu, D., Daly, M. J., Taboni, S., Sahovaler, A., Gilbank, A. N., and Irish, J. C. (2024). "Evaluation of a 3D Printed Silicone Oral Cavity Cancer Model for Surgical Simulations." In: *J. Pers. Med.* 14.5, p. 450. ISSN: 2075-4426. DOI: [10.3390/jpm14050450](https://doi.org/10.3390/jpm14050450).
- Fan, S., Li, Q.-X., Zhang, H.-Q., Liang, M.-J., Tian, T., Wang, Y.-Y., Lin, Z.-Y., Chen, W.-X., Pan, G.-K., Ahn, M. H.-Y., He, L., Sun, S., Wu, B.-H., Yu, P., and Li, J.-S. (2019a). "'Five-point Eight-Line' Anatomic Flap Design for Precise Hemitongue Reconstruction." In: *Head Neck-J. Sci. Spec. Head Neck* 41.5, pp. 1359–1366. ISSN: 1043-3074, 1097-0347. DOI: [10.1002/hed.25571](https://doi.org/10.1002/hed.25571).
- Fan, S., Zhang, H., Li, Q., Liang, F., Bai, Z., Chen, W., Sun, S., Yu, P., and Li, J. (2019b). "A Novel Anatomy-Based Five-Points Eight-Line-Segments Technique for Precision Subtotal Tongue Reconstruction: A Pilot Study." In: *Oral Oncol.* 89, pp. 1–7. ISSN: 13688375. DOI: [10.1016/j.oraloncology.2018.12.004](https://doi.org/10.1016/j.oraloncology.2018.12.004).
- Gewirtz, J. I., Zhao, S., Brock, G., Luttrull, M. D., Sethuraman, S., Kang, S. Y., Vankoevering, K. K., and Seim, N. B. (2024). "A Pilot Study: Free Flap Atrophy in Tongue Reconstruction Using 3D Volumetric Analysis." In: *Ann. Otol. Rhinol. Laryngol.* 133.3, pp. 253–260. ISSN: 0003-4894, 1943-572X. DOI: [10.1177/00034894231204720](https://doi.org/10.1177/00034894231204720).
- Gordon, A. R., Schreiber, J. E., Patel, A., and Tepper, O. M. (2021). "3D Printed Surgical Guides Applied in Rhinoplasty to Help Obtain Ideal Nasal Profile." In: *Aesth Plast Surg* 45.6, pp. 2852–2859. ISSN: 0364-216X, 1432-5241. DOI: [10.1007/s00266-021-02350-z](https://doi.org/10.1007/s00266-021-02350-z).
- Herederó, S., Falguera, M. I., Gómez, V., and Sanjuan-Sanjuan, A. (2024). "Customized Soft Tissue Free Flaps in Head and Neck Reconstruction." In: *Oral Maxillofac Surg Clin North Am.* ISSN: 1042-3699. DOI: [10.1016/j.coms.2024.07.009](https://doi.org/10.1016/j.coms.2024.07.009).

- Idris, S., Logan, H., Tabet, P., Osswald, M., Nayar, S., and Seikaly, H. (2022). “The Accuracy of 3D Surgical Design and Simulation in Prefabricated Fibula Free Flaps for Jaw Reconstruction.” In: *J. Pers. Med.* 12.11, p. 1766. ISSN: 2075-4426. DOI: [10.3390/jpm12111766](https://doi.org/10.3390/jpm12111766).
- Idris, S., Mlynarek, A. M., Ansari, K., Harris, J. R., Rizk, N., Cote, D., O’Connell, D. A., Allen, H., Dziegielewski, P., and Seikaly, H. (2018). “Multi-Dimensional Analysis of Oral Cavity and Oropharyngeal Defects Following Cancer Extirpation Surgery, a Cadaveric Study.” In: *J. Otolaryngol. Head Neck Surg.* 47, p. 27. ISSN: 1916-0216. DOI: [10.1186/s40463-018-0276-9](https://doi.org/10.1186/s40463-018-0276-9).
- Isazadeh, A. R., Seikaly, H., Westover, L., and Aalto, D. (2024). “Algorithmically Designed Flaps in Tongue Reconstruction: A Feasibility Analysis.” In: *Int. J. Comput. Assist. Radiol. Surg.* 19, pp. 735–746. ISSN: 1861-6410, 1861-6429. DOI: [10.1007/s11548-024-03062-w](https://doi.org/10.1007/s11548-024-03062-w).
- Isazadeh, A. R., Zenke, J., Westover, L., Seikaly, H., and Aalto, D. (2025b). “Patient-Specific Virtual Surgical Planning for Tongue Reconstruction: Evaluating Hyperelastic Inverse FEM with Four Simulated Tongue Cancer Cases.” In: *Phys. Med. Biol.* 70, p. 145017. ISSN: 0031-9155, 1361-6560. DOI: [10.1088/1361-6560/adebd8](https://doi.org/10.1088/1361-6560/adebd8).
- Ishihara, S., Nakao, S., Nakauma, M., Funami, T., Hori, K., Ono, T., Kohyama, K., and Nishinari, K. (2013). “Compression Test of Food Gels on Artificial Tongue and Its Comparison with Human Test.” In: *J. Texture Stud.* 44.2, pp. 104–114. ISSN: 0022-4901, 1745-4603. DOI: [10.1111/jtxs.12002](https://doi.org/10.1111/jtxs.12002).
- Ishikawa, Y., Nabae, H., Gunji, M., Endo, G., and Suzumori, K. (2025). “Pig Tongue Soft Robot Mimicking Intrinsic Tongue Muscle Structure.” In: *Front. Robot. AI* 11, p. 1511422. ISSN: 2296-9144. DOI: [10.3389/frobt.2024.1511422](https://doi.org/10.3389/frobt.2024.1511422).
- Kappert, K. D. R., Voskuilen, L., Smeele, L. E., Balm, A. J. M., Jasperse, B., Nederveen, A. J., and van der Heijden, F. (2021a). “Personalized Biomechanical Tongue Models Based on Diffusion-Weighted MRI and Validated Using Optical Tracking of Range of Motion.” In: *Biomech. Model. Mechanobiol.* 20.3, pp. 1101–1113. ISSN: 1617-7959, 1617-7940. DOI: [10.1007/s10237-021-01435-7](https://doi.org/10.1007/s10237-021-01435-7).
- Kappert, K., Connesson, N., Elahi, S., Boonstra, S., Balm, A., Van Der Heijden, F., and Payan, Y. (2021b). “In-Vivo Tongue Stiffness Measured by Aspiration: Resting vs General Anesthesia.” In: *J. Biomech.* 114, p. 110147. ISSN: 00219290. DOI: [10.1016/j.jbiomech.2020.110147](https://doi.org/10.1016/j.jbiomech.2020.110147).

- Koumoullis, H., Burley, O., and Kyzas, P. (2020). "Patient-Specific Soft Tissue Reconstruction: An IDEAL Stage I Report of Hemiglossectomy Reconstruction and Introduction of the PANSOFOS Flap." In: *Br. J. Oral Maxillofac. Surg.* 58.6, pp. 681–686. ISSN: 0266-4356. DOI: [10.1016/j.bjoms.2020.04.017](https://doi.org/10.1016/j.bjoms.2020.04.017).
- Krauel, L., Valls-Esteve, A., Tejo-Otero, A., and Fenollosa-Artés, F. (2021). "3D-Printing in Surgery: Beyond Bone Structures. A Review." In: *Ann. 3D Print. Med.* 4, p. 100039. ISSN: 26669641. DOI: [10.1016/j.stlm.2021.100039](https://doi.org/10.1016/j.stlm.2021.100039).
- Kurlander, D. E., Garvey, P. B., Largo, R. D., Yu, P., Chang, E. I., Hanasono, M. M., and Mericli, A. F. (2023). "The Cost Utility of Virtual Surgical Planning and Computer-Assisted Design/Computer-Assisted Manufacturing in Mandible Reconstruction Using the Free Fibula Osteocutaneous Flap." In: *J Reconstr Microsurg* 39.03, pp. 221–230. ISSN: 0743-684X, 1098-8947. DOI: [10.1055/s-0042-1755260](https://doi.org/10.1055/s-0042-1755260).
- Lam, L. and Samman, N. (2013). "Speech and Swallowing Following Tongue Cancer Surgery and Free Flap Reconstruction - A Systematic Review." In: *Oral Oncol.* 49.6, pp. 507–524. ISSN: 1368-8375, 1879-0593. DOI: [10.1016/j.oraloncology.2013.03.001](https://doi.org/10.1016/j.oraloncology.2013.03.001).
- Liu, M.-D., Xue, X.-M., Al-Aroomi, M. A., Xiong, J.-J., Liu, S., Sun, C.-F., and Liu, F.-Y. (2023b). "A Novel Flap Design Technique for Subtotal Tongue Reconstruction with an "Individualized and Convenient Tongue Model"." In: *Oral Oncol.* 145, p. 106531. ISSN: 1368-8375. DOI: [10.1016/j.oraloncology.2023.106531](https://doi.org/10.1016/j.oraloncology.2023.106531).
- Logan, H., Wolfaardt, J., Boulanger, P., Hodgetts, B., and Seikaly, H. (2013). "Exploratory Benchtop Study Evaluating the Use of Surgical Design and Simulation in Fibula Free Flap Mandibular Reconstruction." In: *J. Otolaryngol. Head Neck Surg.* 42.1, p. 42. ISSN: 1916-0216. DOI: [10.1186/1916-0216-42-42](https://doi.org/10.1186/1916-0216-42-42).
- McHutchion, L. and Aalto, D. (2021). "Simulation of Tissue-Prosthesis Margin Interface by Using Surface Scanning and Digital Design for Auricular Prostheses." In: *J. Prosthet. Dent.* 125.2, pp. 361–372. ISSN: 00223913. DOI: [10.1016/j.prosdent.2020.01.045](https://doi.org/10.1016/j.prosdent.2020.01.045).

- Nayar, S., Logan, H., Wolfaardt, J., Osswald, M., O’Connell, D., Grosvenor, A., and Seikaly, H. (2024). “Alberta Reconstructive Technique (ART): An Innovative Approach Using Digital Surgical Design and Simulation in Advanced Jaw Reconstruction with Occlusion-Based Prefabricated Vascularized Fibular Flaps and Primary Osseointegrated Implant Installation.” In: *Head Neck-J. Sci. Spec. Head Neck* 46.8, pp. 2098–2101. ISSN: 1043-3074, 1097-0347. DOI: [10.1002/hed.27792](https://doi.org/10.1002/hed.27792).
- Papadopoulos-Nydam, G., Wolfaardt, J., Seikaly, H., O’Connell, D., Harris, J., Osswald, M., Nayar, S., and Rieger, J. (2017). “Comparison of Speech and Resonance Outcomes across Three Methods of Treatment for Maxillary Defects.” In: *Int. J. Maxillofac. Prosthet.* 1.1, pp. 2–8. ISSN: 2432-3993. DOI: [10.26629/ijmp.2017.02](https://doi.org/10.26629/ijmp.2017.02).
- Podolsky, D. J. (2017). “Development and Evaluation of a High Fidelity Cleft Palate Simulator for Surgical Training and for Development of a Robotic Approach to Infant Cleft Palate Surgery.” PhD thesis. Canada: University of Toronto (Canada). ISBN: 978-0-355-53509-9.
- Podolsky, D. J., Fisher, D. M., Wong, K. W., Looi, T., Drake, J. M., and Forrest, C. R. (2017). “Evaluation and Implementation of a High-Fidelity Cleft Palate Simulator.” In: *Plast. Reconstr. Surg.* 139.1, 85e–96e. ISSN: 0032-1052. DOI: [10.1097/PRS.0000000000002923](https://doi.org/10.1097/PRS.0000000000002923).
- Podolsky, D. J., Murphy, B. D., Ahmad, J., Fisher, D. M., Wong Riff, K. W. Y., Drake, J. M., and Forrest, C. R. (2024a). “Development and Evaluation of a High-Fidelity Rhinoplasty Simulator.” In: *Plast. Reconstr. Surg.* 153.3, pp. 619–622. ISSN: 0032-1052. DOI: [10.1097/PRS.0000000000010636](https://doi.org/10.1097/PRS.0000000000010636).
- Robertson, E., Boulanger, P., Kwan, P., Louie, G., and Aalto, D. (2023). “Improving Cranial Vault Remodeling for Unilateral Coronal Craniosynostosis—Introducing Automated Surgical Planning.” In: *Craniofacial Trauma & Reconstruction* 17, pp. 203–213. ISSN: 1943-3875, 1943-3883. DOI: [10.1177/19433875231178912](https://doi.org/10.1177/19433875231178912).
- Rodby, K. A., Turin, S., Jacobs, R. J., Cruz, J. F., Hassid, V. J., Kolokythas, A., and Antony, A. K. (2014). “Advances in Oncologic Head and Neck Reconstruction: Systematic Review and Future Considerations of Virtual Surgical Planning and Computer Aided Design/Computer Aided Modeling.” In: *J Plast Reconstr Aesthet Surg* 67.9, pp. 1171–1185. ISSN: 17486815. DOI: [10.1016/j.bjps.2014.04.038](https://doi.org/10.1016/j.bjps.2014.04.038).

- Sanders, I. and Mu, L. (2013). “A Three-Dimensional Atlas of Human Tongue Muscles.” In: *Anat Rec (Hoboken)* 296.7, pp. 1102–1114. ISSN: 1932-8494. DOI: [10.1002/ar.22711](https://doi.org/10.1002/ar.22711).
- Seikaly, H., Rieger, J., O’Connell, D., Ansari, K., AlQahtani, K., and Harris, J. (2009). “Beavertail Modification of the Radial Forearm Free Flap in Base of Tongue Reconstruction: Technique and Functional Outcomes.” In: *Head Neck* 31.2, pp. 213–219. ISSN: 1043-3074, 1097-0347. DOI: [10.1002/hed.20953](https://doi.org/10.1002/hed.20953).
- Serrano, C., Van Den Brink, H., Pineau, J., Prognon, P., and Martelli, N. (2019). “Benefits of 3D Printing Applications in Jaw Reconstruction: A Systematic Review and Meta-Analysis.” In: *J. Cranio-MaxilloFac. Surg.* 47.9, pp. 1387–1397. ISSN: 10105182. DOI: [10.1016/j.jcms.2019.06.008](https://doi.org/10.1016/j.jcms.2019.06.008).
- Sparks, J. L., Vavalle, N. A., Kasting, K. E., Long, B., Tanaka, M. L., Sanger, P. A., Schnell, K., and Conner-Kerr, T. A. (2015). “Use of Silicone Materials to Simulate Tissue Biomechanics as Related to Deep Tissue Injury.” In: *Adv. Skin Wound Care*. 28.2, pp. 59–68. ISSN: 1527-7941. DOI: [10.1097/01.ASW.0000460127.47415.6e](https://doi.org/10.1097/01.ASW.0000460127.47415.6e).
- Suhaym, O., Moles, L., and Callahan, N. (2024). “Cutting Guides in Mandibular Tumor Ablation: Are We as Accurate as We Think?” In: *The Saudi Dental Journal* 36.2, pp. 340–346. ISSN: 10139052. DOI: [10.1016/j.sdentj.2023.11.017](https://doi.org/10.1016/j.sdentj.2023.11.017).
- Sullivan, C. and Kaszynski, A. (2019). “PyVista: 3D Plotting and Mesh Analysis through a Streamlined Interface for the Visualization Toolkit (VTK).” In: *J. Open Source Softw.* 4.37, p. 1450. ISSN: 2475-9066. DOI: [10.21105/joss.01450](https://doi.org/10.21105/joss.01450).
- Tarsitano, A., Battaglia, S., Cipriani, R., and Marchetti, C. (2016). “Microvascular Reconstruction of the Tongue Using a Free Anterolateral Thigh Flap: Three-dimensional Evaluation of Volume Loss after Radiotherapy.” In: *J. Cranio-MaxilloFac. Surg.* 44.9, pp. 1287–1291. ISSN: 1010-5182. DOI: [10.1016/j.jcms.2016.04.031](https://doi.org/10.1016/j.jcms.2016.04.031).
- Urken, M. L. and Biller, H. F. (1994). “A New Bilobed Design for the Sensate Radial Forearm Flap to Preserve Tongue Mobility Following Significant Glossectomy.” In: *Arch. Otolaryngol. Head Neck Surg.* 120.1, pp. 26–31. ISSN: 0886-4470. DOI: [10.1001/archotol.1994.01880250022002](https://doi.org/10.1001/archotol.1994.01880250022002).
- Urken, M. (2012). *Multidisciplinary Head and Neck Reconstruction: A Defect-Oriented Approach*. Lippincott Williams & Wilkins. ISBN: 978-1-4511-5337-8.

- Vincent, A., Kohlert, S., Lee, T., Inman, J., and Ducic, Y. (2019). “Free-Flap Reconstruction of the Tongue.” In: *Seminars in Plastic Surgery* 33.01, pp. 038–045. ISSN: 1535-2188, 1536-0067. DOI: [10.1055/s-0039-1677789](https://doi.org/10.1055/s-0039-1677789).
- Wrench, A. A. (2024). “The Compartmental Tongue.” In: *J Speech Lang Hear Res* 67.10S, pp. 3887–3913. ISSN: 1092-4388, 1558-9102. DOI: [10.1044/2024_JSLHR-23-00125](https://doi.org/10.1044/2024_JSLHR-23-00125).
- Xu, J., Lai, F., Liu, Y., Tan, Z., Zheng, C., Wang, J., Guo, H., Jiang, L., Ge, X., Lan, X., Chen, C., and Ge, M. (2022). “Novel Computer-Aided Reconstruction of Soft Tissue Defects Following Resection of Oral and Oropharyngeal Squamous Cell Carcinoma.” In: *World J. Surg. Oncol.* 20.1, p. 196. ISSN: 1477-7819. DOI: [10.1186/s12957-022-02654-7](https://doi.org/10.1186/s12957-022-02654-7).
- Yen, C.-I., Zelken, J. A., Chang, C.-S., Lo, L.-J., Yang, J.-Y., Chuang, S.-S., Araniego, C. A., and Hsiao, Y.-C. (2019). “Computer-Aided Design and Three-Dimensional Printing Improves Symmetry in Heminasal Reconstruction Outcomes.” In: *J. Plast. Reconstr. Aesthet. Surg.* 72.7, pp. 1198–1206. ISSN: 17486815. DOI: [10.1016/j.bjps.2019.03.012](https://doi.org/10.1016/j.bjps.2019.03.012).

Chapter 7

Discussion & Conclusion

7.1 Introduction

The reconstruction of the human tongue following cancer resection is one of the most formidable challenges in surgical oncology. Historically, this procedure has been a craft-based art, a domain where outcomes are influenced by the surgeon's intuition, experience, and freehand dexterity. While this approach has yielded remarkable successes in survival and patient outcomes, its inherent variability means that optimal functional results remain difficult to predict and reproduce. This thesis was founded on a central premise: the path to more predictable and consistent outcomes lies in transforming tongue reconstruction from a traditional art into a data-driven science. The spirit of this thesis, therefore, is to establish the core scientific and technological pillars of a new paradigm that leverages patient-specific data, biomechanical principles, modeling, and guided execution to make the entire surgical process more predictable.

This thesis argues that we are now closer to the clinical adoption of Virtual Surgical Planning (VSP) for tongue reconstruction by advancing three fundamental pillars. The first pillar is the **Biomechanically Optimized Free Flap Design**. The development of the hiFEM algorithm (Chapters 3 and 4) represents a shift in the principle of free-flap design from geometric defect matching to optimizing for desired biomechanical outcomes. Such a design minimizes the tissue stretch

and tension required to conform to the defect, alleviating the biomechanical drivers of common post-operative tissue complications and tongue tethering. This pillar lays the foundation for a plan that is not just anatomically matched but biomechanically sound.

The second pillar is the **Outcome Estimation Framework**. A virtual plan is of limited use if the surgeon cannot estimate the long-term consequences of critical, interdependent surgical decisions. The surgical simulation framework serves as a virtual testbed, a tool for gaining granular insight that is ethically and logistically challenging to acquire through clinical trials. This framework serves a dual purpose: it is an engine for research discoveries, as demonstrated in Chapter 5, and it could be used as a powerful tool for empowering the surgeon to make critical, patient-tailored surgical decisions. By transforming complex, experience-based judgments—such as donor site selection and the degree of flap overbulking—into quantifiable, physics-based trade-offs, it enables the development of informed, patient-optimized surgical strategies.

The final pillar is the **Surgical Toolkit**. A superior plan is rendered clinically irrelevant if it cannot be translated accurately to the dynamic environment of the operating room. To bridge this crucial virtual-physical gap, an additively manufactured surgical toolkit was designed and is detailed in Chapter 6. This system provides a physical map that translates the digital blueprint for resection and flap harvesting into precise, repeatable surgical actions and demonstrates that the “Execution Challenge” in soft tissue can be met.

Collectively, these studies establish the feasibility of a cohesive, three-pillar VSP workflow for tongue reconstruction. This chapter, therefore, synthesizes these individual contributions and explores the overarching themes that emerge from this framework. It will critically situate these advancements within the literature, identify remaining challenges, and articulate a vision for the future of data-driven soft tissue reconstruction.

7.2 Surgical Simulation: New Scientific and Clinical Paradigm

The history of VSP for tongue reconstruction is a story of navigating a fundamental tension between two approaches: clinically pragmatic but biomechanically naive tools, and scientifically sophisticated but clinically non-feasible research models. On one end of the spectrum, clinical practice

has evolved from generic templates to patient-specific, image-guided approaches. Landmark-based measurement techniques (Fan et al., 2019a; Liu et al., 2023b) improved personalization, and the algorithmic flattening method in the CARST workflow (Xu et al., 2022) represented a major leap into digital planning. Yet these methods remain fundamentally geometric and lack biomechanical considerations. On the other end, the research community has pursued ever-higher fidelity. The powerful dynamic models (Stavness et al., 2011; Stavness et al., 2014a; Anderson et al., 2017; Kappert et al., 2021a) provided profound insights but were hampered by immense computational cost, notorious numerical fragility, and an unreasonable labor requirement for a clinical workflow, making them unsuitable for day-to-day clinical use. Harandi and colleagues (2017b; 2017a) made crucial strides in automating patient-specific model generation for speech pathology research and established a foundation that this thesis builds upon and adapts for the distinct application of surgical simulation.

This thesis posits that the path to clinical adoption lies not in choosing one pole over the other, but in developing a pragmatic and robust ecosystem of simulation tools in which the level of fidelity is intelligently matched to the specific task. First, to address the need for rapid planning, we developed the hiFEM algorithm (Chapters 3 and 4). It retains the goals of speed and automation while integrating the core principle of high-fidelity models: minimizing the biomechanical cost associated with tissue stretching. The rapid optimization framework provides a robust tool for the design phase. Second, to address the need for a framework that estimates outcomes of surgical decisions, we drew on the procedural simulation concepts pioneered by Kappert et al. (2019) and the patient-specific modeling work of Harandi and colleagues (2017b; 2017a). We presented a patient-specific surgical simulation framework (Chapter 5) that advances their work by modeling free-flap tongue reconstruction while being engineered from the ground up for automation and stability.

Achieving this level of automation and numerical reliability required substantial engineering effort, as simulations involving hyperelastic materials, large deformations, and complex contact mechanics are notoriously prone to failure. Through systematic exploration of multiple technical strategies (detailed in Chapter 5)—including evaluation of different element types, rigorous mesh

convergence analysis, systematic tuning of contact penalty parameters, and implementation of uncoupled material formulations—the framework ultimately completed an automated 120-simulation factorial study without numerical failures. This achievement transforms surgical simulation from a temperamental research instrument into a reliable engineering tool, demonstrating that high-fidelity biomechanical modeling can bridge the gap between the simulation lab and the clinical workflow.

7.3 Shifting the Flap Planning Objective from Defect-Fitting to Biomechanical Optimization

This thesis argues for a fundamental reframing of free-flap design: move from geometric pattern-matching toward biomechanical energy optimization. This theme directly supports our primary objective of developing a clinically relevant method for patient-specific flap planning. The hiFEM algorithm was engineered to answer a fundamentally different question than its predecessors. It departs from approaches centered on landmark-based defect matching (Fan et al., 2019a; Liu et al., 2023b) or digital surface flattening (Xu et al., 2022), which implicitly ask, “What 2D shape best unfolds from the 3D defect?” Instead, hiFEM poses a more surgically pertinent question: “What 2D shape requires the least physical work to deform *into* the 3D defect?” This distinction is critical because it aligns the optimization objective with the physical reality of the surgical procedure. A planar flap is harvested and then actively stretched and shaped to conform to the 3D defect. The clinical success of a reconstruction is therefore governed not by the abstract elegance of its geometric unfolding, but by the physical consequences of the required deformation.

The clinical relevance of excessive tissue tension in free-flap reconstruction is well established. Excessive tension can impede blood flow, increasing the risk of ischemia and tissue necrosis, while tethering the residual native tongue and restricting the mobility crucial for speech and swallowing (Buchaillard et al., 2007; Rui et al., 2022; Tepole et al., 2014b; Al-Zanoon et al., 2024). This evidence motivated the development of hiFEM, which seeks to minimize stored strain energy—a direct physical proxy for intrinsic tissue stretch and tension within the reconstructed flap. Chapter 4 demonstrated hiFEM’s success in achieving this objective. This biomechanical focus was further

reinforced in Chapter 5, where surgical simulations revealed that tissue stiffness is a dominant predictor of the final neotongue state. By targeting the root cause of these adverse effects at the planning stage, hiFEM proposes a physics-based VSP.

While hiFEM represents a significant advance, its current implementation is an initial step toward higher fidelity. The algorithm presently employs a literature-validated but simplified isotropic Ogden hyperelastic model for the skin, which does not account for known complexities of skin biomechanics such as anisotropy, time-dependence, and pre-stress. The next frontier for planning algorithms is to integrate these properties to design flaps that are also optimized for directional stiffness. However, this raises a critical question about the trade-off between theoretical perfection and practical execution. The initial shift from geometric flattening to biomechanical optimization already yielded millimeter-level geometric differences, with Hausdorff distances between hiFEM and NURBS-flattening designs ranging from 0.9 mm to 2.3 mm. We hypothesize that the *additional* geometric refinement from incorporating anisotropy would be smaller, likely on the sub-millimeter scale. Pursuing higher model fidelity must therefore remain grounded in tangible surgical benefits.

7.4 Clinically Feasible Virtual Framework for Surgical Decision-Making

This thesis introduces a new methodology for surgical research and preoperative planning by establishing a feasible virtual testbed for surgical decision-making. The novel Surgery Simulation framework allows for the systematic, factorial investigation of critical surgical variables, such as donor site and overbulking, which would be ethically and logistically prohibitive in prospective randomized clinical trials. Its immediate impact is the transformation of complex, experience-based judgments into quantifiable, physics-based relationships. For the first time, the trade-off between restoring anatomical form via overbulking and the resulting biomechanical strain imposed on the native tongue can be explored preoperatively for an individual patient. This framework represents an advance over foundational simulation studies that either modified material properties in a generic mesh (Buchaillard et al., 2007; Al-Zanoon et al., 2024) or simulated simpler primary closure surgeries (Kappert et al., 2019). It is the first to simulate the entire free-flap reconstruction

workflow—from the virtual suturing of a distinct flap body to long-term biological atrophy—with the robustness required for large-scale computational experiments. This unique capability, demonstrated by the successful 120-simulation factorial study, provides surgeons with a methodology to de-risk and quantitatively compare surgical strategies and makes patient-specific optimization computationally feasible.

The computational robustness and scalability of this framework could also address the fundamental barrier to applying machine learning in this domain: data scarcity. By functioning as a high-throughput data factory, this framework can generate the thousands of synthetic reconstruction scenarios required to pretrain Physics-Informed Neural Networks (PINNs). This unlocks a Hybrid Digital Twin workflow, in which models pretrained on our synthetic physics data are subsequently fine-tuned using expensive and sparse clinical preoperative and postoperative data. This fine-tuning step is particularly promising because it allows the system to bridge the gap between idealized physics simulation and clinical reality; by fine-tuning the model with actual patient outcomes, the AI can implicitly learn to correct for complex, patient-specific nonlinearities—such as scarring and radiation-induced fibrosis—that are not incorporated in the physical simulation. Thus, the simulation framework presented here is not only a decision-support tool in its own right but also an enabling engine for future data-driven surgical intelligence.

7.5 Fidelity in Modeling the Flap-Tongue Interface, Reconstruction, and Flap Atrophy

This thesis introduces a new standard for procedural fidelity in surgical simulation by modeling reconstruction as a dynamic sequence of surgical actions and subsequent biological responses. The framework models the free flap as an independent, multi-component biomechanical entity. To achieve this, technical innovations were engineered to better reflect clinical reality. The act of suturing was simulated using time-activated tension elements that draw the free flap and native tongue together, a method that adapts concepts pioneered for primary closure (Kappert et al., 2019) to the more complex free-flap scenario. This is followed by a simulation of the immediate viscoelastic stress relaxation in the tissue to establish a stable postoperative state. Furthermore, a

“sticky” contact formulation at the flap–tongue interface was introduced, which computationally enforces the desired surgical goal of complete tissue apposition. This multi-stage simulation provides a novel and robust platform for predicting the immediate postoperative state resulting from the physical actions of the surgeon.

Another significant advancement lies in the simulation of long-term biological remodeling, a critical factor governing the final anatomical outcome and degree of tissue tethering. By implementing a clinically informed volumetric atrophy of the free flap, the model provides a biomechanical rationale for the surgical practice of overbulking. Advancing prior studies that examined the consequences of volume loss (Wang et al., 2019), our approach simulates the mechanics of the atrophy process itself and estimates the final resting anatomy and the internal strain state imposed on the native tongue. The model, while powerful, makes key simplifications for clinical feasibility. The atrophy is modeled as a uniform, isotropic process, and the distinct biomechanical effects of radiation-induced fibrosis and scar contracture are not yet included. A future iteration could build upon recent work by Al-Zanoon et al. (2024) and implement a time-dependent constitutive model for the elements. This would simulate the progressive stiffening and contraction of scar tissue to capture another critical component of the post-reconstruction biomechanical environment.

7.6 Patient-Specific Parameters

The findings from Chapter 5 demonstrate that the predictive power of the model is critically dependent on an individual’s unique biomechanical and biological characteristics. First, the study quantified the downstream effects of patient-specific tissue properties. It revealed that the anatomical benefit of overbulking (a surgeon’s primary tool for compensating long-term atrophy) is fundamentally coupled to the flap’s tissue pliability. The results showed a significant interaction between overbulking and donor-site selection: the same increase in volume produced a significantly better anatomical outcome when applied to a more compliant forearm flap compared to a stiffer thigh flap. This provides a physics-based explanation for the clinical concept of “pliability” and suggests that an accurate, patient-specific stiffness measurement is essential for determining the clinical

effectiveness of a surgical maneuver. These findings highlight the role of patient-specific parameterization and elevate the importance of incorporating non-invasive, in vivo measurement techniques into the pre-operative workflow. For example, methods such as MR elastography or suction-based devices could provide patient-specific data on tongue stiffness (Cheng et al., 2011; Kappert et al., 2021b), while indentation or micro-robotic devices can characterize the mechanical properties of the donor-site skin before surgery (Flynn et al., 2011).

Similarly, the simulation establishes a direct mathematical relationship between the assumed rate of postoperative atrophy and the required overbulking percentage needed to achieve long-term volume restoration (Figure 5.5). Using 40% atrophy in the model was a necessary simplification for a systematic study, but it was guided by clinical literature showing this value is highly variable, with reported volume loss ranging from 20% to nearly 60%. This variability depends on factors such as adjuvant radiotherapy, flap composition, patient age, and BMI (Gewirtz et al., 2024; Razavi et al., 2022; Tarsitano et al., 2016; Sakakibara et al., 2019). This highlights a clear direction for future work: integrating predictive clinical models of atrophy directly into the simulation pipeline. The goal is to evolve the simulation's input from a single, population-averaged estimate to a patient-specific probability calculated from individual risk factors.

While these patient-specific parameters are crucial for accurately predicting the final passive anatomy, the ultimate clinical goal is to restore dynamic function. To bridge this gap, the next essential advancement is to incorporate active muscle dynamics into the simulation framework. This requires two critical extensions to the current model. First, the tongue's complex muscle architecture must be represented through anisotropic material formulations that capture the directional stiffness and contractile capacity conferred by muscle fibers. Patient-specific muscle fiber orientations can be mapped using diffusion-weighted MRI (DW-MRI), a technique pioneered for tongue modeling by Kappert et al. (Kappert et al., 2021a). Second, the framework must account for motor adaptation and compensatory behaviors that emerge during functional recovery. Recent work by Al-Zanoon et al. (2025) demonstrates that patients develop sophisticated compensatory strategies—altering muscle activation patterns and coordination—to overcome structural limitations imposed by the reconstruction. Incorporating such neuromuscular adaptation models would enable the simulation

to predict not only the immediate biomechanical constraints but also the patient's capacity for functional rehabilitation. This integrated approach would empower surgeons to optimize reconstructions for their long-term impact on speech and swallowing and ultimately guide targeted rehabilitation strategies tailored to the patient's specific neotongue anatomy and residual motor capacity.

7.7 The Path to Clinical Impact: Implementation, Validation, and Human Factors

The full clinical potential of a biomechanically superior flap design is contingent upon the ability to execute the surgical plan with millimeter-level precision. The hiFEM design provides a plan optimized to minimize tissue tension, but these benefits are only realized if the virtual plan can be translated to the dynamic environment of the operating room with high fidelity. Chapter 6 provides the first crucial demonstration that this level of accuracy is feasible: a guided reconstruction on a deformable phantom can attain geometric accuracy comparable to the gold standard for mandibular and maxillary reconstruction. However, the toolkit's key innovation lies not just in its precision but in its introduction of a dynamic intraoperative feedback loop. This transforms the surgical workflow from a rigid, linear "Plan-Execute" sequence into a more clinically pragmatic "Plan-Execute-Inspect-Adapt" paradigm. The Resection Volume Inspection Guide, for instance, allows the surgeon to intraoperatively inspect the excised tissue against the plan and make informed adjustments to the flap's volume. This integration paves the way for future innovations, such as designing a discrete set of guides with pre-calculated alternative margins (e.g., +5 mm, +10 mm) to provide surgeons with immediate options if unexpected tumor spread is found.

Beyond procedural execution, the framework offers a powerful platform for addressing the human factors integral to surgical success through preoperative rehearsal and surgeon training. Patient-specific, high-fidelity phantoms serve a dual purpose. First, they enable preoperative rehearsal, which allows the surgeon to physically practice the entire guided procedure on a replica of the patient's anatomy. This can build muscle memory, reduce cognitive load in the operating room, and allow anticipation of case-specific challenges. Second, these phantoms provide a safe, repeatable environment for surgical training, where residents and fellows can learn complex

reconstructive techniques (Eu et al., 2024). Future studies can evaluate the learning curve for surgeons in training using this toolkit versus freehand methods to quantify its impact on skill acquisition and standardization. As multi-material 3D printing technology becomes more accessible, the field is moving toward creating phantoms with even greater biomechanical fidelity to enhance training realism.

Finally, translating this framework from the lab to routine clinical practice requires a deliberate validation pathway. While Chapter 6 established benchtop feasibility by demonstrating high geometric accuracy, this remains a surrogate for functional outcome improvement. This focus on anatomical restoration is not merely a convenience; it is grounded in extensive clinical evidence establishing a direct link between the quality of anatomical reconstruction and recovery of function. The literature consistently shows that achieving a more natural neotongue structure through patient-specific planning directly improves functional outcomes (Fan et al., 2019b; Fan et al., 2019a; Xu et al., 2022; Liu et al., 2023b; Liu et al., 2023c; Liu et al., 2023a; Idris et al., 2018; Baskin et al., 2019; Urken, 2012; Leymarie et al., 2012; Amer, 2021). Therefore, the immediate next step—a pilot clinical case series validating in-vivo safety and geometric accuracy—serves as a critical confirmation that the toolkit can achieve this clinically relevant anatomical goal. A subsequent prospective randomized clinical trial is required to examine the strength of the association between reconstructed form and function and to determine whether the framework’s enhanced precision yields a measurable functional benefit over existing techniques. Endpoints will focus on speech, swallowing, and patient-reported quality of life.

7.8 Limitations of the Thesis

The simulations and tools presented in this thesis operate within a computational feasibility framework, which is a necessary precursor to, but distinct from, clinical predictive validity. Closing the gap to the realities of clinical outcomes requires progression through multiple validation levels. The first is constitutive modeling and parameter validation. Although the hyperelastic constitutive models and material parameters used here were rigorously adopted from the experimental literature, whether they capture patient-specific biological nuances remains unknown (Hermant et al., 2017).

True patient-specificity requires complex parameter identification, including the incorporation of individual muscle fiber architecture (Kappert et al., 2021a; Kappert et al., 2021b). Second is the construct validity. In simulation, this research isolated and quantified the impact of flap parameters and postoperative atrophy, but omitted phenomena such as active muscle tone, heterogeneity, anisotropy, radiation-induced fibrosis, and scar contracture, whose combined effects may further drive deviation from reality. Similarly, the physical tongue phantoms utilized silicone materials as analogues, yet they lack physiological features such as perfusion, heterogeneity, anisotropy, and the higher viscosity and nonlinearity observed in the real tongue. Randomized controlled trials (RCTs) involving surgeons are therefore needed to evaluate haptic fidelity and in vivo representativeness.

The third is anatomical predictive validity, which applies equally to the surgical simulation outcomes and the physically guided execution. Establishing clinical validity requires quantitative comparison of the predicted resting morphology with longitudinal postoperative imaging of actual patients. Fourth is functional correspondence: determining whether the biomechanical proxies established in this thesis—effective strain and tissue tethering—correlate statistically with clinical metrics such as speech intelligibility and swallowing efficiency, consistent with the validation precedent set by Xu et al. (2022). Finally, intraoperative usability must be evaluated to confirm that the toolkit’s utility is maintained under the practical challenges of live surgery, accounting for variables such as fluid obscuration, tissue mobility, saliva, and restricted access. This thesis primarily develops computational infrastructure; the experimental and clinical data needed to close the loop between virtual prediction and clinical reality remain the next frontier. What follows dives deeper into the limitations within each study.

Generalizability and Fidelity of the Foundational Models

The entire computational framework, from the hiFEM algorithm to the surgical simulation and toolkit design, is built upon the anatomy of a single healthy male volunteer. This choice limits the immediate generalizability of the numerical findings to a broader patient population with diverse ages, sexes, and pathologies. Furthermore, the performance of the VSP framework on larger defects, such as total or subtotal glossectomies, remains untested.

Additionally, the models employ literature-derived isotropic material properties for soft tissues. This necessary simplification does not capture patient-specific biomechanical complexities such as the anisotropy introduced by muscle fibers, the pre-stress inherent in skin, or the altered tissue properties in post-radiation patients or in scarred tissue. The biological remodeling in Chapter 5, while innovative, also relies on a single population-averaged atrophy value (40%) and models it as a uniform, isotropic process. This approach does not account for the heterogeneous, direction-dependent stiffening and contracture that occurs with scar tissue formation at the flap–tongue interface, a critical component of the final biomechanical environment.

Static Simulation Scope: Predicting Anatomical Form Over Dynamic Function

The surgical simulation focuses on estimating the passive, resting anatomy of the neotongue and its final biomechanical state. The simulation framework does not incorporate active muscle contraction, which drives critical tongue functions such as speech and swallowing. While the framework provides novel insights into how surgical decisions affect tissue tethering and resting morphology (an established and important proxy for function), it cannot directly estimate functional outcomes. For example, it can quantify the strain that may limit mobility, but it cannot simulate the act of speech articulation itself. Therefore, the conclusions drawn concern the anatomical and biomechanical potential for function, rather than a direct simulation of functional performance. Addressing this, alongside the computationally unexplored domain of sensory recovery, represents a necessary next step for the research.

Reliance on Pre-Clinical Demonstrations

Current validation of the presented VSP workflow is limited to pre-clinical studies using surrogate endpoints, with geometric accuracy as the primary measure. This thesis is built upon the clinically supported premise that improved anatomical restoration leads to better function. While the general principle is well supported, this thesis cannot, within its pre-clinical scope, definitively establish this causal link. That confirmation awaits the rigorous clinical trials discussed previously.

The benchtop validation environment, by its nature, introduces its own limitations. The homogeneous silicone phantoms used in Chapter 6, while sufficient for a feasibility analysis, simplify the tongue’s complex, layered, and anisotropic structure. This simplification reduces the realism

of simulated surgical tasks such as dissection and suturing. Furthermore, as non-biological surrogates, the phantoms cannot model key biological phenomena such as bleeding, saliva production, or postoperative atrophy. As a result, we were unable to test an important clinical utility of the proposed surgical toolkit: operationalizing a pre-planned overbulking strategy to address expected postoperative atrophy.

Procedural and Intraoperative Limitations of the Proposed Toolkit

The physical toolkit, as demonstrated in Chapter 6, presents practical limitations for intraoperative use. A primary challenge is the system’s reliance on referential and inspection guidance rather than active, constraining guidance. The resection marking guide delineates the 2D incision boundary, but guiding the cut’s depth and angle relies on the surgeon’s tactile reference to a separate model rather than on a physical constraint that controls the instrument’s trajectory.

Furthermore, the system’s fidelity is entirely contingent upon the stable and accurate registration of a rigid guide onto the mobile, deformable tongue. In the landmark-poor environment of the tongue, minor misalignments risk propagating significant errors throughout the reconstruction. Finally, the workflow is only partially robust to unexpected intraoperative findings. The need for significantly wider oncologic margins than anticipated could render the pre-fabricated, patient-specific guides unusable. Multiple surgical guides, each with incrementally increasing resection margins, could be pre-manufactured to provide some flexibility, but this remains a logistical challenge.

Practical Barriers to Clinical Integration and Workflow Automation

While this thesis establishes the scientific and technical feasibility of a comprehensive VSP framework, it does not fully address the remaining practical and logistical hurdles to routine clinical adoption. A major unresolved challenge is the “turnaround time” of the VSP pipeline. The process of manually segmenting a patient’s tongue anatomy from MRI scans remains labor-intensive and requires significant expertise, potentially taking hours. For VSP to become a standard-of-care tool, this initial step must be significantly streamlined, likely through the development and validation of semi-automated or AI-based segmentation algorithms such as Biomedisa (Lösel et al., 2020).

7.9 Conclusion and Future Vision

This thesis presented a novel, end-to-end computational framework for patient-specific tongue reconstruction. The framework integrates the three critical pillars of biomechanics-based surgical planning, outcome estimation, and guided surgical execution. This work pursued a single, unifying goal: to begin the systematic transformation of this complex procedure from a craft-based art reliant on surgical intuition into a data-driven science grounded in patient-specific biomechanics and predictive modeling.

The clinical impact of this approach is not to replace the surgeon's expertise but to augment it with predictive power and procedural precision. By shifting the objective of flap design from geometric matching to biomechanical optimization, the hiFEM algorithm provides a plan engineered to mitigate the root causes of tethering and impaired mobility. The surgical simulation framework transforms the qualitative art of surgical judgment into a quantitative science of decision analysis. It converts the challenge of navigating interdependent variables from a process of reactive intra-operative problem-solving to one of proactive preoperative strategy optimization. The additively manufactured toolkit then closes the crucial loop between plan and action and demonstrates that the biomechanical advantages conceived in the virtual world can be delivered with high fidelity in the complex, deformable environment of the operating room. These pillars together offer a pathway to standardize a variable procedure, reduce surgical guesswork, and improve the predictability of functional outcomes for patients.

The tools and workflows developed in this thesis are not an endpoint but a foundation for the future of soft-tissue reconstructive surgery. The next step is rigorous clinical validation to translate the demonstrated geometric accuracy into proven functional benefits in speech and swallowing. This requires prospective trials and the refinement of our models from population-averaged estimates to simulations tuned to an individual's physiology. Further, the scalability of this framework requires automating the final manual bottleneck: the segmentation of medical images. The integration of AI-driven segmentation, such as Biomedisa (Lösels et al., 2020), will transform this VSP workflow from a specialized research tool into a routine, accessible component of standard clinical care.

This foundation supports a future where a “surgical digital twin” is created for every patient. On this virtual replica, surgeons will not only optimize a flap design but also simulate the entire procedure, predict its long-term functional consequences, and rehearse complex maneuvers before making a single incision. The patient-specific toolkit presented here represents the first generation of physical interfaces between this digital twin and the patient. Its evolution, combined with advancements in intraoperative navigation and robotics, will allow optimized surgical plans to be executed with a precision beyond the limits of the human hand. This thesis is a definitive step toward that future of predictable, personalized, and functionally superior soft-tissue reconstruction.

Bibliography

- Al-Zanoon, N., Cummine, J., Jeffery, C. C., and Aalto, D. (2025). “Compensatory Behaviours after Oral Cancer: Stretch Reflex Improves Simulated Tongue Protrusion.” In: *Computer Methods in Biomechanics and Biomedical Engineering*, pp. 1–11. ISSN: 1025-5842, 1476-8259. DOI: [10.1080/10255842.2025.2482128](https://doi.org/10.1080/10255842.2025.2482128).
- Al-Zanoon, N., Cummine, J., Jeffery, C. C., Westover, L., and Aalto, D. (2024). “The Effect of Simulated Radiation Induced Fibrosis on Tongue Protrusion.” In: *Biomech. Model. Mechanobiol.* 23, pp. 1649–1660. ISSN: 1617-7940. DOI: [10.1007/s10237-024-01860-4](https://doi.org/10.1007/s10237-024-01860-4).
- Amer, I. A. (2021). “New Modification of Flap Design for Total and Near-Total Mobile Tongue Reconstruction Using Anterolateral Thigh Flap.” In: *Egypt. J. Surg.* 40.1, pp. 330–341. ISSN: 1110-1121, 1687-7624. DOI: [10.4103/ejs.ejs_350_20](https://doi.org/10.4103/ejs.ejs_350_20).
- Ammam, I., Guillermin, A., Ouillon, L., Vargiolu, R., Perrot, J.-L., and Zahouani, H. (2024). “Coupling Tensile Test with LC-OCT and Ultrasound Imaging: Investigation of the Skin Sub-layers Mechanical Behaviour.” In: *R. Soc. Open Sci.* 11.6, p. 231712. ISSN: 2054-5703. DOI: [10.1098/rsos.231712](https://doi.org/10.1098/rsos.231712).
- Anderson, P., Fels, S., Harandi, N. M., Ho, A., Moisik, S., Sánchez, C. A., Stavness, I., and Tang, K. (2017). “FRANK: A Hybrid 3D Biomechanical Model of the Head and Neck.” In: *Biomechanics of Living Organs*. Ed. by Y. Payan and J. Ohayon. Vol. 1. Translational Epigenetics. Oxford: Academic Press, pp. 413–447. DOI: [10.1016/B978-0-12-804009-6.00020-1](https://doi.org/10.1016/B978-0-12-804009-6.00020-1).
- Ansarin, M., Bruschini, R., Navach, V., Giugliano, G., Calabrese, L., Chiesa, F., Medina, J. E., Kowalski, L. P., and Shah, J. P. (2019). “Classification of GLOSSECTOMIES: Proposal for Tongue Cancer Resections.” In: *Head Neck-J. Sci. Spec. Head Neck* 41.3, pp. 821–827. ISSN: 1043-3074, 1097-0347. DOI: [10.1002/hed.25466](https://doi.org/10.1002/hed.25466).

- Arias, E., Huang, Y.-H., Zhao, L., Seelaus, R., Patel, P., and Cohen, M. (2019). “Virtual Surgical Planning and Three-Dimensional Printed Guide for Soft Tissue Correction in Facial Asymmetry.” In: *J. Craniofac. Surg.* 30.3, pp. 846–850. ISSN: 1049-2275, 1536-3732. DOI: [10.1097/SCS.0000000000005204](https://doi.org/10.1097/SCS.0000000000005204).
- Bashorun, O. H., Johnson, R. M., and Johnson, E. A. (2023). “Flap Debulking and Secondary Revisions in Head and Neck Reconstruction: A Systematic Review with Clinical Applications.” In: *Semin Plast Surg* 37.01, pp. 073–082. ISSN: 1535-2188, 1536-0067. DOI: [10.1055/s-0042-1760444](https://doi.org/10.1055/s-0042-1760444).
- Baskin, R. M., Seikaly, H., Sawhney, R., Danan, D., Burt, M., Idris, S., Shama, M., Boyce, B., and Dziegielewska, P. T. (2019). “Tongue Reconstruction: Rebuilding Mobile Three-Dimensional Structures from Immobile Two-Dimensional Substrates, a Fresh Cadaver Study.” In: *Head Neck-J. Sci. Spec. Head Neck* 41.10, pp. 3693–3699. ISSN: 1043-3074, 1097-0347. DOI: [10.1002/hed.25889](https://doi.org/10.1002/hed.25889).
- Benjamin, M., Aliano, K., Davenport, T., and Frank, D. (2020). “Functional Outcomes Regarding Facial Artery Musculomucosal Flap for Reconstruction of Partial Glossectomy Defects.” In: *Ann. Plast. Surg.* 85, S76–S79. ISSN: 0148-7043, 1536-3708. DOI: [10.1097/SAP.0000000000002346](https://doi.org/10.1097/SAP.0000000000002346).
- Bibb, R., Eggbeer, D., Paterson, A., and Mohammed, M. I. (2024). *Medical Modeling: The Application of Advanced Design and Additive Manufacturing Techniques in Medicine*. Elsevier. ISBN: 978-0-323-95734-2.
- Bielecki, K. (2017). “Are Randomized Clinical Trials Necessary in Cancer Surgery?” In: *Nowotw. J. Oncol.* 66.6, pp. 494–498. ISSN: 2300-2115, 0029-540X. DOI: [10.5603/NJO.2016.0088](https://doi.org/10.5603/NJO.2016.0088).
- Bonet, J. and Wood, R. D. (2008). *Nonlinear Continuum Mechanics for Finite Element Analysis*. 2nd ed. Cambridge University Press. ISBN: 978-0-521-83870-2 978-0-511-75544-6. DOI: [10.1017/CB09780511755446](https://doi.org/10.1017/CB09780511755446).
- Bostan Shirin, M. and Assempour, A. (2014). “Some Improvements on the Unfolding Inverse Finite Element Method for Simulation of Deep Drawing Process.” In: *Int J Adv Manuf Technol* 72.1, pp. 447–456. ISSN: 1433-3015. DOI: [10.1007/s00170-014-5663-4](https://doi.org/10.1007/s00170-014-5663-4).
- Brückler, H. and Campen, M. (2023). “Collapsing Embedded Cell Complexes for Safer Hexahedral Meshing.” In: *ACM Trans. Graph.* 42.6, pp. 1–24. ISSN: 0730-0301, 1557-7368. DOI: [10.1145/3618384](https://doi.org/10.1145/3618384).

- Buchaillard, S., Brix, M., Perrier, P., and Payan, Y. (2007). “Simulations of the Consequences of Tongue Surgery on Tongue Mobility: Implications for Speech Production in Post-Surgery Conditions.” In: *Int. J. Med. Robot. Comput. Assist. Surg.* 3.3, pp. 252–261. ISSN: 1478-5951, 1478-596X. DOI: [10.1002/rcs.142](https://doi.org/10.1002/rcs.142).
- Buchaillard, S., Perrier, P., and Payan, Y. (2009). “A Biomechanical Model of Cardinal Vowel Production: Muscle Activations and the Impact of Gravity on Tongue Positioning.” In: *J. Acoust. Soc. Am.* 126.4, pp. 2033–2051. ISSN: 0001-4966, 1520-8524. DOI: [10.1121/1.3204306](https://doi.org/10.1121/1.3204306).
- Cai, Y.-C., Li, C., Zeng, D.-f., Zhou, Y. Q., Sun, R. H., Shui, C. Y., Pei, J., Liu, W., Wang, X., Jiang, Z. H., Tang, Z. Q., Jiang, J., and Wang, W. (2019). “Comparative Analysis of Radial Forearm Free Flap and Anterolateral Thigh Flap in Tongue Reconstruction after Radical Resection of Tongue Cancer.” In: *ORL-J. Oto-Rhino-Laryngol. Head Neck Surg.* 81.5-6, pp. 252–264. ISSN: 0301-1569, 1423-0275. DOI: [10.1159/000502151](https://doi.org/10.1159/000502151).
- Carrion, K., Akhter, M. F., Bernal, C., Cox, E., Krakauer, K., Mohan, A. T., Nazerali, R. S., Lee, G. K., and Nguyen, D. (2024). “A Systematic Review of 3D Printing in Autologous Breast Reconstruction.” In: *Plast Aesthet Res* 11, N–A. ISSN: 2349-6150. DOI: [10.20517/2347-9264.2024.56](https://doi.org/10.20517/2347-9264.2024.56).
- Chabanas, M., Luboz, V., and Payan, Y. (2003). “Patient Specific Finite Element Model of the Face Soft Tissues for Computer-Assisted Maxillofacial Surgery.” In: *Medical Image Analysis. Mathematical Methods in Biomedical Image Analysis - MMBIA 2001* 7.2, pp. 131–151. ISSN: 1361-8415. DOI: [10.1016/S1361-8415\(02\)00108-1](https://doi.org/10.1016/S1361-8415(02)00108-1).
- Chae, M. P., Hunter-Smith, D. J., Chung, R. D., Smith, J. A., and Rozen, W. M. (2021). “3D-printed, Patient-Specific DIEP Flap Templates for Preoperative Planning in Breast Reconstruction: A Prospective Case Series.” In: *Gland Surg* 10.7, pp. 2192–2199. ISSN: 2227684X, 22278575. DOI: [10.21037/gs-21-263](https://doi.org/10.21037/gs-21-263).
- Chagnon, G., Rebouah, M., and Favier, D. (2015). “Hyperelastic Energy Densities for Soft Biological Tissues: A Review.” In: *J Elast* 120.2, pp. 129–160. ISSN: 0374-3535, 1573-2681. DOI: [10.1007/s10659-014-9508-z](https://doi.org/10.1007/s10659-014-9508-z).

- Chang, A. M. V., Kim, S. W., Duvvuri, U., Johnson, J. T., Myers, E. N., Ferris, R. L., Gooding, W. E., Seethala, R. R., and Chiosea, S. I. (2013). “Early Squamous Cell Carcinoma of the Oral Tongue: Comparing Margins Obtained from the Glossectomy Specimen to Margins from the Tumor Bed.” In: *Oral Oncol.* 49.11, pp. 1077–1082. ISSN: 13688375. DOI: [10.1016/j.oraloncology.2013.07.013](https://doi.org/10.1016/j.oraloncology.2013.07.013).
- Chartier, C., Watt, A., Lin, O., Chandawarkar, A., Lee, J., and Hall-Findlay, E. (2022). “BreastGAN: Artificial Intelligence-Enabled Breast Augmentation Simulation.” In: *Aesthet Surg J Open Forum.* 4, ojab052. ISSN: 2631-4797. DOI: [10.1093/asjof/ojab052](https://doi.org/10.1093/asjof/ojab052).
- Chattrairat, A., Aimmanee, S., Kandare, E., Tran, P., and Das, R. (2023). “Modelling and Characterisation of the Dynamic Behaviours of Silicone-Based Composite Skin Simulant with Short Polyethylene Fibres and Bioactive Glass Particles.” In: *Mechanics of Materials* 184, p. 104740. ISSN: 01676636. DOI: [10.1016/j.mechmat.2023.104740](https://doi.org/10.1016/j.mechmat.2023.104740).
- Chaudhry, H. R., Bukiet, B., Siegel, M., Findley, T., Ritter, A. B., and Guzelsu, N. (1998). “Optimal Patterns for Suturing Wounds.” In: *J. Biomech.* 31.7, pp. 653–662. ISSN: 0021-9290. DOI: [10.1016/S0021-9290\(98\)00069-4](https://doi.org/10.1016/S0021-9290(98)00069-4).
- Cheng, S., Gandevia, S., Green, M., Sinkus, R., and Bilston, L. (2011). “Viscoelastic Properties of the Tongue and Soft Palate Using MR Elastography.” In: *Journal of Biomechanics* 44.3, pp. 450–454. ISSN: 00219290. DOI: [10.1016/j.jbiomech.2010.09.027](https://doi.org/10.1016/j.jbiomech.2010.09.027).
- Chepeha, D. B., Teknos, T. N., Shargorodsky, J., Sacco, A. G., Lyden, T., Prince, M. E., Bradford, C. R., and Wolf, G. T. (2008). “Rectangle Tongue Template for Reconstruction of the Hemiglossectomy Defect.” In: *Arch. Otolaryngol. Head Neck Surg.* 134.9, pp. 993–998. ISSN: 0886-4470. DOI: [10.1001/archotol.134.9.993](https://doi.org/10.1001/archotol.134.9.993).
- Choi, J.-W., Alshomer, F., and Kim, Y.-C. (2022). “Current Status and Evolution of Microsurgical Tongue Reconstructions, Part I.” In: *Arch Craniofac Surg* 23.4, pp. 139–151. ISSN: 2287-1152, 2287-5603. DOI: [10.7181/acfs.2022.00654](https://doi.org/10.7181/acfs.2022.00654).
- Chuka, R., Abdullah, W., Rieger, J., Nayar, S., Seikaly, H., Osswald, M., and Wolfaardt, J. (2017). “Implant Utilization and Time to Prosthetic Rehabilitation in Conventional and Advanced Fibular Free Flap Reconstruction of the Maxilla and Mandible.” In: *Int. J. Prosthodont.* 30.3, pp. 289–294. ISSN: 0893-2174, 1942-4426. DOI: [10.11607/ijp.5161](https://doi.org/10.11607/ijp.5161).

- Darmont Araya, F. and Radhakrishnan, P. (2021). “Investigating the Design and Manufacture of PneuNet Actuators as a Prosthetic Tongue for Mimicking Human Deglutition.” In: *J. Eng. Sci. Med. Diagn. Ther.* 4.4, p. 041003. ISSN: 2572-7958, 2572-7966. DOI: [10.1115/1.4051665](https://doi.org/10.1115/1.4051665).
- De Berardinis, R., Tagliabue, M., Belloni, P., Gandini, S., Scaglione, D., Maffini, F., Margherini, S., Riccio, S., Giugliano, G., Bruschini, R., Chu, F., and Ansarin, M. (2022). “Tongue Cancer Treatment and Oncological Outcomes: The Role of Glossectomy Classification.” In: *Surg. Oncol.-Oxf.* 42, p. 101751. ISSN: 0960-7404, 1879-3320. DOI: [10.1016/j.suronc.2022.101751](https://doi.org/10.1016/j.suronc.2022.101751).
- Duchscherer, J., Aalto, D., and Westover, L. (2021). “Evaluation of Facial Symmetry after Jaw Reconstruction Surgery.” In: *Comput. Methods Biomech. Biomed. Engin.* 24.11, pp. 1212–1220. ISSN: 1025-5842. DOI: [10.1080/10255842.2020.1870965](https://doi.org/10.1080/10255842.2020.1870965).
- Dzidek, B. M., Adams, M. J., Andrews, J. W., Zhang, Z., and Johnson, S. A. (2017). “Contact Mechanics of the Human Finger Pad under Compressive Loads.” In: *J. R. Soc. Interface.* 14.127, p. 20160935. ISSN: 1742-5689, 1742-5662. DOI: [10.1098/rsif.2016.0935](https://doi.org/10.1098/rsif.2016.0935).
- Dziegielewski, P. T., Rieger, J., Shama, M. A., O’Connell, D. A., Harris, J. R., and Seikaly, H. (2019). “Beavertail Modification of the Radial Forearm Free Flap in Total Oral Glossectomy Reconstruction: Technique and Functional Outcomes.” In: *Oral Oncol.* 96, pp. 71–76. ISSN: 1368-8375, 1879-0593. DOI: [10.1016/j.oraloncology.2019.07.004](https://doi.org/10.1016/j.oraloncology.2019.07.004).
- Dzioba, A., Aalto, D., Papadopoulos-Nydam, G., Seikaly, H., Rieger, J., Wolfaardt, J., Osswald, M., Harris, J. R., O’Connell, D. A., Lazarus, C., Urken, M., Likhterov, I., Chai, R. L., Rauscher, E., Buchbinder, D., Okay, D., Happonen, R.-P., Kinnunen, I., Irjala, H., Soukka, T., and Laine, J. (2017). “Functional and Quality of Life Outcomes after Partial Glossectomy: A Multi-Institutional Longitudinal Study of the Head and Neck Research Network.” In: *J. Otolaryngol. Head Neck Surg.* 46, p. 59. ISSN: 1916-0216. DOI: [10.1186/s40463-017-0236-9](https://doi.org/10.1186/s40463-017-0236-9).
- Efanov, J. I., Roy, A.-A., Huang, K. N., and Borsuk, D. E. (2018). “Virtual Surgical Planning: The Pearls and Pitfalls.” In: *Plast. Reconstr. Surg. Glob. Open* 6.1, e1443. ISSN: 2169-7574. DOI: [10.1097/GOX.0000000000001443](https://doi.org/10.1097/GOX.0000000000001443).

- Engel, H., Huang, J. J., Lin, C.-Y., Lam, W., Kao, H.-K., Gazyakan, E., and Cheng, M.-H. (2010). “A Strategic Approach for Tongue Reconstruction to Achieve Predictable and Improved Functional and Aesthetic Outcomes.” In: *Plast. Reconst. Surg.* 126.6, p. 1967. ISSN: 0032-1052. DOI: [10.1097/PRS.0b013e3181f44742](https://doi.org/10.1097/PRS.0b013e3181f44742).
- Eu, D., Daly, M. J., Taboni, S., Sahoaler, A., Gilbank, A. N., and Irish, J. C. (2024). “Evaluation of a 3D Printed Silicone Oral Cavity Cancer Model for Surgical Simulations.” In: *J. Pers. Med.* 14.5, p. 450. ISSN: 2075-4426. DOI: [10.3390/jpm14050450](https://doi.org/10.3390/jpm14050450).
- Fan, S., Li, Q.-X., Zhang, H.-Q., Liang, M.-J., Tian, T., Wang, Y.-Y., Lin, Z.-Y., Chen, W.-X., Pan, G.-K., Ahn, M. H.-Y., He, L., Sun, S., Wu, B.-H., Yu, P., and Li, J.-S. (2019a). ““Five-point Eight-Line” Anatomic Flap Design for Precise Hemitongue Reconstruction.” In: *Head Neck-J. Sci. Spec. Head Neck* 41.5, pp. 1359–1366. ISSN: 1043-3074, 1097-0347. DOI: [10.1002/hed.25571](https://doi.org/10.1002/hed.25571).
- Fan, S., Zhang, H., Li, Q., Liang, F., Bai, Z., Chen, W., Sun, S., Yu, P., and Li, J. (2019b). “A Novel Anatomy-Based Five-Points Eight-Line-Segments Technique for Precision Subtotal Tongue Reconstruction: A Pilot Study.” In: *Oral Oncol.* 89, pp. 1–7. ISSN: 13688375. DOI: [10.1016/j.oraloncology.2018.12.004](https://doi.org/10.1016/j.oraloncology.2018.12.004).
- Flynn, C., Taberner, A., and Nielsen, P. (2011). “Modeling the Mechanical Response of In Vivo Human Skin Under a Rich Set of Deformations.” In: *Ann. Biomed. Eng.* 39.7, pp. 1935–1946. ISSN: 0090-6964, 1573-9686. DOI: [10.1007/s10439-011-0292-7](https://doi.org/10.1007/s10439-011-0292-7).
- Gewirtz, J. I., Zhao, S., Brock, G., Luttrull, M. D., Sethuraman, S., Kang, S. Y., Vankoevering, K. K., and Seim, N. B. (2024). “A Pilot Study: Free Flap Atrophy in Tongue Reconstruction Using 3D Volumetric Analysis.” In: *Ann. Otol. Rhinol. Laryngol.* 133.3, pp. 253–260. ISSN: 0003-4894, 1943-572X. DOI: [10.1177/00034894231204720](https://doi.org/10.1177/00034894231204720).
- Gordon, A. R., Schreiber, J. E., Patel, A., and Tepper, O. M. (2021). “3D Printed Surgical Guides Applied in Rhinoplasty to Help Obtain Ideal Nasal Profile.” In: *Aesth Plast Surg* 45.6, pp. 2852–2859. ISSN: 0364-216X, 1432-5241. DOI: [10.1007/s00266-021-02350-z](https://doi.org/10.1007/s00266-021-02350-z).
- Griffin, M. F., Leung, B. C., Premakumar, Y., Szarko, M., and Butler, P. E. (2017). “Comparison of the Mechanical Properties of Different Skin Sites for Auricular and Nasal Reconstruction.” In: *Journal of Otolaryngology - Head & Neck Surgery* 46.1, p. 33. ISSN: 1916-0216, 1916-0216. DOI: [10.1186/s40463-017-0210-6](https://doi.org/10.1186/s40463-017-0210-6).

- Groves, R., Coulman, S., Birchall, J., and Evans, S. (2013). “An Anisotropic, Hyperelastic Model for Skin: Experimental Measurements, Finite Element Modelling and Identification of Parameters for Human and Murine Skin.” In: *J. Mech. Behav. Biomed. Mater.* 18, pp. 167–180. ISSN: 1878-0180. DOI: [10.1016/j.jmbbm.2012.10.021](https://doi.org/10.1016/j.jmbbm.2012.10.021).
- Han, S.-W., Hong, X.-W., Li, H., and Han, W. (2022). “Utility of a Palatal Speech Appliance Combined with the Five-Point Eight-Line Segment Technique in Total Tongue Reconstruction.” In: *Int. J. Oral Maxillofac. Surg.* 51.8, pp. 1016–1021. ISSN: 0901-5027, 1399-0020. DOI: [10.1016/j.ijom.2022.01.001](https://doi.org/10.1016/j.ijom.2022.01.001).
- Hanubal, K. S., Reschly, W. J., Conrad, D., Festa, B. M., Weiss, J. P., Shama, M., Danan, D., Hughley, B., and Dziegielewski, P. T. (2023). “The Beavertail Modified Radial Forearm Free Flap: Retrospective Review of a Versatile Technique to Increase Flap Bulk in the Head and Neck.” In: *Microsurgery* 43.8, pp. 767–774. ISSN: 0738-1085, 1098-2752. DOI: [10.1002/micr.31028](https://doi.org/10.1002/micr.31028).
- Harandi, N., Stavness, I., Woo, J., Stone, M., Abugharbieh, R., and Fels, S. (2017a). “Subject-Specific Biomechanical Modelling of the Oropharynx: Towards Speech Production.” In: *Comput. Methods Biomech. Biomed. Eng. Imaging Vis.* 5.6, pp. 416–426. ISSN: 2168-1163. DOI: [10.1080/21681163.2015.1033756](https://doi.org/10.1080/21681163.2015.1033756).
- Harandi, N., Woo, J., Stone, M., Abugharbieh, R., and Fels, S. (2017b). “Variability in Muscle Activation of Simple Speech Motions: A Biomechanical Modeling Approach.” In: *The Journal of the Acoustical Society of America* 141.4, pp. 2579–2590. ISSN: 0001-4966. DOI: [10.1121/1.4978420](https://doi.org/10.1121/1.4978420).
- Hashida, N., Shamoto, H., Maeda, K., Wakabayashi, H., Suzuki, M., and Fujii, T. (2017). “Rehabilitation and Nutritional Support for Sarcopenic Dysphagia and Tongue Atrophy after Glossectomy: A Case Report.” In: *Nutrition* 35, pp. 128–131. ISSN: 08999007. DOI: [10.1016/j.nut.2016.11.003](https://doi.org/10.1016/j.nut.2016.11.003).
- Heredero, S., Falguera, M. I., Gómez, V., and Sanjuan-Sanjuan, A. (2024). “Customized Soft Tissue Free Flaps in Head and Neck Reconstruction.” In: *Oral Maxillofac Surg Clin North Am.* ISSN: 1042-3699. DOI: [10.1016/j.coms.2024.07.009](https://doi.org/10.1016/j.coms.2024.07.009).
- Hermant, N., Perrier, P., and Payan, Y. (2017). “Human Tongue Biomechanical Modeling.” In: *Biomechanics of Living Organs*. Ed. by Y. Payan and J. Ohayon. Vol. 1. Translational Epigenetics. Oxford: Academic Press, pp. 395–411. DOI: [10.1016/B978-0-12-804009-6.00019-5](https://doi.org/10.1016/B978-0-12-804009-6.00019-5).

- Holzapfel, G. A. (2002). *Nonlinear Solid Mechanics: A Continuum Approach for Engineering Science*. Kluwer Academic Publishers Dordrecht. ISBN: 0025-6455.
- Hu, Y., Schneider, T., Wang, B., Zorin, D., and Panozzo, D. (2020). “Fast Tetrahedral Meshing in the Wild.” In: *ACM Trans. Graph.* 39.4. ISSN: 0730-0301, 1557-7368. DOI: [10.1145/3386569.3392385](https://doi.org/10.1145/3386569.3392385).
- Huang, P., Gu, L., Xu, H., Dong, J., Liu, J., Zhang, J., Song, J., Yu, X., Chen, W., Zhou, H., Li, B., and Pei, W. (2007). “Image-Guided Surgery Planning for Breast Reconstruction Flap Design.” In: *IEEE Nucl. Sci. Symp. Conf. Rec.* New York: IEEE. ISBN: 978-1-4244-0922-8. DOI: [10.1109/NSSMIC.2007.4436997](https://doi.org/10.1109/NSSMIC.2007.4436997).
- Idris, S., Logan, H., Tabet, P., Osswald, M., Nayar, S., and Seikaly, H. (2022). “The Accuracy of 3D Surgical Design and Simulation in Prefabricated Fibula Free Flaps for Jaw Reconstruction.” In: *J. Pers. Med.* 12.11, p. 1766. ISSN: 2075-4426. DOI: [10.3390/jpm12111766](https://doi.org/10.3390/jpm12111766).
- Idris, S., Mlynarek, A. M., Ansari, K., Harris, J. R., Rizk, N., Cote, D., O’Connell, D. A., Allen, H., Dziegielewska, P., and Seikaly, H. (2018). “Multi-Dimensional Analysis of Oral Cavity and Oropharyngeal Defects Following Cancer Extirpation Surgery, a Cadaveric Study.” In: *J. Otolaryngol. Head Neck Surg.* 47, p. 27. ISSN: 1916-0216. DOI: [10.1186/s40463-018-0276-9](https://doi.org/10.1186/s40463-018-0276-9).
- Iivarinen, J. T., Korhonen, R. K., Julkunen, P., and Jurvelin, J. S. (2011). “Experimental and Computational Analysis of Soft Tissue Stiffness in Forearm Using a Manual Indentation Device.” In: *Med. Eng. Phys.* 33.10, pp. 1245–1253. ISSN: 13504533. DOI: [10.1016/j.medengphy.2011.05.015](https://doi.org/10.1016/j.medengphy.2011.05.015).
- Isazadeh, A. R. (2025). *VSP-BioSim: Patient-Specific Virtual Surgical Planning via Biomechanical Simulations*. DOI: [10.5281/zenodo.17438091](https://doi.org/10.5281/zenodo.17438091).
- Isazadeh, A. R., Seikaly, H., Westover, L., and Aalto, D. (2024). “Algorithmically Designed Flaps in Tongue Reconstruction: A Feasibility Analysis.” In: *Int. J. Comput. Assist. Radiol. Surg.* 19, pp. 735–746. ISSN: 1861-6410, 1861-6429. DOI: [10.1007/s11548-024-03062-w](https://doi.org/10.1007/s11548-024-03062-w).
- Isazadeh, A. R., Shamloofard, M., and Assempour, A. (2021). “Some Improvements on the One-Step Inverse Isogeometric Analysis by Proposing a Multi-Step Inverse Isogeometric Methodology in Sheet Metal Stamping Processes.” In: *Applied Mathematical Modelling* 91, pp. 476–492. ISSN: 0307904X. DOI: [10.1016/j.apm.2020.09.032](https://doi.org/10.1016/j.apm.2020.09.032).

- Isazadeh, A. R., Zenke, J., Westover, L., Seikaly, H., and Aalto, D. (2025a). *hiFEM: Hyperelastic Inverse Finite Element Method*. DOI: [10.5281/zenodo.17427896](https://doi.org/10.5281/zenodo.17427896).
- (2025b). “Patient-Specific Virtual Surgical Planning for Tongue Reconstruction: Evaluating Hyperelastic Inverse FEM with Four Simulated Tongue Cancer Cases.” In: *Phys. Med. Biol.* 70, p. 145017. ISSN: 0031-9155, 1361-6560. DOI: [10.1088/1361-6560/adebd8](https://doi.org/10.1088/1361-6560/adebd8).
- Ishihara, S., Nakao, S., Nakauma, M., Funami, T., Hori, K., Ono, T., Kohyama, K., and Nishinari, K. (2013). “Compression Test of Food Gels on Artificial Tongue and Its Comparison with Human Test.” In: *J. Texture Stud.* 44.2, pp. 104–114. ISSN: 0022-4901, 1745-4603. DOI: [10.1111/jtxs.12002](https://doi.org/10.1111/jtxs.12002).
- Ishikawa, Y., Nabae, H., Gunji, M., Endo, G., and Suzumori, K. (2025). “Pig Tongue Soft Robot Mimicking Intrinsic Tongue Muscle Structure.” In: *Front. Robot. AI* 11, p. 1511422. ISSN: 2296-9144. DOI: [10.3389/frobt.2024.1511422](https://doi.org/10.3389/frobt.2024.1511422).
- Jacquet, E., Chambert, J., Pauchot, J., and Sandoz, P. (2017). “Intra- and Inter-individual Variability in the Mechanical Properties of the Human Skin from in Vivo Measurements on 20 Volunteers.” In: *Skin Research and Technology* 23.4, pp. 491–499. ISSN: 0909-752X, 1600-0846. DOI: [10.1111/srt.12361](https://doi.org/10.1111/srt.12361).
- Jeong, W. S. and Oh, T. S. (2016). “Oral and Oropharyngeal Reconstruction with a Free Flap.” In: *Arch Craniofac Surg* 17.2, p. 45. ISSN: 2287-1152, 2287-5603. DOI: [10.7181/acfs.2016.17.2.45](https://doi.org/10.7181/acfs.2016.17.2.45).
- Ji, X., Wen, G., Gong, H., Sun, R., and Li, H. (2024). “Three-Dimensional Wound Flattening Method for Mapping Skin Mechanical Properties Based on Finite Element Method.” In: *Comput. Methods Biomech. Biomed. Eng.* 27.2, pp. 237–250. ISSN: 1025-5842, 1476-8259. DOI: [10.1080/10255842.2023.2183347](https://doi.org/10.1080/10255842.2023.2183347).
- Kappert, K. D. R., van Alphen, M. J. A., van Dijk, S., Smeele, L. E., Balm, A. J. M., and van der Heijden, F. (2019). “An Interactive Surgical Simulation Tool to Assess the Consequences of a Partial Glossectomy on a Biomechanical Model of the Tongue.” In: *Comput. Methods Biomech. Biomed. Eng.* 22.8, pp. 827–839. ISSN: 1025-5842, 1476-8259. DOI: [10.1080/10255842.2019.1599362](https://doi.org/10.1080/10255842.2019.1599362).

- Kappert, K. D. R., Voskuilen, L., Smeele, L. E., Balm, A. J. M., Jasperse, B., Nederveen, A. J., and van der Heijden, F. (2021a). “Personalized Biomechanical Tongue Models Based on Diffusion-Weighted MRI and Validated Using Optical Tracking of Range of Motion.” In: *Biomech. Model. Mechanobiol.* 20.3, pp. 1101–1113. ISSN: 1617-7959, 1617-7940. DOI: [10.1007/s10237-021-01435-7](https://doi.org/10.1007/s10237-021-01435-7).
- Kappert, K., Connesson, N., Elahi, S., Boonstra, S., Balm, A., Van Der Heijden, F., and Payan, Y. (2021b). “In-Vivo Tongue Stiffness Measured by Aspiration: Resting vs General Anesthesia.” In: *J. Biomech.* 114, p. 110147. ISSN: 00219290. DOI: [10.1016/j.jbiomech.2020.110147](https://doi.org/10.1016/j.jbiomech.2020.110147).
- Kim, M.-S., Oh, K. H., Cho, J.-G., Baek, S.-K., Woo, J.-S., Jung, K.-Y., and Kwon, S. Y. (2020). “Assessment of Chronological Volume Changes in Radial Forearm Free Flaps for Tongue Cancer.” In: *ORL* 82.1, pp. 40–46. ISSN: 0301-1569, 1423-0275. DOI: [10.1159/000504196](https://doi.org/10.1159/000504196).
- Koumoullis, H., Burley, O., and Kyzas, P. (2020). “Patient-Specific Soft Tissue Reconstruction: An IDEAL Stage I Report of Hemiglossectomy Reconstruction and Introduction of the PANSOFOS Flap.” In: *Br. J. Oral Maxillofac. Surg.* 58.6, pp. 681–686. ISSN: 0266-4356. DOI: [10.1016/j.bjoms.2020.04.017](https://doi.org/10.1016/j.bjoms.2020.04.017).
- Krauel, L., Valls-Esteve, A., Tejo-Otero, A., and Fenollosa-Artés, F. (2021). “3D-Printing in Surgery: Beyond Bone Structures. A Review.” In: *Ann. 3D Print. Med.* 4, p. 100039. ISSN: 26669641. DOI: [10.1016/j.stlm.2021.100039](https://doi.org/10.1016/j.stlm.2021.100039).
- Kreiser, J., Meuschke, M., Mistelbauer, G., Preim, B., and Ropinski, T. (2018). “A Survey of Flattening-Based Medical Visualization Techniques.” In: *Computer Graphics Forum* 37.3, pp. 597–624. ISSN: 0167-7055, 1467-8659. DOI: [10.1111/cgf.13445](https://doi.org/10.1111/cgf.13445).
- Kurlander, D. E., Garvey, P. B., Largo, R. D., Yu, P., Chang, E. I., Hanasono, M. M., and Mericli, A. F. (2023). “The Cost Utility of Virtual Surgical Planning and Computer-Assisted Design/Computer-Assisted Manufacturing in Mandible Reconstruction Using the Free Fibula Osteocutaneous Flap.” In: *J Reconstr Microsurg* 39.03, pp. 221–230. ISSN: 0743-684X, 1098-8947. DOI: [10.1055/s-0042-1755260](https://doi.org/10.1055/s-0042-1755260).
- Lam, L. and Samman, N. (2013). “Speech and Swallowing Following Tongue Cancer Surgery and Free Flap Reconstruction - A Systematic Review.” In: *Oral Oncol.* 49.6, pp. 507–524. ISSN: 1368-8375, 1879-0593. DOI: [10.1016/j.oraloncology.2013.03.001](https://doi.org/10.1016/j.oraloncology.2013.03.001).

- Laurent, A., Mistretta, F., Bottiglioli, D., Dahel, K., Goujon, C., Nicolas, J. F., Hennino, A., and Laurent, P. E. (2007). “Echographic Measurement of Skin Thickness in Adults by High Frequency Ultrasound to Assess the Appropriate Microneedle Length for Intradermal Delivery of Vaccines.” In: *Vaccine* 25.34, pp. 6423–6430. ISSN: 0264-410X. DOI: [10.1016/j.vaccine.2007.05.046](https://doi.org/10.1016/j.vaccine.2007.05.046).
- Lee, T., Gosain, A. K., Billionis, I., and Tepole, A. B. (2019). “Predicting the Effect of Aging and Defect Size on the Stress Profiles of Skin from Advancement, Rotation and Transposition Flap Surgeries.” In: *Journal of the Mechanics and Physics of Solids* 125, pp. 572–590. ISSN: 00225096. DOI: [10.1016/j.jmps.2019.01.012](https://doi.org/10.1016/j.jmps.2019.01.012).
- Leymarie, N., Karsenti, G., Sarfati, B., Rimareix, F., and Kolb, F. (2012). “Modification of Flap Design for Total Mobile Tongue Reconstruction Using a Sensitive Antero-Lateral Thigh Flap.” In: *J. Plast. Reconstr. Aesthet. Surg.* 65.7, e169–e174. ISSN: 1748-6815. DOI: [10.1016/j.bjps.2011.11.012](https://doi.org/10.1016/j.bjps.2011.11.012).
- Li, M., Ferguson, Z., Schneider, T., Langlois, T., Zorin, D., Panozzo, D., Jiang, C., and Kaufman, D. M. (2020). “Incremental Potential Contact: Intersection-and Inversion-Free, Large-Deformation Dynamics.” In: *ACM Trans. Graph.* 39.4. ISSN: 0730-0301, 1557-7368. DOI: [10.1145/3386569.3392425](https://doi.org/10.1145/3386569.3392425).
- Liu, M.-D., Liu, D.-Z., Al-Arooomi, M. A., Xiong, J.-J., Liu, S., Sun, C.-F., and Liu, F.-Y. (2023a). “Reconstruction of Large Defects of Anterior Floor of Mouth with Free Flaps Using a Novel Individualized Flap Design Method.” In: *J. Cranio-MaxilloFac. Surg.* 51.9, pp. 574–579. ISSN: 1010-5182, 1878-4119. DOI: [10.1016/j.jcms.2023.08.006](https://doi.org/10.1016/j.jcms.2023.08.006).
- Liu, M.-D., Xue, X.-M., Al-Arooomi, M. A., Xiong, J.-J., Liu, S., Sun, C.-F., and Liu, F.-Y. (2023b). “A Novel Flap Design Technique for Subtotal Tongue Reconstruction with an “Individualized and Convenient Tongue Model”.” In: *Oral Oncol.* 145, p. 106531. ISSN: 1368-8375. DOI: [10.1016/j.oraloncology.2023.106531](https://doi.org/10.1016/j.oraloncology.2023.106531).
- Liu, S., Zhang, S., Su, Y.-X., Zhou, X., Gong, Z. J., and Wu, H. J. (2023c). “Optimization of Total Tongue Functional Reconstruction with the Sushi Roll Technique and Its Application in Pectoralis Major Myocutaneous Flaps.” In: *Int. J. Oral Maxillofac. Surg.* 52.8, pp. 839–846. ISSN: 0901-5027, 1399-0020. DOI: [10.1016/j.ijom.2023.01.002](https://doi.org/10.1016/j.ijom.2023.01.002).

- Liu, Y.-J., Zhang, D.-L., and Yuen, M. M.-F. (2010). “A Survey on CAD Methods in 3D Garment Design.” In: *Computers in Industry* 61.6, pp. 576–593. ISSN: 01663615. DOI: [10.1016/j.compind.2010.03.007](https://doi.org/10.1016/j.compind.2010.03.007).
- Lloyd, J. E., Stavness, I., and Fels, S. (2012). “ArtiSynth: A Fast Interactive Biomechanical Modeling Toolkit Combining Multibody and Finite Element Simulation.” In: *Soft Tissue Biomechanical Modeling for Computer Assisted Surgery*. Ed. by Y. Payan. Berlin, Heidelberg: Springer, pp. 355–394. ISBN: 978-3-642-29014-5. DOI: [10.1007/8415_2012_126](https://doi.org/10.1007/8415_2012_126).
- Lloyd, R. A., Dissanayake, E., Jugé, L., Bolsterlee, B., Ball, I. K., and Bilston, L. E. (2025). “How Mandibular and Hyoid Morphology Alters Tongue Muscle Architecture in Healthy Adults: An Anatomical Atlas and Statistical Shape Model of the Tongue.” In: *Comput. Biol. Med.* 189, p. 110006. ISSN: 00104825. DOI: [10.1016/j.compbiomed.2025.110006](https://doi.org/10.1016/j.compbiomed.2025.110006).
- Logan, H., Wolfaardt, J., Boulanger, P., Hodgetts, B., and Seikaly, H. (2013). “Exploratory Benchtop Study Evaluating the Use of Surgical Design and Simulation in Fibula Free Flap Mandibular Reconstruction.” In: *J. Otolaryngol. Head Neck Surg.* 42.1, p. 42. ISSN: 1916-0216. DOI: [10.1186/1916-0216-42-42](https://doi.org/10.1186/1916-0216-42-42).
- Lösel, P. D., Van De Kamp, T., Jayme, A., Ershov, A., Faragó, T., Pichler, O., Tan Jerome, N., Aadepe, N., Bremer, S., Chilingaryan, S. A., Heethoff, M., Kopmann, A., Odar, J., Schmelzle, S., Zuber, M., Wittbrodt, J., Baumbach, T., and Heuveline, V. (2020). “Introducing Biomedisa as an Open-Source Online Platform for Biomedical Image Segmentation.” In: *Nat Commun* 11.1, p. 5577. ISSN: 2041-1723. DOI: [10.1038/s41467-020-19303-w](https://doi.org/10.1038/s41467-020-19303-w).
- Lott-Crumpler, D. A. and Chaudhry, H. R. (2001). “Optimal Patterns for Suturing Wounds of Complex Shapes to Foster Healing.” In: *J. Biomech.* 34.1, pp. 51–58. ISSN: 0021-9290. DOI: [10.1016/S0021-9290\(00\)00160-3](https://doi.org/10.1016/S0021-9290(00)00160-3).
- Maas, S. A., Ellis, B. J., Ateshian, G. A., and Weiss, J. A. (2012). “FEBio: Finite Elements for Biomechanics.” In: *J. Biomech. Eng.* 134.1, p. 011005. ISSN: 0148-0731, 1528-8951. DOI: [10.1115/1.4005694](https://doi.org/10.1115/1.4005694).
- Maas, S. A., Ellis, B. J., Rawlins, D. S., and Weiss, J. A. (2016). “Finite Element Simulation of Articular Contact Mechanics with Quadratic Tetrahedral Elements.” In: *J. Biomech.* 49.5, pp. 659–667. ISSN: 00219290. DOI: [10.1016/j.jbiomech.2016.01.024](https://doi.org/10.1016/j.jbiomech.2016.01.024).

- Magnenat Thalmann, N., Choi, H. F., and Thalmann, D. (2014). “Towards Effective Diagnosis and Prediction via 3D Patient Model: A Complete Research Plan.” In: *3D Multiscale Physiological Human*. Ed. by N. Magnenat-Thalmann, O. Ratib, and H. F. Choi. London: Springer, pp. 3–22. ISBN: 978-1-4471-6275-9. DOI: [10.1007/978-1-4471-6275-9_1](https://doi.org/10.1007/978-1-4471-6275-9_1).
- Mallet, Y., El Bedoui, S., Penel, N., Ton Van, J., Fournier, C., and Lefebvre, J. L. (2009). “The Free Vascularized Flap and the Pectoralis Major Pedicled Flap Options: Comparative Results of Reconstruction of the Tongue.” In: *Oral Oncol.* 45.12, pp. 1028–1031. ISSN: 1368-8375. DOI: [10.1016/j.oraloncology.2009.05.639](https://doi.org/10.1016/j.oraloncology.2009.05.639).
- Mannelli, G., Arcuri, F., Agostini, T., Innocenti, M., Raffaini, M., and Spinelli, G. (2018). “Classification of Tongue Cancer Resection and Treatment Algorithm.” In: *J. Surg. Oncol.* 117.5, pp. 1092–1099. ISSN: 0022-4790, 1096-9098. DOI: [10.1002/jso.24991](https://doi.org/10.1002/jso.24991).
- Marra, C., Pinto, V., Benanti, E., De Maria, F., Pinelli, M., Spaggiari, A., and De Santis, G. (2023). “Radial Forearm Flap versus Antero-Lateral Thigh Flap in Oral Soft Tissue Reconstruction: Update and Statistical Analysis on Our 77 Consecutive Cases for an Objective Selection Criteria.” In: *Acta Biomed* 94.5, e2023252. ISSN: 2531-6745. DOI: [10.23750/abm.v94i5.15174](https://doi.org/10.23750/abm.v94i5.15174).
- Martin, L., Jain, P., Ferguson, Z., Gholamalizadeh, T., Moshfeghifar, F., Erleben, K., Panozzo, D., Abramowitch, S., and Schneider, T. (2024). “A Systematic Comparison between FEBio and PolyFEM for Biomechanical Systems.” In: *Computer Methods and Programs in Biomedicine* 244, p. 107938. ISSN: 01692607. DOI: [10.1016/j.cmpb.2023.107938](https://doi.org/10.1016/j.cmpb.2023.107938).
- Maxwell, J. H., Thompson, L. D. R., Brandwein-Gensler, M. S., Weiss, B. G., Canis, M., Purgina, B., Prabhu, A. V., Lai, C., Shuai, Y., Carroll, W. R., Morlandt, A., Duvvuri, U., Kim, S., Johnson, J. T., Ferris, R. L., Seethala, R., and Chiosea, S. I. (2015). “Early Oral Tongue Squamous Cell Carcinoma: Sampling of Margins From Tumor Bed and Worse Local Control.” In: *JAMA Otolaryngol Head Neck Surg* 141.12, p. 1104. ISSN: 2168-6181. DOI: [10.1001/jamaoto.2015.1351](https://doi.org/10.1001/jamaoto.2015.1351).
- McHutchion, L. and Aalto, D. (2021). “Simulation of Tissue-Prosthesis Margin Interface by Using Surface Scanning and Digital Design for Auricular Prostheses.” In: *J. Prosthet. Dent.* 125.2, pp. 361–372. ISSN: 00223913. DOI: [10.1016/j.prosdent.2020.01.045](https://doi.org/10.1016/j.prosdent.2020.01.045).

- Mitchell, D., Kanatas, A., Murphy, C., Chengot, P., Smith, A., and Ong, T. (2018). “Margins and Survival in Oral Cancer.” In: *British Journal of Oral and Maxillofacial Surgery* 56.9, pp. 820–829. ISSN: 02664356. DOI: [10.1016/j.bjoms.2018.06.021](https://doi.org/10.1016/j.bjoms.2018.06.021).
- Miyota, S., Kobayashi, T., Abé, T., Miyajima, H., Nagata, M., Hoshina, H., Kobayashi, T., Takagi, R., and Saku, T. (2014). “Intraoperative Assessment of Surgical Margins of Oral Squamous Cell Carcinoma Using Frozen Sections: A Practical Clinicopathological Management for Recurrences.” In: *BioMed Research International* 2014, pp. 1–9. ISSN: 2314-6133, 2314-6141. DOI: [10.1155/2014/823968](https://doi.org/10.1155/2014/823968).
- Mostafavi Yazdi, S. and Baqersad, J. (2022). “Mechanical Modeling and Characterization of Human Skin: A Review.” In: *J. Biomech.* 130. ISSN: 0021-9290. DOI: [10.1016/j.jbiomech.2021.110864](https://doi.org/10.1016/j.jbiomech.2021.110864).
- Nayar, S., Logan, H., Wolfaardt, J., Osswald, M., O’Connell, D., Grosvenor, A., and Seikaly, H. (2024). “Alberta Reconstructive Technique (ART): An Innovative Approach Using Digital Surgical Design and Simulation in Advanced Jaw Reconstruction with Occlusion-Based Prefabricated Vascularized Fibular Flaps and Primary Osseointegrated Implant Installation.” In: *Head Neck-J. Sci. Spec. Head Neck* 46.8, pp. 2098–2101. ISSN: 1043-3074, 1097-0347. DOI: [10.1002/hed.27792](https://doi.org/10.1002/hed.27792).
- Ní Annaidh, A., Bruyère, K., Destrade, M., Gilchrist, M. D., and Otténio, M. (2012). “Characterization of the Anisotropic Mechanical Properties of Excised Human Skin.” In: *J. Mech. Behav. Biomed. Mater.* 5.1, pp. 139–148. ISSN: 17516161. DOI: [10.1016/j.jmbbm.2011.08.016](https://doi.org/10.1016/j.jmbbm.2011.08.016).
- Ogawa, R., Okai, K., Tokumura, F., Mori, K., Ohmori, Y., Huang, C., Hyakusoku, H., and Akaishi, S. (2012). “The Relationship between Skin Stretching/Contraction and Pathologic Scarring: The Important Role of Mechanical Forces in Keloid Generation.” In: *Wound Repair Regen.* 20.2, pp. 149–157. ISSN: 1067-1927, 1524-475X. DOI: [10.1111/j.1524-475X.2012.00766.x](https://doi.org/10.1111/j.1524-475X.2012.00766.x).
- Padilla, P. L., Mericli, A. F., Largo, R. D., and Garvey, P. B. (2021). “Computer-Aided Design and Manufacturing versus Conventional Surgical Planning for Head and Neck Reconstruction: A Systematic Review and Meta-Analysis.” In: *Plast. Reconst. Surg.* 148.1, pp. 183–192. ISSN: 0032-1052. DOI: [10.1097/PRS.00000000000008085](https://doi.org/10.1097/PRS.00000000000008085).

- Papadopoulos-Nydam, G., Wolfaardt, J., Seikaly, H., O'Connell, D., Harris, J., Osswald, M., Nayar, S., and Rieger, J. (2017). "Comparison of Speech and Resonance Outcomes across Three Methods of Treatment for Maxillary Defects." In: *Int. J. Maxillofac. Prosthet.* 1.1, pp. 2–8. ISSN: 2432-3993. DOI: [10.26629/ijmp.2017.02](https://doi.org/10.26629/ijmp.2017.02).
- Podolsky, D. J., Lupu, A., Wong Riff, K. W., and Fisher, D. M. (2025). "Surgical Anatomy of Cleft Palate: A Three-Dimensional Computer Model." In: *Plast. Reconstr. Surg.* ISSN: 0032-1052, 1529-4242. DOI: [10.1097/PRS.00000000000012383](https://doi.org/10.1097/PRS.00000000000012383).
- Podolsky, D. J. (2017). "Development and Evaluation of a High Fidelity Cleft Palate Simulator for Surgical Training and for Development of a Robotic Approach to Infant Cleft Palate Surgery." PhD thesis. Canada: University of Toronto (Canada). ISBN: 978-0-355-53509-9.
- Podolsky, D. J. and Borschel, G. H. (2019). "Preoperative Modeling for Mirror Hand: Simplifying a Difficult Problem Using 3-Dimensional Printing and Simulation." In: *Plast. reconstr. surg., Glob. open.* 7.1, e1929. ISSN: 2169-7574. DOI: [10.1097/GOX.0000000000001929](https://doi.org/10.1097/GOX.0000000000001929).
- Podolsky, D. J., Fisher, D. M., Wong, K. W., Looi, T., Drake, J. M., and Forrest, C. R. (2017). "Evaluation and Implementation of a High-Fidelity Cleft Palate Simulator." In: *Plast. Reconstr. Surg.* 139.1, 85e–96e. ISSN: 0032-1052. DOI: [10.1097/PRS.0000000000002923](https://doi.org/10.1097/PRS.0000000000002923).
- Podolsky, D. J., Fisher, D. M., Wong Riff, K. W., Looi, T., Drake, J. M., and Forrest, C. R. (2021). "Robotic Cleft Palate Surgery and Simulation." In: *Robotics in Plastic and Reconstructive Surgery*. Ed. by J. C. Selber. Cham: Springer International Publishing, pp. 71–88. ISBN: 978-3-030-74243-0 978-3-030-74244-7. DOI: [10.1007/978-3-030-74244-7_7](https://doi.org/10.1007/978-3-030-74244-7_7).
- Podolsky, D. J., Murphy, B. D., Ahmad, J., Fisher, D. M., Wong Riff, K. W. Y., Drake, J. M., and Forrest, C. R. (2024a). "Development and Evaluation of a High-Fidelity Rhinoplasty Simulator." In: *Plast. Reconstr. Surg.* 153.3, pp. 619–622. ISSN: 0032-1052. DOI: [10.1097/PRS.00000000000010636](https://doi.org/10.1097/PRS.00000000000010636).
- Podolsky, D. J., Wong, C., and Hopper, R. A. (2024b). "Simulation in Craniosynostosis Surgery." In: *Plast. Reconstr. Surg.* 153.1, 220e–222e. ISSN: 0032-1052. DOI: [10.1097/PRS.00000000000010879](https://doi.org/10.1097/PRS.00000000000010879).
- Pucci, R., Weyh, A., Smotherman, C., Valentini, V., Bunnell, A., and Fernandes, R. (2020). "Accuracy of Virtual Planned Surgery versus Conventional Free-Hand Surgery for Reconstruction of the Mandible with Osteocutaneous Free Flaps." In: *International Journal of Oral and Maxillofacial Surgery* 49.9, pp. 1153–1161. ISSN: 0901-5027. DOI: [10.1016/j.ijom.2020.02.018](https://doi.org/10.1016/j.ijom.2020.02.018).

- Razavi, C. R., Hostetter, J., Shukla, A., Cheng, Z., Aygun, N., Boahene, K., Byrne, P. J., Richmon, J., Quon, H., and Desai, S. C. (2022). “Predictors of Free Flap Volume Loss in Nonosseous Reconstruction of Head and Neck Oncologic Defects.” In: *ENT-Ear Nose Throat J.* 101.1, pp. 48–53. ISSN: 0145-5613, 1942-7522. DOI: [10.1177/0145561320938903](https://doi.org/10.1177/0145561320938903).
- Reynolds, T. (1999). “Why Randomized Surgical Oncology Trials Are So Scarce.” In: *JNCI Journal of the National Cancer Institute* 91.14, pp. 1182–1183. ISSN: 0027-8874, 1460-2105. DOI: [10.1093/jnci/91.14.1182](https://doi.org/10.1093/jnci/91.14.1182).
- Robertson, E., Boulanger, P., Kwan, P., Louie, G., and Aalto, D. (2023). “Improving Cranial Vault Remodeling for Unilateral Coronal Craniosynostosis—Introducing Automated Surgical Planning.” In: *Craniomaxillofacial Trauma & Reconstruction* 17, pp. 203–213. ISSN: 1943-3875, 1943-3883. DOI: [10.1177/19433875231178912](https://doi.org/10.1177/19433875231178912).
- Rodby, K. A., Turin, S., Jacobs, R. J., Cruz, J. F., Hassid, V. J., Kolokythas, A., and Antony, A. K. (2014). “Advances in Oncologic Head and Neck Reconstruction: Systematic Review and Future Considerations of Virtual Surgical Planning and Computer Aided Design/Computer Aided Modeling.” In: *J Plast Reconstr Aesthet Surg* 67.9, pp. 1171–1185. ISSN: 17486815. DOI: [10.1016/j.bjps.2014.04.038](https://doi.org/10.1016/j.bjps.2014.04.038).
- Rui, X., Huang, Z., Zuo, J., Wang, Y., Liang, Q., Jin, T., Wang, J., Chang, S., and Huang, Z. (2022). “Application of an L-shaped Anterolateral Thigh Flap in Reconstruction after Hemiglossectomy.” In: *BMC Surg.* 22.1, p. 32. ISSN: 1471-2482. DOI: [10.1186/s12893-022-01473-7](https://doi.org/10.1186/s12893-022-01473-7).
- Sakakibara, A., Kusumoto, J., Sakakibara, S., Hasegawa, T., Akashi, M., Minamikawa, T., Furudoi, S., Hashikawa, K., and Komori, T. (2019). “Long-Term Effects on Volume Change in Musculocutaneous Flaps after Head and Neck Reconstruction.” In: *J. Reconstr. Microsurg.* 35.4, pp. 235–243. ISSN: 0743-684X, 1098-8947. DOI: [10.1055/s-0038-1672134](https://doi.org/10.1055/s-0038-1672134).
- Salinero, L., Boczar, D., Barrow, B., Berman, Z. P., Diep, G. K., Trilles, J., Howard, R., Chaya, B. F., Rodriguez Colon, R., and Rodriguez, E. D. (2022). “Patient-Centred Outcomes and Dental Implant Placement in Computer-Aided Free Flap Mandibular Reconstruction: A Systematic Review and Meta-Analysis.” In: *British Journal of Oral and Maxillofacial Surgery* 60.10, pp. 1283–1291. ISSN: 0266-4356. DOI: [10.1016/j.bjoms.2022.09.006](https://doi.org/10.1016/j.bjoms.2022.09.006).
- Sanders, I. and Mu, L. (2013). “A Three-Dimensional Atlas of Human Tongue Muscles.” In: *Anat Rec (Hoboken)* 296.7, pp. 1102–1114. ISSN: 1932-8494. DOI: [10.1002/ar.22711](https://doi.org/10.1002/ar.22711).

- Sawhney, R. and Crane, K. (2018). “Boundary First Flattening.” In: *ACM Trans. Graph.* 37.1, pp. 1–14. ISSN: 0730-0301, 1557-7368. DOI: [10.1145/3132705](https://doi.org/10.1145/3132705).
- Schneider, T., Hu, Y., Gao, X., Dumas, J., Zorin, D., and Panozzo, D. (2022). “A Large-Scale Comparison of Tetrahedral and Hexahedral Elements for Solving Elliptic PDEs with the Finite Element Method.” In: *ACM Trans. Graph.* 41.3, pp. 1–14. ISSN: 0730-0301, 1557-7368. DOI: [10.1145/3508372](https://doi.org/10.1145/3508372).
- Seabold, S. and Perktold, J. (2010). “Statsmodels: Econometric and Statistical Modeling with Python.” In: *9th Python Sci. Conf.*
- Seikaly, H., Rieger, J., O’Connell, D., Ansari, K., AlQahtani, K., and Harris, J. (2009). “Beavertail Modification of the Radial Forearm Free Flap in Base of Tongue Reconstruction: Technique and Functional Outcomes.” In: *Head Neck* 31.2, pp. 213–219. ISSN: 1043-3074, 1097-0347. DOI: [10.1002/hed.20953](https://doi.org/10.1002/hed.20953).
- Seikaly, H., Rieger, J., Wolfaardt, J., Moysa, G., Harris, J., and Jha, N. (2003). “Functional Outcomes After Primary Oropharyngeal Cancer Resection and Reconstruction With the Radial Forearm Free Flap.” In: *Laryngoscope* 113.5, pp. 897–904. ISSN: 0023-852X, 1531-4995. DOI: [10.1097/00005537-200305000-00023](https://doi.org/10.1097/00005537-200305000-00023).
- Sepúlveda, L., Peralta J., G., Ampuero E., V., and Fernández A., F. (2025). “Comparison of Functional Outcomes of Radial Forearm and Anterolateral Thigh Free Flaps after Hemiglossectomy in Patients with Tongue Cancer: A Systematic Review and Meta-Analysis.” In: *Rev. Otorrinolaringol. Cir. Cabeza Cuello* 85.1, pp. 95–104. ISSN: 0718-4816. DOI: [10.4067/s0718-48162025000100549](https://doi.org/10.4067/s0718-48162025000100549).
- Serrano, C., Van Den Brink, H., Pineau, J., Prognon, P., and Martelli, N. (2019). “Benefits of 3D Printing Applications in Jaw Reconstruction: A Systematic Review and Meta-Analysis.” In: *J. Cranio-MaxilloFac. Surg.* 47.9, pp. 1387–1397. ISSN: 10105182. DOI: [10.1016/j.jcms.2019.06.008](https://doi.org/10.1016/j.jcms.2019.06.008).
- Sinha, P., Bylapudi, B. P., Puranik, P., Subash, A., and Rao, V. (2022). “3D Patient-Specific Biomechanical Model of the Tongue for the Management of Tongue Tumors: Conceptualization to Reality.” In: *Med. Bull. Sisli Etfal Hosp.* 56.4, pp. 559–563. ISSN: 1302-7123, 1308-5123. DOI: [10.14744/SEMB.2022.37039](https://doi.org/10.14744/SEMB.2022.37039).

- Sorouri, K., Podolsky, D. J., Wang, A. M., Fisher, D. M., Wong, K. W., Looi, T., Drake, J. M., and Forrest, C. R. (2018). “Utilization of a Robotic Mount to Determine the Force Required to Cut Palatal Tissue.” In: *Journal of the Mechanical Behavior of Biomedical Materials* 86, pp. 433–439. ISSN: 17516161. DOI: [10.1016/j.jmbbm.2018.06.010](https://doi.org/10.1016/j.jmbbm.2018.06.010).
- Sparks, J. L., Vavalle, N. A., Kasting, K. E., Long, B., Tanaka, M. L., Sanger, P. A., Schnell, K., and Conner-Kerr, T. A. (2015). “Use of Silicone Materials to Simulate Tissue Biomechanics as Related to Deep Tissue Injury.” In: *Adv. Skin Wound Care*. 28.2, pp. 59–68. ISSN: 1527-7941. DOI: [10.1097/01.ASW.0000460127.47415.6e](https://doi.org/10.1097/01.ASW.0000460127.47415.6e).
- Stavness, I. (2010). “Byte Your Tongue: A Computational Model of Human Mandibular-Lingual Biomechanics for Biomedical Applications.” PhD thesis. University of British Columbia. DOI: [10.14288/1.0071518](https://doi.org/10.14288/1.0071518).
- Stavness, I., Lloyd, J. E., Payan, Y., and Fels, S. (2011). “Coupled Hard–Soft Tissue Simulation with Contact and Constraints Applied to Jaw–Tongue–Hyoid Dynamics.” In: *Numer Methods Biomed Eng* 27.3, pp. 367–390. ISSN: 2040-7939, 2040-7947. DOI: [10.1002/cnm.1423](https://doi.org/10.1002/cnm.1423).
- Stavness, I., Nazari, M. A., Flynn, C., Perrier, P., Payan, Y., Lloyd, J. E., and Fels, S. (2014a). “Coupled Biomechanical Modeling of the Face, Jaw, Skull, Tongue, and Hyoid Bone.” In: *3D Multiscale Physiological Human*. Ed. by N. Magnenat-Thalmann, O. Ratib, and H. F. Choi. London: Springer, pp. 253–274. ISBN: 978-1-4471-6275-9. DOI: [10.1007/978-1-4471-6275-9_11](https://doi.org/10.1007/978-1-4471-6275-9_11).
- Stavness, I., Sánchez, C. A., Lloyd, J., Ho, A., Wang, J., Fels, S., and Huang, D. (2014b). “Unified Skinning of Rigid and Deformable Models for Anatomical Simulations.” In: *SIGGRAPH Asia 2014 Tech. Briefs*. SA ’14. New York, NY, USA: Association for Computing Machinery, pp. 1–4. ISBN: 978-1-4503-2895-1. DOI: [10.1145/2669024.2669031](https://doi.org/10.1145/2669024.2669031).
- Suhaym, O., Moles, L., and Callahan, N. (2024). “Cutting Guides in Mandibular Tumor Ablation: Are We as Accurate as We Think?” In: *The Saudi Dental Journal* 36.2, pp. 340–346. ISSN: 10139052. DOI: [10.1016/j.sdentj.2023.11.017](https://doi.org/10.1016/j.sdentj.2023.11.017).
- Sullivan, C. and Kaszynski, A. (2019). “PyVista: 3D Plotting and Mesh Analysis through a Streamlined Interface for the Visualization Toolkit (VTK).” In: *J. Open Source Softw.* 4.37, p. 1450. ISSN: 2475-9066. DOI: [10.21105/joss.01450](https://doi.org/10.21105/joss.01450).

- Sun, Z., Gepner, B. D., Cottler, P. S., Lee, S.-H., and Kerrigan, J. R. (2021a). “In Vitro Mechanical Characterization and Modeling of Subcutaneous Adipose Tissue: A Comprehensive Review.” In: *J. Biomech. Eng.* 143.7, p. 070803. ISSN: 0148-0731, 1528-8951. DOI: [10.1115/1.4050286](https://doi.org/10.1115/1.4050286).
- Sun, Z., Gepner, B. D., Lee, S.-H., Rigby, J., Cottler, P. S., Hallman, J. J., and Kerrigan, J. R. (2021b). “Multidirectional Mechanical Properties and Constitutive Modeling of Human Adipose Tissue under Dynamic Loading.” In: *Acta Biomaterialia* 129, pp. 188–198. ISSN: 17427061. DOI: [10.1016/j.actbio.2021.05.021](https://doi.org/10.1016/j.actbio.2021.05.021).
- Sun, Z., Lee, S.-H., Gepner, B. D., Rigby, J., Hallman, J. J., and Kerrigan, J. R. (2021c). “Comparison of Porcine and Human Adipose Tissue Loading Responses under Dynamic Compression and Shear: A Pilot Study.” In: *J. Mech. Behav. Biomed. Mater.* 113, p. 104112. ISSN: 17516161. DOI: [10.1016/j.jmbbm.2020.104112](https://doi.org/10.1016/j.jmbbm.2020.104112).
- Tarsitano, A., Battaglia, S., Cipriani, R., and Marchetti, C. (2016). “Microvascular Reconstruction of the Tongue Using a Free Anterolateral Thigh Flap: Three-dimensional Evaluation of Volume Loss after Radiotherapy.” In: *J. Cranio-MaxilloFac. Surg.* 44.9, pp. 1287–1291. ISSN: 1010-5182. DOI: [10.1016/j.jcms.2016.04.031](https://doi.org/10.1016/j.jcms.2016.04.031).
- Tepole, A. B., Gosain, A. K., and Kuhl, E. (2014a). “Computational Modeling of Skin: Using Stress Profiles as Predictor for Tissue Necrosis in Reconstructive Surgery.” In: *Comput. Struct.* 143, pp. 32–39. ISSN: 0045-7949, 1879-2243. DOI: [10.1016/j.compstruc.2014.07.004](https://doi.org/10.1016/j.compstruc.2014.07.004).
- Tepole, A. B., Steinberg, J. P., Kuhl, E., and Gosain, A. K. (2014b). “Application of Finite Element Modeling to Optimize Flap Design with Tissue Expansion.” In: *Plast. Reconst. Surg.* 134.4, pp. 785–792. ISSN: 0032-1052. DOI: [10.1097/PRS.0000000000000553](https://doi.org/10.1097/PRS.0000000000000553).
- Then, C., Stassen, B., Depta, K., and Silber, G. (2017). “New Methodology for Mechanical Characterization of Human Superficial Facial Tissue Anisotropic Behaviour in Vivo.” In: *Journal of the Mechanical Behavior of Biomedical Materials* 71, pp. 68–79. ISSN: 17516161. DOI: [10.1016/j.jmbbm.2017.02.022](https://doi.org/10.1016/j.jmbbm.2017.02.022).
- Tonutti, M., Gras, G., and Yang, G.-Z. (2017). “A Machine Learning Approach for Real-Time Modelling of Tissue Deformation in Image-Guided Neurosurgery.” In: *Artif. Intell. Med.* 80, pp. 39–47. ISSN: 09333657. DOI: [10.1016/j.artmed.2017.07.004](https://doi.org/10.1016/j.artmed.2017.07.004).

- Urken, M. L. and Biller, H. F. (1994). “A New Bilobed Design for the Sensate Radial Forearm Flap to Preserve Tongue Mobility Following Significant Glossectomy.” In: *Arch. Otolaryngol. Head Neck Surg.* 120.1, pp. 26–31. ISSN: 0886-4470. DOI: [10.1001/archotol.1994.01880250022002](https://doi.org/10.1001/archotol.1994.01880250022002).
- Urken, M. (2012). *Multidisciplinary Head and Neck Reconstruction: A Defect-Oriented Approach*. Lippincott Williams & Wilkins. ISBN: 978-1-4511-5337-8.
- Vaidya, A. J. and Wheatley, B. B. (2020). “An Experimental and Computational Investigation of the Effects of Volumetric Boundary Conditions on the Compressive Mechanics of Passive Skeletal Muscle.” In: *Journal of the Mechanical Behavior of Biomedical Materials* 102, p. 103526. ISSN: 17516161. DOI: [10.1016/j.jmbbm.2019.103526](https://doi.org/10.1016/j.jmbbm.2019.103526).
- van Alphen, M. J. A., Kreeft, A. M., van der Heijden, F., Smeele, L. E., and Balm, A. J. M. (2013). “Towards Virtual Surgery in Oral Cancer to Predict Postoperative Oral Functions Preoperatively.” In: *Br. J. Oral Maxillofac. Surg.* 51.8, pp. 747–751. ISSN: 0266-4356, 1532-1940. DOI: [10.1016/j.bjoms.2013.06.012](https://doi.org/10.1016/j.bjoms.2013.06.012).
- Van De Water, L., Varney, S., and Tomasek, J. J. (2013). “Mechanoregulation of the Myofibroblast in Wound Contraction, Scarring, and Fibrosis: Opportunities for New Therapeutic Intervention.” In: *Adv. Wound Care.* 2.4, pp. 122–141. ISSN: 2162-1918, 2162-1934. DOI: [10.1089/wound.2012.0393](https://doi.org/10.1089/wound.2012.0393).
- Vincent, A., Kohlert, S., Lee, T., Inman, J., and Ducic, Y. (2019). “Free-Flap Reconstruction of the Tongue.” In: *Seminars in Plastic Surgery* 33.01, pp. 038–045. ISSN: 1535-2188, 1536-0067. DOI: [10.1055/s-0039-1677789](https://doi.org/10.1055/s-0039-1677789).
- Wang, J., Ho, A. K., Papadopoulos-Nydam, G., Rieger, J., Inamoto, Y., Fels, S., Saitoh, E., Guo, C., and Aalto, D. (2019). “Simulated Volume Loss in the Base of Tongue in a Virtual Swallowing Model.” In: *Comp. Meth. Biomech. Biomed. Eng.* 7.4, pp. 389–395. ISSN: 2168-1163, 2168-1171. DOI: [10.1080/21681163.2017.1382392](https://doi.org/10.1080/21681163.2017.1382392).
- Wang, L., Liu, K., Shao, Z., and Shang, Z. (2016). “Individual Design of the Anterolateral Thigh Flap for Functional Reconstruction after Hemiglossectomy: Experience with 238 Patients.” In: *Int. J. Oral. Maxillofac. Surg.* 45.6, pp. 726–730. ISSN: 0901-5027. DOI: [10.1016/j.ijom.2015.11.020](https://doi.org/10.1016/j.ijom.2015.11.020).

- Wang, X., Yan, G., Zhang, G., Li, J., Liu, J., and Zhang, Y. (2013). “Functional Tongue Reconstruction with the Anterolateral Thigh Flap.” In: *World J Surg Onc* 11.1, p. 303. ISSN: 1477-7819. DOI: [10.1186/1477-7819-11-303](https://doi.org/10.1186/1477-7819-11-303).
- Warnakulasuriya, S. (2009). “Global Epidemiology of Oral and Oropharyngeal Cancer.” In: *Oral Oncol*. 45.4-5, pp. 309–316. ISSN: 13688375. DOI: [10.1016/j.oraloncology.2008.06.002](https://doi.org/10.1016/j.oraloncology.2008.06.002).
- Wei, H., Zhang, A., Tao, C., Li, X., Li, Q., Shen, C., Wang, P., and Jin, P. (2022). “Three-Dimensional Rapid Printing-Assisted Individualized Total Nasal Reconstruction Based on a Database of Normal External Noses.” In: *Plast. Reconstr. Surg.* 150.4, pp. 888–898. ISSN: 0032-1052, 1529-4242. DOI: [10.1097/PRS.00000000000009566](https://doi.org/10.1097/PRS.00000000000009566).
- Wrench, A. A. (2024). “The Compartmental Tongue.” In: *J Speech Lang Hear Res* 67.10S, pp. 3887–3913. ISSN: 1092-4388, 1558-9102. DOI: [10.1044/2024_JSLHR-23-00125](https://doi.org/10.1044/2024_JSLHR-23-00125).
- Xu, J., Lai, F., Liu, Y., Tan, Z., Zheng, C., Wang, J., Guo, H., Jiang, L., Ge, X., Lan, X., Chen, C., and Ge, M. (2022). “Novel Computer-Aided Reconstruction of Soft Tissue Defects Following Resection of Oral and Oropharyngeal Squamous Cell Carcinoma.” In: *World J. Surg. Oncol.* 20.1, p. 196. ISSN: 1477-7819. DOI: [10.1186/s12957-022-02654-7](https://doi.org/10.1186/s12957-022-02654-7).
- Yen, C.-I., Zelken, J. A., Chang, C.-S., Lo, L.-J., Yang, J.-Y., Chuang, S.-S., Araniego, C. A., and Hsiao, Y.-C. (2019). “Computer-Aided Design and Three-Dimensional Printing Improves Symmetry in Heminasal Reconstruction Outcomes.” In: *J. Plast. Reconstr. Aesthet. Surg.* 72.7, pp. 1198–1206. ISSN: 17486815. DOI: [10.1016/j.bjps.2019.03.012](https://doi.org/10.1016/j.bjps.2019.03.012).
- Yi, C. R., Jeong, W. S., Oh, T. S., Koh, K. S., and Choi, J.-W. (2020). “Analysis of Speech and Functional Outcomes in Tongue Reconstruction after Hemiglossectomy.” In: *J. Reconstr. Microsurg.* 36.07, pp. 507–513. ISSN: 0743-684X, 1098-8947. DOI: [10.1055/s-0040-1709493](https://doi.org/10.1055/s-0040-1709493).
- Zhou, M., González, P. J., Van Haasterecht, L., Soylu, A., Mihailovski, M., Van Zuijlen, P., and Groot, M. L. (2024). “Uniaxial Mechanical Stretch Properties Correlated with Three-Dimensional Microstructure of Human Dermal Skin.” In: *Biomech Model Mechanobiol* 23.3, pp. 911–925. ISSN: 1617-7959, 1617-7940. DOI: [10.1007/s10237-023-01813-3](https://doi.org/10.1007/s10237-023-01813-3).

- Zhou, X., He, Z.-J., Su, Y.-X., Zhang, S., Gong, Z. J., and Wu, H. J. (2020). “”Sushi Roll” Technique for Precise Total Tongue Functional Reconstruction Using a Pre-Sutured Femoral Anterolateral Myocutaneous Flap.” In: *Oral Oncol.* 110, p. 104866. ISSN: 1368-8375, 1879-0593. DOI: [10.1016/j.oraloncology.2020.104866](https://doi.org/10.1016/j.oraloncology.2020.104866).
- Zirk, M., Niknazemi, M., Riekert, M., Kreppel, M., Linz, C., and Lentzen, M.-P. (2024). “Comparative Analysis of Volumetric Changes between Resection Volume of Oral Tongue Cancer and Post Operative Volume of Radial Forearm Flaps.” In: *Clin Oral Invest* 28.9, p. 498. ISSN: 1436-3771. DOI: [10.1007/s00784-024-05885-y](https://doi.org/10.1007/s00784-024-05885-y).